



Setting The
Standard for
Seafood



ASC Farm Interpretation Manual

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Version control, available language(s), and copyright notice

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It is the responsibility of the user of the document to use the latest version as published on the ASC website.

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Review and revision

ASC will review this Interpretation Manual periodically as and when required.

If review determines that revision is necessary, ASC will revise the document accordingly, publishing the new version on the ASC website.

Available language(s)

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Principle 1

The UoC operates legally and applies effective business management



Criterion 1.1 – Legal Compliance

Indicator:	Requirement:
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1.1.1 The UoC shall hold all applicable legal licences and permits.

How do I interpret this requirement?

- The UoC should be responsible for:
 - Having a good understanding of local licensing and permit systems that apply.
 - Obtaining all necessary documentation for siting, constructing and operating their facilities. Examples of these can include:
 - business/operational licences,
 - land deeds,
 - leases or concession agreements,
 - construction permits,
 - water use permits,
 - effluent permits,
 - landfill operation permits,
 - fire safety certificate,
 - building fitness certificate,
 - permit or licence to store fuel (such as oil and/or gas),
 - licences or permits to house workers in provided accommodation,
 - changes to labour legislation, regulations and laws are reflected in workers' contracts.
 - Paying attention to:
 - pending licence renewal for applications of, e.g. relocation of sites, biomass increase;
 - conditions set as part of the licence/permits.

Evidence of implementation should be maintained, e.g.:

- Up-to-date documentation for siting, constructing and operating, including any applications in progress.

Auditors should confirm:

- Documentation confirming required legal licences and permits are in place.
- Conditions required as part of business licence applications are adhered to. For example:
 - In some parts of the world, a business licence is not issued unless mandatory certificates of building fitness and emergency preparedness for natural disasters are obtained.

Useful Resources

- Assistance in determining necessary licences and permits can be sought from governmental agencies dealing with business and the environment and communities.

Indicator:	Requirement:
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- 1.1.2** The UoC shall comply with all applicable legal requirements, including those relevant to the Criteria within the ASC Farm Standard.

How do I interpret this requirement?

- 'Applicable legal requirements' refers to any type of law or regulation that is relevant to the scope of the ASC Farm Standard.
- The UoC should be responsible for:
 - having a good understanding of laws and regulations including their applicability to the local context; and
 - facilitating any inspections by authorities.

Evidence of implementation should be maintained, e.g.:

- A register for legal compliance, for example, listing:
 - relevant applicable laws and regulations, including, for labour-related laws and regulations, those within the home country of any migrant employees; and
 - compliance status, and any future actions needed to remain compliant; and
 - any measures needed to correct potential non-compliance with a law or regulation.
- Certificates or other records to confirm compliance with legal requirements.
- Documented outcomes and records of any inspections by authorities.

Auditors should confirm:

- Existence of any legal documentation, outcomes of inspections, or records kept by UoC to demonstrate legal compliance.
- Management understands the legal context in which they are operating.

Useful Resources

- The International Labour Organization (ILO) maintains NATLEX (www.ilo.org/dyn/natlex), a database of national labour, social security and related human rights legislation across 196 countries and more than 160 territories and subdivisions. NATLEX is a free source of information for all users.

Criterion 1.2 – Management System

Indicator:	Requirement:
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- 1.2.1** The UoC shall develop, implement and maintain a management system to support the implementation of the ASC Requirements that is appropriate to the size and complexity of the UoC.

How do I interpret this requirement?

- The scope of the management system should suit the operation that it is designed for. For example:
 - A site situated at a remote distance at sea might not need additional elements in the management system to work with communities, or
 - A site operating a RAS system might not need additional elements in the management system to manage the risks of escapes.
- A management system should include:
 - Policies: explain why a site is operating in such a way.
 - they can be beneficial for employees to understand their rights and duties.
 - it is not necessary to have a policy for every topic in the ASC Farm Standard and sometimes topics are grouped together in one policy.
 - for example, forced labour, child labour, discrimination, collective bargaining and freedom of association and others can all be addressed together through the human rights policy.
 - policies are most valuable for topics which are not straightforward in the local context.
 - for example, where employees have an inconsistent or incomplete understanding of an issue, a policy might be needed.
 - Procedures: explain how a site operates in a certain way and specify how to put policies into operation to ensure the outcome intended.
 - Records: provide an essential source of information for both internal self-assessments and external audits, documenting that policies and procedures have been followed in specific instances.

- sites can use recording templates to guide the user through all of the steps of a record and provide accurate information in a set format.
- records should be adequately maintained and controlled, including ensuring only up-to-date versions of documents are used.

Evidence of implementation should be maintained, e.g.:

- o Documents of the management system, including policies, procedures and records.
- o Evidence that demonstrates management's ownership of the management system (see Indicator 1.2.2).

Auditors should confirm:

- o Existence of up-to-date policies, procedures and records.
- o Management's ownership of the management system:
 - attendance at management reviews (see Indicator 1.2.4).
 - active role in the development, implementation and maintenance of the management system.
- demonstrable during interviews with the management.

Useful Resources

- o The ISO 9001:2015 management system standards: <https://www.iso.org/management-system-standards.html>
- o Plan Do Check Act Cycle: <https://asq.org/quality-resources/pdca-cycle>

Indicator:	Requirement:
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1.2.2 The UoC shall appoint at least one member of management responsible for the implementation of the ASC Requirements.

How do I interpret this requirement?

- The responsible person may be one person or several persons, covering the wide range of issues addressed by the ASC Requirements.
 - For example, a human resources manager may be responsible for the requirements on labour rights and working conditions, and a fish health and welfare manager may be responsible for the requirements on fish health;
 - senior management should be responsible for seeking more specialised knowledge when needed.
- The responsibility should be embedded in the organisational structure documentation to ensure clear accountability.

Evidence of implementation should be maintained, e.g.:

- Records:
 - that evidence the responsible person(s) were designated by the management.
 - role description of the responsible person(s) for the implementation of ASC Requirements.

Auditors should confirm:

- The management designated responsible person(s) for the implementation of the ASC Requirements.
 - through records and interviews with employees and the management.
- Role description of the responsible person(s) for the implementation of the ASC requirements.

Useful Resources

N/A

Indicator:	Requirement:
------------	--------------

1.2.3

The UoC shall ensure that relevant employees have the necessary skills and training to implement the ASC Requirements.

How do I interpret this requirement?

- At the site, it should be clear who is responsible for what tasks and management should ensure all employees are competent for the role that they will be carrying out. Employees should have:
 - Knowledge,
 - Experience,
 - Training (theory and on the job),
 - Capacity, and
 - Decision-making permissions.
- 'Relevant employees' and 'necessary competencies' refer to an employee's function and area of responsibility:
 - For example, a Health and Safety Manager is able to carry out and regularly update the Health and Safety risk assessment and define mitigating measures, but might not have the experience to treat injuries. Or the person responsible for first aid can take immediate action in the case of an injury and has access to and an overview of all first aid supplies, but might not have the experience to carry out a Health and Safety risk assessment.

Evidence of implementation should be maintained, e.g.:

- Job descriptions of all employees.
- Organisational chart with clarity on who is responsible for what.
- Records and documents of training.

Auditors should confirm:

- Employees and management are clear on their roles and responsibilities in the implementation of ASC Requirements.
- Employees have the necessary competencies, knowledge and experience to implement the ASC Requirements.

Useful Resources

N/A

Indicator:	Requirement:
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- 1.2.4** The UoC shall annually review the management system and revise it where necessary to ensure its continued suitability, adequacy and effectiveness in implementing the ASC Requirements.

How do I interpret this requirement?

- The UoC's management system review should provide the opportunity to assess its suitability, adequacy and effectiveness in continuously ensuring conformance with the ASC requirements.
- The review should consider:
 - Progress and effectiveness of changes implemented following the previous management review.
 - Financial and personnel resources necessary to maintain the management system.
 - Changes in the internal and external context of the site which may give rise to previously unanticipated issues.
 - Performance review since the previous management review.
 - Internal and external audit outcomes.
 - Grievances raised.
 - Progress towards longer-term ASC targets, e.g. an increase in wage payments.
 - Progress towards site commitments.
 - A mechanism for employees' feedback on the effectiveness of the management system
- The outcome of the review should include addressing any need for changes and opportunities for improvement in the management system, which is documented and communicated to relevant employees for implementation.

Evidence of implementation should be maintained, e.g.:

- Records:
 - of the review and its outcomes.
 - meeting minutes.
-

Auditors should confirm:

- Review is conducted at least annually.
 - Records.
-

Useful Resources

N/A

Criterion 1.3 – Business Ethics

Indicator:	Requirement:
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1.3.1	The UoC shall discourage all forms of corruption and shall encourage ethical behaviour and integrity in their business operations.
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How do I interpret this requirement?

- Corruption can:
 - Happen anywhere;
 - Involve anyone;
 - Take many forms¹. For example:
 - extortion
 - embezzlement
 - bribery
- Examples of corruption are:
 - A manager buys supplies from a friend and in return, the friend gives 20% of the profits to the manager. This may go as far as the manager excluding other appropriate suppliers from the system of approved suppliers.
 - A piece of infrastructure has an unsafe mechanism and the owner pays the manager cash to not disclose this at the next audit.
- Examples of extortion are:
 - Employee A threatens to provide information to the manager which could get Employee B into trouble or fired, unless Employee A is paid a requested amount of money.
 - A site owner threatens to hurt the owner of the neighbouring plot of land unless the land-owner agrees to give up some of his land so that the site can build an additional building.
- Examples of embezzlement are:
 - A site manager getting cash from the first point of sale in exchange for under-declaring harvest volumes (i.e. the purchaser receiving more product than they are paying for).

¹ <https://www.transparency.org/en/what-is-corruption>

- A sourcing manager purchases cheaper non-eligible feed but registers them as eligible and provides documentation falsely declaring it as more costly eligible feed, as financial sourcing performance increases the sourcing manager's bonus.
- Examples of bribery are:
 - A supplier buys a new car for a site manager, in exchange for the site manager giving the sourcing contract to that supplier.
 - A site pays cash to customs officials to speed up the passage of their feed through the import and export system.
 - A sustainability manager offers auditors excessive accommodation and lunches to reduce the audit effort or to turn a blind eye to non-conformities.

Evidence of implementation should be maintained, e.g.:

- Policies preventing corruption, including extortion, embezzlement and bribery.
 - Terms of dismissal should employees engage in these practices.
-

Auditors should confirm:

- Management and employees are aware of:
 - what these practices involve and how to recognise them.
 - the policies in place for preventing corruption, including extortion, embezzlement and bribery.
-

Useful Resources

- Transparency International: <https://www.transparency.org/en>

Indicator:	Requirement:
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1.3.2 The UoC shall ensure that information provided as evidence of conformance to the ASC Requirements is not falsified or misrepresented.

How do I interpret this requirement?

- Records are documented information that provides permanent evidence about past events. They should:

- Be completed and maintained in a way that ensures information is accurate and preserved;
 - Be genuine;
 - they should be completed at the time of the checks or shortly after.
 - Capture and illustrate information when processes do not perform as expected.
 - for example, if the cooling system ceases to work, it is important that temperatures above the acceptable range according to internal requirements are recorded so that management can learn from these events.
 - recording temperatures only within the acceptable range will lead to management not understanding that a backup temperature regulator might be needed.
 - Be legible;
 - Be complete;
 - which may include a form of authorisation through the initials of the supervisor verifying records are complete;
 - Be stored appropriately:
 - paper records should be stored in a way that prevents damage.
 - where records are stored electronically, this storage should be backed up.
- If incidences of falsification of records or misrepresentation of information are identified, corrective action and actions to prevent recurrence should be taken.

Evidence of implementation should be maintained, e.g.:

- Records:
 - that demonstrate genuineness.
 - corrective action and actions to prevent recurrence when incidences of falsification of records or misrepresentation of information are identified.

Auditors should confirm:

- records and their genuineness
- effectiveness of corrective action and actions to prevent

Useful Resources N/A

Criterion 1.4 – Traceability and Disclosure

Indicator:	Requirement:
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1.4.1 The UoC shall identify, evaluate and mitigate risks of mixing and substituting ASC certified products with non-certified products at all sites.

1.4.2 The UoC shall develop, implement and maintain a traceability system, covering all sites, to ensure that the following conditions are met:

- only larvae/juveniles that conform with the requirements in Criterion 1.5 are used to produce certified product; and
- all stocked animals are traceable back to the supplier; and
- all batches of ASC certified and non-certified product are identifiable and segregated at all sites and at all times; and
- all batches of ASC certified product are traceable at all times to:
 - the site(s) used ; and
 - the production unit(s) used; and
- corrective actions are taken if non-conforming product is identified, sold or dispatched; and
- accompanying documentation shall identify product sold or dispatched as ASC certified or non-certified.

How do I interpret this requirement?

- The UoC is responsible for ensuring it mitigates risks of substitution and mixing of certified and non-certified products.
- The UoC may use the Risk Assessment Method outlined in Appendix 3 of the ASC Farm Standard, Section 3.2, to identify, evaluate and mitigate risks of mixing and substituting ASC certified products with non-certified products at all sites (Indicator 1.4.1).
- Certified and non-certified products shall be identifiable and segregated at all sites and at all times within the UoC (Indicator 1.4.2(c)).
 - This includes during production while fish are alive.
- Certified products shall be traceable at all sites and at all times within the UoC (Indicator 1.4.2(d)).

- The UoC may decide to sell products from the UoC as non-certified (Indicator 1.4.2(f)).
 - In this case, the products shall be identified as non-certified when sold or dispatched.
- The requirements of these indicators establish a consistent, streamlined approach for traceability, identification and segregation at all ASC certified sites.
 - They relieve the need for separate Chain of Custody certification to cover farming activities.

Evidence of implementation should be maintained, e.g.:

- Records:
 - risk assessment and any resulting actions.
 - use or verification of the effectiveness of the traceability system, for example:
 - outcome of input-output exercises and traceability tests.

Auditors should confirm:

- Risks have been adequately assessed and mitigated.
- Effective traceability and segregation systems are in place.
- Outcome of any input-output exercises and traceability tests.

Useful Resources

N/A

Indicator:	Requirement:
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1.4.3 *Indicator applicability: every UoC using compound feed*

The UoC shall demonstrate that feed products, subject to conformance with Indicator 2.12.1, are accompanied by the following information:

- a. the originating feed mill holds valid certification against the ASC Feed Standard with the appropriate scope of certification in relation to the ASC Production Model (i.e. Mass Balance Production Model or Segregation Production Model product); and
- b. feed is traceable as ASC-conforming throughout all steps between the feed mill and site; and

- c. documentation accompanying the feed identifies the feed as ASC-conforming; and
- d. documentation accompanying the feed identifies the production model applied (i.e. Mass Balance Production Model or Segregation Production Model); and
- e. for ASC-conforming Segregation Production Model feed, the product is identifiable by a distinct feed name in formal documentation accompanying the product and is listed on the schedule to the feed mill certificate.

How do I interpret this requirement?

- Indicator 2.12.1 requires the UoC to only use ASC-conforming compound feed.
 - Use of non-conforming feed will be treated as a non-conformity against Indicator 2.12.1.
- For product which is sold under the segregation model, identifying the feed production model and distinct feed name is required by the ASC Feed Standard, Indicator 3.3.2.

Evidence of implementation should be maintained, e.g.:

- Process to confirm the originating feed mill holds valid certification against the ASC Feed Standard.
- Records:
 - traceability system and its verification, for example:
 - use of a digital traceability system
 - outcome of any input-output exercises and traceability tests.
 - accompanying documents incorporating information required by 1.4.3(c), (d) and (e).

Auditors should confirm:

- Process to confirm the originating feed mill holds valid certification against the ASC Feed Standard.
- Effective traceability and segregation systems are in place.
- Outcome of any input-output exercises and traceability tests.

Useful Resources

N/A

Indicator:	Requirement:
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1.4.4 *Indicator applicability: every UoC using compound feed*

The UoC shall disclose to all buyers of their ASC certified product when a product's feed contains >0.9%² Genetically Modified Organisms (GMO) or macro ingredients produced from GMO.

How do I interpret this requirement?

- The required disclosure is to the UoC's buyers of ASC certified products.
- A public disclosure is possible but not required.
- The UoC shall ensure buyers are informed when a product's feed contains or is produced from GMO.
 - Disclosure may be provided to buyers individually or collectively.

Evidence of implementation should be maintained, e.g.:

- Records:
 - disclosure to buyers.

Auditors should confirm:

- Evidence of disclosure to buyers.
-

Useful Resources

N/A

² European Council. (2003). Regulation (EC) No 1831/2003 concerning the traceability and labelling of genetically modified organisms and the traceability of food and feed products produced from genetically modified organisms and amending Directive 2001/18/EC. European Parliament, Council of the European Union.

Indicator:	Requirement:
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1.4.5

The UoC shall not sell or dispatch the following as ASC certified:

- a. crustaceans treated with antibiotics at the UoC, or
- b. any product treated with antimicrobials listed as Critically Important Antimicrobials for Human Medicine by the World Health Organisation (WHO), at any stage in the lifecycle of the product.

How do I interpret this requirement?

- This requirement addresses situations where a product may conform to the Standard, however it cannot be sold or dispatched as ASC certified.
- The UoC should:
 - Maintain detailed records of any antibiotic treatments applied prior to the UoC (see Indicator 1.5.2).
 - Implement a process to ensure that the product under 1.4.5(a) or (b) is not sold or dispatched as ASC certified.

Evidence of implementation should be maintained, e.g.:

- Records:
 - antibiotic treatments, including those applied prior to the UoC.

Auditors should confirm:

- UoC's process to ensure that the product under 1.4.5(a) or (b) is not sold or dispatched as ASC certified.

Useful Resources

N/A

Indicator:	Requirement:
------------	--------------

- 1.4.6** The UoC shall disclose to buyers of their ASC certified product, upon request, a list of all veterinary therapeutants applied to the product.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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- 1.4.7** The UoC shall report to ASC the production volumes and sales/dispatch volumes of ASC certified and non-certified products originating from each site, and submit specified traceability data (when requested), according to ASC data submission procedures.

No additional guidance is provided for this Indicator.

Criterion 1.5 - Supply Chain

Indicator:	Requirement:
------------	--------------

1.5.1 The UoC shall map their supply chain using the ASC Supply Chain Mapping Template, provided on the ASC website.

1.5.2 The UoC shall only stock ASC certifiable product if accompanied by a completed ASC Stock Status Record³, provided on the ASC website.

How do I interpret this requirement?

- Traceability in the supply chain is essential for ensuring transparency, improving product quality and safety, and promptly identifying and addressing issues.
- The indicators allow for information related to the status of ASC certified products to be passed along the supply chain.
- The scope of the supply chain that needs to be mapped includes all sites supplying certifiable products, from broodstock onwards.

Evidence of implementation should be maintained, e.g.:

- Completed ASC Supply Chain Mapping Template listing the supply chain from broodstock sites onwards.
- Completed ASC Stock Status Record.

Auditors should confirm:

- ASC Supply Chain Mapping Template is completed listing the supply chain from broodstock sites onwards.
 - Crosscheck mapping with purchase records.
- ASC Stock Status Record completed.

Useful Resources N/A

³ The ASC Stock Status Record holds batch-specific information on products moving through the supply chain. This includes information on origin of broodstock, treatments, feed, and ASC certification status.

Indicator:	Requirement:
------------	--------------

1.5.3

Indicator applicability: freshwater trout, flatfish, seabass, seabream, meagre and salmon

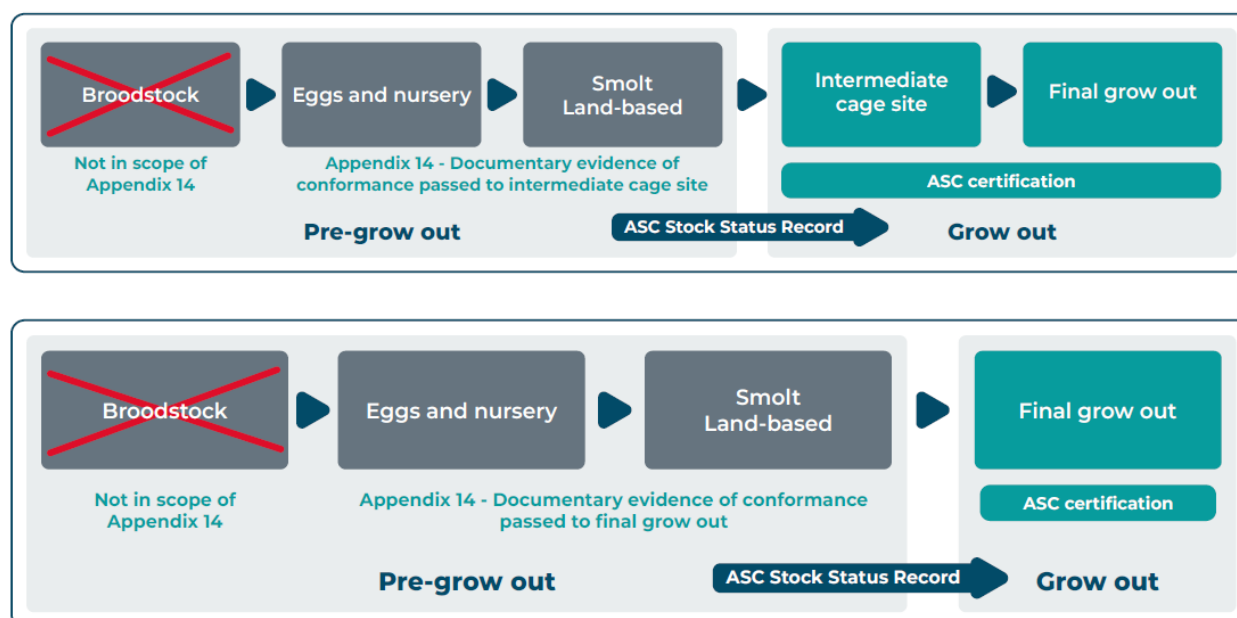
The UoC shall⁴ only stock ASC certifiable product if it can be demonstrated that all sites⁵ in the supply chain conform with the requirements in **Appendix 14**.

An exception applies for salmon sites that are supplied with product from other cage site(s), in which case all cage site(s) in the supply chain shall be ASC certified.

How do I interpret this requirement?

- Appendix 14 (14.1) establishes that documentation is required to be maintained from the sites in the supply chain to demonstrate that these sites conform with the requirements of this appendix.
 - The documentation should provide information to the extent necessary to ensure that the supplier sites conform with the requirements of Appendix 14 (14.1).
- Different applicability examples are provided below:

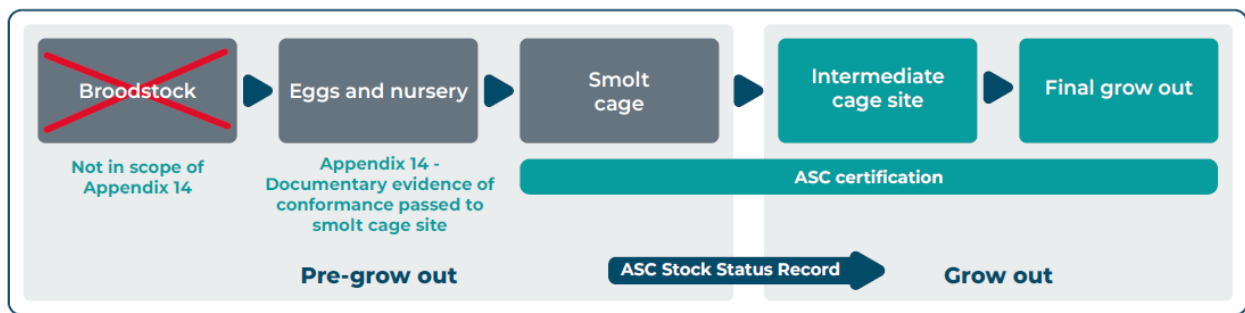
**Salmon with and without intermediate cage sites.
Land based smolts production.**



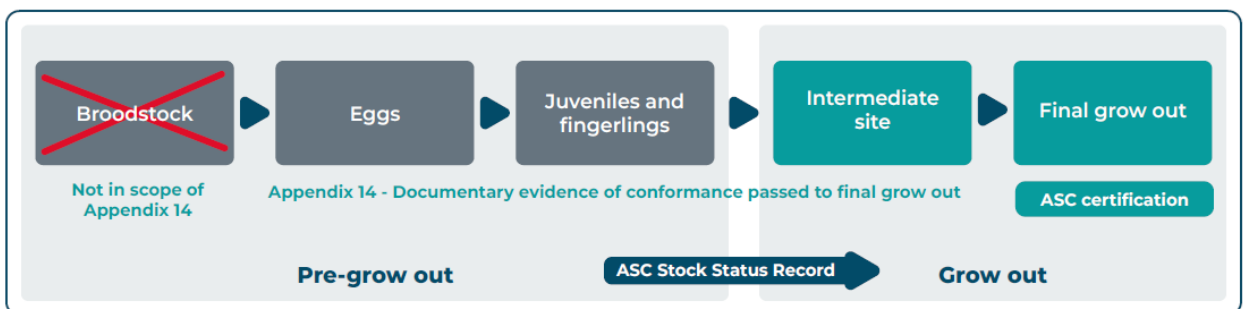
⁴ An exemption applies to sites in the supply chain that hold valid ASC certification. These sites do not need to comply with Indicator 1.5.3.

⁵ An exemption applies to broodstock sites.

Salmon with and without intermediate cage sites. Cage smolts production.



Flatfish, freshwater trout and SSM with and without intermediate sites



Evidence of implementation should be maintained, e.g.:

- Documentation that demonstrates suppliers of the site conform with the requirements of Appendix 14 (14.1).

Auditors should confirm:

- Documentation that demonstrates the suppliers of the site conform with the requirements of Appendix 14 (14.1)

Useful Resources

N/A



Principle 2

The aquaculture industry
demonstrates environmental
stewardship



Criterion 2.1 - Environmental Risk Assessment

Indicator:	Requirement:
------------	--------------

2.1.1 The UoC shall conduct an Environmental Risk Assessment (ERA) per site, following the method outlined in **Appendix 3 (3.3)**.

How do I interpret this requirement?

- The Environmental Risk Assessment (ERA) determines the risk level of environmental risks resulting from operational activities in the vicinity of the site. A default set of environmental risks are listed in Appendix 3.1.
- The outcome of the ERA will be used to determine compliance and impacts on Ecologically Important Habitats (Criterion 2.2) and Wildlife Interactions (Criterion 2.3).

Evidence of implementation should be maintained, e.g.:

- An ERA has been conducted, following the method outlined in Appendix 3.1, and records are available to demonstrate that actions are taken to reduce risks with risk level 'medium' or 'high'.

Auditors should confirm:

- The UoC has presented the ERA, which was conducted in accordance with the method in Appendix 3.1.

Useful Resources

- European Food Safety Authority: Environmental Risk Assessment: [Environmental risk assessment | EFSA](#).

Indicator:

Requirement:

2.1.2

The UoC shall ensure that the ERA (see Indicator 2.1.1) is conducted by a person with the necessary competencies, experience and training that is appropriate to the size and complexity of the site.

For larger, more complex sites, an independent person shall conduct the ERA. This person shall:

- a. have an academic background in Biology, Ecology, Environmental Science, or an equivalent field of study; and
- b. have a minimum of 5 years of experience in conducting ERA, including survey design, data collection, data analysis and interpretation.

In cases where legal requirements are defined, the UoC shall ensure that the person conducting the ERA complies with these requirements.

How do I interpret this requirement?

- An independent person is required to conduct the ERA for larger, more complex sites. However, small-scale or extensive aquaculture sites are not expected to contract an independent person to conduct the ERA.
- To demonstrate competence, personnel conducting the ERA for small-scale or extensive aquaculture sites should be able to:
 - Analyse environmental risks.
 - Conduct data analysis.
 - Record, review and authorise the release of results.
 - Contribute to or produce reports that can clearly and concisely follow the method outlined in Appendix 3.1.
 - Have undergone training, or have experience, relating to ERAs.

Evidence of implementation should be maintained, e.g.:

- Documents outlining the person conducting the ERA has experience and competences that are relevant to the size and complexity of the site.

- For larger, more complex sites, documents outlining the qualifications and experience of the independent person conducting the ERA.

Auditors should confirm:

- Evidence and documents have been presented to demonstrate the person conducting the ERA has the necessary experience, competencies and qualifications.
- Auditor discretion is permitted to determine whether the experience, competences and qualifications are suitable for the size of the site.

Useful Resources

N/A

Criterion 2.2 - Ecologically Important Habitats

Indicator:	Requirement:
------------	--------------

- 2.2.1** The UoC shall⁶ ensure that the site is not located in **Protected Areas** (PA). An exception applies where conditional siting is allowed according to **Appendix 4**.

How do I interpret this requirement?

- Conditional allowances to the siting in Protected Areas (PA) have been taken from the ASC species standards while work is ongoing to develop updated conditional allowances.

Evidence of implementation should be maintained, e.g.:

- Whether the site is in a PA and if so, provide evidence that the PA meets the conditional allowances specified in Appendix 4. This includes the demonstration of the Protected Areas Management Plan.

Auditors should confirm:

- Whether the site is in a PA. If so, evidence has been provided to demonstrate that the site meets the conditional allowances in Appendix 4 (4.2).

Useful Resources

- Protected Planet: World Database on Protected Areas (WDPA): [Explore the World's Protected Areas](#).

⁶ An exemption applies for the following species groups: tilapia, pangasius, bivalve and abalone.

Indicator:	Requirement:
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2.2.2 The UoC shall ensure that the site has a low risk of negative impact on **critical habitats**, as determined through the ERA (see Indicator 2.1.1).

How do I interpret this requirement?

- The Environmental Risk Assessment (ERA) will determine whether there are critical habitats within the vicinity of the site and the risk drivers that increase the likelihood of negative impacts on any critical habitats.

Evidence of implementation should be maintained, e.g.:

- An ERA report has been completed, which assesses the risk of negative impacts on any relevant critical habitats.
- The ERA report demonstrates whether the risk of negative impacts on critical habitats is 'low'.
 - In cases where the risk level is not 'low', the ERA report identifies the risk drivers.
 - In cases where the risk level is 'medium' or 'high', the site demonstrates the actions taken to reduce the risk level to 'low'.

Auditors should confirm:

- Whether there are critical habitats in the vicinity of the site. If so, evidence has been provided in the ERA that the risk of negative impacts on the critical habitats is minimal.

Useful Resources

N/A

Indicator:	Requirement:
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2.2.3 The UoC shall preserve any critical habitat identified within the site.

How do I interpret this requirement?

- In accordance with Indicator 2.2.2, the ERA report will identify whether critical habitats are located within the boundaries of the site.
 - The size and biodiversity characteristics of critical habitats should either remain the same since the ERA was conducted or have increased/improved.

Evidence of implementation should be maintained, e.g.:

- The ERA report has identified the presence of critical habitats within the boundaries of the site.
- Documented measures and outcomes, which demonstrate critical habitats have been preserved, and that their size and biodiversity has not decreased.

Auditors should confirm:

- Whether critical habitats have been identified in the ERA report.
- If critical habitats are present, evidence has been provided to demonstrate measures to preserve the documented critical habitats.
 - Outputs of preservation measures should also be available where appropriate.

Useful Resources

- GIS portal: [GIS portal - ASC International \(asc-aqua.org\)](https://asc-aqua.org)

Indicator:	Requirement:
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2.2.4 The UoC shall ensure that the site has a low risk of negative impact on **sensitive habitats**, as determined through the ERA (see Indicator 2.1.1).

How do I interpret this requirement?

- The Environmental Risk Assessment (ERA) will determine whether there are sensitive habitats within the vicinity of the site and the risk drivers that increase the likelihood of negative impacts on any sensitive habitats.
 - The ERA will determine whether the risk of negative impacts on the identified sensitive habitats is minimal.
- Appendix 3.3 outlines the risk factors that shall be considered as a minimum when conducting the ERA.
 - Appendix 3.3 also outlines the types of sensitive habitats that shall be considered as a minimum.
 - Other types of sensitive habitats may be identified in the ERA.

Evidence of implementation should be maintained, e.g.:

- An ERA report has been completed, which assesses the risk of negative impacts on any relevant sensitive habitats.
- The ERA report demonstrates whether the risk of negative impacts on sensitive habitats is 'low'.

Auditors should confirm:

- An ERA has been completed, which includes at a minimum the risk factors outlined in Appendix 3.3.
- The ERA determined whether sensitive habitats are located within the vicinity of the site.
 - If sensitive habitats were identified, the ERA determined whether the risk of negative impacts on the sensitive habitats is minimal.

Useful Resources

- GIS portal: [GIS portal - ASC International \(asc-aqua.org\)](https://asc-aqua.org)

Indicator:	Requirement:
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2.2.5 *Indicator applicability: land-based systems*

The UoC shall maintain any natural **wildlife corridors** within the site, as determined through the ERA (see Indicator 2.1.1).

How do I interpret this requirement?

- Maintaining wildlife corridors mitigates impacts on biodiversity and allows for wildlife movement across larger landscapes within their habitats or natural ranges.
 - Examples include ensuring corridors are maintained to allow the movement of wildlife to and from feeding or breeding sites either side of the site.
- The ERA shall determine whether any wildlife corridors are present within the site and whether the site has taken adequate measures to maintain them.

Evidence of implementation should be maintained, e.g.:

- The ERA demonstrates whether wildlife corridors are present within the site and whether the corridors are adequately maintained.
- Spatial data and maps over a relevant timescale can determine if wildlife corridors are maintained.

Auditors should confirm:

- Whether the ERA report has identified the presence of wildlife corridors within the site.
- If present, the ERA determines the site is maintaining the wildlife corridors.
- The UoC has presented maps or spatial data to demonstrate wildlife corridors have been maintained.

Useful Resources

- GIS portal: [GIS portal - ASC International \(asc-aqua.org\)](https://asc-aqua.org)

Indicator:	Requirement:
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2.2.6 *Indicator applicability: land-based systems*

The UoC shall maintain a **riparian zone** between the site and any freshwater body, with a minimum width of either 15m or as determined through the ERA (see Indicator 2.1.1), whichever is stricter.

How do I interpret this requirement?

- Riparian zones provide a buffer between natural waterbodies and adjacent terrestrial ecosystems.
 - A 15m buffer zone is the minimum width to provide habitats for wildlife, minimise erosion and run-off, improve water quality and stabilise the ecosystem around the water source.
- The ERA will determine the minimum width of the riparian zone which needs to be maintained to ensure the functioning of the zone.
 - The ERA should also assess the functioning of the riparian zone, for example species diversity.
- The UoC shall ensure the width of the riparian zone is at least 15m, or greater if the ERA determines the buffer zone needs to be wider.

Evidence of implementation should be maintained, e.g.:

- The ERA report demonstrates the required width of the riparian zone.
- The site ensures the riparian zone is maintained and is at least 15m wide. The buffer zone shall be wider if determined by the ERA.
 - Spatial data and maps over a relevant timescale can demonstrate the width of the riparian zone is maintained.

Auditors should confirm:

- The ERA report has assessed the suitable width and functioning of the riparian zone.
- The width of the riparian zone is at least 15m, or wider as determined by the ERA.
- Evidence has been presented, e.g. spatial data and maps, demonstrating the riparian zone is maintained at 15m or greater (as determined by the ERA).

Useful Resources

- GIS portal: [GIS portal - ASC International \(asc-aqua.org\)](https://asc-aqua.org)
- United States Environmental Protection Agency (2005): Riparian Buffer Width, Vegetative Cover, and Nitrogen Removal Effectiveness: A Review of Current Science and Regulations: [Riparian buffer width, vegetative cover, and nitrogen removal effectiveness: A review of current science and regulations](#)
- Climate ADAPT (2023): Establishment and restoration of riparian buffers: [Establishment and restoration of riparian buffers](#).
- National Institute of Water and Atmospheric Research (2005): Review of Riparian Buffer Zone Effectiveness: [Review of Riparian Buffer Zone Effectiveness](#).

Indicator:	Requirement:
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2.2.7 *Indicator applicability: land-based systems*

The UoC shall maintain a **coastal barrier** between the site and any marine or brackish waterbody, with a minimum width of either 100m or as determined through the ERA (see Indicator 2.1.1), whichever is stricter.

How do I interpret this requirement?

- Coastal barriers can be either artificial or natural and mitigate concerns relating to storm or flood risks. The ERA will assess the required width of the coastal barrier to mitigate these risks.
 - An additional benefit of coastal barriers is that they control environmental events directly linked to escapes and disease transfer.
 - Communities can also access marine resources through the maintenance of a coastal barrier.
- The UoC shall ensure the width of the coastal barrier is at least 100m, or greater if the ERA determines the coastal barrier needs to be wider.

Evidence of implementation should be maintained, e.g.:

- The ERA has determined the suitable width of the coastal barrier.
- The maintained coastal barrier has a minimum width of 100m or greater, as determined by the ERA.

- Spatial data and maps can provide evidence of the width of the coastal barrier and its maintenance.

Auditors should confirm:

- The ERA has assessed the width of the coastal barrier.
 - The coastal barrier is 100m or greater, as determined by the ERA.
 - Spatial data or maps are presented to demonstrate the width of the coastal barrier and that it is maintained.
-

Useful Resources

N/A

Indicator:	Requirement:
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2.2.8 *Indicator applicability: land-based systems*

The UoC shall have successfully **restored** a minimum of 50% of **natural wetland** converted before May 1999.

2.2.9 *Indicator applicability: land-based and suspended systems*

The UoC shall not convert natural wetland after May 1999. An exception applies where conversion is needed for establishing and maintaining pumping stations, water pipes or canals, in which case the UoC shall successfully restore 100% of the converted area.

2.2.10 *Indicator applicability: land-based and suspended systems*

The UoC shall report to ASC the area of the restored wetland, according to ASC data submission procedures.

How do I interpret this requirement?

- Rehabilitation projects should be located within or adjacent to farming activities, to restore ecosystem function close to where it has originally been lost.
- If this is not feasible, the UoC may contribute to regional rehabilitation activities.

- Restoration activities should ensure habitats and biodiversity are equivalent to habitat and biodiversity lost through conversion.

Evidence of implementation should be maintained, e.g.:

- Spatial mapping of the converted and restored areas:
 - Time series maps showing changes in the cover of natural wetlands.
 - Polygons or measurements of the converted and restored areas showing the surface areas of conversion and rehabilitation.
 - This evidence shall be submitted to ASC in accordance with Indicator 2.2.10.
- The location of rehabilitation projects. If local rehabilitation efforts were not possible, evidence should be presented demonstrating why they were not possible.

Auditors should confirm:

- The appropriate area of natural wetland has been successfully rehabilitated.
- Polygons or measurement of the areas showing conversion and rehabilitation have been presented by the UoC.

Useful Resources

- Best Practice Guidelines for Mangrove Restoration developed by the Global Mangrove Alliance and the Blue Carbon Initiative: [Best-Practice-Guidelines-for-Mangrove-Restoration_v4.pdf](#).
 - The guidelines lay out the steps for rehabilitation project design, setting goals and assessing feasibility, project implementation, monitoring, evaluation and adaptive management.
- Spatial mapping tools:
 - GIS portal: [GIS portal - ASC International \(asc-aqua.org\)](#)
 - Clark Labs Coastal Habitat Mapping: Mangrove and Pond Aquaculture Conversion [Coastal Habitat Mapping: Mangrove and Pond Aquaculture Conversion - Clark Labs](#).
 - Global Mangrove Watch country-level information on mangrove habitat extent and change: [Global Mangrove Watch](#).
- Additional rehabilitation tools:
 - The Climate-Smart Mangrove Tool (CSMT) is designed to aid conservation practitioners in selecting appropriate “climate-smart” actions to reduce the current and future impacts of climate change on mangrove ecosystems: <https://www.worldwildlife.org/publications/the-climate-smart-mangrove-tool-excel-based-workbook>.
- Where to restore mangrove habitats:

- The Mangrove Restoration Potential Mapping Tool provides information on the ecosystem services provided in restoration of regions within a country
<https://maps.coastalresilience.org/mangrove-restoration/>.
- SFP Technical Paper: How coastal shrimp farming can support effective mangrove regeneration: [FINAL - Shrimp Farming and Mangroves Technical Paper Dec2022.pdf - Google Drive](#).
- To help the farmed shrimp industry more effectively identify and protect mangrove habitat, SFP and Longline Environment developed the Aquascope mangrove-shrimp pond mapping tool to show where shrimp farming regions and mangrove habitat overlap in India, Indonesia, Thailand, and Vietnam. Register to use the tool at: [Mangrove mapping tool - Sustainable Fisheries Partnership](#).

Indicator:	Requirement:
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2.2.11 Indicator applicability: bivalve culture

The UoC shall⁷ ensure that total bivalve culture in a waterbody does not result in phytoplankton depletion either by:

- demonstrating that the waterbody has a ratio of **Clearance Time** (CT) over **Retention Time** (RT) > 1, following the method outlined in **Appendix 5**. Where CT over RT is <1, demonstrate that the waterbody has a ratio of CT over **Primary Production Time** (PPT) > 3, following the method outlined in **Appendix 5**; or
- conducting a modelling study that demonstrates that the carrying capacity of the waterbody is not exceeded.

How do I interpret this requirement?

- The ratio of Clearance Time (CT) and Retention Time (RT) demonstrate whether the carrying capacity of a waterbody for bivalve production has been exceeded.
- Primary Production Time (PPT) is calculated when cultured bivalves may negatively impact the carrying capacity of a waterbody.

⁷ An exemption to Indicator 2.2.11 applies for situations where the total surface area of aquaculture within a waterbody is less than 10% of the total surface area of the waterbody.

Evidence of implementation should be maintained, e.g.:

- The method used to calculate CT, RT and PPT, in accordance with Appendix 5.
 - Maps demonstrating whether the site is in an easily definable waterbody. This will support the selection of the correct method in Appendix 5 (5.2) regarding estimating the zone of influence.
- Results demonstrating the ratio of CT over RT, or CT over PPT.
- A modelling study has been conducted to demonstrate the carrying capacity of the waterbody, if the method outlined in Appendix 5 has not been followed.

Auditors should confirm:

- Methods and results have been presented demonstrating the ratio of CT over RT, or CT and PPT depending on whether CT over RT > 1, in accordance with Indicator 2.11.
- A modelling study has been conducted to demonstrate the carrying capacity of the waterbody, if the method outlined in Appendix 5 has not been followed.
 - The modelling study shall demonstrate carrying capacity has not been exceeded.

Useful Resources

N/A

Criterion 2.3 - Wildlife Interactions

Indicator:	Requirement:
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2.3.1 The UoC shall ensure the site has a low risk of Human-Wildlife Conflict within the site and in the area surrounding the site, as determined through the ERA (see Indicator 2.1.1).

How do I interpret this requirement?

- Indicator 2.1.1 and Appendix 3.1 outline the site-specific Environmental Risk Assessment (ERA) following the Standardised Risk Methodology.
- Mitigation measures may need to be implemented to manage Human-Wildlife Conflict within the site and in the area surrounding the site.
- The risk level and identified mitigation measures (if applicable) should be identified in the ERA.

Evidence of implementation should be maintained, e.g.:

- The ERA assesses the risk of Human-Wildlife Conflict. Evidence should be presented to demonstrate whether the risk is low.
- Mitigation measures implemented to manage Human-Wildlife Conflict to maintain the low risk level.

Auditors should confirm:

- The ERA has assessed the risk of Human-Wildlife Conflict and determined the risk level is low.
- Mitigation measures are implemented (if required and recommended by the ERA) to maintain the low risk level.

Useful Resources

N/A

Indicator:	Requirement:
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2.3.2 The UoC shall not cause the mortality of any **Threatened Species**.

An exception applies for occasional incidents that do not affect the population status, and where a written statement from a **registered veterinarian** or the responsible authorities is available to confirm that:

- injured animals are unlikely to recover; or
- immediate human safety is threatened; or
- legal requirements mandate emergency euthanasia.

2.3.3 The UoC shall not injure or cause mortality of mammals, elasmobranchs or reptiles (excluding **vermin**).

An exception applies for occasional incidents where:

- injured animals are unlikely to recover; or
- immediate human safety is threatened; or
- legal requirements mandate emergency euthanasia.

How do I interpret this requirement?

- Mortalities include animals intentionally killed through lethal action, as well as accidental deaths through entanglement or other means.
- Occasional incidents of injury or mortality will be rare and only occur in exceptional circumstances which are necessary for the reasons outlined in Indicators 2.3.2 and 2.3.3. All other feasible measures should be pursued prior to causing injuries or mortality.

Evidence of implementation should be maintained, e.g.:

- A list of Threatened Species, mammals, elasmobranchs and reptiles which are present in the vicinity of the site should be presented.
- Records of any interactions, injuries and mortalities to/of these species should be maintained.
- Where there are occasional incidents of interactions with or mortalities of Threatened Species, evidence has been maintained to demonstrate that the incidents will not affect the population status.

- A written statement has also been obtained from a registered veterinarian or the responsible authorities to confirm at least one of the three points (a-c) in Indicator 2.3.2 has been met.
- Records demonstrating that if there are occasional injuries or mortalities of mammals, elasmobranchs or reptiles, the incidents meet the exception in Indicator 2.3.3.

Auditors should confirm:

- Records of interactions with Threatened Species, mammals, elasmobranchs and reptiles have been maintained. Specifically, records of any injuries or mortality.
- Written statements are present from a registered veterinarian or the responsible authorities, in accordance with Indicators 2.3.2, if mortalities of Threatened Species have occurred.
 - The statement confirms that at least one of the three points (a-c) in Indicator 2.3.2 have been met.
- Records have been presented to demonstrate that if there are occasional injuries or mortalities of mammals, elasmobranchs or reptiles, the incidents meet the exception in Indicator 2.3.3.

Useful Resources

N/A

Indicator:	Requirement:
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2.3.4	The UoC shall investigate all wildlife mortality incidents to determine the root cause and shall implement appropriate corrective actions to prevent recurrence.
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How do I interpret this requirement?

N/A

Evidence of implementation should be maintained, e.g.:

- Records showing that the UoC has investigated each mortality incident and how the investigation has led to the implementation of appropriate actions to prevent recurrence.
- Documents demonstrating the actions have been implemented and prevented recurrence.

Auditors should confirm:

- Investigations have taken place following each mortality event.
- Actions have been implemented to prevent recurrence of wildlife mortalities.

Useful Resources

N/A

Indicator:	Requirement:
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2.3.5

The UoC shall only use **aquatic acoustic deterrence** where scientific evidence suggests that it does not result in significant stress or injury to, or mortality of, the affected species or its population(s).

How do I interpret this requirement?

- Acoustic deterrence includes any device or method used in which acoustic ('sound'/'noise') is used to deter wildlife. This includes, but is not limited to, the use of acoustic deterrent devices (ADDs) and acoustic harassment devices (AHDs).

Evidence of implementation should be maintained, e.g.:

- Records of the use of any aquatic acoustic deterrence, which includes the frequency and duration and duration of the deterrence.
- If aquatic acoustic deterrence is used, scientific reports/publications are obtained which demonstrate that the deterrence does not result in significant stress, injuries to, or mortality of, the species impacted/targeted by the deterrence.

Auditors should confirm:

- Records are provided to demonstrate whether any aquatic acoustic deterrence has been used.
- If aquatic acoustic deterrence has been used, scientific reports/publications have been presented by the UoC to demonstrate that the deterrence does not result in significant stress, injuries to, or mortality of, the species impacted/targeted by the deterrence.

Useful Resources N/A

Indicator:	Requirement:
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2.3.6 *Indicator applicability: all fish species*

The UoC shall only stock fish that is hatchery-raised.

An exception applies where the UoC can demonstrate the following for *Seriola/Cobia*, in which case wild caught fish are permitted to be stocked:

- the species is not listed as a Threatened Species; and
- the originating fishery is certified to an ASC-recognised fisheries certification scheme⁸; and
- traceability back to the originating fishery is provided.

How do I interpret this requirement?

N/A

Evidence of implementation should be maintained, e.g.:

- Batch records or other documentations are maintained to demonstrate stocked fish are hatchery-raised.
- For *Seriola/Cobia*, where wild caught fingerlings are stocked, documentation demonstrating the species is not listed as a Threatened Species and records demonstrating the traceability of the fish.
 - The IUCN Red List can be used to check whether the species is listed as Threatened.
 - Documentation demonstrating the fish was sourced from an ASC-recognised fisheries certification scheme can facilitate traceability back to the fishery if chain of custody documentation has been maintained.
 - Traceability documentation should include the species, catch method and area in which the fish was caught.

Auditors should confirm:

- Batch records or other documentation are presented to demonstrate stocked fish are hatchery-raised.

⁸ Refer to the ASC website for details of ASC-recognised fisheries certification schemes.

- For *Seriola/Cobia* stocked from a wild fishery, adequate documentation is presented to demonstrate traceability back to the fishery and the certification of the originating fishery (to an ASC-recognised fisheries certification scheme).

Useful Resources

- IUCN: The IUCN Red List of Threatened Species: [IUCN Red List of Threatened Species](#).

Indicator:	Requirement:
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2.3.7 The UoC shall not use Threatened Species for wild broodstock.

How do I interpret this requirement?

N/A

Evidence of implementation should be maintained, e.g.:

- Documentation (e.g. batch records) demonstrating the wild broodstock species is not listed as a Threatened Species.
 - The IUCN Red List can be used to check whether the wild broodstock species is listed as Threatened.
 - Traceability documentation can be used to demonstrate the origin of the wild broodstock species. Traceability should be back to the source fishery.

Auditors should confirm:

- Evidence has been presented that wild broodstock is not a Threatened Species.
 - This may include verifying traceability documentation, which demonstrate traceability back to the source fishery.

Useful Resources

- IUCN: The IUCN Red List of Threatened Species: [IUCN Red List of Threatened Species](#).

Indicator:	Requirement:
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2.3.8 *Indicator applicability: all crustacean species*

The UoC shall only stock larvae that is hatchery-raised.

An exception applies where the UoC can demonstrate the following for extensive crustacean production, in which case wild larvae are permitted to be stocked:

- a. the larvae are collected passively; and
- b. the production system is a non-fed, low-intensive pond production system; and
- c. inflow of larvae occurs through natural tidal flow into the ponds.

How do I interpret this requirement?

N/A

Evidence of implementation should be maintained, e.g.:

- Hatchery permits and import licences which demonstrate the source of the larvae.
- If applicable, documentation that confirms crustacean production is extensive.
 - Documentation demonstrating points a-c under the exception in Indicator 2.3.8 are met if wild larvae are sourced and stocked for extensive crustacean production.
 - If the UoC is not vertically integrated, larvae suppliers should provide documentation to the UoC to verify how wild larvae are collected.

Auditors should confirm:

- Hatchery permits and import licences are presented which demonstrate wild larvae is hatchery-raised.
- If crustacean production is extensive and wild larvae are stocked, documents are presented demonstrating the exception in Indicator 2.3.8 is met.

Useful Resources

N/A

Indicator:	Requirement:
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2.3.9 *Indicator applicability: hatchery-raised bivalve seed*

The UoC shall only source from hatcheries that have procedures to address genetic concerns specific to the species and geographical region where the seed will be stocked.

How do I interpret this requirement?

- The translocation of wild bivalve seed can pose risks to the genetic diversity of wild populations.
 - Genetic risks to wild populations therefore need to be assessed and procedures need to be implemented to address any genetic risks. These will be based on the species and geographic region where the seed will be stocked.
- Efforts should be made to address genetic concerns specific to the species and the geographic region where the seed will be stocked.
 - This may include preserving diversity of broodstock and seed by:
 - Using local broodstock,
 - Rotation of broodstock within spawning seasons and between years, and
 - Avoiding the use of hatchery-propagated stock in the hatchery as broodstock.
 - The scale of farming and the reproductive potential of crops (e.g. whether diploid or triploid, or considering age at harvest and age at first maturation) are well below the size and reproductive potential of the natural population within a reasonable 'dispersal kernel' from the site.
 - Sterile seed may be produced for stocking.

Evidence of implementation should be maintained, e.g.:

- An assessment of genetic risks presented from the translocation of wild bivalve seed, based on the species and geographic region where the seed will be stocked.
- The hatchery has procedures and measures in place to address any genetic concerns. For example:
 - Documentation of the use of local, wild broodstock to address genetic concerns specific to the species and the geographic region where the seed will be out-planted.

- Documentation of the scale of farming activities and the reproductive potential of crops (e.g. whether diploid or triploid, or considering age at harvest and age at first maturation) are well below the size and reproductive potential of the natural population within a reasonable 'dispersal kernel' from the site.
- Documentation on the production of sterile seed for out-planting from breeding programmes that intentionally alter wild stocks for improved culture traits, such as growth, yield, survival and morphology.
- Provide documentation of cooperation with restoration efforts in the geographic region using out-planting that involves the intentional divergence from wild stocks to produce disease resistant wild populations.

Auditors should confirm:

- Evidence has been presented that genetic risks to wild populations have been assessed and the hatchery has implemented procedures and measures to address these risks. For example:
 - Verify that efforts have been made to address genetic concerns specific to the species and the geographic region where the seed will be out-planted.
 - Document the site's use of the local, wild population for the broodstock with appropriate rotation of spawned individuals. If there is uncertainty about whether a broodstock is of local origin, record source locality.
 - Verify that the reproductive potential (e.g. whether diploid or triploid, or considering age at harvest and age at first maturation) of the hatchery produced seed is well below the size and reproductive potential of the natural population within a reasonable 'dispersal kernel' from the site.
 - Verify the existence of a breeding programme that produces altered sterile seed for out-planting.
 - Verify that the site's breeding programme is in coordination with existing restoration efforts in the geographic region.

Useful Resources

N/A

Indicator:	Requirement:
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2.3.10 *Indicator applicability: all bivalve species*

The UoC shall⁹ only use wild caught seed if:

- the species is not listed as a Threatened Species; and
- the seed is **harvested** from a regulated source.

How do I interpret this requirement? N/A

Evidence of implementation should be maintained, e.g.:

- Documentation (e.g. batch records) demonstrating that the species used for wild caught seed is not listed as a Threatened Species.
 - The IUCN Red List can be used to check whether the species used for wild caught seed is listed as Threatened.
 - Traceability documentation can be used to demonstrate the origin of the species used for wild caught seed.
 - Traceability should be back to the source and demonstrate it is regulated and not from an open-access, unregulated source.

Auditors should confirm:

- Evidence has been presented that the species used for wild caught seed is not a Threatened Species.
 - This may include verifying traceability documentation, demonstrating traceability back to the source.
 - The evidence demonstrates the source is well regulated and not from an open-access unregulated source.

Useful Resources

- IUCN: The IUCN Red List of Threatened Species: [IUCN Red List of Threatened Species](#).

⁹ An exemption applies to wild caught larvae.

Indicator:	Requirement:
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2.3.11

Indicator applicability: all abalone farming in regions where the farmed species is Threatened

The UoC shall only use seed originating from native wild broodstock (not selectively bred animals).

How do I interpret this requirement?

- Wild abalone populations can be impacted through selectively bred animals recruiting into, and breeding with, wild abalone populations.
- Potential genetic impacts to at-risk populations can be mitigated by means of minimising escapes from abalone sites and by farming with stock originating from the same genetic stock as the local wild populations surrounding the site.

Evidence of implementation should be maintained, e.g.:

- Documentation demonstrating the origin of seed and that it originates from native wild broodstock.

Auditors should confirm:

- Documentation has been presented demonstrating the origin of seed and that it originates from native wild broodstock.

Useful Resources

N/A

Indicator:	Requirement:
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2.3.12	The UoC shall publicly disclose all wildlife mortalities per site within 30 days of each incident.
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How do I interpret this requirement?

N/A

Evidence of implementation should be maintained, e.g.:

- Wildlife mortalities are disclosed and are easily and publicly available.
- Hardcopies of mortalities are available to stakeholders, if requested.
- Records are kept demonstrating when mortalities have been disclosed.

Auditors should confirm:

- Sampling results are easily and publicly available. If applicable, confirm that the site has made hard copies of test results easily available to stakeholders.
- Mortalities have been disclosed per site within 30 days of each incident.

Useful Resources

N/A

Indicator:	Requirement:
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2.3.13	The UoC shall report to ASC on wildlife mortalities, according to ASC data submission procedures.
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How do I interpret this requirement?

N/A

Evidence of implementation should be maintained, e.g.:

- Records of data submissions to ASC.

Auditors should confirm:

- Data on wildlife mortalities has been submitted to ASC and is accessible.

Useful Resources

N/A

Criterion 2.4 - Escapes and Native Biodiversity

Indicator:	Requirement:
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2.4.1 The UoC shall not stock **transgenic** animals.

How do I interpret this requirement?

- The stocking and culture of transgenic animals is not permitted on ASC certified sites.
 - Transgenic animals are not common in aquaculture, but transgenic stocks of Atlantic salmon, rainbow trout, channel catfish and Nile tilapia are known to exist.

Evidence of implementation should be maintained, e.g.:

- Evidence that non-transgenic animals are stocked should be presented to the auditor. This could consist of a declaration from the hatcheries that non-transgenic animals are used.
 - In case of doubt regarding origin, submitting animals for DNA-testing could be considered.

Auditors should confirm:

- All stock is from non-transgenic origin.

Useful Resources

N/A

Indicator:	Requirement:
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- | | |
|--------------|--|
| 2.4.2 | The UoC shall implement escape prevention measures that include the installation, inspection and maintenance of containment structures suitable for the species, life stages and environmental conditions under which the site is operating. |
|--------------|--|

How do I interpret this requirement?

- This Indicator requires the UoC to prevent the escape of cultured animals. It focuses on three key aspects:
 - Installation of containment structure; these structures must be appropriate for:
 - The species; consider size and behaviour
 - Different life stages; consider that smaller animals require different mesh sizes compared to bigger animals
 - Environmental conditions; consider currents, tides, floodings, floating debris, natural predators that can damage the containment structure.
 - Inspection of containment structures to check for structure integrity, including:
 - Signs of damage or weaknesses; consider especially moving elements of the construction for 'wear and tear' signs
 - Environmental conditions that impact the structure (e.g. erosion)
 - Species-specific behaviours (e.g. net-biting, burrowing)
 - Maintenance of containment structure:
 - Regular maintenance needs to be conducted so that containment structures are kept in, or restored to, a good state, or to prepare for known severe weather events (e.g. forecasted storms, flooding, etc.).

Evidence of implementation should be maintained, e.g.:

- Escape prevention measures are installed according to construction specifications that incorporate environmental conditions and species-specific characteristics.
- Documents demonstrating inspection and maintenance is occurring regularly and when needed (e.g. after a storm).

Auditors should confirm:

- Escape prevention measures have been installed according to construction specifications and are appropriate to environmental conditions and species-specific characteristics.
- Documents have been presented which demonstrate inspection and maintenance has occurred regularly and as required.

Useful Resources

- <https://www.gov.scot/publications/aquaculture-code-practice-containment-prevention-escape-fish-fish-farms-relation-marine-mammal-interactions/pages/1/>

Indicator:	Requirement:
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- 2.4.3** The UoC shall carry out a site-specific handling and containment risk assessment, approved by a member of management, following the method outlined in **Appendix 3**.

How do I interpret this requirement?

- This Indicator requires the UoC to conduct a site-specific risk assessment to proactively identify and mitigate risks associated with handling and containment of cultured animals. The Indicator emphasises the need for tailored evaluation rather than a generic approach as site-specific circumstances might require different responses. Management approval ensures that the assessment is formally reviewed, acknowledged and integrated into wider management structures.

Evidence of implementation should be maintained, e.g.:

- Documentational evidence of a risk assessment, the identification of appropriate risk levels and any mitigation actions taken (including the implementation thereof). The evidence should also include proof of management approval and periodic review of the risk assessment to ensure that it remains relevant.

Auditors should confirm:

- Presence of a documented risk assessment, which:
 - Is specific to the conditions of the site.

- Covers all handling and containment risks.
- That the risk assessment identifies:
 - Risk factors and the associated risk levels.
 - Mitigation actions and records the implementation, including the desired outcome (i.e. reduction of risk level).
- That the risk assessment is approved (e.g. signed off) by management and shows evidence of periodic review to keep it relevant.

Useful Resources

N/A

Indicator:	Requirement:
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2.4.4	The UoC shall ensure that any person involved in handling activities is trained on handling procedures to minimise escapes.
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How do I interpret this requirement?

- Training and handling procedures includes:
 - An understanding of which staff are involved in animal handling (including external services providers that conduct activities at the site).
 - Coverage of relevant training content, e.g. correct handling process, correct use of equipment and emergency protocols.
 - Appropriate means of training delivery relevant to the site-specific conditions (e.g. considering whether a workshop is sufficient or whether 'live training' on-site is more appropriate).

Evidence of implementation should be maintained, e.g.:

- Training documents and records of staff attendance.
- Presence of relevant procedures/protocols.

Auditors should confirm:

- Training documents and records of staff attendance.

- Presence of relevant procedures/protocols.
- Interviews with staff.

Useful Resources

N/A

Indicator:	Requirement:
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2.4.5	The UoC shall log all handling and containment incidents and near misses.
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How do I interpret this requirement?

- This indicator requires the UoC to log (i.e. register) all handling and containment incidents, including near misses, in order to inform the risk assessment (see Indicator 2.4.3) and the training (2.4.4) on situations that could result, or have resulted, in registered escapes.

Evidence of implementation should be maintained, e.g.:

- A documented register of all handling and containment incidents and near misses.

Auditors should confirm:

- Presence of the register.
- Interviews with employees to confirm that employees are aware of the register and that it is being used.

Useful Resources

N/A

Indicator:	Requirement:
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2.4.6 *Indicator applicability: bivalve culture*

The UoC shall make efforts to address genetic concerns specific to the species and geographic region where hatchery-raised seed will be out-planted.

How do I interpret this requirement?

- This Indicator requires the UoC to take efforts to mitigate genetic risks associated with releasing hatchery-raised bivalve seed into the environment.
 - The emphasis is on species-specific and region-specific genetic concerns, meaning the UoC must consider the local ecosystem and population genetics before out-planting.

Evidence of implementation should be maintained, e.g.:

- Any of the following:
 - Documentation of the use of local, wild broodstock to address genetic concerns specific to species and the geographic region where the seed will be out-planted;
 - Documentation of the scale of farming activities and the reproductive potential of crops (e.g. whether diploid or triploid, or considering age at harvest and age at first maturation) are well below the size and reproductive potential of the natural population within a reasonable 'dispersal kernel' from the site;
 - Documentation of the production of sterile seed for out-planting from breeding programmes that intentionally alter wild stocks for improved culture traits, such as growth, yield, survival and morphology;
 - Documentation of cooperation with restoration efforts in the geographic region using out-planting that involves the intentional divergence from wild stocks to produce disease-resistant wild populations.

Auditors should confirm:

- In relationship to the items listed under 'Evidence of implementation':
 - Verify that the site uses local, wild population for the broodstock with appropriate rotation of spawned individuals;

- Verify that the reproductive potential (e.g. whether diploid or triploid, or considering age at harvest and age at first maturation) of the hatchery produced seed is well below the size and reproductive potential of the natural population within a reasonable 'dispersal kernel' from the site;
- Verify the existence of a breeding programme that produces altered sterile seed for out-planting;
- Verify that the site's breeding programme is in coordination with existing restoration efforts in the geographic region.

Useful Resources

N/A

Indicator:	Requirement:
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2.4.7	The UoC shall report to ASC the escape data, according to ASC data submission procedures.
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How do I interpret this requirement?

- This Indicator requires the UoC to report the relevant data to ASC, as per data submission procedure.

Evidence of implementation should be maintained, e.g.:

- Evidence that data has been submitted and received by ASC.

Auditors should confirm:

- Verify that UoC holds evidence that data has been submitted to ASC.

Useful Resources

N/A

Indicators 2.4.8 - 2.4.13: Additional impact prevention measures

Indicator:	Requirement:
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2.4.8 *Indicator applicability: species with an EICAT Category MO*

The UoC shall not exceed a maximum number of **known escapes** per production cycle or year, whichever is stricter, of the following:

- marine salmonids: 300 individuals
- other fish species: 6%

If the maximum number of known escapes is exceeded more than once in a six-year period, the UoC shall, by the next working day:

- submit a formal notification to the CAB for cancellation of certification; and
- notify ASC of the exceedance of known escapes, according to ASC data submission procedures.

How do I interpret this requirement?

- Note: Indicator 2.4.8 only applies for species/regions that are classified as 'Category Moderate (MO)' in the EICAT Species Factsheet.
- This Indicator requires a strict limit on the number of known escapes allowed per production cycle or per year, whichever threshold is stricter.
 - 'Known escapes' refers to escape incidents that are either observed directly or detected through monitoring. It does not include unverified or speculative losses.
- Escape limits:
 - Marine salmonids (e.g. Atlantic salmon, trout): A hard limit of 300 individuals per production cycle. The formula for calculation can be found in Appendix 6.1.
 - Other fish species: A percentage-based limit of 6% of the total stocked population per production cycle or year (whichever is stricter). The formula for calculation can be found in Appendix 6.1.
- If the UoC exceeds the set limit, a formal notification to the CAB shall be submitted by the next working day.

Evidence of implementation should be maintained, e.g.:

- The UoC has a register of known escapes in which exceedance of the set limit is monitored.

Auditors should confirm:

- Records of known escapes are maintained.
- Escape limits remain below the set threshold and that calculations are made correctly.

Useful Resources

N/A

Indicator:	Requirement:
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2.4.9 *Indicator applicability: species with an EICAT Category MO*

The UoC shall not exceed one **Mass Escape Event** in a six-year period, defined as:

- marine salmonids: >5000 individuals
- other fish species: >6% Total Escape Count

If a mass escape event occurs more than once in a six-year period, the UoC shall, by the next working day:

- submit a formal notification to the CAB for cancellation of certification; and
- notify ASC of the mass escape event, according to ASC data submission procedures.

How do I interpret this requirement?

- Note: Indicator 2.4.9 only applies for species/regions that are classified as 'Category Moderate (MO)' in the EICAT Species Factsheet.
- This Indicator requires a strict limit on mass escape events over a six-year period to ensure that large-scale fish escapes are rare and properly managed. It also requires certification cancellation if the limit is exceeded.
- An event counts as a 'mass escape' if the thresholds are exceeded, regardless of cause (e.g. storm damage, net failure, human error). The thresholds are:

- For marine salmonids: more than 5,000 individuals escaping at once. The formula for calculation can be found in Appendix 6.1.
- For other fish species: More than 6% of the stocked population escaping at once. The formula for calculation can be found in Appendix 6.1.
- The UoC is allowed only one mass escape event in a rolling six-year period. If a second mass escape event occurs within six years, the UoC must notify the CAB the next working day and request cancellation of certification.
- The UoC should have a system, or method, to detect and document mass escape events, including the identification of root cause(s) and determination of corrective action(s) to prevent re-occurrence.

Evidence of implementation should be maintained, e.g.:

- The UoC has a register of known escapes in which exceedance of the set limit is monitored.
-

Auditors should confirm:

- Records of mass escape events are maintained.
 - Escape limits remain below the set threshold and that calculations are made correctly.
-

Useful Resources

N/A

Indicator:	Requirement:
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2.4.10 *Indicator applicability: species with an EICAT Category MO*

The UoC shall implement appropriate mitigation measures according to the contextualisation of the EICAT category, following the method outlined in **Appendix 6 (6.2)**.

How do I interpret this requirement?

- Note: Indicator 2.4.9 only applies for species/regions that are classified as 'Category Moderate (MO)' in the EICAT Species Factsheet.

- This Indicator requires that the UoC implements suitable actions to minimise the potential harm to biodiversity by a species, depending on its context as determined in Appendix 6 (following the decision-tree in this Appendix).
- For situations in which either the species, the production region or the culture system results in a low likelihood of contributing to biodiversity harm, the UoC is not required to implement actions.
- For situations in which either the species, the production region or the culture system contributes to biodiversity harm, the UoC is required to implement actions that reduce this risk. This could mean, e.g.:
 - Installation of net pens as per technical standards in order to further reduce the likelihood of escapes occurring.
 - Installation of additional screens or barriers in pond/tank culture to further reduce the likelihood of escapes occurring.

Evidence of implementation should be maintained, e.g.:

- Documents demonstrating to what extent the decision tree in Appendix 6 is applicable, and evidence of a reduction in the harm category (where relevant).
- In cases where a reduction in the harm category is not possible, the UoC should demonstrate which additional measures have been taken, including the intended outcome of these measures.

Auditors should confirm:

- Confirmation of the decision-tree results in Appendix 6.
 - In cases where a reduction in the harm category is not possible, confirmation that additional measures have been taken, including the intended outcome of these measures.

Useful Resources

N/A

Indicator:	Requirement:
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2.4.11 *Indicator applicability: species with an EICAT Category MR/MV*

The UoC shall calculate the number of known escapes per production cycle or year, whichever is stricter, following the method outlined in **Appendix 6 (6.1)**.

The UoC shall not exceed a maximum number of known escapes per production cycle or year, whichever is stricter, of the following:

- marine salmonids: 300 individuals
- other fish species: 6%

If the maximum number of known escapes is exceeded, the UoC shall, by the next working day:

- submit a formal notification to the CAB for cancellation of certification; and
- notify ASC of the exceedance of known escapes, according to ASC data submission procedures.

How do I interpret this requirement?

- Note: Indicator 2.4.11 only applies for species/regions that are classified as 'Category Major/Massive (MR/MV)' in the EICAT Species Factsheet.
- This Indicator requires a strict limit on the number of known escapes allowed per production cycle or per year, whichever threshold is stricter. 'Known escapes' refers to escape incidents that are either observed directly or detected through monitoring. It does not include unverified or speculative losses.
- Escape limits:
 - Marine salmonids (e.g., Atlantic salmon, trout): A hard limit of 300 individuals per production cycle. The formula for calculation can be found in Appendix 6 (6.1).
 - Other fish species: A percentage-based limit of 6% of the total stocked population per production cycle or year (whichever is stricter). The formula for calculation can be found in Appendix 6 (6.1).
- If the UoC exceeds the set limit, a formal notification to the CAB shall be submitted by the next working day.

Evidence of implementation should be maintained, e.g.:

- The UoC has a register of known escapes in which exceedance of the set limit is monitored.

Auditors should confirm:

- Records of known escapes are maintained.
- Escape limits are to remain below the set threshold and that calculations are made correctly.

Useful Resources

N/A

Indicator:	Requirement:
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2.4.12 *Indicator applicability: fish species with an EICAT Category MR/MV*

The UoC shall not have any mass escape events. If a mass escape event occurs, the UoC shall, by the next working day:

- submit a formal notification to the CAB for cancellation of certification; and
- notify ASC of the mass escape event, according to ASC data submission procedures.

How do I interpret this requirement?

- Note: Indicator 2.4.12 only applies for species/regions that are classified as 'Category Major/Massive (MR/MV)' in the EICAT Species Factsheet.
- This indicator requires a zero limit on mass escape. It also requires certification cancellation if the limit is exceeded.
- An event counts as a 'mass escape' if the thresholds are exceeded, regardless of cause (e.g. storm damage, net failure, human error). The thresholds are:
 - For marine salmonids: more than 5,000 individuals escaping at once. The formula for calculation can be found in Appendix 6 (6.1).
 - For other fish species: More than 6% of the stocked population escaping at once. The formula for calculation can be found in Appendix 6 (6.1).

- If a mass escape event occurs, the UoC must notify the CAB the next working day and request cancellation of certification.
- The UoC should have a system, or method, to detect and document mass escape events, including the identification of root cause(s) and determination of corrective action(s) to prevent re-occurrence.

Evidence of implementation should be maintained, e.g.:

- The UoC has a register of known escapes in which exceedance of the set limit is monitored.
-

Auditors should confirm:

- Records of mass escape events are maintained.
 - Escape limits remain below the set threshold and that calculations are made correctly.
-

Useful Resources

N/A

Indicator:	Requirement:
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2.4.13 *Indicator applicability: fish species with an EICAT Category MR/MV*

The UoC shall implement appropriate mitigation measures according to the contextualisation of the EICAT category, following the method outlined in **Appendix 6 (6.2)**.

How do I interpret this requirement?

- Note: Indicator 2.4.9 only applies for species/regions that are classified as 'Category Major/Massive (MR/MV)' in the EICAT Species Factsheet.
- This Indicator requires that the UoC implements suitable actions to minimise the potential harm to biodiversity by a species, depending on its context as determined in Appendix 6 (following the decision-tree in this Appendix).
- For situations in which either the species, the production region or the culture system results in a low likelihood of contributing to biodiversity harm, the UoC is not required to implement actions.

- For situations in which either the species, the production region or the culture system contributes to biodiversity harm, the UoC is required to implement actions that reduce this risk. This could mean, e.g.:
 - Installation of net pens as per technical standards in order to further reduce the likelihood of escapes occurring.
 - Installation of additional screens or barriers in pond/tank culture to further reduce the likelihood of escapes occurring.

Evidence of implementation should be maintained, e.g.:

- Documents demonstrating to what extent the decision tree in Appendix 6 is applicable, and evidence of a reduction in the harm category (where relevant).
- In cases where a reduction in the harm category is not possible, the UoC should demonstrate which additional measures have been taken, including the intended outcome of these measures.

Auditors should confirm:

- The outcome of the decision-tree results in Appendix 6.
- In cases where a reduction in the harm category is not possible, confirmation that additional measures have been taken, including the intended outcome of these measures.

Useful Resources

N/A

Indicators 2.4.14 - 2.4.15: Additional escape prevention measures

Indicator:	Requirement:
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2.4.14 *Indicator applicability: species with an EICAT Category MO or MR/MV*

The UoC shall count fish with a minimum of 98% accuracy.

How do I interpret this requirement?

- This Indicator requires the UoC to ensure that fish counting procedures achieve a minimum accuracy of 98%, meaning that the error margin must not exceed 2% and applies to both manual and automated counting methods.
- The total number of fish at key stages of production (e.g. stocking and harvesting) needs to be counted.
- The actual count should be within 2% of the true number (e.g. if 100,000 fish are stocked, the count must be between 98,000 and 102,000).
- Acceptable counting methods include e.g.:
 - Automated systems: optical counters, AI-based image recognition or volumetric counting.
 - Manual counting: hand counting, sample-based extrapolation or visual verification.

Evidence of implementation should be maintained, e.g.:

- The UoC shall demonstrate that fish are counted at key stages of production (e.g. stocking and harvesting) using a method that achieves 98% accuracy.
 - If automated systems are used, ensure they are regularly calibrated.
- Documentation of counting procedures, accuracy checks and error rates.

Auditors should confirm:

- Results of fish counts at key stages of production have been presented.
- That calibration of the counting method is conducted.
- Documents are maintained to record counting procedures, accuracy checks and error rates.

Useful Resources N/A

Indicator:	Requirement:
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2.4.15 *Indicator applicability: tilapia with an EICAT Category MO or MR/MV*

The UoC shall use all-male stock.

How do I interpret this requirement?

- This Indicator requires that the UoC only stocks tilapia that is 'all-male' if the EICAT Category is set at Moderate (MO), Major (MR) or Massive (MV).

Evidence of implementation should be maintained, e.g.:

- Documents demonstrating that the UoC is aware of the EICAT Category.
- If the EICAT Category is set at Moderate (MO), Major (MR) or Massive (MV) documents demonstrate that the stocked tilapia is 'all-male'.

Auditors should confirm:

- The correct EICAT categorisation is applied.
- If the EICAT Category is set at Moderate (MO), Major (MR) or Massive (MV), documents have been presented to demonstrate all stocked tilapia are 'all-male'.

Useful Resources

N/A

Additional requirements for UoCs stocking Salmon Smolts produced in open cages culture operating in freshwater (e.g. lochs, lakes, reservoirs)

Indicator:	Requirement:
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2.4.16 The UoC shall only stock smolts produced in cage culture if the supplier operates in a region where native salmonids of the same species are present.

How do I interpret this requirement?

- This Indicator requires that smolts produced in cage culture are only sourced from suppliers operating in regions where native populations of the same species exist to prevent the potential competition with, or displacement of, native species in regions without natural salmonid populations.

Evidence of implementation should be maintained, e.g.:

- Demonstrate that the supplier of smolts operates in a region where native salmonids of the same species are present.

Auditors should confirm:

- Verify that the supplier of smolts operates in a region where native salmonids of the same species are present.

Useful Resources

N/A

Indicator:	Requirement:
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2.4.17 The UoC shall communicate each **escape incident** to the local fishery trust(s).

How do I interpret this requirement?

- This indicator requires the UoC to notify local fishery trusts whenever an escape incident occurs to ensure transparency, facilitate environmental monitoring and support mitigation efforts.

- For each escape incident, the UoC must report key details such as:
 - Date and time of the escape.
 - Estimated number of escaped fish.
 - Species involved.
- The UoC should identify the relevant local fishery trust(s) and establish clear communication channels. Fishery trusts are typically regional organisations responsible for managing wild fish populations and aquatic ecosystems. They may include government agencies, conservation groups or local fisheries management bodies.

Evidence of implementation should be maintained, e.g.:

- In case of escapes, the UoC shall maintain records of communications with fishery trusts (emails, reports or acknowledgments).

Auditors should confirm:

- Verify that, in case of escapes, the UoC has communicated each escape incident to the fishery trusts.

Useful Resources

N/A

Indicator:	Requirement:
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2.4.18 The UoC shall collaborate with fishery trust(s) in conducting native salmonid monitoring programme(s) in the waterbody where it operates.

How do I interpret this requirement?

- This Indicator requires the UoC to collaborate with local fishery trusts in conducting native salmonid monitoring programmes within the waterbody where it operates.
 - The purpose is to track the status of native salmonid populations and assess potential impacts from aquaculture activities. 'Collaborate' would imply involvement e.g. beyond providing funding or passive support.

- Collaboration may include:
 - Sharing data on site operations and potential interactions with native salmonids.
 - Assisting in monitoring efforts (e.g. providing access to sites, participating in surveys).
 - Supporting scientific research or environmental assessments.
- The monitoring programme should address as minimum:
 - Tracking population health, abundance and migration patterns of wild salmonids.
 - Assessing genetic integrity (to detect any impact from escaped farmed fish).

Evidence of implementation should be maintained, e.g.:

- The UoC maintains records (e.g. agreements, partnerships, etc.) of collaboration with fishery trust(s).

Auditors should confirm:

- Records have been presented to demonstrate collaboration with fishery trust(s).
-

Useful Resources

N/A

Indicator:	Requirement:
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2.4.19

The UoC shall conduct, in collaboration with the local fishery trust(s), a scientific baseline study to determine the genetic composition of the contemporary native and farmed salmonid population(s) in the waterbody where it operates. The baseline study shall:

- a. use credible methodologies and analysis and undergo peer review; and
- b. include known historical farmed and native salmonid genetic profiles; and
- c. determine if changes in the genetic composition of the contemporary native salmonid population(s) have occurred; and
- d. if changes in the genetic composition of the contemporary native salmonid population(s) are detected, seek to determine if these have occurred through introgression of farmed salmon or through other mechanisms; and
- e. be updated every three years.

How do I interpret this requirement?

- This Indicator requires the UoC to conduct a collaborative scientific baseline study on the genetic composition of both native and farmed salmonid populations in the waterbody where it operates.
 - The goal is to assess potential genetic changes in wild populations, particularly from interbreeding (introgression) with farmed salmon, and ensure ongoing monitoring.
- The study should establish a genetic baseline for comparison over time and should follow rigorous scientific methods and undergo peer review for credibility.
- The study should use established genetic analysis techniques (e.g. DNA sequencing, microsatellite markers, SNP analysis).
- The study should, if available, include known historical genetic profiles of both native and farmed salmonid populations in the waterbody.
- The study should determine if genetic changes have occurred in native populations and identify if this is the result from introgression from farmed fish or other evolutionary pressures.
- The study must be updated every three years.

Evidence of implementation should be maintained, e.g.:

- The UoC maintains records of the conducted study or demonstrates that the study is under development.

Auditors should confirm:

- Verify that the study has been concluded and has been conducted according to the requirements or is under development.

Useful Resources

N/A

Indicator:	Requirement:
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2.4.20 The UoC shall publicly disclose the scientific baseline study (see Indicator 2.4.19).

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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2.4.21 The UoC shall time following periods to occur outside the out-migration period of native salmonids.

How do I interpret this requirement?

- This indicator requires the UoC to schedule following periods outside the out-migration period of native salmonids.
- The UoC should
 - Identify the local out-migration period of native salmonids in its operating region.
 - Schedule following periods so that no fish are present in cages during this time.
 - Work with scientists, local fishery trusts, and regulatory bodies to align following timing with migration patterns.

Evidence of implementation should be maintained, e.g.:

- The out-migration period of native salmonids has been determined.
- Documents demonstrating fallowing periods have been scheduled outside the out-migration period.

Auditors should confirm:

- Fallowing periods occur outside the out-migration period of native salmonids.

Useful Resources

N/A

Criterion 2.5 – Benthic Impact

Indicator:	Requirement:
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2.5.1

Indicator applicability: cage culture and suspended marine mollusc systems

The UoC shall¹⁰ determine benthic status, following the method outlined in **Appendix 7 (7.2.1, 7.3.1, 7.4.1, 7.7)**.

2.5.2

Indicator applicability: cage culture and suspended marine mollusc systems

The UoC shall demonstrate an 'Acceptable' benthic status¹¹, following the method outlined in **Appendix 7 (7.2.2, 7.3.2, 7.4.2)**.

2.5.3

Indicator applicability: cage culture and suspended marine mollusc systems

The UoC shall ensure that the person conducting sediment sampling and analysis (see Indicator 2.5.1) is independent and has the necessary competencies, experience and training. In cases where regulatory requirements are defined, the UoC shall ensure that the person conducting the sampling and analysis complies with these requirements.

How do I interpret this requirement?

- The method for benthic monitoring for cage culture and suspended marine mollusc systems is composed of:
 - Sediment sampling zones and sampling locations across transects.
 - Benthic quality categories and thresholds for defining those categories and the benthic status.

Sampling approach

- The method uses a three-level sampling approach:
 - The number of transects and sampling locations, as well as the complexity of analysis, may increase based on Level 1 or Level 2 monitoring results:

¹⁰ An exemption applies to sites situated in areas classified as having a 'hard bottom'. Evidence is required to support the classification 'hard bottom'.

¹¹ An exemption applies to cage sites situated in lakes and reservoirs, where determination of benthic status is required (Indicator 2.5.1) but demonstration that the status is 'Acceptable' is not required for the first three years of the ASC Farm Standard being effective.

- Level 1 requires sampling along one transect in the direction of the predominant current. The samples are to be analysed for redox potential and total free sulphide.
 - Direction of the predominant current is the compass direction in which the water flows the majority of the time.
- Level 2 requires sampling along three additional transects. The samples are to be analysed for redox potential and total free sulphide.
- Level 3 requires sampling as per Level 2 (at a minimum, on the two transects in the direction of the predominant current). The samples are to be analysed for three (of the 14) biotic indicators selected from Table 2 of Appendix 7 (7.1).
 - Sites should select biotic indicators that are more appropriate based on their location or based on regulatory requirements.
 - In Scotland, sites should select AZTI's Marine Biotic Index (AMBI) as one of the biotic indicators since it is a component of the Infaunal Quality Index (IQI) required by the regulation in that country.
- Sites can choose to commence monitoring at any of the three levels.
 - Previous benthic impact results can provide an indication of the level at which monitoring should commence.
- The same sediment samples collected for Level 1 or Level 2 sampling can be used for the analysis required by Level 3 sampling.
 - Sites are not required to collect new sediment samples.
- Transects:
 - In cases where a transect overlaps with another site, the direction of the overlapping transect should be adjusted.
 - The same applies to any transect/sampling station overlapping with land.
 - Transect directions should also be adjusted to avoid sampling in areas where water depth changes rapidly along the transect.
 - In all cases, each transect should be as close to 90 degrees from each other as possible.
 - In cases where multiple cages are located in close proximity, transects should be located at the edge of the last cage site.
- Given the low complexity associated with Level 1 and Level 2 sampling and analysis, it is advisable for sites to conduct monitoring not only at the required timings by the indicator but also prior to those timings

- The results can indicate the need for Level 3 sampling at the timings required by the indicator, thereby assisting in monitoring planning.

Ecological Quality Status (EQS) categories

- o Ecological Quality Status (EQS) categories are used to determine the benthic status (see more details on the categories in the Useful Resources section).

Compliance Framework

- o A numerical scoring method that combines all indicator data collected from each sampling zone is used to determine the EQS Category and benthic status:
 - The method requires calculating the mean value for each indicator in each sampling zone to determine the Indicator EQS Score for that zone.
 - Next, an overall Zone EQS Score is calculated by combining the Indicator EQS Scores.
 - The overall Zone EQS Score is used to determine the final EQS Category for each sampling zone and the 'acceptable' benthic status.
 - In the case of cages, the EQS Category of the Reference Zone will determine the EQS Categories for 'acceptable' benthic status of the sampling zones.

Sampling Timing

- o Sampling is required to be conducted during the period when the benthic impact is expected to be highest, i.e. at peak impact, and uses proxies for determining peak impact:
 - Peak biomass (cage culture and suspended marine mollusc systems).
 - Maximum annual bottom water temperature (cage culture and suspended marine mollusc systems).
 - Peak feeding (cage culture only).
- By considering these proxies, cage culture sites shall (see Appendix 7 (7.2.1 and 7.3.1)):
 - Provide the CAB with information on when the highest impact on the benthos is predicted to occur.
 - Determine the timing sampling that applies.
 - o For cage sites with multiple peaks in feeding/biomass occurring in any year and for suspended marine mollusc systems, the requirements allow for reducing the sampling frequency as long as there have been no significant changes to farming practices. Examples of significant changes to farming practices are:

- Modification of the disposition of cages/long lines.
 - Addition of cages/long lines.
 - Increase of biomass.
 - Change in species produced.
- An independent person is a person who is not an employee of the ASC Certificate Holder.
 - To demonstrate competence, the person conducting sediment sampling and analysis should be able to:
 - Perform requested sampling and/or analysis.
 - Operate equipment
 - Develop, verify and validate methods.
 - Use suitable reference standards.
 - Analyse sampling and calibration results, including statements of conformity and interpretations.
 - Record, review and authorise the release of results.
 - Contribute to, or produce, reports that can clearly and concisely explain how the results are able to meet or otherwise not meet the requirement.
 - Perform internal audits on the sampling, analysis and results

Evidence of implementation should be maintained, e.g.:

- Production overview information:
 - Biomass (projected and current).
 - Feed delivered (projected and current).
 - Harvest (biomass, time of harvesting).
- Bottom water temperature data.
- Visual imagery confirming hard bottom.
- Records:
 - Sampling activity, including:
 - Timing and person conducting sampling.

- GPS sampling locations.
- Competence, experience and training of the person conducting sampling and analysis.
- Sampling analysis results.

Auditors should confirm:

- Sampling was conducted according to the method and timing stated in Appendix 7 (7.2, 7.3, 7.4), by cross-checking production overview information and records.
 - Competence, experience and training of the person conducting sampling.
 - Adequate level of personnel competence should be verified by observations and interviews.
 - The sampling analysis results that demonstrate an acceptable EQS.
-

Useful Resources

The benthic monitoring method for cage culture and suspended marine mollusc systems uses five Ecological Quality Status (EQS) categories to define abiotic and biotic qualities based on the seabed macrofaunal communities. The EQS categories are outlined in Table 1 below.

Table 1. Descriptions of benthic macrofauna assemblages for each of the five Ecological Quality Status (EQS) categories of the benthic monitoring method for cage culture and suspended marine mollusc systems:

EQS Category	Description
High	No or very minor disturbance: Species abundance, richness and diversity is high and sensitive taxa dominate. Opportunistic taxa are absent or of negligible abundance. Geochemical quality elements indicate aerobic conditions with low free sulphide toxicity.
Good	Slight disturbance: The level of diversity and abundance of invertebrate taxa is slightly reduced. Most of the sensitive taxa are present but slightly reduced. Opportunistic taxa are present but negligible. Geochemical quality elements indicate aerobic sediment conditions with a slight increase in free sulphide levels.
Moderate	Moderate disturbance: The level of diversity and abundance of invertebrate taxa is moderately reduced. Sensitive taxa have negligible abundance or are absent. Tolerant and first-order opportunistic taxa co-dominate in abundance. Geochemical quality elements indicate a moderate increase in anaerobic conditions with free sulphide levels known to be lethal to sensitive and indifferent taxa.

Poor

Major disturbance: Evidence of major alterations to the values of the biological quality elements. Diversity is greatly reduced with sensitive and indifferent taxa showing negligible abundance or are absent. Tolerant taxa are sub-dominant to first-order opportunistic taxa. Geochemical quality elements indicate a major increase in anaerobic conditions and sulphide concentrations lethal to most taxa.

Bad

Severe disturbance: Evidence of severe alterations to the values of the biological quality elements and in which large portions of the relevant biological communities normally associated with undisturbed conditions are absent. First-order opportunistic taxa dominate but are greatly reduced in abundance. Geochemical quality elements indicate a severe increase in sulphide concentrations that are lethal to all taxa.

Indicator:	Requirement:
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2.5.4

Indicator applicability: freshwater systems discharging into rivers

The UoC shall demonstrate that the benthic status does not deteriorate downstream from the site's effluent point, following the method outlined in **Appendix 7 (7.5)**.

2.5.5

Indicator applicability: freshwater systems discharging into rivers

The UoC shall ensure that:

- the person conducting sediment sampling (see Indicator 2.5.3) has the necessary competencies, experience and training; and
- the laboratory conducting the analysis of the samples is accredited and has approved the sampling methodology.

In cases where regulatory requirements are defined, the UoC shall ensure that the person conducting the sampling and/or the laboratory conducting the analysis complies with these requirements.

How do I interpret this requirement?

- The benthic monitoring method for freshwater systems discharging into rivers:
 - Uses sediment sampling upstream and downstream from the site outflow point to:
 - Determine the presence, composition, abundance and diversity of invertebrate fauna; and

- Categorise the benthic status upstream and downstream as a measure of assessing the aquatic ecosystem health.

Sampling Timing

- o Should consider the seasonality of the presence of the macroinvertebrate species, e.g. insects in their larval stage of the life cycle.
 - It is recommended that sampling is conducted during summer and winter.
 - In geographical regions like Scandinavia, spring and autumn are recommended as the best times for sampling.

Location

- o Practical considerations should be taken into account regarding the sampling locations midstream (e.g. strong currents, particularly in wide, deep rivers).
 - In cases where it is impossible to collect samples in locations midstream, sites should ensure sampling all available substrate types present along the bank and in areas with slower water flow.

Gear

- o Standard equipment such as a surber sampler, handnet and grab should be used.

Sampling analysis

- o ISO/IEC 17025 is an example of laboratory accreditation.
- o For more guidance, see the section Useful Resources below.

Evidence of implementation should be maintained, e.g.:

- o Records:
 - Sampling activity, including:
 - Scope of the sampling
 - Timing and frequency of the sampling and the person conducting sampling.
 - GPS sampling locations.
 - Classification system.
 - Analysis results.
 - Competence, experience and training of the person conducting sampling and analysis.

- Accreditation of the laboratory conducting the analysis of the samples.

Auditors should confirm:

- Sampling was conducted as per the minimum requirements in Appendix 7 (7.5) by cross-checking records.
 - Sampling analysis results demonstrate benthic status downstream from the effluent discharge similar to, or better than, upstream.
 - Sampling frequency agrees with the frequency required by Appendix 7 (7.5).
-

Useful Resources

- [ISO 10870:2012, Water quality — Guidelines for the selection of sampling methods and devices for benthic macroinvertebrates in fresh waters.](#)

Indicator:	Requirement:
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2.5.6 *Indicator applicability: cage culture, suspended marine mollusc systems*

The UoC shall report to ASC the benthic status, according to ASC data submission procedures.

No additional guidance is provided for this Indicator.

Criterion 2.6 – Water Quality

Indicator:	Requirement:
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2.6.1 *Indicator scope: fed systems*

The UoC shall adhere to species-specific limits on Total Nitrogen (TN) and Total Phosphorus (TP) release, according to **Appendix 2(2.1)**.

2.6.2 *Indicator scope: fed systems*

Reporting The UoC shall annually report to ASC TN and TP release (Indicator 2.6.1), according to ASC data submission procedure.

How do I interpret this requirement?

- TN and TP in the feed are required for the calculations. TN and TP content in the feed should be obtained from the feed supplier.
- The suitability of the TN and TP limits per tonne of production should be reviewed if the UoC is in a Type A waterbody and needs to implement corrective actions, in accordance with Indicator 2.6.13.
- The annual limits may need to be reduced to prevent shifts in trophic status or correct increases in limiting nutrients.
- Equipment used to measure TN and TP per tonne of production should be calibrated according to the technical specifications. The equipment used will be based on the formula used in Appendix 8 (8.2):
 - Equipment is required for Formula 2 to measure TN and TP concentration in the influent and effluent water and the volume of the effluent water.
 - Equipment is required for Formula 3 to measure TN and TP concentration in the influent and effluent water.

Evidence of implementation should be maintained, e.g.:

- The formula (outlined in Appendix 8 (8.1)) used to determine the TN and TP load per tonne of production.
- Results of the TN and TP load per tonne of production based on the relevant calculation.

- Equipment to measure TN and TP load per tonne of production has been calibrated according to the manufacturer's instructions, based on the formula used in Appendix 8 (8.2).

Auditors should confirm:

- TN and TP load per tonne of production has been determined.
 - Species-specific limits have been adhered to (where applicable) over the previous 24-month period.
 - TN and TP have been determined and reported where species-specific limits are not available.
 - The UoC has calibrated the equipment used to determine the TN and TP load per tonne of production, based on the formula used in Appendix 8 (8.2).
-

Useful Resources

N/A

Indicator:	Requirement:
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2.6.3 *Indicator scope: fed systems with feed of a diameter $\geq 3\text{mm}$.*

The UoC shall ensure that feed used contains <1% fines, following the method outlined in **Appendix 8 (8.3)**.

How do I interpret this requirement?

- Indicator 2.6.3 complements the other water quality indicators by addressing the direct release of uneaten feed in the form of fines into the environment.
- Setting a maximum percentage of fines in the feed addresses the efficient and proper transport, storage and physical delivery of feed pellets to the site.
- Poor performance in any of the above phases of feed handling will result in a higher percentage of fines (fine particles of feed) and potentially greater impacts on water quality, due to an increase in suspended organic particles and nutrients released into the environment.
- Sampling of feed lots includes feed delivered as material in bulk, big bags and small bags to the site.

- Testing by the feed manufacturer to the UoC is permissible where there is no significant risk of feed deterioration during subsequent handling, storage or application.
 - The UoC should demonstrate that deterioration in feed does not take place during the processes outlined above.
 - The UoC should report the schedule and location of testing of feed if it is not conducted on-site.
 - Rationale for not testing on site should also be reported.

Evidence of implementation should be maintained, e.g.:

- The method used to determine the percentage of fines in feed, in accordance with Appendix 8 (8.3).
- The determination and documentation of a schedule and location for the testing of feed.
- If testing takes place prior to delivery to the site, the rationale should be documented for not testing on site.
- Calibration of sieving machines (if used) in accordance with the manufacturer's recommendations.

Auditors should confirm:

- The rationale for the timing and location of feed sampling.
- The UoC has appropriate testing technology on site and that, if applicable, it is calibrated as required.
- The method used to determine the percentage of fines in feed, in accordance with Appendix 8 (8.3).
- Review testing results and confirm that results demonstrate feed contains <1% of fines.

Useful Resources

N/A

Indicator:	Requirement:
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2.6.4 *Indicator scope: cages*

The UoC shall ensure that cages are sited under at least one of the following conditions:

- In water that is at least twice the cage depth; or,
- $\geq 10\text{m}$ above the bottom of the waterbody; or,
- During periods at $>75\%$ peak biomass, the velocity of the current below the cages is $>0.1\text{ m/s}$, measured according to **Appendix 8 (8.4)**¹².

How do I interpret this requirement?

- The UoC should initially determine the depth of the waterbody in comparison to the depth of the cage.
 - Measurements should take place at the lowest water level and from the bottom of the cage.
 - Depth should also be measured from the outer circumference of the culture containment space, i.e. excluding predator nets, mortality collection chambers etc.
 - In tidal systems, the depth should be measured at mean low water spring tide (MLWS).
- If the depth of the water column (surface to seafloor) is less than 10m or less than double the cage depth, the UoC should measure the current velocity below the cage to determine whether the $>0.1\text{ m/s}$ threshold is met.

Evidence of implementation should be maintained, e.g.:

- Results of measurements which demonstrate that all cages in the UoC meet the depth/current velocity requirements in Indicator 2.6.4.

Auditors should confirm:

- The UoC has provided evidence demonstrating that all cage systems meet the siting requirements outlined in Indicator 2.6.4.

¹² An exemption applies to sites situated in areas classified as having a “hard bottom”. Evidence is required to support the classification “hard bottom”.

- If mean current velocity is used to meet the requirement, the UoC has provided evidence that measurements were taken during periods with >75% peak biomass.

Useful Resources

- An example of how to measure current velocity below cages can be found under Section 2.1 of the following document: Marine Scotland (2015): [A Technical Standard for Scottish Finfish Aquaculture](#).

Indicator:	Requirement:
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2.6.5 The UoC shall ensure that the Receiving Waterbody (RW) has been classified per site as Type A, B or C, following the method outlined in **Appendix 8 (8.5)**.

How do I interpret this requirement?

- The UoC should initially define the boundaries of the Receiving Waterbody (RW) based on hydrological characteristics, which determines its sensitivity to nutrient enrichment.
 - Hydrodynamically isolated embayments (HIEs) should be treated as separate waterbodies during the classification process.
- Marine waterbodies include:
 - Off-shore and inshore coastal systems.
 - More enclosed waterbodies susceptible to seawater intrusion through tidal exchange or storm surges.
- Exposed near-shore waterbodies (i.e. not in a bay or channelised system) and offshore waterbodies will be classified as Type C.
- Once the boundaries of the RW are defined, HRT should be calculated unless the RW is an exposed near-shore or offshore waterbody.
 - The area and average depth of the waterbody need to be determined.
 - The cross-sectional area of the waterbody outflow or channel is required to calculate flow rate.

- A transect should be determined from one side of the outflow/channel to the other side.
- The length (in metres) of the transect is multiplied by the average depth of the transect to determine the cross-sectional area.
- The cross-sectional area is then multiplied by the flow velocity (V). The result of this calculation is the flow rate (Q).
- If a marine waterbody has HRT ≥ 5 days, it is a Type A, if it is ≤ 5 days, it is Type C.
- Determining flow rate and levels of Total Suspended Solids (TSS) will distinguish between Type B and C freshwater bodies.
- Flow rate can be obtained from regulatory bodies. Low flow then needs to be calculated by plotting the cumulative distribution of the data and calculating the 25th percentile of the cumulative distribution.
- This can be calculated in Microsoft Excel using the cumulative distribution function. The 25th percentile (or quartile) can then be calculated.
- Waterbody types designated under national legislation should be used only if the following are met:
 - The classification includes area-based water quality monitoring.
 - Targets are used to determine good ecological status. An example is the Water Framework Directive.

Evidence of implementation should be maintained, e.g.:

- A map of the Receiving Waterbody, which outlines:
 - The area and average depth of the waterbody; and,
 - The location where flow rate (Q) was determined; or,
 - That the site is in a RW which is an exposed near-shore or off-shore marine RW (Type C only).
- The waterbody classification, which includes outputs from HRT, flow rate and TSS calculations.
 - The methods used to measure flow rate and TSS, in accordance with Appendix 8 (8.5 and 8.12).
 - If applicable, evidence that demonstrates alternative methodologies to calculate TSS or flow rate are equivalent to the ones in Appendix 8 (8.5).

Auditors should confirm:

- A map has been presented by the UoC which outlines the RW, and which supports the classification of the RW.
- The results of the HRT calculation and flow rate/TSS measurements conducted by the UoC.
 - These results will support the RW classification determined by the UoC.
- Evidence presented by the UoC, demonstrating the methods used to measure flow rate and TSS, in accordance with Appendix 8 (8.5 and 8.12).

Useful Resources**Steps to determine whether the RW is a HIE are as follows:**

1. If the site is in a waterbody that has been identified as a HIE by a regulatory body, this will form the boundaries of the RW.
2. Review the characteristics of the RW and determine if at least one of the following is identified:
 - The RW has poor connectivity to deep offshore waters, and/or;
 - The RW is sheltered from energetic conditions e.g., inflows, currents, exposure to onshore or monsoonal winds, and/or;
 - The hypolimnion is periodically or permanently poorly flushed.

Where the RW shares these characteristics, it is possible to demonstrate the RW is not a HIE. The steps to demonstrate that the RW is not a HIE are as follows:

1. Identify the narrowest point in the embayment opening (red line in Figure 1).
2. Establish two transects of length, which start at the narrowest point and extend in and out of the embayment (red dashed line in Figure 1).
3. Collect water samples at 0.5m depth from an equal number of equally spaced sites in each transect.
4. Analyse water samples for concentrations of chloride, conductivity, salinity, TN or TP. Analytical methods for TN and TP can be found in Appendix 8 (8.12).
5. Plot the concentrations of the parameter analysed in step 4 against their position on the transects.
6. Based on the results from step 5, determine whether there is a statistically significant increasing or decreasing trend in concentrations moving from the furthest offshore to the most enclosed site.

7. An embayment is hydrologically isolated when a statistically significant increasing or decreasing trend in concentrations has been determined.

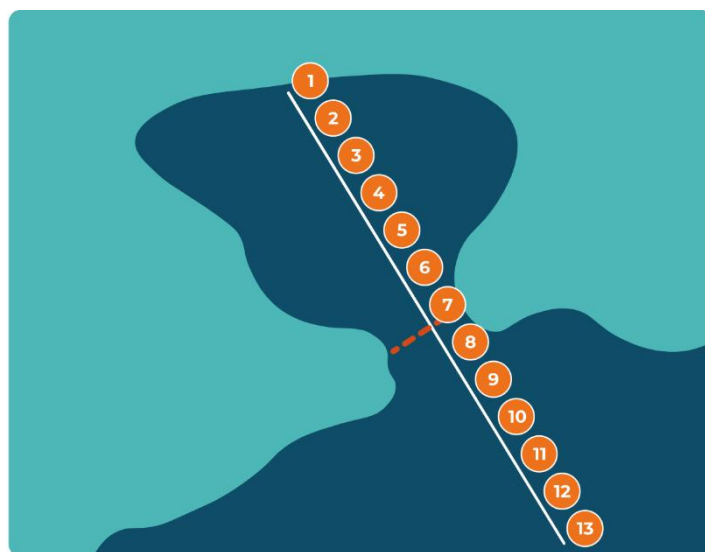


Figure 1. Tracer sampling points along a transect for HIE determination.

- If measurements of HRT or flow rates are not available, publicly accessible global databases can be used to obtain annual outflow estimates. Data sources include:
 - HydroAtlas, accessed November 22, 2023: <https://www.hydrosheds.org/hydroatlas>
 - G-RUN: Global Runoff Reconstruction, accessed November 2023: <https://doi.org/10.6084/m9.figshare.9228176>
- Google Earth Pro can be used to estimate the area and average depth of a waterbody to help calculate the HRT, accessed May 2024: [Earth Versions – Google Earth](#) and [Measure distance & elevation - Google Earth Help](#)
- Examples of credible secondary sources for flow rates include government flow monitoring and modelling, as well as publicly available databases, including:
 - WWF Risk Filter Suite, accessed November 22, 2023: <https://riskfilter.org/>
 - Water Risk Atlas, accessed November 22, 2023: <https://www.wri.org/aqueduct>
- Guides on how to calculate cumulative distribution (NORMDIST function) and the 25th percentile/quartile (QUARTILE function) for the flow rate data from secondary sources, using Microsoft Excel:
 - NORMDIST function, accessed November 27, 2024: [NORMDIST function - Microsoft Support](#)
 - QUARTILE function, accessed November 27, 2024: [QUARTILE function - Microsoft Support](#)

Indicator:	Requirement:
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2.6.6

The UoC shall define the **Waterbody Unit of Management (WUM)**, following the method outlined in **Appendix 8 (8.6)**.

An exception applies where a WUM is already defined, in which case the UoC shall provide evidence of the defined WUM.

How do I interpret this requirement?

- The following characteristics should be used to define the Waterbody Unit of Management (WUM):
 - The WUM reflects a logical geographic scope, for example, a lake and its catchment area.
 - The boundaries of the waterbody reflect the ability to realistically manage eutrophication risk within it.
 - Boundary setting should consider hydro-morphology, bathymetry, water movement and stratification characteristics.
 - Areas with high risk of adverse impacts due to cumulative effects, for example HIEs in larger waterbodies, should be a separate WUM.
 - Areas established for water quality monitoring at a national or regional level.
 - The Water Quality Advisory Committee (WQAC) can resolve issues with defining the WUM where the WUM boundaries are not clear.

Evidence of implementation should be maintained, e.g.:

- A map of the WUM which includes the components outlined in Appendix 8 (8.6).
 - The map should be updated on an annual basis where required.
- The results of measurements to confirm if the WUM is a HIE (if applicable).

Auditors should confirm:

- A map of the WUM has been developed and presented by the UoC, according to the method outlined in Appendix 8 (8.6).

Useful Resources N/A

Indicator:	Requirement:
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2.6.7

The UoC shall ensure that the person conducting water quality monitoring has the necessary competencies, experience and training.

The UoC shall ensure that the person defining the WUM has the necessary competencies. In cases where regulatory requirements are defined, the UoC shall ensure that the person defining the WUM complies with these requirements.

How do I interpret this requirement?

- To demonstrate competence, personnel responsible for conducting water quality analysis and defining the WUM should be able to:
 - Perform the required sampling and/or analysis.
 - Operate sampling/analytical equipment.
 - Develop, change, verify and validate methods.
 - Use suitable reference standards.
 - Analyse sampling and calibration results, including statements of conformity and interpretations.
 - Record, review and authorise the release of results.
 - Contribute to or produce reports that can clearly and concisely explain whether the results meet the requirements.
 - Have completed relevant training modules.

Evidence of implementation should be maintained, e.g.:

- The personnel responsible for conducting water quality analysis and defining the WUM have competencies to do so, for example a suitable level of education and/or experience.
 - Where applicable, the personnel meet national requirements/qualifications for the spatial definition and/or management of waterbodies.
- Relevant training modules have been completed by the personnel responsible for conducting water quality analysis and defining the WUM.

Auditors should confirm:

- The personnel responsible for conducting water quality analysis and defining the WUM have the necessary competencies.
 - Where applicable, national requirements/qualifications are met by the personnel defining the WUM.
- Evidence has been presented that relevant training modules have been completed.

Useful Resources

N/A

Indicator:	Requirement:
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2.6.8

The UoC shall establish a Water Quality Monitoring Network (WQMN), following the method outlined in **Appendix 8 (8.7)**.

An exception applies where a WQMN is already established, in which case the UoC shall participate in the WQMN.

How do I interpret this requirement?

- Evidence of spatial management at a national/regional level for water quality can be used by the UoC to demonstrate that the intent of Indicator 2.6.8 has been met. To use this evidence, the following should be met:
 - The boundaries of waterbodies are defined.
 - Waterbodies are managed to ensure water quality remains at a 'Good' or 'Very Good' status.
 - Where required, corrective actions are implemented by the regulator and complied with to ensure water quality status improves.
 - Examples of spatial management at a national/regional level include the European Union Water Framework Directive.
- It is the responsibility of the UoC to conduct outreach to ASC Certified sites and non-certified sites belong to ASC Certificate Holders if a Water Quality Monitoring Network (WQMN) has not been established.

- Examples of outreach can include:
 - Copies of correspondence (written and email).
 - Agendas and minutes of meetings.
 - Third-party stakeholder testimony.
- As per Appendix 8, section 8.7, a Terms of Reference (ToR) document shall be drafted and agreed upon by all members of the WQMN upon formation, which includes the following:
 - The membership list and commitments from members to meet the WQMN requirements, as outlined in Appendix 8 (8.7.1).
 - The assignment of a focal point, who will coordinate data sharing, the drafting of the ToR and agreement to the ToR by all members of the WQMN.
 - The focal point should ideally be a rotating position, though determination is left to discretion of the WQMN membership.
 - A commitment to review the ToR on a regular basis (for example every 12 months) or as and when required, such as when changes in membership of the WQMN occur, or when the waterbody demonstrates evidence of excessive nutrient loading, eutrophication or trophic status shifts.

Evidence of implementation should be maintained, e.g.:

- An agreed map of the WUM by members of the WQMN, which includes the components outlined in Appendix 8 (8.7).
 - The map should be updated on an annual basis where required.
- Evidence of outreach efforts and responses (where applicable) to ASC Certified site and non-certified sites belong to ASC Certificate Holders if a WQMN needs to be established.
- If a WQMN has been established, the UoC presents evidence that they have joined the WQMN or the UoC can demonstrate the spatial management of water quality at a national/regional level.
- A ToR has been agreed upon by all members of the WQMN, which includes commitments to the WQMN requirements, as outlined in Appendix 8 (8.7.1).

Auditors should confirm:

- A map of the WUM has been developed by the WQMN and presented by the UoC, which includes the components outlined in Appendix 8 (8.7).

- Evidence of outreach efforts and responses to ASC Certified site and non-certified sites belong to ASC Certificate Holders if a WQMN needs to be established.
 - Auditor discretion should be used to determine if sufficient levels of outreach has been conducted, when such outreach has not been reciprocated.
- Evidence that the UoC has formed/joined a WQMN, or that the UoC presents evidence that water quality is managed at a national/regional level.
- A ToR has been presented, which has been agreed upon by all members of the WQMN. The ToR includes commitments to the WQMN requirements, as outlined in Appendix 8 (8.7.1).

Useful Resources

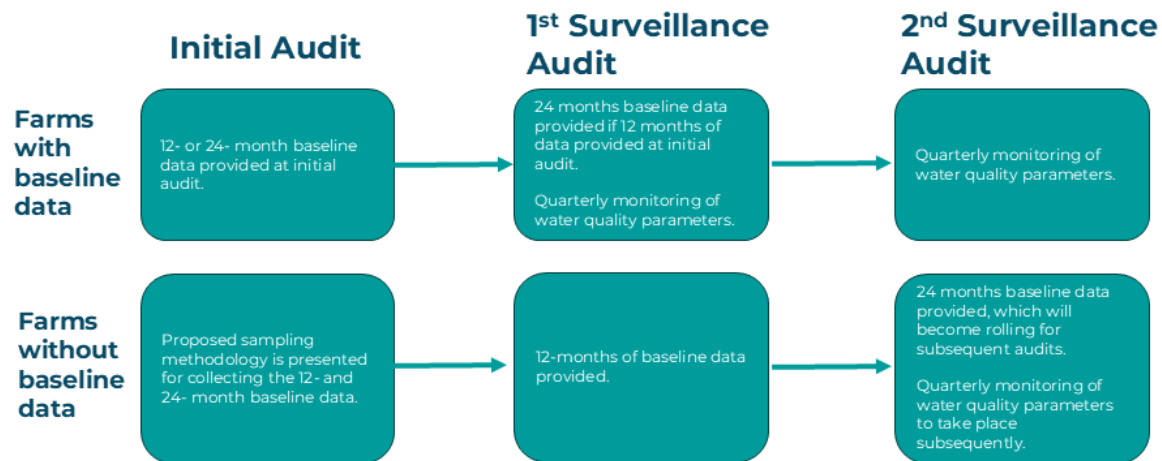
N/A

Indicator:	Requirement:
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2.6.9 The UoC shall conduct a water quality survey and determine the water quality baseline, including the trophic status, following the method outlined in **Appendix 8 (8.8)**.

How do I interpret this requirement?

- The data collected as part of the 24-month site level monitoring survey is submitted by the UoC to the ASC Water Quality Calculator.
- The water quality baseline is based on 24-months of monitoring data from the site level monitoring survey. However, the following phased approach is permitted when sites do not have sufficient data prior to the initial audit:



- The water quality baseline is characterised by the ASC Water Quality Calculator, in accordance with Appendix 8 (8.8).
- The baseline should be conducted per site and per waterbody. The results should then be shared with members of the WQMN.
- The baseline determines whether primary production in the waterbody is limited by nitrogen (N), phosphorous (P), or if it is co-limited. Other factors (than N or P) may be limiting primary productivity in a waterbody.
 - The UoC should present evidence of these other limiting factors (e.g. temperature or altitude) where applicable.
- Harmful algal blooms (HABs) and Adverse Turnover Events (ATEs) are required to be reported as part of the baseline characterisation. The number of ATEs and HABs that have occurred can be recorded by conducting the following:
 - Reviewing news media and consulting local stakeholders. Alternatively, reviewing data collected by regulatory bodies where available.
 - Reporting the number of ATEs and HABs which have resulted in fish mortality events (farmed or wild) or the mortality of other aquatic species.
 - Reporting instances where fish mortality events or loss of other aquatic species can be attributed to natural phenomena. These instances will not count towards the frequency limits for ATEs and HABs.
 - Reporting the number of ATEs and HABs, as determined by national or regional regulatory/scientific organisations.

Evidence of implementation should be maintained, e.g.:

- The methodology and analytical methods used to conduct site level monitoring survey are in accordance with Appendix 8 (8.8).
- The sampling methodology for collecting 24-months of data for the site level monitoring survey, if this is not available at the initial audit.
- The results of the site level monitoring survey, as determined by the ASC Water Quality Calculator.
- The results of the 24-month baseline characterisation, as determined by the ASC Water Quality Calculator.
- Records of HABs and ATEs from the last 10 years.

Auditors should confirm:

- A water quality monitoring survey has been conducted by the UoC. Results are presented from the ASC Water Quality Calculator.
- A sampling methodology has been presented for collecting 24-months of data for the site level monitoring survey, if this is not available at the initial audit.
- The data collected for the site level monitoring survey followed the methodology and analytical measures outlined in Appendix 8 (8.8).
- A water quality baseline has been characterised. Results are presented from the ASC Water Quality Calculator.
- The results of the review by the UoC into adverse turnover events and harmful algal blooms in the waterbody.

Useful Resources

- Vollenweider, R.A., Rinaldi, A. and Montanari, G. (1992) Eutrophication, structure and dynamics of a marine coastal system: results of ten-year monitoring along the Emilia-Romagna Coast (Northwest Adriatic Sea). *Marine Coastal Eutrophication, Proceedings of an International Conference, Bologna, Italy, 21–24 March 1990, Pages 63-106*: <https://doi.org/10.1016/B978-0-444-89990-3.50014-6>
- More information about the equations used for TSI assessment, including the relevant peer-reviewed scientific literature, is provided in Carlson, R.E. and Simpson, J. (1996) A Coordinator's Guide to Volunteer Lake Monitoring Methods. *North American Lake Management Society*, 96 p.

Indicator:	Requirement:
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2.6.10 The UoC shall, after the determination of the water quality baseline, conduct monitoring of the water quality parameters listed below, following the method outlined in **Appendix 8 (8.8)**¹³:

- Total Nitrogen (TN); and,
- Total Phosphorous (TP); and,
- Dissolved Oxygen (DO); and,
- Secchi Disk (SD) depth; and,
- Chlorophyll-a (Chl-a).

How do I interpret this requirement?

- Quarterly monitoring ensures the UoC can capture any seasonal or inter-annual variations in the data collected.
- Measurements should be timed to coincide with, or immediately follow, annual turnover events when the waterbody is not permanently stratified (holomictic).
- The UoC should calibrate equipment used to measure DO, TN, TP, SD and Chl-a according to the technical specifications of the equipment.
- The UoC should conduct measurements of the parameters in-situ.
 - Samples may be sent for laboratory analysis, however variability with sample maintenance during transit can lead to inconsistent results. This is particularly evident with DO and Chl-a samples.

Evidence of implementation should be maintained, e.g.:

- Documentation outlining the sampling plan to conduct quarterly monitoring and that is in accordance with Appendix 8 (8.8).
- Quarterly monitoring data has been collected and submitted to the ASC Water Quality Calculator. Data should be submitted at least annually.

¹³ An exemption applies if the mean of Secchi Disk (SD) depth calculated over any rolling two-year period remains $\geq 10\text{m}$.

- Evidence that quarterly monitoring data has been timed with annual turnover events (where applicable).
- That equipment to measure DO, TN, TP, SD and Chl-a has been calibrated according to technical specifications.

Auditors should confirm:

- The sampling methodology to conduct quarterly monitoring is in accordance with Appendix 8 (8.8).
 - The UoC has conducted quarterly monitoring of the water quality parameters and submitted data to the ASC Water Quality Calculator at least annually.
 - Evidence has been provided demonstrating the UoC has calibrated equipment to measure DO, TN, TP, SD and Chl-a in accordance with the technical specifications.
-

Useful Resources

- Paerl, H.W., Scott, J.T., McCarthy, M.J., Newell, S.E., Gardner, W.S., Havens, K.E., Hoffman, D.K., Wilhelm, S.W. and Wurtsbaugh, W.A., 2016. It takes two to tango: when and where dual nutrient (N & P) reductions are needed to protect lakes and downstream ecosystems. *Environmental science & technology*, 50(20), pp.10805-10813.
- Information on the appropriate use of Secchi Disks can be found here: 'What is a Secchi disk', accessed November 22, 2023: <https://www.nalms.org/secchidipin/monitoring-methods/the-secchi-disk/what-is-a-secchi-disk/>
- Example of a method for Chl-a measurements made using ethanol extraction followed by spectrophotometric measurement at wavelengths 665 and 750nm (using accredited methods: EPA method 445.0: Method 445.0 In Vitro Determination of Chlorophyll a and Pheophytin in Marine and Freshwater Algae by Fluorescence, accessed November 22, 2023: https://cfpub.epa.gov/si/si_public_record_report.cfm?Lab=NERL&dirEntryId=309417
- Cost effective example for measuring using in-situ Chl-a using a field spectrofluorometer include the FluoroSense handheld fluorometer manufactured by Turner Designs and the YSI handheld meter with EXO1 (#599080) or EXO2 (#599201) sondes.
- Hamdhani, H., Eppehimer, D.E., Walker, D. and Bogan, M.T. (2021) Performance of a Handheld Chlorophyll-a Fluorometer: Potential Use for Rapid Algae Monitoring. *Water*, 13, 1409: <https://doi.org/10.3390/w13101409>
- Kuha, J., Järvinen, M., Salmi, P. and Karjalainen, J. (2020). Calibration of in situ chlorophyll fluorometers for organic matter. *Restoration of Eutrophic Lakes*, 847, 4377-4387: <https://doi.org/10.1007/s10750-019-04086-z>

Indicator:	Requirement:
2.6.11	The UoC shall, based on the monitoring of the water quality parameters, ensure that there is no decline in trophic status compared to the water quality baseline determined, following the method outlined in Appendix 8 (8.8) .
2.6.12	The UoC shall, based on the monitoring of the water quality parameters, ensure that there is no upward rate of change >25% over the previous 24 months, for any of the water quality parameters listed below, following the method outlined in Appendix 8 (8.8): ○ TN; and, ○ TP; and, ○ Chl-a; and, ○ Biological Oxygen Demand (BOD).
2.6.13	The UoC shall implement a nutrient reduction plan when at least one of the following is identified: ○ TN, TP or Chl-a has increased by ≥ 5 index points according to Table 2 in Appendix 8 (8.8), compared to the water quality baseline; or, ○ TN, TP or Chl-a increased by $\geq 15\%$ over the past 24 months; or, ○ The distance from the surface to the zone of oxygen depletion or to the zone of anoxia has decreased by $\geq 25\%$ over the past 24 months; or, ○ BOD has increased by $\geq 15\%$ over the past 24 months; or, ○ ≥ 1 adverse turnover event or ≥ 1 harmful algal bloom within the past 10 years.

How do I interpret this requirement?

- Quarterly monitoring of TN, TP and Chl-a will determine whether there has been a change in the status of the limiting nutrients or a decline in trophic status. BOD will also need to be calculated annually. The calculated data will be compared to the water quality baseline.
- Quarterly monitoring data will be submitted to the ASC Water Quality Calculator at least annually. The ASC Water Quality Calculator will determine rates of change in the limiting nutrient(s) and trophic status compared to the baseline.
- A decline in trophic status will be determined when the next Trophic Status Index (TSI) breakpoint has been exceeded, in accordance with Table 2 in Appendix 8 (8.8).

- Exceedance of the next TSI breakpoint will occur when one parameter falls into the next TSI category.
- The ASC Water Quality Calculator also determines whether one of the scenarios in Indicator 2.6.13 has occurred based on the quarterly monitoring data.
 - If one of scenarios has occurred, an action plan needs to be implemented to reduce nutrients released by the site and/or to have improved the efficiency of the feed used.
 - The action plan should be specific, measurable, achievable, relevant and time bound.

Evidence of implementation should be maintained, e.g.:

- Monitoring data has been submitted annually to the ASC Water Quality Calculator and results have been obtained.
- Results will determine the trophic status and status of TN, TP, Chl-a and BOD, in accordance with Indicator 2.6.12. Results will also be compared to the water quality baseline after the first annual submission.
 - Any upward transitions in trophic status at the compared to the water quality baseline will be determined based on the results from the ASC Water Quality Calculator.
 - Results from the ASC Water Quality will determine rate of change for TN, TP, Chl-a and BOD compared to the water quality baseline, specifically whether this exceeds 25% over the previous 24 months.
- Results from the ASC Water Quality Calculator will be obtained to determine whether one of the scenarios in Indicator 2.6.13 has occurred.
 - This will be based on the TN, TP, Chl-a and BOD data submitted to the ASC Water Quality Calculator.
 - It will also be based on submissions of the number of HABs and ATEs to the ASC Water Quality Calculator.
 - If one of the scenarios has occurred, an action plan will have been documented which will reduce nutrient nutrients released by the site and/or to have improved the efficiency of the feed used.

Auditors should confirm:

- The UoC presents evidence demonstrating monitoring data has been submitted annually to the ASC Water Quality Calculator.

- The UoC presents the outputs from the ASC Water Quality Calculator, which demonstrates whether there has been a decline in trophic status or upward rate of change >25% for TN, TP, Chl-a and BOD.
- The UoC presents evidence which determines whether one or more of the scenarios in Indicator 2.6.13 have occurred.
- A specific, measurable, achievable, relevant and time bound action plan has been documented by the UoC, and has been presented, if one of the scenarios in Indicator 2.6.13 has occurred.

Useful Resources

- Trophic status is determined using the Carlson Index values (0-100), with 'breakpoints' set at minimum intervals of 10 index points.
 - Carlson's Index is used to classify Type A waterbodies and consists of four separate normalised indices: Chlorophyll-a, Total Phosphorous, Total Nitrogen and Secchi Disk transparency measurements.
 - Normalised TSI values range from 0 (ultra-oligotrophic) to 100 (hypereutrophic). A TSI of less than 40 is indicative of oligotrophic conditions, while TSI values greater than 50 are indicative of eutrophic conditions. TSI values between 40 and 50 are indicative of mesotrophic conditions.
- More information about the equations used for TSI assessment, including the relevant peer-reviewed scientific literature, is provided in Carlson, R.E. and Simpson, J. (1996) A Coordinator's Guide to Volunteer Lake Monitoring Methods. *North American Lake Management Society*, 96 p.

Indicator:	Requirement:
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2.6.14 The UoC shall report annually to ASC the WUM hydraulic characteristics, according to ASC data submission procedures.

2.6.15 The UoC shall report annually to ASC the water quality monitoring results and the BOD value, according to ASC data submission procedures.

How do I interpret this requirement?

- ASC data collection spreadsheets will be completed with data used to characterise the baseline of the RW, in accordance with Appendix 8 (8.8.4).

- The data reported will facilitate the implementation of the requirements for sites located in Type A waterbodies.
- Auditors will have access to the reported data to support with the auditing process.

Evidence of implementation should be maintained, e.g.:

- Complete data collection spreadsheets and results from the ASC Water Quality Calculator that confirm data has been submitted.

Auditors should confirm:

- Data has been submitted by the UoC and results have been obtained from the ASC Water Quality Calculator.

Useful Resources

N/A

Indicator:	Requirement:
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2.6.16 The UoC shall use a treatment system prior to the discharge of effluent water that ensures:

- ≥65% of **Total Suspended Solids** (TSS) are captured; and,
- concentration of **settleable solids** in effluent water is <3.3ml/L.

How do I interpret this requirement?

- Methods to measure Total Suspended Solids (TSS) can be found in Appendix 8 (8.12).
- Concentration of settleable solids (ml/L) can be measured by calculating the volume of solids that settles to the bottom of a conical cone (Imhoff cone) in one hour.
- More sophisticated methods such as spectrophotometer readings can also be used to measure settleable solids.

Evidence of implementation should be maintained, e.g.:

- Details and location of the system used to treat effluent prior to discharge.

- Evidence regular testing of TSS and settleable solids has taken place. Quarterly testing would demonstrate best practice.
- Results from the testing of point source releases which demonstrate $\geq 65\%$ of TSS are captured and the concentration of settleable solids in effluent water is consistently $< 3.3\text{ml/L}$.

Auditors should confirm:

- A treatment is in place and located prior to the point of discharge.
 - Results of testing from the UoC which confirm the treatment system captures $\geq 65\%$ of suspended solids and that the concentration of settleable solids in effluent water is $< 3.3\text{ml/L}$.
-

Useful Resources

N/A

Indicator:	Requirement:
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2.6.17 *Indicator scope: waterbody Type B*

The UoC shall conduct monitoring of TN, TP and TSS upstream from the inflow point and downstream from the outflow point, following the method outlined in **Appendix 8 (8.9)**, if one of the following conditions applies:

- The site effluent flow contribution to the receiving water flow is $> 10\%$, as calculated following the method outlined in **Appendix 8 (8.10)**; or,
- The site effluent enters a river or channel $< 1\text{km}$ upstream of a Type A waterbody.

2.6.18 *Indicator scope: waterbody Type B*

The UoC shall, if Indicator 2.6.17 applies, ensure that there is $< 25\%$ increase in TN, TP and TSS upstream from the inflow point and downstream from the outflow point.

How do I interpret this requirement?

- The methods for measuring TN, TP and TSS can be found under Appendix 8 (8.12).
- The UoC can measure flow contribution, TN, TP and TSS upstream from the inflow point of the first site and immediately downstream of the outflow point of the last site in a waterbody.

- This may reduce sampling costs when there are multiple sites discharging point source effluent releases to a Type B waterbody near each other.

Evidence of implementation should be maintained, e.g.:

- Maps demonstrating the sampling locations are upstream from the inflow point and downstream from the outflow point.
 - Annual calculation of site effluent flow contribution to the receiving water flow based on the methodology in Appendix 8 (8.10).
 - Results of the site effluent flow contribution calculation, which demonstrate whether the contribution is greater than or less than 10%.
 - If Indicator 2.6.17 applies, results of measurements of TN, TP and TSS upstream from the inflow point and downstream from the outflow point.
 - Results of calculations to demonstrate there is <25% increase in site effluent flow between upstream and downstream sample sites.
-

Auditors should confirm:

- A map has been presented which demonstrates the sampling locations are upstream from the inflow point and downstream from the outflow point.
 - The UoC has conducted an annual calculation of the site effluent flow contribution to the receiving water flow.
 - Whether the site effluent flow contribution calculation presented by the UoC exceeds 10%.
 - If Indicator 2.6.17 applies, the UoC has presented quarterly results of TN, TP and TSS measurements from sampling locations upstream from the inflow point and downstream from the outflow point, whilst still following the sampling method in Appendix 8 (8.8 and 8.9).
 - Results of calculations presented to demonstrate there is <25% increase in site effluent flow between upstream and downstream sample sites.
-

Useful Resources

N/A

Indicator:	Requirement:
2.6.19	The UoC shall conduct monitoring of DO concentration and DO saturation, following the method outlined in Appendix 8 (8.11) .
2.6.20	The UoC shall demonstrate that the weekly average of daily DO saturation is $\geq 65\%$ in freshwater and $\geq 70\%$ in seawater ¹⁴ .
2.6.21	The UoC shall demonstrate that $\leq 5\%$ of the weekly averages of daily DO concentrations are $< 2\text{mg/l}$. An exception applies if the UoC cannot demonstrate this, in which case the UoC shall: ○ Demonstrate $\geq 70\%$ saturation by conducting continuous monitoring using an electronic probe and recorder with a 15-minute sampling frequency for at least one week; or, ○ Ensure percentage saturation results are consistent with results from an 'upstream, near-field' non-impacted reference location, at least 500m upstream of the site.

How do I interpret this requirement?

- For both diffuse-source and point-source effluent releases, the UoC should monitor and record percent saturation of DO.
- All measurements should be taken using a calibrated oxygen meter or equivalent method.
- The percent saturations should then be calculated based on the monitoring data.
- The analytical methods for DO sampling can be found in Appendix 8 (8.12).
- Any missed samples or deviations in sampling time should be accompanied by a written justification.
- The UoC can measure DO concentration and saturation immediately downstream of the last site in a waterbody whilst still following the sampling method in Appendix 8 (8.11).
 - This may reduce sampling costs when there are multiple sites discharging point source effluent releases to a Type B waterbody near each other.

¹⁴Waterbodies with salinity levels 30-49 ppt (parts per thousand) shall be considered as seawater for this Indicator.

Evidence of implementation should be maintained, e.g.:

- The oxygen meter has been calibrated according to the technical specifications.
- Records of measurement locations, demonstrating monitoring took place immediately downstream from the outflow point, or in the receiving waterbody downstream from the site for diffuse source releases.
- Records of daily monitoring of DO concentration and saturation, which result in weekly averages of daily DO saturation.
- $\leq 5\%$ of the weekly averages of daily DO concentrations are 2mg/l.
- Written justifications are maintained for any missed samples or deviations in sampling times.

Auditors should confirm:

- The UoC has calibrated the oxygen meter according to the technical specifications.
- Measurement locations were immediately downstream from the outflow point, or in the receiving waterbody downstream from the site for diffuse source releases.
- Weekly averages of daily DO saturation are presented by the UoC and that the averages are $\geq 65\%$ for freshwater and $\geq 70\%$ for seawater.
- The UoC has presented evidence that $\leq 5\%$ of weekly samples fall under 2 mg/L DO.
- Written justifications are provided for any missed samples or deviations in sampling times.

Useful Resources

N/A

Indicator:	Requirement:
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2.6.22 *Indicator scope: point source systems in waterbody types A and B*

The UoC shall conduct DO monitoring downstream from the outflow point and shall demonstrate that daily diurnal DO (DDDO) fluctuation is $\leq 65\%$ saturation, following the method outlined in **Appendix 8 (8.11)**.

How do I interpret this requirement?

- For UoCs located in tidal systems, timing of measurements should correspond with high and low tides, to reflect variations related to the tidal regime.
- The UoC can conduct DO monitoring immediately downstream of the last site in a waterbody whilst still following the sampling method in Appendix 8 (8.11).
 - This may reduce sampling costs when there are multiple sites discharging point source effluent releases to a Type B waterbody near each other.
- The methods outlined in Appendix (8.11.2 and 8.11.3) shall be used to record DO concentrations used to determine daily diurnal DO (DDDO).
- Any missed samples or deviations in sampling time should be accompanied by a written justification.

Evidence of implementation should be maintained, e.g.:

- Records that DO measurements correspond to high and low tides in tidal systems.
- The location of the sampling point, demonstrating that measurements have taken place downstream from the outflow point.
- Results of daily DO monitoring and DDDO fluctuation calculations.

Auditors should confirm:

- DO measurements have taken place at high and low tides in tidal systems.
- DO measurements have taken place downstream from the outflow point.
- The UoC has presented results of the DDDO calculations based on DO measurements collected in accordance with the methods outlined in Appendix 8.11.1 and 8.11.2.
- Confirm the fluctuation level based on the DDDO calculation is $\leq 65\%$ saturation level.

- Written justifications were provided if there were any missed samples or deviations in the sampling time.

Useful Resources

N/A

Criterion 2.7 - Salinisation

Indicator:	Requirement:
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2.7.1 *Indicator applicability: brackish or saltwater pond culture*

The UoC shall¹⁵ use low permeable liners.

How do I interpret this requirement?

- Brackish water has a salinity of 0.5-29.9 ppt, while seawater has a salinity of > 29.9 ppt.
- Low permeable liners include liners made from:
 - Plastic,
 - Imported clays,
 - Ripping and re-compaction of in situ clays, mixing with bentonite,
 - Geo-membranes, and
 - Composite liners.

Evidence of implementation should be maintained, e.g.:

- Records demonstrating the installation and maintenance of low permeable liners in ponds.

Auditors should confirm:

- Low permeable liners have been installed and maintained in ponds.

Useful Resources

N/A

¹⁵ An exemption applies where ponds are located in naturally brackish or saline environments.

Indicator:	Requirement:
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2.7.2 The UoC shall not discharge brackish or saltwater effluent over land.

How do I interpret this requirement?

N/A

Evidence of implementation should be maintained, e.g.:

- The location and/or route of the discharge of brackish or saltwater effluent to demonstrate discharge does not take place over land.
 - Examples of evidence may include maps/plans of the discharge locations/routes.

Auditors should confirm:

- Evidence has been presented, such as maps/plans, to demonstrate brackish or saltwater effluent is not discharged over land.

Useful Resources

N/A

Indicator:	Requirement:
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2.7.3 The UoC shall only discharge effluent water of a salinity level equal to or lower than the salinity level of the receiving waterbody or natural wetland. An exception applies to waterbodies and natural wetlands that have naturally periodic varying salinity levels, in which case the salinity level of the effluent water shall be within range of the natural variation.

How do I interpret this requirement?

- Salinity levels are: Freshwater (limnetic; <0.5 ppt), slightly saline (oligohaline; 0.5-4.9 ppt), moderate saline (mesohaline; 5-17.9 ppt), highly saline (polyhaline; 18-29.9 ppt) and seawater (euhaline; 30-34.9 ppt).

- River estuaries are an example of a waterbody that have periodic varying salinity levels. These can also be referred to as poikilohaline waterbodies.
- Salinity testing of effluent water in the receiving waterbody or natural wetlands should be conducted.
 - Salinity levels in the receiving waterbody or natural wetlands may be obtained from credible sources, e.g. results of government monitoring.
- To comply with the exception in the indicator, salinity testing in the waterbody or natural wetland should take place. Alternatively, test results can be obtained from credible sources, e.g. scientific or government body.
 - Results should then be compared to the salinity level of the effluent water to determine whether it is within the range of natural variation for the receiving waterbody or natural wetlands.

Evidence of implementation should be maintained, e.g.:

- Results of salinity testing for effluent water and in the receiving waterbody or natural wetlands.
- If applicable, reports or measurements determining whether waterbodies and natural wetlands have naturally varying salinity levels.
 - Reports or measurements determining whether the salinity level of the effluent water is within the range of the natural variation for the receiving waterbody or natural wetlands.

Auditors should confirm:

- Salinity test results have been presented which determine one of the following:
 - Salinity level of the effluent water is equal to or lower than the salinity level of the receiving waterbody or natural wetlands.
 - Salinity level of the effluent water is within the range of natural variation for the receiving waterbody or natural wetlands that have naturally varying salinity levels.

Useful Resources

N/A

Indicator:	Requirement:
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2.7.4

The UoC shall ensure that freshwater-wells used for culture water do not exceed a conductivity of 1,500 micro ohms/cm or chloride concentration of 300mg/L.

An exception applies where well-monitoring is not permitted by applicable law, in which case regulatory records shall be obtained to provide evidence of conductivity levels.

How do I interpret this requirement?

- Freshwater wells with a conductivity or chloride concentration above the thresholds in Indicator 2.7.4 will demonstrate that salinisation has taken place.
- Specific conductance or chloride concentration should be monitored at a frequency adapted to fluctuations due to natural factors, such as seasonal rains.

Evidence of implementation should be maintained, e.g.:

- Records of specific conductance of freshwater wells on, or in proximity to the site (within 1km radius).
 - Records should demonstrate monitoring has taken place at least every six months and considered natural factors, such as seasonal rains.
- Regulatory records obtained where applicable, to demonstrate well-monitoring is not permitted by law.
 - Records shall contain evidence of conductivity testing.

Auditors should confirm:

- Monitoring records are available for freshwater wells in proximity to the site, which demonstrate either conductivity or chloride concentration is below the relevant threshold specified in Indicator 2.7.4.
 - Monitoring has taken place at least every six months and considered natural factors, such as seasonal rains.
- Regulatory records are available for instances where well-monitoring is not permitted by law.
- The records demonstrate conductivity testing has taken place.

Useful Resources

- Hand-held refractometers can be used to measure salinity. These devices are appropriate for salinities of approximately 2 or 3 ppt, but they are not sensitive enough to determine if sites are causing the salinisation of freshwater bodies.
- The quickest and easiest method to evaluate salinity of water is to measure specific conductance with a conductivity meter. An alternative method is a chloride test kit, which is typically cheaper.

Criterion 2.8 - Biosolids

Indicator:	Requirement:
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2.8.1 The UoC shall ensure that **biosolids** are appropriately contained.

How do I interpret this requirement?

- The UoC should develop and implement a biosolids management plan, which includes procedures relating to the containment of biosolids.
- The appropriate containment of biosolids should consider the risk of runoff during anticipated and/or recurring extreme weather events, such as tropical storms or flooding.
 - This risk is higher when biosolids are removed from culture systems, canals and/or treatment systems.

Evidence of implementation should be maintained, e.g.:

- Documents demonstrating the appropriate containment of biosolids. An example may be a biosolids management plan.
 - The management plan may include a flow diagram that outlines key steps for the containment of biosolids.

Auditors should confirm:

- Documents, for example a biosolids management plan, have been presented which demonstrate the appropriate containment of biosolids.
- Conformance should be confirmed through a visual inspection of the site.

Useful Resources

N/A

Indicator:	Requirement:
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2.8.2 The UoC shall ensure that biosolids are not disposed or released into receiving waterbodies, natural wetlands or other ecosystems.

How do I interpret this requirement?

- This requirement will mitigate impacts on water quality and the risk of salinisation, therefore supporting the implementation of Criterion 2.6 and 2.7 respectively.

Evidence of implementation should be maintained, e.g.:

- Records which demonstrate:
 - Information on the disposal and/or release of biosolids.
 - The location where biosolids are disposed and/or released, which confirm they are not disposed of and/or released into receiving waterbodies, natural wetlands or other ecosystems.

Auditors should confirm:

- Records have been presented which demonstrate biosolids are not disposed of and/or released into receiving waterbodies, natural wetlands or other ecosystems.
- There is no evidence of the discharge and/or release of biosolids into receiving waterbodies, natural wetlands or other ecosystems during the audit.
 - This may be carried out through a visual inspection of the site and any surrounding waterbodies.

Useful Resources

N/A

Indicator:	Requirement:
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2.8.3 The UoC shall only re-use uncontaminated biosolids.

How do I interpret this requirement?

- The following outlines examples of the purposes for uncontaminated biosolids re-use:
 - Fertiliser in agriculture.
 - Maintenance and building of dykes.
 - Maintenance of roads or infrastructure.
 - Biogas.

Evidence of implementation should be maintained, e.g.:

- Documents demonstrating the appropriate treatment and re-use of biosolids.
 - An example may be a biosolids management plan.
 - The management plan may include a flow diagram that outlines key steps for the treatment of biosolids prior to re-use to ensure they are uncontaminated.

Auditors should confirm:

- Documents, for example a biosolids management plan, have been presented which demonstrate only uncontaminated biosolids are re-used.

Useful Resources

N/A

Indicator:	Requirement:
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2.8.4 The UoC shall dispose of biosolids responsibly if they cannot be re-used.

How do I interpret this requirement?

- The responsible disposal of biosolids mitigates impacts on natural ecosystems and waterbodies.
 - Impacts may include salinisation or eutrophication.
- The UoC should develop and implement a biosolids management plan which procedures relating to the responsible disposal of biosolids.

Evidence of implementation should be maintained, e.g.:

- Documents outlining the process and location of biosolids disposal.
 - The documents should demonstrate biosolids are disposed of responsibly. An example may be a biosolids management plan.
 - The management plan may include a flow diagram that outlines key steps for the disposal of biosolids.
- If biosolids are transferred to a third party, evidence should be available to confirm that the third party has disposed of biosolids responsibly.

Auditors should confirm:

- Documents, for example a biosolids management plan, have been presented which demonstrate the responsible disposal of biosolids.
- If biosolids are transferred to a third-party, evidence has been presented to confirm that the third-party has responsibly disposed of biosolids.

Useful Resources

N/A

Criterion 2.9 - Freshwater Use

Indicator:	Requirement:
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2.9.1 The UoC shall develop, implement and maintain a Water Efficiency Management Plan (WEMP) in areas identified as having 'high' or 'extremely high' water stress according to the Aqueduct 'Water Risk Atlas'¹⁶, with the intent to reduce water consumption.

2.9.2 The UoC shall ensure the WEMP is annually reviewed and revised as necessary.

How do I interpret this requirement?

- The Aqueduct 'Water Risk Atlas' measures all water-related risks. Higher values indicate higher water risk.
 - 'Water Stress' is an indicator under 'Physical Risks Quantity' and measures the ratio of total water demand to available renewable surface and groundwater supplies.
 - The site needs to check whether its location is classified as being in an area of 'high' or 'extremely high' water stress.

Evidence of implementation should be maintained, e.g.:

- The 'Water Stress' classification of the area in which the site is located. This must match the coordinates of where the site is located.
- A Water Efficiency Management Plan (WEMP) has been developed, implemented and maintained if the site is in an area identified as having 'high' or 'extremely high' water stress.
 - The WEMP should have specific, measurable, achievable, realistic and time-bound objectives and targets to reduce water consumption.
- Documents demonstrating the WEMP has been reviewed annually, to determine whether it is meeting the objective of reducing water consumption.
- Data or documents to demonstrate the WEMP is effective at reducing water consumption.

¹⁶ https://www.wri.org/applications/aqueduct/water-risk-atlas/#/?advanced=false&basemap=hydro&indicator=w_awr_def_tot_cat&lat=-14.445396942837744&lng=-142.85354599620152&mapMode=view&month=1&opacity=0.5&ponderation=DEF&predefined=false&projection=absolute&scenario=optimistic&scope=baseline&timeScale=annual&year=baseline&zoom=2

- Revisions to the WEMP if it has not been deemed effective at reducing water consumption.

Auditors should confirm:

- The risk of water stress according the 'Water Risk Atlas'.
 - Evidence that a WEMP has been developed, implemented and maintained, if the site is in an area identified as having 'high' or 'extremely high' water stress.
 - Objectives and targets are in place in the WEMP which aim to reduce water consumption.
 - Evidence of an annual review of the WEMP.
 - Evidence has been presented to demonstrate the WEMP is effective at reducing water consumption.
 - Revisions to the WEMP have been made if the WEMP has not been effective at reducing water consumption.
-

Useful Resources

- Aqueduct Water Risk Atlas: [Aqueduct Water Risk Atlas](#).

Indicator:	Requirement:
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2.9.3

The UoC shall monitor and record water levels in groundwater wells, and shall include the results in the WEMP.

An exception applies where monitoring is not permitted by applicable law, in which case regulatory records shall be obtained to provide evidence of recording of water levels.

How do I interpret this requirement?

- Water levels in groundwater wells should be tested and recorded at similar times during the year.
- Monitoring of water levels in groundwater wells determines whether water is being replenished in the ground. When water abstraction increases beyond the rate of replenishment, the result is a reduction in the water table.
 - A reduction in the water table would demonstrate that the WEMP is not effective at reducing water consumption.

Evidence of implementation should be maintained, e.g.:

- Results from the monitoring of water levels in groundwater wells.
- Documents demonstrating the monitoring results have been included in, and informed revisions of, the WEMP (where required).
- Regulatory records which provide evidence of the recording of water levels, if monitoring is not permitted by law.

Auditors should confirm:

- Monitoring results have been presented which demonstrate the recording of water levels in groundwater wells.
 - Results can either be from testing conducted by the site or a regulatory body if monitoring is not permitted by law.
- Monitoring results have informed the drafting or revision of the WEMP.

Useful Resources

N/A

Indicator:	Requirement:
2.9.4	The UoC shall not use freshwater to reduce the salinity of culture water. An exception applies if discharged effluent water is desalinated.
2.9.5	The UoC shall not use salt to increase the salinity of fresh culture water. An exception applies if discharged effluent water is desalinated.

No additional guidance is provided for these indicators.

Indicator:	Requirement:
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2.9.6 The UoC shall not abstract >50% volume of surface water. An exception applies if the UoC can demonstrate that the water abstraction is based on a minimum **Vital Flow** determined by a regulatory permit or scientific study.

2.9.7 The UoC shall return >90% of the abstracted surface water volume to its source.

How do I interpret this requirement?

- Abstraction of water refers to the process of taking water from a natural source, in this case surface water.
- Responsible abstraction, or a minimum vital flow, is important for mitigating impacts on freshwater ecosystems and water quality.

Evidence of implementation should be maintained, e.g.:

- Records demonstrating the volume of water abstracted from surface water and the volume returned to the source from which the water was originally abstracted.
- A minimum vital flow has been determined by a regulatory permit or scientific study, if the exception in Indicator 2.9.6 applies.

Auditors should confirm:

- Records have been presented which demonstrate the volume of water extracted from surface water and the volume returned to the source from which the water was originally extracted.
- Evidence of a minimum vital flow determined by a regulatory permit or scientific body, if the exception in Indicator 2.9.6 applies.

Useful Resources N/A

Criterion 2.10 - Energy Use and Greenhouse Gas (GHG) Emissions

Indicator:	Requirement:
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2.10.1 The UoC shall calculate the energy use of production following the method outlined in **Appendix 9 (9.3)**.

How do I interpret this requirement?

- Tracking use of energy inputs is a necessary step in calculating the GHG emissions associated with aquaculture production and products.
 - Energy inputs include all those sources of energy used for the purpose of production, including electricity and energy carriers such as diesel or natural gas.
- To the extent possible, Certificate Holders are encouraged to calculate energy use specific to the certified site. However, in some cases, this may be difficult to do. For example, fuel purchased may be used to supply vessels servicing many sites, potentially both certified and non-certified sites or even multiple sites or companies.
 - In these cases, a portion of the overall input may be allocated to the certified site based on a logical allocation basis.
 - For example, dividing energy inputs between sites based on the relative number of vessel trips or using the site's relative share of production among the group of sites serviced.
- Energy inputs to hatcheries are not included in calculations of on-site energy use. However, in cases where the juvenile stage of the supply chain is integrated with grow-out, sites may choose to include this stage in their calculations to simplify calculations.

Evidence of implementation should be maintained, e.g.:

- Documentation of energy intensity factors (e.g., MJ of energy per L of diesel) used if they are different to the default values used in the ASC Farm Standard Energy Use Data Submission Template.
- Evidence to support the quantity of each energy input specified in calculations is available on request, along with any allocation of energy inputs between multiple sites if applicable.
 - This may be, for example, fuel purchase invoices or electricity use records.

Auditors should confirm:

- The energy use value is calculated and reported in MJ of energy per tonne of live weight production.
- Energy use of production is calculated for either (a) the most recent completed production cycle; or (b) three consecutive years of production.
- Documentation is provided to support input variables and calculations.
- The ASC Farm Standard Energy Use Data Submission Template is used to demonstrate results.

Useful Resources

- Certificate Holders are encouraged to use the ASC Farm Standard Energy Use Data Submission Template for calculating the energy intensity of production after energy inputs are quantified.
- Additional energy intensity factors for a range of energy inputs are available from the UK Department of Energy Security and Net Zero¹⁷.

Indicator:	Requirement:
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2.10.2 The UoC shall calculate the feed and farm-related GHG emissions of production following the method outlined in **Appendix 9 (9.4)**.

How do I interpret this requirement?

- Consistent and transparent measurement of GHG emissions following a life cycle assessment approach is necessary to underpin effective GHG reduction and management and to inform broader efforts to understand and reduce the GHG impact of global food systems.
 - This indicator ensures that all ASC certified sites undertake this important first step in contributing to efforts to combat climate change.
- ASC does not require the use of any particular GHG calculation framework or tool or conformance with any particular GHG methodology or standard. However, some

¹⁷ DESNZ (2022). Greenhouse gas reporting: Conversion factors 2022. United Kingdom Department for Energy Security and Net Zero. Available at <https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2022>

methodological requirements are stipulated in Appendix 9 to ensure a sufficient level of consistency, completeness and accuracy of GHG calculations.

- Certificate Holders should consider the reporting format of the ASC Farm Standard GHG Data Submission Template in undertaking their calculations.
 - This includes the characterisation of GHG emissions in three distinct ways:
 - By scope (based on Scope 1, Scope 2, and Scope 3 GHG emissions as defined in the GHG Protocol),
 - By category (e.g., land use change emissions, if known; if unknown, emissions may be characterised as 'unspecified'), and
 - By input (e.g., broad categories of feed ingredients, energy inputs, etc.).
- To the extent possible, Certificate Holders are encouraged to calculate GHG emissions specific to the certified site. However, in some cases, this may be difficult to do. For example, fuel purchased may be used to supply vessels servicing many sites, potentially both certified and non-certified sites, or even multiple sites or companies.
 - In these cases, a portion of the overall input may be allocated to the certified site based on a logical allocation basis.
 - For example, dividing energy inputs between sites based on the relative number of vessel trips or using the site's relative share of production among the group of sites serviced.
 - For feed inputs, because the certified site's feed use records are needed for certification, data used for GHG calculations should be specific to the certified site.
- GHG profiles from hatcheries are not required.
 - The hatchery or juvenile stage of the aquaculture supply chain typically accounts for a very small percentage of overall GHG emissions, and due to the varied nature of sourcing juveniles, requiring their inclusion in calculations could result in confusion and unnecessary burden.
 - Instead, sites may account for juvenile inputs as described in Appendix 9.
 - GHG calculations may optionally still include hatchery-specific data, and efforts to reduce GHG emissions at the hatchery stage may also be included to demonstrate efforts in the site's GHG management plan for Indicator 2.10.4.

- GHG emissions should first be calculated relative to live weight production, as kg CO₂-equivalent (CO₂e) emissions per tonne of live weight production. As a final step, emissions are converted to an edible weight basis.
 - This step is facilitated by the ASC Farm Standard GHG Data Submission Template.
 - By default, the muscle fraction of the animal is the conversion factor, such that a muscle fraction of 0.5 results in an edible weight GHG value double that of the live weight basis value.
 - If, however, a portion of the non-edible fraction of the animal is used for other purposes (e.g., by-product meal), a portion of emissions may be allocated to the non-edible product.
 - In these cases, the allocation is undertaken directly within the ASC Farm Standard GHG Data Submission Template following both allocation methods: dividing emissions between edible and non-edible product on the basis of their relative mass, and on the basis of their relative economic value as expressed by their revenue generation.
 - This allocation step is not applicable to bivalve and abalone shells; for these species, the muscle fraction alone is used for converting results to an edible weight basis.

Evidence of implementation should be maintained, e.g.:

- GHG profiles for feeds used, specific to the feed supplier and species or species group and updated on an annual basis, as required to be shared by the feed supplier according to Criterion 1.21 of the ASC Feed Standard and using the format of the ASC Feed Standard GHG Data Submission Template.
- *For marine salmonids only*: GHG profiles for any supplying grow-out site(s) as required to be shared by the previous grow-out site(s).
- Documentation of the source of any GHG emission factors used.
 - This may be documentation of calculation tools (e.g., the ASC GHG Calculator) or documentation of specific emission factors and their source if calculating emissions manually.
- Documentation of the input variables used (e.g., quantity of electricity).
 - Inputs of energy should reflect those calculated in Indicator 2.10.1. All input variables should be supported by evidence such as invoices or feed use records if requested.
- Documentation of conversion factor and, if applicable, allocation factors, used to convert GHG calculations from a live weight basis to an edible weight basis.

- Documentation of power purchase agreements, if applicable, to demonstrate use of electricity from specific dedicated sources rather than the national or regional electricity grid.

Auditors should confirm:

- Records of GHG profiles from feed suppliers and (for marine salmonids) prior grow-out sites are available.
 - GHG emissions are calculated in kg CO₂e relative to live weight production and edible weight product and recorded using the ASC Farm Standard GHG Data Submission Template.
 - If using a tool other than the ASC GHG Calculator, documentation showing how the tool conforms to the methodological requirements outlined in Appendix 9;
 - This may be a signed declaration from the tool provider or highlighted elements in the tool's user documentation.
 - Values input into the ASC GHG Calculator, another tool, or manual calculations are supported with evidence (e.g., fuel purchase invoices).
 - If undertaking calculations manually, demonstration of the calculations undertaken, including the use of GHG data from feed suppliers and (for marine salmonids) prior grow-out sites in conformance with the requirements of Appendix 9.
 - The scope of included activities reflects the requirements outlined in Appendix 9.
 - GHG emissions are not adjusted using carbon credits or offsets.
-

Useful Resources

- Certificate Holders are encouraged to use the ASC GHG Calculator, available at <https://ghg.asc-aqua.org>, to ensure compliance with regards to calculation scope, methods, data sources and reporting format. Use of this tool is voluntary; other tools, consultants, or internal procedures may be used if documentation is available to demonstrate compliance with the requirements in Appendix 9.
- Recommended factors for converting GHG emissions from a live weight basis to an edible weight basis may be sourced from supplemental material of Gephart et al. (2021¹⁸). Other conversion factors may be used if they reflect the actual edible yield (i.e., muscle fraction) of the animal.

¹⁸ Gephart, J. A., Henriksson, P. J., Parker, R. W., Shepon, A., Gorospe, K. D., Bergman, K., ... & Troell, M. (2021). Environmental performance of blue foods. *Nature*, 597(7876), 360-365. <https://doi.org/10.1038/s41586-021-03889-2>

- A range of methodological frameworks and standards are available to guide life cycle assessments and GHG accounting. Use of any of these is optional, provided that calculations are still in compliance with the requirements of Appendix 9. Examples include:
 - General life cycle assessment methods in ISO 14040¹⁹ and ISO 14044²⁰
 - The GHG Protocol from the World Resources Institute²¹
 - ISO 14067²² and marine fish product category rules in ISO 22948²³
 - The Product Environmental Footprint (PEF) marine fish product category rules²⁴
 - PAS 2050²⁵ and seafood product category rules in PAS 2050-2²⁶

Indicator:	Requirement:
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2.10.3 The UoC shall develop, implement and maintain a GHG Management Plan, and shall ensure the plan is annually reviewed, and revised as necessary, following the targets set out in **Appendix 9 (9.5)**.

How do I interpret this requirement?

- After calculating GHG emissions and identifying major contributors ('hot spots') to overall emissions, a Management Plan allows a site to identify actions to reduce, minimise and/or avoid GHG emissions.
- Actions proposed in a GHG Management Plan should demonstrably relate to emissions identified during GHG calculations. Actions may also relate to GHG emissions beyond the

¹⁹ ISO. (2006). ISO 14040 — Environmental management — Life cycle assessment — Principles and framework. Geneva, Switzerland: ISO.

²⁰ ISO. (2006). ISO 14044 — Environmental management — Life cycle assessment — Requirements and guidelines. Geneva, Switzerland: ISO.

²¹ WRI. (2004). The greenhouse gas protocol: A corporate accounting and reporting standard. Washington, D.C., USA: World Resources Institute.

²² ISO. (2013). ISO 14067 — Greenhouse gases — Carbon footprint of products — Requirements and guidelines for quantification and communication. Geneva, Switzerland: ISO.

²³ ISO. (2020). ISO 22948 — Carbon footprint for seafood — Product category rules (CFP-PCR) for finfish. Geneva, Switzerland: ISO.

²⁴ Marine Fish PEF CR. (2024). Product Environmental Footprint Category Rules (PEFCR) for unprocessed Marine Fish Products. In draft available at: <https://www.marinefishpefcr.eu/>

²⁵ BSI. (2011). PAS 2050:2011 - Specification for the assessment of the life cycle greenhouse gas emissions of goods and services. London, UK: British Standards Institution.

²⁶ BSI. (2012). PAS 2050-2:2012 - Assessment of life cycle greenhouse gas emissions. Supplementary requirements for the application of PAS 2050:2011 to seafood and other aquatic food products. London, UK: British Standards Institution.

scope of ASC's GHG calculation requirements if evidence is shown that these emissions are substantial.

- For example, efforts to reduce GHG emissions from product transport can have a significant effect in some cases, particularly if air freight is used.
- Actions proposed in a GHG Management Plan should be accompanied with means to track implementation and effectiveness.
- Not all GHG emission reduction efforts require major changes to operations or large new infrastructure investments.
 - Incremental improvements to feed conversion ratios, mortality rates, and energy efficiency can also be effective ways to reduce GHG emissions.
- GHG targets, relative to both live weight production and edible weight product, are provided in Appendix 9.
 - These targets are not intended as thresholds for standard conformance.
 - Rather, they provide a guide for Certificate Holders to assess their performance against established benchmarks, support efforts in their GHG Management Plan and facilitate tracking over time.
- By calculating and reporting emissions on an annual basis, sites will be able to track their GHG performance over time and assess the effectiveness of actions taken in their GHG Management Plan.
 - If emissions relative to production are increasing, the GHG Management Plan should be reviewed and, if needed, revised.
- Reporting of GHG emissions to ASC, with a consistent approach across certified sites, will also support ASC's efforts to provide guidance back on the types of actions that are achieving significant improvements in GHG performance.
- While calculation and reporting of GHG emissions requires the use of two allocation methods to promote transparency and consistency across certified sites, Certificate Holders may select either allocation method as the basis for identifying actions in their GHG Management Plan and tracking emissions over time.
- For many sites, emissions may come primarily from the production and use of feed.
 - Coordinating with feed suppliers may be an effective way to target improvement of GHG emissions in these cases.

- This may take the form of identifying lower GHG ingredients, transitioning to ingredients associated with lower rates of deforestation and land conversion, or improving feed conversion ratios through improved feed compositions.
- o The GHG Management Plan may be site-specific or may be applicable across many sites or developed at a corporate level, provided that the GHG tracking and actions taken are demonstrated to be relevant to the site and to reflect identified hot spots of emissions.

Evidence of implementation should be maintained, e.g.:

- o Copies of current and previous versions of the GHG Management Plan.

Auditors should confirm:

- o Presence of a GHG Management Plan.
- o The GHG Management Plan is maintained by a designated member of staff and is regularly reviewed and updated.
- o Actions proposed in the GHG Management Plan reflect identified sources of GHG emissions.
- o The GHG Management Plan includes tracking of GHG emissions over time relative to the targets set out in Appendix 9.

Useful Resources

N/A

Indicator:	Requirement:
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2.10.4 The UoC shall report annually to ASC the energy use and GHG emissions per site according to ASC data submission procedures.

How do I interpret this requirement?

- o Public disclosure of energy use and GHG data are not required by the ASC Farm Standard. Instead, reporting to ASC is required.

- This ensures that energy use and GHG performance of ASC certified production can be tracked and communicated in an aggregated form and also supports assessment of the requirements and identification of common issues, errors or need for improved guidance.
 - Data submitted to ASC is used in accordance with ASC's data use policies.
- There are two data submission templates: one for energy use and one for GHG emissions.
- Data submitted to ASC should reflect the same calculation results as demonstrated during the audit in compliance with Indicators 2.10.1 and 2.10.2.
- Data submission for both energy use and GHG emissions is on an annual basis. However, in some cases, sites may not have new data to submit in a given year.
 - If the site is calculating values relative to a production cycle and a new production cycle has not completed since the prior year's data submission. In this case, data submission is still required but may reflect the same submission as the prior year.
 - If calculating values for three consecutive years of production and previous records are not available, data submissions in the first and second years may reflect one or two years of production, respectively.

Evidence of implementation should be maintained, e.g.:

- Demonstration of successful data submission on an annual basis.

Auditors should confirm:

- Successful annual data submissions using the ASC Farm Standard Energy Use Data Submission Template and the ASC Farm Standard GHG Data Submission Template.
 - Data reported in data submissions reflect results shown in audit for Indicators 2.10.1 and 2.10.2.
-

Useful Resources

N/A

Indicator:	Requirement:
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2.10.5 *Indicator applicability: Marine salmonids*

The UoC shall provide GHG emissions data to any subsequent ASC certified site to which it supplies fish for grow-out.

How do I interpret this requirement?

- In the case of marine salmonids, grow-out sites that supply fish to subsequent grow-out sites are required to be certified under the ASC Farm Standard.
 - This necessitates that they calculate and report GHG emissions as per Indicator 2.10.2.
 - This facilitates the sharing of data with subsequent grow-out sites to ensure that GHG calculations at the point of harvest include relevant inputs from the prior site.
- Specific data on the energy inputs and feeds used from the prior site are not required. Only the GHG profile, reflecting the format of the ASC Farm Standard GHG Data Submission Template, is needed.
 - This does not apply to other species which do not require full certification of prior sites.
- GHG profiles from hatcheries are not required, as explained above for Indicator 2.10.2.

Evidence of implementation should be maintained, e.g.:

N/A

Auditors should confirm:

- Subsequent grow-out sites are provided a GHG profile reflecting the format of the ASC Farm Standard GHG Data Submission Template either as part of regular process or upon request.

Useful Resources

N/A

Criterion 2.11 – Material Use, Waste and Pollution Control

Indicator:	Requirement:
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2.11.1 The UoC shall assess and, where feasible, implement measures to reduce, reuse and recycle waste materials.

How do I interpret this requirement?

- 'Reducing' refers to using fewer materials, like fewer plastic bottles or buying materials, like products or equipment, with less packaging.
 - This will cut down on waste from the start.
- 'Reusing' refers to using materials again instead of disposing of them.
 - This produces economic savings and reduces the amount of materials that end up as waste.
- 'Recycling' refers to collecting and processing materials to be turned into new materials.
 - As a means of giving materials a new use instead of disposing of them as waste in landfills.
- Sites should identify the materials used and assess whether it is reasonably practical to prioritise reduction, reuse and recycling over disposal.
 - The assessment should consider, for example, the availability of recycling facilities in the area.
 - If applicable, the outcome of the assessment should be turned into policies or procedures.

Evidence of implementation should be maintained, e.g.:

- Identification of materials used and an assessment of the feasibility of reducing, reusing and recycling.
- Policies and procedures outlining the steps to handle materials deemed feasible for reduction, reuse and recycling.
 - Employees should be aware of these policies or procedures.

Auditors should confirm:

- The site identified the material used and assessed the feasibility of reducing, reusing and recycling.
- Materials identified as feasible to reduce, reuse or recycle are effectively and appropriately reduced, reused or recycled.
- Employees' awareness of the policies and procedures that outline the steps to handle materials deemed feasible for reduction, reuse and recycling.
 - This may be demonstrated through training conducted by the site or the presence of signs/posters regarding the appropriate reduction, reuse and recycling of materials.

Useful Resources

N/A

Indicator:	Requirement:
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2.11.2 *Indicator applicability: point-source discharge*

The UoC shall capture non-biological solid waste, including plastics, prior to effluent discharge.

How do I interpret this requirement?

- Filters or screen barriers installed in the effluent pipe/channel prior to discharge are examples of methods for capturing non-biological solid waste.
- Based on their context, sites should identify and assess non-biological solid waste that could end up in the effluent to determine the most appropriate method for its capture.

Evidence of implementation should be maintained, e.g.:

- Identification (e.g. a list) of non-biological solid waste generated at the site.
- An assessment of the characteristics of non-biological solid waste to determine the most appropriate method for capturing it in case it ends up in the effluent.
- Records of maintenance of the method used to capture non-biological solid waste (if applicable, e.g. filter or screen barriers).

Auditors should confirm:

- The site has identified and assessed non-biological solid waste generated.
- Through site inspection, the method to capture non-biological solid waste prior to effluent discharges and its suitability.

Useful Resources

N/A

Indicator:	Requirement:
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2.11.3 The UoC shall dispose of waste responsibly.

How do I interpret this requirement?

- Sites should identify and classify waste based on where it will be disposed of. This includes the following types of waste:
 - Recyclables: e.g. plastics, metals and paper.
 - Hazardous: e.g. chemicals, batteries and medical waste.
 - Non-recyclable: e.g. biological and non-biological waste
- Based on the classification, sites should determine the facilities to use for waste disposal, for example:
 - Recycling facility.
 - Landfill.
 - Waste management company.

Evidence of implementation should be maintained, e.g.:

- Identification and classification of waste.
- Records confirming that waste is disposed of in appropriate facilities according to its classification and the legality of those facilities.

Auditors should confirm:

- The site:
 - Has identified and classified the waste it generates.
 - Uses facilities that are appropriate for the type of waste and that they legally operate.

Useful Resources

N/A

Indicator:	Requirement:
2.11.4	The UoC shall not use pesticides that are banned or restricted by any of the following conventions or organisations: ○ The Rotterdam Convention on Prior Informed Consent (PIC)²⁷ ○ The Stockholm Convention on Persistent Organic Pollutants (POPs)²⁸ ○ The World Health Organisation (WHO), for products classed as 'extremely hazardous' or 'highly hazardous' (classes Ia and Ib).²⁹
2.11.5	The UoC shall only use net antifouling agents containing biocides that are approved by local regulations in either Australia, the European Union, Japan or the United States of America.

How do I interpret this requirement?

- Pesticides: The Rotterdam Convention on Prior Informed Consent (PIC) defines pesticides as chemicals used to destroy, prevent, or control pests, including vectors of human or animal disease, and unwanted species of plants or animals causing harm during or otherwise interfering with the production, processing, storage, transport, or marketing of food, agricultural commodities, wood, and wood products or animal feedstuffs.
- Biocides: The Biocidal Products Regulation (BPR, Regulation (EU) 528/2012) defines biocides as any substance or mixture, in the form in which it is supplied to the user, consisting of,

²⁷ <http://www.pic.int/TheConvention/Chemicals/AnnexIIIChemicals>

²⁸ <http://chm.pops.int/TheConvention/ThePOPs/ListingofPOPs/tabid/2509/Default.aspx>

²⁹ <https://www.who.int/publications/i/item/9789240005662>

containing or generating one or more active substances, with the intention of destroying, deterring, rendering harmless, preventing the action of, or otherwise exerting a controlling effect on, any harmful organism by any means other than mere physical or mechanical action.

Evidence of implementation should be maintained, e.g.:

- List of pesticides and biocides used by the site.
 - Including purchase records.
 - Copy of the list of banned or restricted pesticides by the conventions or organisations stated by the indicator.
 - Copy of the list of biocides approved by local regulations in either Australia, the European Union, Japan or the United States of America.
-

Auditors should confirm:

- Pesticides used by the site are not on the list of banned or restricted pesticides by the conventions or organisations stated by the indicator.
 - Biocides used by the site are not on the list of net antifouling agents containing biocides approved by local regulations in either Australia, the European Union, Japan or the United States of America.
-

Useful Resources

- [REGULATION \(EU\) No 528/2012 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 22 May 2012 concerning the making available on the market and use of biocidal products.](#)

Indicator:	Requirement:
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2.11.6 The UoC shall hold water containing hormones for at least 48 hours or as per product specifications, whichever is greater, prior to discharge.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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2.11.7 The UoC shall ensure that all processing effluent is treated to minimise biosecurity and environmental risks.

How do I interpret this requirement?

- An appropriate effluent treatment should be able to eliminate or reduce to acceptable levels the discharge of (amongst others):
 - Organic matter: measured, for example, as the reduction of Biological Oxygen Demand (BOD), Chemical Oxygen Demand (COD) or Total Suspended Solids (TSS) after treatment.
 - Nutrients: measured, for example, as the reduction of nitrogen and phosphorous after treatment.
 - Pathogenic microorganisms: measured, for example, as the reduction of microbial counts after treatment.
 - Oils and greases.

Evidence of implementation should be maintained, e.g.:

- An effluent monitoring plan appropriate for the type and amount of processing effluent discharged.
 - Including the parameters, frequency of monitoring and acceptable limits.
- Analysis results of the parameters included in the effluent monitoring plan.
- Corrective actions when analysis results exceed acceptable levels.

Auditors should confirm:

- Effluent monitoring plan has been developed and implemented.
 - This includes monitoring analysis results.
- Corrective actions taken when analysis results exceed acceptable levels.

Useful Resources

- [V. Venugopal, A. Sasidharan \(2021\). Seafood industry effluents: Environmental hazards, treatment and resource recovery. Journal of Environmental Chemical Engineering.](#)

Indicator:	Requirement:
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2.11.8

Indicator applicability: cage culture and suspended culture not using copper-treated gear or infrastructure

The UoC shall clean **aquaculture gear or infrastructure** in land-based facilities only if the facility treats effluents.

2.11.9

Indicator applicability: cage culture and suspended culture using copper-treated gear or infrastructure

The UoC shall not treat or clean aquaculture gear or infrastructure in situ.

2.11.10

Indicator applicability: cage culture and suspended culture using copper-treated gear or infrastructure

The UoC shall clean aquaculture gear or infrastructure in land-based facilities that treat effluents, including the capturing of copper.

How do I interpret this requirement?

N/A

Evidence of implementation should be maintained, e.g.:

- Documentation confirming land-based facilities treat effluents.
 - Including verification that the facility captures copper in its effluent treatment, if applicable.

Auditors should confirm:

- Documentation confirming land-based facilities treat effluents.

Useful Resources

N/A

Indicator:	Requirement:
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2.11.11 *Indicator applicability: cage culture in marine/brackish waters using copper nets or copper-treated nets*

The UoC shall demonstrate that copper levels in the sediment are ≤ 34 mg Cu/kg dry sediment weight, following the method outlined in **Appendix 7 (7.6)**.

An exception applies where copper levels in sediment > 34 mg Cu/kg dry sediment weight, in which case the UoC shall demonstrate that the levels fall within the range of background concentrations measured following the method outlined in **Appendix 7 (7.6)**.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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2.11.12 The UoC shall handle, use, store and dispose of chemicals according to the Safety Data Sheets (SDS) provided by the supplier.

2.11.13 The UoC shall not discharge effluent containing chemicals without adequate treatment.

How do I interpret this requirement?

- The adequacy of a treatment depends on the types of chemicals found in the effluent, such as heavy metals, organic compounds, acids and bases.
 - To determine the most adequate treatment, it is essential to understand the following characteristics of the effluent:
 - Chemical composition (organic vs. inorganic, hazardous vs. non-hazardous).
 - pH levels (acidic, neutral or alkaline).
 - Solubility (dissolved, suspended or emulsified).
 - Toxicity (bioaccumulative, carcinogenic, etc.).
 - Based on the above characteristics, examples of treatments that might be used are:

- Physical-Chemical Treatment: pH neutralisation, coagulation and flocculation, chemical precipitation, adsorption, membrane filtration, electrochemical treatment, solvent extraction.
- Advanced Oxidation Processes (AOPs): ozonation, ultraviolet + hydrogen peroxide.
- Biological Treatment: aerobic and anaerobic treatment, bioaugmentation.

Evidence of implementation should be maintained, e.g.:

- Presence of treatment method for effluents containing chemicals.
 - Including the rationale for determining the most adequate method.
-

Auditors should confirm:

- Through site inspection the presence of a treatment method for effluents containing chemicals.
 - Including the rationale for determining the most adequate method
-

Useful Resources

N/A

Indicator:	Requirement:
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2.11.14 The UoC shall adequately contain **hazardous materials** to prevent spillage, contamination and pollution.

2.11.15 The UoC shall implement spill prevention and emergency response procedures for hazardous materials.

How do I interpret this requirement?

- Sites should consider their context and external circumstances when determining the most adequate containment method.
 - Examples of external circumstances are natural disasters and sabotage.

- Containment should be:
 - Impervious.
 - Capable of containing at least 110% of the volume of the largest container stored within it.
- The spill prevention and emergency response procedures for hazardous materials should:
 - Outline the preventative maintenance of equipment destined to prevent and respond to spills from, for example, vehicles, winches, cranes, and mechanical equipment on land and water.
 - Be tested for adequacy at established frequencies.
 - Depending on the test results, the plan should be revised.
- Employees should be trained on the spill prevention and emergency response procedures.

Evidence of implementation should be maintained, e.g.:

- List of hazardous materials present or used on the site.
- Records of employee training on the spill prevention and emergency response procedures.

Auditors should confirm:

- List of hazardous materials present or used on the site.
- Employees have been trained on the spill prevention and emergency response procedures.
 - Cross-check during employees' interviews.
- Through site inspection:
 - That hazardous materials are adequately contained.

Useful Resources

N/A

Indicator:	Requirement:
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- 2.11.16** The UoC shall adequately store perishable materials to minimise spoilage, spillage, contamination or pest damage.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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- 2.11.17** The UoC shall adequately maintain machinery, aquaculture gear and infrastructure to minimise performance degradation and failure.

How do I interpret this requirement?

- The site should develop a maintenance plan that identifies the schedule for regular maintenance of machinery, aquaculture gear and infrastructure.
- The person(s) conducting maintenance should have an adequate level of competence based on the type of maintenance required.

Evidence of implementation should be maintained, e.g.:

- Maintenance plan that identifies the schedule for regular maintenance of machinery, aquaculture gear and infrastructure.
- Records to demonstrate maintenance was performed according to the schedule outlined in the maintenance plan.
 - Written justification should be in place in cases of deviation from the schedule outlined in the maintenance plan.

Auditors should confirm:

- The site has a maintenance plan covering regular maintenance of machinery, aquaculture gear and infrastructure.
- Records that the schedule has been adhered to.

Useful Resources

N/A

Indicator:	Requirement:
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2.11.17 *Indicator applicability: water-based UoC*

The UoC shall mark and record substantial aquaculture gear.

2.11.19 *Indicator applicability: water-based UoC*

The UoC shall retrieve lost aquaculture gear.

How do I interpret this requirement?

- Intent of Indicator 2.11.18 is that substantial aquaculture gear is identifiable as belonging to the site.
 - So, it can be attributable to the site in case of gear loss and facilitate retrieval (see Indicator 2.11.9).
- Substantial gear includes:
 - Floats.
 - Cages.
 - Bags.
 - Predator nets.
 - Racks.
- The site should maintain equipment and/or mechanisms for retrieving lost aquaculture gear, for example, scoop nets and grapple hooks (Indicator 2.11.19).

Evidence of implementation should be maintained, e.g.:

- A record, or log, of substantial aquaculture gear should be maintained.
 - At a defined frequency, the substantial aquaculture gear record should be contrasted with the physical inventory present at the site to determine whether any gear has been lost.

- Policies or procedures for locating and retrieving substantial aquaculture gear.

Auditors should confirm:

- Record, or log, of substantial aquaculture gear.
 - Through site inspection:
 - That substantial aquaculture gear is attributable to the site and recorded.
 - That the site maintains equipment and/or mechanisms for retrieving lost aquaculture gear.
 - Policies or procedures are in place to locate and retrieve substantial aquaculture gear.
-

Useful Resources

- Global Ghost Gear Initiative (2021) Best Practice Framework for the Management of Aquaculture Gear. Prepared by Huntington, T. of Poseidon Aquatic Resources Management Ltd. for GGGI. 81 pp. plus appendices:
[GGGI_Best_Practice_Framework_for_the_Management_of_Aquaculture_Gear.pdf](#)

Indicator:	Requirement:
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2.11.20 The UoC shall clean up waste from the receiving shoreline or land surrounding the site.

How do I interpret this requirement?

N/A

Evidence of implementation should be maintained, e.g.:

- Records of regular litter collection activities and the responsible disposal of any items recovered.
-

Auditors should confirm:

- Regular litter collection activities have taken place.

- Alternatively, conduct a visual inspection to demonstrate low levels of waste are present on the receiving shoreline or land surrounding the site.

Useful Resources

- Global Ghost Gear Initiative (2021) Best Practice Framework for the Management of Aquaculture Gear. Prepared by Huntington, T. of Poseidon Aquatic Resources Management Ltd. for GGGI. 81 pp. plus appendices:
[GGGI_Best_Practice_Framework_for_the_Management_of_Aquaculture_Gear.pdf](#)

Indicator:	Requirement:
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2.11.21 *Indicator applicability: water-based UoC*

The UoC shall ensure that visible aquaculture infrastructure is uniform in positioning and orientation.

2.11.22 *Indicator applicability: water-based UoC*

The UoC shall not use any floats made from polystyrene foam.

2.11.23 *Indicator applicability: Bivalve producers*

The UoC shall ensure that surface floats are uniform in colour.

How do I interpret this requirement?

- Aquaculture infrastructure should not impede navigation.
- Legal requirements might impose the use of specific colour floats, for example, for navigational purposes.

Evidence of implementation should be maintained, e.g.:

N/A

Auditors should confirm:

- Through site inspection:

- That visible aquaculture infrastructure is uniform in positioning and orientation.
- That floats are not made from polystyrene foam.
- That surface floats are uniform in colour.

Useful Resources

N/A

Criterion 2.12 – Feed

Indicator:	Requirement:
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2.12.1 *Indicator scope: every UoC using **compound feed***

The UoC shall only use ASC-conforming compound feed.

How do I interpret this requirement?

- ASC-conforming compound feed is sourced from feed mills certified to the ASC Feed Standard.
 - It includes feeds produced under the mass balance model, as well as feeds produced under the segregation model.
- This requirement does not apply to non-certifiable species (e.g., cleaner fish).

Evidence of implementation should be maintained, e.g.:

- Documented process to confirm feed is ASC-conforming that includes verification of (see Indicator 1.4.3):
 - The feed mill certification against the ASC Feed Standard.
 - The documentation that identifies:
 - The feed as ASC-conforming.
 - The feed production model applied (i.e., mass balance or segregation).
 - In the case of segregated feed, a distinct feed name which is also listed on the schedule to the feed mill certificate.
 - Whether the feed is traceable as ASC-conforming throughout all steps between the feed mill and site.
- Records:
 - Feed purchased and type.
 - Feed delivery notes - confirming traceability between feed mill and site.
 - Feeding records.

Auditors should confirm:

- Documented process to confirm feed product is ASC-conforming that includes verification of (see Indicator 1.4.3):
 - The feed mill certification against the ASC Feed Standard.
 - The documentation identifies:
 - The feed as ASC-conforming.
 - The feed production model applied (i.e., mass balance or segregation).
 - In the case of segregated feed, a distinct feed name which is also listed on the schedule to the feed mill certificate.
 - Whether the feed is traceable as ASC-conforming throughout all steps between the feed mill and site.
- Records
 - Feed purchased and type.
 - Feed delivery notes - confirming traceability between feed mill and site.
 - Feeding records.

Useful Resources

N/A

Indicator:	Requirement:
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2.12.2 *Indicator applicability: every UoC using seaweed as a direct feed source.*

The UoC shall only use seaweed from regulated sources.

How do I interpret this requirement?

- Regulated source means that there are national/international laws, agreements or policies governing the harvesting or farming of seaweed and that these are complied with, or the seaweed is from a certification scheme which addresses the responsible management of seaweed, whether wild-harvested or farmed.

- Seaweed certified to the following standards meet this ASC requirement on regulated sources:
 - [ASC-MSC Seaweed Standard](#)
 - [FairWild Standard](#)
 - Standards approved in the [IFOAM Family of Standards Organic Standards](#)

Evidence of implementation should be maintained, e.g.:

- Legislation which stipulates responsible harvesting or farming.
- Supplier harvesting or farming permits.
- Certification documentation.
- Records:
 - Harvest.
 - Transport to the UoC.

Auditors should confirm:

- Legislation which stipulates responsible harvesting or farming.
- Presence and validity of permits.
- Certification documentation.
- Records.
 - Harvest.
 - Transport to the UoC.

Useful Resources

N/A

Indicator:	Requirement:
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- 2.12.3** The UoC shall not use feed containing protein derived from the same genus as the species being farmed. An exception applies where it can be demonstrated that same-species protein hydrolysates are verified to be <10,000 daltons, in which case use is allowed.

How do I interpret this requirement?

N/A

Evidence of implementation should be maintained, e.g.:

- Records:
 - Feed purchased (including information on species/genus used as marine ingredients for fishmeal).
 - Feed fed.

Auditors should confirm:

- Records:
 - Feed purchased (including information on species/genus used as marine ingredients for fishmeal).
 - Feed fed.

Useful Resources

- Although the science behind prion diseases, infectivity and pathogenicity in fish in association with Transmissible Spongiform Encephalopathies (TSEs) is not very well developed, ASC has taken a precautionary approach and followed the guidelines of FAO (2011) Aquaculture Development. 5. Use of Wild Fish as Feed in Aquaculture Principle 9.1 “For biosecurity reasons, intraspecies recycling is an unacceptable practice that should be prohibited”³⁰.

³⁰ [FAO. Aquaculture development. 5. Use of wild fish as feed in aquaculture. FAO Technical Guidelines for Responsible Fisheries. No. 5, Suppl. 5. Rome, FAO. 2011. 79p.](#)

- Protein hydrolysates verified to <10,000 daltons are allowed as controlling the molecular weight provides a good verification of the appropriateness of an applied transformation process. This is in line with Regulation (EU) 142/2011 regarding the use of animal by-products.

Indicator:	Requirement:
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2.12.4 The UoC shall not feed **wet feedstuffs, moist pellets,** or **uncooked/ unprocessed marine raw materials.**

How do I interpret this requirement?

- Wet feedstuffs used as binders for probiotics and other feed additives in top-coating or top-dressing applications are allowed.
- Wet feedstuffs can be used as carbon sources for water quality control i.e., anaerobic fermentation or biofloc technology.

Evidence of implementation should be maintained, e.g.:

- Records:
 - Feed purchased.
 - Feed fed.

Auditors should confirm:

- Wet feedstuffs, moist pellets, or uncooked/ unprocessed fish are not used by the UoC by cross-checking records with observations and personnel interviews.
- Records:
 - Feed purchased.
 - Feed fed.

Useful Resources

N/A

Indicator:	Requirement:
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2.12.5 The UoC shall adhere to the species-specific limits on feed efficiency, according to **Appendix 2 (2.2)**.

How do I interpret this requirement?

- To facilitate compliance with the feed efficiency requirements the UoC should consider:
 - Using feed formulations:
 - With a high percentage of marine by-products and trimmings.
 - That includes alternatives to fishmeal/fish oil.
 - Implementing on-site feed management measures, for example:
 - The use of:
 - Feed trays.
 - Cameras.
 - Pellet sensors.
 - Staff training on the use of feed management measures.
 - Verification process to confirm management measures are operational and continuously meet their intent, for example, documented records of visual feed response.

Evidence of implementation should be maintained, e.g.:

- Records:
 - Feed purchased (including % of whole fish ingredients, EPA and DHA).
 - Feed fed.
 - Feed efficiency calculations.
 - Visual feed response.
 - Staff training.
 - Stocking biomass.
 - Production (weight harvested).

Auditors should confirm:

- Feed efficiency metrics are met.
- Records:
 - Feed purchased (including % of whole fish ingredients, EPA and DHA).
 - Feed fed.
 - Feed efficiency calculations.
 - Visual feed response.
 - Staff training.
 - Stocking biomass.
 - Production (weight harvested).

Useful Resources

N/A

Indicator:	Requirement:
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2.12.6 The UoC shall annually report to ASC the feed properties, feed use and calculated feed efficiency values per site, according to ASC data submission procedures.

No additional guidance is provided for this Indicator.



Principle 3

The aquaculture industry is committed to respecting and promoting human rights



Criterion 3.1 – Rights Awareness

Indicator:	Requirement:
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3.1.1

The UoC shall have a written human rights policy, approved by senior management, with the intent of publicly demonstrating its commitment to at least the following:

- freedom of association and the effective recognition of the right to collective bargaining; and
- the elimination of all forms of forced or compulsory labour; and
- the effective abolition of child labour; and
- the elimination of discrimination in respect of employment and occupation; and
- a safe and healthy working environment; and
- being a good neighbour and respecting the rights of surrounding communities.

How do I interpret this requirement?

- A human rights policy:
 - is the “company’s public expression of its commitment to meet its responsibility to respect internationally recognized human rights standards”³¹
 - is appropriate to the strategic direction, context and purpose of the organisation
 - provides a framework for setting quality objectives
 - includes an explicit commitment to respect all other internationally recognised human rights standards³² that apply to people that may be impacted by the UoC’s operations
 - includes stipulations concerning the company’s expectations of personnel, business partners and other relevant parties
 - includes information on how the company will implement its commitment
 - is aligned with the requirements in the ASC Farm Standard, and local and international human rights laws and standards
 - is signed by senior management

³¹ United Nations Global Compact (2015), A Guide for Business: How to Develop a Human Rights Policy. Available at: [How to Develop a Human Rights Policy 20151013.indd \(ohchr.org\)](#)

³² International Bill of Human Rights, A Brief History and the two International Covenants. Available at: [International Bill of Human Rights | OHCHR](#)

- is understood by all employees.
- Each company will need to determine for itself which human rights may need to be prioritised in its policy because this will depend on the risks associated with the:
 - company's location
 - type of farming practices
 - composition of the workforce.
- The content of the human rights policy may vary according to the UoC's size and national legal frameworks, and could also include the following topics:
 - recruitment and termination (see Indicators 3.2.3-3.2.8, 3.3.3-3.3.6, 3.4.1, 3.4.3, 3.4.4, 3.7.1-3.7.4)
 - employee representation (see Indicator 3.1.3)
 - loans and advances (see Indicator 3.2.9)
 - remediation (see Indicators 3.2.2, 3.3.2)
 - anti-discrimination (see Criterion 3.4)
 - training (see Indicators 3.1.4, 3.5.5)
 - freedom of association and collective bargaining (see Criterion 3.6)
 - transparent contracts (see Criterion 3.7)
 - fair wages (see Indicators 3.8.1, 3.8.4, 3.8.5, 3.8.6)
 - working hours (see Criterion 3.9)
 - performance and workplace conduct (see Criterion 3.10)
 - employee accommodation (see Criterion 3.11)
 - grievance mechanisms (see Criterion 3.12)
 - community relations and Indigenous rights (see Criterion 3.13).

Evidence of implementation should be maintained, e.g.:

- human rights policy
 - evidence that senior management have signed it off
-

Auditors should confirm:

- that the human rights policy is in place, approved by a member of senior management;
- where the policy is communicated and available publicly;
- whether the policy is available to all relevant stakeholders;
- that the human rights policy is consistent with other company policies and procedures.

Useful Resources

- Some examples of human rights policies can be found [here](#).
- United Nations (2011), Guiding Principles on Business and Human Rights. Available at: https://www.ohchr.org/sites/default/files/documents/publications/guidingprinciplesbusinesshr_en.pdf
- ILO (2022), ILO Declaration on Fundamental Principles and Rights at Work and its Follow-up. Available at: [wcms_716594.pdf \(ilo.org\)](https://www.ilo.org/public/english/standards/declaration/declaration.htm)
- International Bill of Human Rights, A Brief History and the two International Covenants. Available at: [International Bill of Human Rights | OHCHR](#)
- United Nations Global Compact (2015), A Guide for Business: How to Develop a Human Rights Policy. Available at: [How to Develop a Human Rights Policy 20151013.indd \(ohchr.org\)](#)
- OECD (2023), OECD Guidelines for Multinational Enterprises on Responsible Business Conduct. Available at: <https://www.oecd-ilibrary.org/docserver/81f92357-en.pdf?expires=1729687419&id=id&accname=guest&checksum=62CA2A98BB5F19532709D450D88B4404>

Indicator:	Requirement:
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3.1.2 The UoC shall have at least one named member of management who is responsible for the implementation and maintenance of the human rights policy.

How do I interpret this requirement?

- Implementing and maintaining the human rights policy means that the named member of management is responsible for:

- translating the policy statement into practical actions and initiatives within the UoC and in day-to-day operations
 - monitoring the UoC's human rights performance, including:
 - identifying areas of improvement
 - reporting progress or issues to senior management or relevant stakeholders
 - overseeing continuous improvement
 - overseeing training and awareness programmes aimed at educating employees and stakeholders about human rights principles, policies, and procedures
 - updating the policy regularly or when there are changes at the UoC.
- The named member of management should also commit to working with employee representation in the implementation of the policy, ensuring that employees' views are taken into account.

Evidence of implementation should be maintained, e.g.:

- records that allow for the verification of the named member of management
- plan of actions involved in implementing the policy
- plans for monitoring implementation of the policy
- evidence of any remediation or corrective actions that have taken place
- plans for regularly updating and revising the policy.

Auditors should confirm:

- that the named member of management understands their responsibilities in implementing and maintaining the policy;
- that employees are aware of the named member of management responsible for the human rights policy;
- that the human rights policy is being implemented, which can include checking:
 - progress against any action plans and monitoring plans,
 - if employees are aware of and understand the contents of the policy,
 - evidence of any remediation or corrective actions where there are negative human rights impacts.

Useful Resources

N/A

Indicator:	Requirement:
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3.1.3 The UoC shall facilitate the establishment of employee representation to support the implementation and maintenance of the human rights policy.

How do I interpret this requirement?

- Employee representation:
 - can vary with the size and context of the UoC. It can be a committee or nominated representatives
 - ensures that the voices and perspectives of employees are heard and considered in matters related to human rights.
- A committee structure preferably:
 - represents the composition of employees, enhancing inclusivity
 - holds elections for the employee committee, allowing all employees to nominate and vote for members.
 - Meets on a regular basis and occasionally on a more ad-hoc basis, where need arises.
- A committee's role includes:
 - assisting in the development of strategies, initiatives, and activities that promote human rights within the organisation
 - helping to monitor the UoC's human rights performance
 - identifying areas where the policy might not be effectively implemented or where issues arise
 - engaging in dialogue with management to address concerns, propose improvements, and ensure that the policy remains relevant and effective
 - raising awareness among their colleagues about the UoC's human rights policy, principles, and procedures.

- The UoC should ensure that any employee representative or committee:
 - has knowledge about the human rights policy
 - understands what their role involves. This can include ensuring that they have access to training.

Evidence of implementation should be maintained, e.g.:

- list of members of the employee committee or employee representation
- records of meetings and actions
- records of election procedures.

Auditors should confirm:

- employees are aware of who represents them and how to contact them;
 - the employee representation understands their role and the human rights policy.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.1.4 The UoC shall ensure that employees are trained on the human rights policy.

How do I interpret this requirement?

- The policy is:
 - distributed to all employees
 - available for all employees to access
 - in a format and language that they can understand, with provisions for illiteracy, if necessary.
- Informing employees about their human rights can help:

- prevent issues such as harassment, discrimination, or other human rights abuses within the workplace
 - demonstrate transparency and commitment to the human rights policy to employees and stakeholders
 - develop a more inclusive, equitable and ethical work environment.
- Training for employees on the policy should include:
 - an understanding of the significance of the human rights policy
 - the rights and responsibilities that the policy includes
 - how to report human rights violations or concerns through the UoC's grievance mechanism
 - information in a format that is understandable and accessible to all employees, including those who may have language or literacy barriers.
 - Training should take place annually or when there are new employees
 - Records of training attendance and content should be maintained.

Evidence of implementation should be maintained, e.g.:

- records of the training, including attendees and topics covered
- policy available for employees in an accessible format.

Auditors should confirm:

- employees understand the purpose and contents of the policy;
- training is taking place annually and all employees are trained.

Useful Resources

N/A

Indicator:	Requirement:
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3.1.5 The UoC shall publicly disclose the human rights policy and actively communicate it to all suppliers, buyers and other relevant stakeholders.

How do I interpret this requirement?

- The policy can be published:
 - on the UoC's website
 - in annual reports anywhere where it is easily accessible to anyone interested in learning about the UoC's commitment to human rights
 - in written (printed) materials to distribute to employees, suppliers, and communities
 - through local media or social media
 - through letters to suppliers
 - in collaboration with local organisations (such as NGOs).
- The policy should be:
 - actively communicated to all suppliers, including subcontractors and business partners
 - included in contracts and agreements with suppliers.
- For some specific target groups, such as local communities, it may be necessary to communicate more actively about (parts of) the policy, for example, during stakeholder meetings.
- 'Other relevant stakeholders' can include anyone who is interested in or impacted by the operations of the UoC, for example:
 - anyone in the supply chain
 - local community members
 - local NGOs and community organisations.

Evidence of implementation should be maintained, e.g.:

- publication of the policy

Auditors should confirm:

- that the policy is publicly available and easily accessible;
- that stakeholders are aware of the existence and contents of the policy.

Useful Resources

N/A

Criterion 3.2 – Forced Labour

Indicator:	Requirement:
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3.2.1 The UoC shall not engage in or support forced labour.

How do I interpret this requirement?

- Forced labour can take many forms and can include:
 - work that is exacted from any person under the threat of any penalty:
 - penalties do not need to actually take place for forced labour conditions to occur. The threat itself of the penalty or any other type of retaliation, as well as the penalty itself, constitutes forced labour
 - penalties and other types of retaliation include but are not limited to the following (adapted from ILO Handbook on Forced Labour³³):
 - isolation, restriction of movement, confinement or imprisonment
 - physical and sexual violence against employees, family or close associates
 - abusive working and living conditions
 - excessive overtime
 - financial penalties, withholding of wages
 - shift to even worse working conditions, removal of rights or privileges
 - exclusion from future employment
 - deprivation of food, shelter or other necessities
 - denunciation to authorities such as police or department of immigration, deportation
 - any other type of physical, mental or social harm
 - work for which the person has not offered himself or herself voluntarily, including where employees are not allowed to terminate employment at any time

³³ILO (2015), Combating Forced Labour: A Handbook for Employers and Business. Available at: <https://www.ilo.org/media/441596/download>

- the use of deception or other forms of coercion, including where employees are not informed of working conditions prior to or during employment
- the use of exploitative loans to prevent employees from leaving their jobs. This is also referred to as debt bondage, which can occur when employees:
 - incur debts related to employer loans with high interest rates
 - are forced to pay off recruitment fees or associated costs
 - are forced to pay high portions of their salaries towards employer provided accommodation or supplies purchased at an inflated cost at employer-run stores.
- The wording 'or support' in the indicator means that the UoC:
 - does not use any suppliers, subcontractors or other workers who are in any way involved in a system of forced labour
 - does not engage in forced labour through any other third parties or business relationships.

Evidence of implementation should be maintained, e.g.:

- employee contracts
- licences or certificates of any recruitment agencies used (see Indicator 3.2.3)
- evidence that any recruitment agencies used are complying with relevant regulations (see Indicator 3.2.3)
- records from internal audits.

Auditors should confirm:

- general risk factors for forced labour in a country (see useful resources section);
- NGOs' or other local organisations' knowledge of any reports of forced labour in the sector in this region;
- human resources personnel's awareness of any issues in the workplace;
- worker or union representative's awareness of any issues in the workplace;
- whether or not any migrant workers are working at the UoC whether local or international, as a heightened risk factor for forced labour;
- whether or not recruitment agencies are being used, as a heightened risk factor for forced labour;
- that employees are not threatened with any penalties or retaliation;

- that employees are not in any form of debt bondage;
- that employee contracts are clear and up-to-date and employees understand their terms and conditions of employment;
- consistency with records related to other parts of the ASC Farm Standard that represent risk factors for forced labour, e.g.:
 - recruitment and workplace policies
 - wage and payroll records
 - overtime records
 - employee committee records
 - grievance mechanism records.

Useful Resources

- ILO (2015), Combating Forced Labour: A Handbook for Employers and Business. Available at: <https://www.ilo.org/media/441596/download>
- US State Department country reports for human trafficking: <https://www.state.gov/trafficking-in-persons-report/>

Indicator:	Requirement:
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3.2.2 If forced labour is identified, the UoC shall implement remediation processes that prioritise the best interest of the person, following the method outlined in Appendix 10.

How do I interpret this requirement?

- The remediation framework is set out in Appendix 10, including timelines, corrective action and reporting obligations.
- Remediation of forced labour:
 - will vary depending on the situation and the needs of the employee in question
 - always takes into account the best interest of the employee, respecting their wishes and protecting their privacy

- is ideally carried out in consultation with local experts, such as NGOs.
- When remediating a case of forced labour, there can be different steps in the process, which can include:
 - immediate actions to safeguard the employee, for example removing them to a safe location
 - medium term actions to develop a plan to provide appropriate remedy to the victim, including but not limited to:
 - restoration of original job terms
 - wage repayment or other compensation
 - medical or other care as necessary
 - repatriation or other steps deemed necessary
 - long term actions to enact and monitor the plan, which can take up to several years.
- The UoC should then engage in a process of developing corrective actions to prevent forced labour reoccurring. This can include:
 - performing an analysis of what went wrong in their system to prevent it
 - implementing measures to prevent future recurrence
 - considering working with local experts on this corrective action.
- Given the complexity of human rights issues, partnership with external partners is often necessary to provide effective remediation. They can:
 - provide support for the development of a remediation plan
 - help in implementing the remediation plan
 - help to implement corrective actions to prevent recurrence.
- These potential partners can include:
 - local NGOs
 - individual experts
 - community liaisons/leaders
 - government offices
 - supply chain partners, or others.

- Potential support that these partners can provide could include:
 - training
 - improved access to social services and education
 - advocacy
 - direct financial support or other support for the affected person and their families to find other sources of income.

Organisation Type	Potential Services to Remediate Child Labour or Forced Labour
Human rights organisations or expert consultants	○ Support with identification of the affected person(s) ○ Support for remediation plan development ○ Human rights and worker rights training ○ Addressing root causes of abuse
Unions or worker organisations	○ Support for remediation plan development ○ Remediation plan monitoring ○ Human rights and worker rights training ○ Addressing root causes of abuse
Community-based and religious organisations and leaders	○ Support services to former child or forced labourers ○ Advocacy and awareness raising regarding external drivers of the risk of child or forced labour—for example local advocacy with the government to provide access to schools, or campaigns against child labour as a cultural practice
School leadership and staff	○ Facilitate access to education for former child labourers ○ Support advocacy efforts to address external drivers of child labour with community, parents and children
Healthcare providers	○ Medical support to former child or forced labourers ○ Input into remediation plan when there are health concerns
Government agencies	○ Social support services to former child or forced labourers—for example access to government benefits or social programmes

Buyers and retailers	○ Joint efforts to address root causes ○ Financial support for remediation and prevention efforts
Industry organisations and multistakeholder groups	○ Advocacy for system and national-level change in conditions that lead to child or forced labour or other human rights abuses ○ Joint efforts to address root causes

Evidence of implementation should be maintained, e.g.:

- documented remediation procedures
- records and documents of any remediation process that has taken place
- records and documents of any corrective actions that have been put into place
- documented monitoring system in place.

Auditors should confirm:

- existence of remediation procedures;
 - knowledge and capacity of staff to implement remediation procedures, or availability of external capacity;
 - the ability of the site to monitor any corrective actions within their management system;
 - knowledge and capacity of staff to develop corrective actions, where necessary;
 - amended contracts with recruitment agencies, where relevant.
-

Useful Resources

- Bureau of International Labor Affairs, Forced Labor Remediation Guide. Available at: <https://www.dol.gov/agencies/ilab/comply-chain/steps-to-a-social-compliance-system/step-6-remediate-violations/key-topic-forced-labor-remediation-guide#:~:text=REMEDIATION%20ACTION%3A,-Ensure%20that%20wages&text=Ensure%20that%20payment%20schedules%20are,receive%20their%20full%20wages%20due.>
- ILO (2015), Combating Forced Labour: A Handbook for Employers and Business. Available at: <https://www.ilo.org/media/441596/download>

Criterion 3.3 – Child Labour

Throughout this criterion, when we refer to the age of the child, we are referring to a full Western calendar year, where the child is 0 at birth.

Indicator:	Requirement:
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3.3.1

The UoC shall not engage in child labour. Child labour is work that:

- is mentally, physically, socially or morally dangerous and harmful to children; or
- interferes with their schooling.

How do I interpret this requirement?

- 'No engagement' means that child labour does not occur:
 - within the UoC's production sites and facilities
 - as a result of actions taken by recruitment agencies for the UoC
 - within the UoC's supply chain.
- A child is anyone under the age of 18.
- Child labour is any unacceptable form of work carried out by children and is prohibited by ASC, including the following:
 - all children under the age of 13 taking part in any economic activity;
 - all children aged 13 and 14 years old engaged in more than light work as stipulated by national, regional and/or local regulations;
 - all children under the age of 18 involved in work that is mentally, physically, socially or morally dangerous and harmful to children or interferes with their schooling.
- Table 2 below shows the ages of children and the type of work that they are permitted to engage in:

Table 2. Ages of children and type of work permitted:

Allowed type of work	Hazardous work	Non-hazardous work	Light work
Adult Employee aged 18 or older	Allowed	Allowed	Allowed
Young Employee between 15 – 17 years old (and older than the age of mandatory schooling)	Not allowed	Allowed	Allowed
Child engaged in light work between 13 and 14 years old (and anyone younger than the age of mandatory schooling)	Not allowed	Not allowed	Allowed
Child under 13 years old	Not allowed	Not allowed	Not allowed

- o ILO notes: 'Not all work done by children should be classified as child labour that is to be targeted for elimination. The participation of children or adolescents above the minimum age for admission to employment in work that does not affect their health and personal development or interfere with their schooling, is generally regarded as being something positive. This includes activities such as assisting in a family business or earning pocket money outside school hours and during school holidays. These kinds of activities contribute to children's development and to the welfare of their families; they provide them with skills and experience, and help to prepare them to be productive members of society during their adult life.'³⁴
- o If children are assisting at the UoC as part of it being a family business, this would be covered by the scope of this Criterion.

Evidence of implementation should be maintained, e.g.:

- o age verification records
- o employee contracts

³⁴ ILO, What is Child Labour? Available at: <https://www.ilo.org/international-programme-elimination-child-labour-ipecc/what-child-labour>

- job descriptions of all employees.

Auditors should confirm:

- presence of general risk factors for child labour in a country;
 - NGOs' or other local organisations' knowledge of any reports of child labour in the sector in this region;
 - human resources personnel's awareness of any issues in the workplace;
 - worker or union representatives' awareness of any issues in the workplace;
 - whether or not any children are present at the UoC;
 - that employees and managers have an understanding of what is hazardous work and what is light work;
 - confirm life or work histories of employees, when a suspicion exists that the age of an employee does not match the records.
-

Useful Resources

- What is Child Labour? [What is child labour | International Labour Organization \(ilo.org\)](https://www.ilo.org/what-is-child-labour/)
- Supplier Guidance on Preventing, Identifying and Addressing Child Labour, ILO https://www.ilo.org/ipec/Informationresources/WCMS_792211/lang--en/index.htm
- Roundtable on Sustainable Palm Oil Guidance on Child Rights for Palm Oil Producer https://rspo.org/wp-content/uploads/RSPO-GUI-T08-007_V1_ENG_Guidance_on_Child_Rights_for_PO_Producers.pdf#:~:text=The%20RSPO%20standards%20require%20palm,for%20employees%20and%20their%20families.
- Rainforest Alliance Child Labor Guide <https://www.rainforest-alliance.org/resource-item/child-labor-guide-an-introduction/>
- FAO Guidance on addressing child labour in fisheries and aquaculture: <https://www.fao.org/3/i3318e/i3318e.pdf>

Indicator:	Requirement:
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3.3.2 If child labour is identified, the UoC shall implement remediation processes that prioritise the best interest of the child, following the method outlined in Appendix 10.

How do I interpret this requirement?

- The remediation framework is set out in Appendix 10, including timelines, corrective action and reporting obligations.
- Remediation of child labour:
 - will vary depending on the situation and the needs of the child in question
 - always takes into account the best interest of the child
 - is ideally carried out in consultation with local experts such as NGOs.
- When remediating a case of child labour, there can be different steps in the process, which can include:
 - immediate actions to safeguard the child, for example removing them from the work situation to a safe location and locating their parents;
 - medium-term actions to develop a plan to protect the child both in the short and longer term and provide appropriate remedy to the victim, including but not limited to:
 - medical or other care as necessary
 - ensuring they are being safely looked after
 - access to adequate resources for the child and for their education;
 - long-term actions to enact and monitor the plan, which can take up to several years.
- The UoC should engage in a process of developing corrective actions to prevent child labour reoccurring. This can include:
 - performing an analysis of what went wrong in their system to prevent it
 - implementing measures to prevent future recurrence
 - considering working with local experts on this corrective action.
- Potential measures to prevent child labour include:
 - robust age verification system prior to employment

- facilitate children's access to school
 - employee education about child labour and child rights
 - facilitate access to childcare for employees
 - pay workers a living wage
 - promote community awareness on child labour
 - provide literacy and numeracy training for employees
 - birth registration campaign.
- Given the complexity of human rights issues, partnership with external partners is often necessary to provide effective remediation. They can:
 - provide support for the development of a remediation plan
 - help in implementing the remediation plan
 - help to implement corrective actions to prevent recurrence.
 - These potential partners can include:
 - local NGOs
 - individual experts
 - community liaisons/leaders
 - government offices
 - supply chain partners, or others.
 - Potential support that these partners can provide could include:
 - training
 - improved access to social services and education
 - advocacy
 - directly financial support or other support for the affected person and their families to find other sources of income.

Organisation Type	Potential Services to Remediate Child Labour or Forced Labour
Human rights organisations or expert consultants	○ Support with identification of the affected person(s) ○ Support for remediation plan development ○ Human rights and worker rights training ○ Addressing root causes of abuse
Unions or worker organisations	○ Support for remediation plan development ○ Remediation plan monitoring ○ Human rights and worker rights training ○ Addressing root causes of abuse
Community-based and religious organisations and leaders	○ Support services to former child or forced labourers ○ Advocacy and awareness raising regarding external drivers of the risk of child or forced labour—for example local advocacy with the government to provide access to schools, or campaigns against child labour as a cultural practice
School leadership and staff	○ Facilitate access to education for former child labourers ○ Support advocacy efforts to address external drivers of child labour with community, parents and children
Healthcare providers	○ Medical support to former child or forced labourers ○ Input into remediation plan when there are health concerns
Government agencies	○ Social support services to former child or forced labourers—for example access to government benefits or social programmes
Buyers and retailers	○ Joint efforts to address root causes ○ Financial support for remediation and prevention efforts
Industry organisations and multistakeholder groups	○ Advocacy for system and national-level change in conditions that lead to child or forced labour or other human rights abuses ○ Joint efforts to address root causes

Evidence of implementation should be maintained, e.g.:

- documented remediation procedures
- records and documents of any remediation process that has taken place
- records and documents of any corrective actions that have been put into place
- documented monitoring system in place.

Auditors should confirm:

- existence of effective and appropriate remediation procedures;
- knowledge and capacity of staff to implement remediation procedures;
- the ability of the site to monitor any corrective actions within their management system;
- knowledge and capacity of staff to develop corrective actions, where necessary.

Useful Resources

- Eliminating Child Labor, Best Practice for Remediation, GoodWeave
<https://goodweave.org/best-practice-briefs/#:~:text=The%20GoodWeave%20System%20is%20a,root%20causes%20of%20child%20labor>

Indicator:	Requirement:
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3.3.3 The UoC shall implement an age verification mechanism for all employees and shall maintain records of verified ages.

How do I interpret this requirement?

- An employment process that includes an age verification mechanism helps to confirm that employees are of the age category indicated, thereby enabling the UoC to prevent all forms of child labour.
- Age is verified by legal documentation in a robust age identification process where the UoC requires proof of identity:
 - preferably an official photo ID card such as a national ID or passport, or

- birth documentation.
- If there are concerns that the identification may not be authentic, the employer can interview the applicant including information on their background (for example about schooling history, family and family members) to make sure that the potential employee is being honest about their age, and the information provided is plausible.
- If there is no identification available, the employer can interview family members and community members, using significant events to help to place the year of the person's birth. For example, finding out if the person was born before or after the big flood in the village.

Evidence of implementation should be maintained, e.g.:

- age verification records.

Auditors should confirm:

- how management implements age verification
 - the ages of employees.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.3.4 The UoC shall report to ASC if the UoC is a family farm and if any children work on the site, according to ASC data submission procedures.

How do I interpret this requirement?

- This is a reporting requirement, not a disclosure requirement, which means that all data is held confidentially by ASC, who does not share it publicly.
- Companies should never publicly share personal employee data.
- The information collected through this reporting requirements will enable ASC to:
 - gain insights into family farms where children may help out their parents, which will inform the future development of ASC's child labour requirements

- foster open discussions about the challenges of family farms.

Evidence of implementation should be maintained, e.g.:

- confirmation of data uploaded to MyASC.
-

Auditors should confirm:

- that data uploaded to MyASC is consistent with results of the wage gap analysis.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.3.5 The UoC shall not employ children under the age of 13 under any circumstances.

How do I interpret this requirement?

- Employing children under the age of 13 carries risks for their physical, mental, social and moral development and wellbeing.
 - There are no circumstances in which employing a child under the age of 13 is permitted under the ASC Farm Standard.
-

Evidence of implementation should be maintained, e.g.:

- age verification records
 - employee contracts.
-

Auditors should confirm:

- whether there are any children living at the site or otherwise present;
- whether employee contracts match the age verification records;
- employees' knowledge of any children having been on site.

Useful Resources

N/A

Indicator:	Requirement:
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3.3.6

The UoC ensure any child employed aged 13 and 14, or below the age of completion of mandatory schooling, only conducts light work if permitted by applicable law and where the following conditions are met:

- a) a risk assessment is conducted to determine that the work is non-hazardous; and
- b) the child receives appropriate training prior to work; and
- c) the child receives appropriate supervision; and
- d) parental/guardian consent is given; and
- e) the work does not interfere with schooling; and
- f) the work is for limited hours (see Indicators 3.3.9 and 3.3.14)

How do I interpret this requirement?

- The age of completion of mandatory schooling refers to the national, legal age at which children are allowed to leave school.
- If children are below this age, they cannot be employed as young employees and come under the category of only being allowed to conduct light work. For example, if school is mandatory in a country until the age of 16, a 15-year-old child may not be employed as a young employee, but only employed for light work.
- Allowing children to conduct light work means that there are stricter working conditions in place than for other employees, including:
 - the type of work they are able to do
 - the hours they can work
 - the extra support that they need
 - the place that work must have in the wider context of their lives.
- Children aged 13 and 14 may be employed by the UoC for light work when:

- they are appropriately supervised and trained by adults
 - the work is non-hazardous
 - the work does not interfere with their schooling
 - the work does not take place during normal school hours
 - they have parental/guardian consent to work
 - work is limited to specific times of the day.
- Children helping parents at home with daily chores or participating in other activities unrelated to the function of the UoC are not covered in the scope of the ASC Farm Standard.

Evidence of implementation should be maintained, e.g.:

- age verification records
- job and activity descriptions
- training records
- working hours records
- evidence of parental/guardian consent.

Auditors should confirm:

- the kinds of tasks that all employees are carrying out;
- whether there are any children employed by the UoC;
- whether there are any children living at the UoC with their families;
- who provides supervision of young employees and how;
- young employees' description of their tasks and working hours.

Useful Resources

- United States Department of Labor: The Findings on the Worst Forms of Child Labour report (updated annually): <https://www.dol.gov/agencies/ilab/resources/reports/child-labor/findings>

Indicator:	Requirement:
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3.3.6

The UoC shall ensure any young employees only conduct work where the following conditions are met:

- a) a risk assessment is conducted to determine that the work is non-hazardous; and
- b) the young employee receives appropriate training prior to work; and
- c) the young employee receives appropriate supervision; and
- d) parental/guardian consent is given; and
- a) work is for limited hours (see Indicators 3.3.9, 3.3.10, 3.3.15-3.3.18).

How do I interpret this requirement?

- The UoC:
 - understands whether a child is allowed to be classed as a young employee
 - understands the legal requirements for children to attend school in their country of operation
 - ensures that young employees do not engage in work that is hazardous
 - complies with the working hours limits set out in this criterion.
- Children aged 15, 16 and 17 may be employed by the UoC as young employees, as long as:
 - the work is not hazardous
 - they are appropriately supervised and trained
 - they are above the age of mandatory schooling
 - they have parental/guardian consent to work
 - the national, regional, and/or local law permits children to work.
- Children below the age of completion of mandatory schooling are not allowed to be employed as young employees but may be permitted to conduct light work. For example, if school is mandatory in a country until the age of 16, a 15-year-old child may not be employed as a young employee, but only employed for light work.
- Children are prohibited not only from any work that has been deemed hazardous for adults, but also from work that may not be hazardous for adults but that may be for children, for example:

- a job that requires long periods of isolation without human contact, which can cause mental harm such as stress, anxiety and depression to a child;
- physical work carrying heavy loads, which can lead to physical injury and permanent damage to a body that is still developing.

Evidence of implementation should be maintained, e.g.:

- risk assessment documents
- training records
- age verification records
- job and activity descriptions
- evidence of parental/guardian consent.

Auditors should confirm:

- the kinds of tasks that all employees are carrying out;
- whether there are any children employed by the UoC;
- whether there are any children living at the UoC with their families;
- who provides supervision of young employees and how;
- young employees' description of their tasks and working hours.

Useful Resources

- United States Department of Labor: The Findings on the Worst Forms of Child Labour report (updated annually): <https://www.dol.gov/agencies/ilab/resources/reports/child-labor/findings>

Indicator:	Requirement:
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3.3.8 The UoC shall facilitate school attendance for any children living at the site.

How do I interpret this requirement?

- Ensuring access to education helps prevent child labour.

- No child under 18 working at the UoC may be:
 - deprived of educational opportunities (including attending school during school hours or after school activities)
 - forced to leave, drop out, or otherwise depart school early
 - forced to combine school attendance with long working hours.
- When employees and their families live on site, the UoC supports access to schools for employees' children. This may mean, for example:
 - providing parents the time to drop off and pick up their children
 - providing a bus service or bicycles to reach school when the nearest school is outside walking distance
 - raising awareness among employees about the importance of children attending school.
- If children live off the UoC site, but on another site or at accommodation provided by the UoC, this indicator applies.

Evidence of implementation should be maintained, e.g.:

- age verification records
- information about which children live on the site.

Auditors should confirm:

- whether there are any children living on the site with their families;
- the availability of schools in the local area
- whether children living on site are attending school and how they get there.

Useful Resources

N/A

Indicator:	Requirement:
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3.3.9 The UoC shall ensure that all employees under the age of 18 have at least 48 consecutive hours of rest within a seven-day period.

How do I interpret this requirement?

- Excessive working hours can have a detrimental physical and mental impact on all people, but particularly on children under the age of 18 who have not yet reached adulthood.
- Protecting children from excessive working hours is an important part of the prevention of child labour.

Evidence of implementation should be maintained, e.g.:

- age verification records
- working hours records
- payslips and pay records
- young employee contracts.

Auditors should confirm:

- whether anyone under the age of 18 is employed at the UoC;
- that information in contracts, age verification records and payslips match;
- young employees' description of their working hours.

Useful Resources

N/A

Indicator:	Requirement:
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3.3.10 The UoC shall not allow overtime hours for employees under the age of 18.

How do I interpret this requirement?

- Any hours over and above the standard contracted hours are considered overtime. For example, an employee could be contracted to a standard four hour week and if they were to work the fifth hour, this would be deemed as overtime.
- Employees under the age of 18 are not allowed to work any hours over their standard contracted hours.
- Excessive working hours can have a detrimental physical and mental impact on all people, but particularly on children under the age of 18 who have not yet reached adulthood.
- Protecting children from excessive working hours is an important part of the prevention of child labour.

Evidence of implementation should be maintained, e.g.:

- age verification records
- working hours records
- payslips and pay records
- young employee contracts.

Auditors should confirm:

- whether anyone under the age of 18 is employed at the UoC;
- that information in contracts, age verification records and payslips match;
- young employees' description of their working hours.

Useful Resources

N/A

Indicator:	Requirement:
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3.3.11 *Indicator applicability: children aged 13 or 14, or below the age of completion of mandatory schooling.*

The UoC shall ensure that children conducting light work:

- a) do not work more than 14 hours per week (excluding breaks); and
- b) do not work more than 2 hours per day on a school day (excluding breaks); and
- c) do not work more than 3 hours per day on a non-school day (excluding breaks); and
- d) have a combined time for school, work and transportation to and from school and work totalling less than 8 hours per day (excluding breaks).

How do I interpret this requirement?

- Excessive working hours can have a detrimental physical and mental impact on all people, but particularly on children under the age of 18 who have not yet reached adulthood.
- Protecting children from excessive working hours is an important part of the prevention of child labour.
- Children engaged in light work have restrictions on the number of hours that they can work:
 - on a school day to ensure that work does not interfere with their schooling
 - on a non-school day and across a week to ensure that they:
 - are not working an excessive number of hours
 - have adequate time for education, rest and play.
- Requiring that a child conducting light work spends no more than 8 hours per day at school, work and on transportation is important for allowing them to have adequate rest and leisure time.

Evidence of implementation should be maintained, e.g.:

- age verification records
- working hours records
- payslips and pay records

- young employees' contracts.

Auditors should confirm:

- whether anyone under the age of 15 or below the age of completion of mandatory schooling is employed at the UoC;
 - working hours and rest hours of employees under the age of 15 or below the age of completion of mandatory schooling;
 - that information in contracts, age verification records and payslips match;
 - young employees' description of their working hours.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.3.12

Indicator applicability: children aged 13 or 14, or below the age of completion of mandatory schooling.

The UoC shall ensure that where children conduct light work for more than 2 hours on a non-school day, an additional 0.5-hour break period is included within the shift.

How do I interpret this requirement?

- Excessive working hours can have a detrimental physical and mental impact on all people, but particularly on children under the age of 18 who have not yet reached adulthood.
- Protecting children from excessive working hours is an important part of the prevention of child labour.
- Children engaged in light work should have regular breaks.
- Where a child is conducting light work for a period of more than 2 hours (only applicable on a non-school day under the ASC Farm Standard), they shall be given a 0.5-hour break period.
- Some examples that demonstrate compliance with the ASC Farm Standard:
 - Child A works on a Saturday morning from 9:00 - 11:00. She has a 30-minute break and works again from 11:30 - 12:30.

- Child A's example is compliant because she is working no more than 3 hours on a non-school day, and she is given a break of 0.5 hours within the work shift.
- Child B works on a Tuesday afternoon from 15:00 - 17:00 and then goes home.
 - Child B's example is compliant because he is working no more than 2 hours on a school day.
- Some examples that demonstrate non-compliance with the ASC Farm Standard:
 - Child C works on a Saturday morning from 9:00 - 11:30 and then goes home.
 - Child C's example is non-compliant because she has worked more than 2 hours without a break.
 - Child D works on a Tuesday afternoon from 15:00 - 17:30 and then goes home.
 - Child D's example is non-compliant because he has worked more than 2 hours on a school day.

Evidence of implementation should be maintained, e.g.:

- age verification records
- working hours records
- payslips and pay records
- young employees' contracts.

Auditors should confirm:

- whether anyone under the age of 15 or below the age of completion of mandatory schooling is employed at the UoC;
 - working hours and rest hours of employees under the age of 15 or below the age of completion of mandatory schooling;
 - that information in contracts, age verification records and payslips match;
 - young employees' description of their working hours.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.3.13 *Indicator applicability: children aged 13 or 14, or below the age of completion of mandatory schooling.*

The UoC shall ensure that children conducting light work have a daily rest period of at least 14 consecutive hours within a 24-hour period.

How do I interpret this requirement?

- Excessive working hours can have a detrimental physical and mental impact on all people, but particularly on children under the age of 18 who have not yet reached adulthood.
- Protecting children from excessive working hours is an important part of the prevention of child labour.
- A rest period means a time when children are not working.
- A rest period of 14 consecutive hours in a 24-hour period means that children have a sufficient amount of time to rest and sleep in between times when they are conducting light work.

Evidence of implementation should be maintained, e.g.:

- age verification records
- working hours records
- payslips and pay records
- young employees' contracts.

Auditors should confirm:

- whether anyone under the age of 15 or below the age of completion of mandatory schooling is employed at the UoC;
- working hours and rest hours of employees under the age of 15 or below the age of completion of mandatory schooling;
- that information in contracts, age verification records and payslips match;
- young employees' description of their working hours.

Useful Resources N/A

Indicator:	Requirement:
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3.3.14 *Indicator applicability: children aged 13 or 14, or below the age of completion of mandatory schooling.*

The UoC shall ensure that children conducting light work do not work between 8pm and 6am.

How do I interpret this requirement?

- Working at night can be hazardous, particularly for children:
 - they can be tired and therefore more susceptible to accidents
 - working conditions at night are often more dangerous and not suitable for children.

Evidence of implementation should be maintained, e.g.:

- age verification records
- working hours records
- payslips and pay records
- young employees' contracts.

Auditors should confirm:

- whether anyone under the age of 15 or below the age of completion of mandatory schooling is employed at the UoC;
- working hours and rest hours of employees under the age of 15 or below the age of completion of mandatory schooling;
- that information in contracts, age verification records and payslips match;
- young employees' description of their working hours.

Useful Resources

N/A

Indicator:	Requirement:
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3.3.15 *Indicator applicability: young employees aged 15 or above and above the age of completion of mandatory schooling.*

The UoC shall ensure that young employees:

- do not work more than 40 hours per week; and
- do not work more than 8 hours per day; and
- have a combined time for school, work and transportation to and from school and work totalling less than 10 hours per day.

How do I interpret this requirement?

- Excessive working hours can have a detrimental physical and mental impact on all people, but particularly on children under the age of 18 who have not yet reached adulthood.
- Protecting children from excessive working hours is an important part of the prevention of child labour.
- Young employees have restrictions on the number of hours that they can work to ensure that they:
 - are not working an excessive number of hours
 - have adequate time for education, rest and play.
- Requiring that a young employee spends no more than 10 hours per day at school, work and on transportation is important for allowing them to have adequate rest and leisure time.

Evidence of implementation should be maintained, e.g.:

- age verification records
 - working hours records
 - payslips and pay records
 - young employees' contracts.
-

Auditors should confirm:

- whether anyone under the age of 18 is employed at the UoC;

- working hours and rest hours of employees under the age of 18;
- that information in contracts, age verification records and payslips match;
- young employees' description of their working hours.

Useful Resources

N/A

Indicator:	Requirement:
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3.3.16 *Indicator applicability: young employees aged 15 or above and above the age of completion of mandatory schooling.*

The UoC shall ensure that young employees have a total break time of at least 1 hour within 8 consecutive hours of work, with at least 0.5 hours break within any 4.5 consecutive hours of work.

How do I interpret this requirement?

- Excessive working hours can have a detrimental physical and mental impact on all people, but particularly on children under the age of 18 who have not yet reached adulthood.
- Protecting children from excessive working hours is an important part of the prevention of child labour.
- Young employees should have regular breaks.
- A young employee should have at least 1 hour break within 8 hours of work, and at least 0.5 hours break within 4.5 hours of work.
- Some examples that demonstrate compliance with the ASC Farm Standard:
 - A works on a Monday from 9:00 - 17:00 and has a break for lunch from 12:00 -13:00.
 - A's example is compliant because her workday is no more than 8 hours long and within those 8 hours, she has 1 hour break.
 - B works on a Thursday afternoon from 10:00 - 14:30 and has a break from 12:00 - 12:30.
 - B's example is compliant because he is working 4.5 hours with a 0.5-hour break in that time.

- Some examples that demonstrate non-compliance with the ASC Farm Standard:
 - C works on a Friday from 9:00 - 18:00 and has a break for lunch from 12:00 -13:00.
 - C's example is non-compliant because, even though she has a 1-hour break, her workday is 9 hours long.
 - Child D works on a Tuesday afternoon from 13:00 - 18:00 and then goes home.
 - Child D's example is non-compliant because he has worked more than 4.5 hours without a break.

Evidence of implementation should be maintained, e.g.:

- age verification records
- working hours records
- payslips and pay records
- young employees' contracts.

Auditors should confirm:

- whether anyone under the age of 18 is employed at the UoC;
 - working hours and rest hours of employees under the age of 18;
 - that information in contracts, age verification records and payslips match;
 - young employees' description of their working hours.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.3.17 *Indicator applicability: young employees aged 15 or above and above the age of completion of mandatory schooling.*

The UoC shall ensure that young employees have a daily rest period of at least 12 consecutive hours within a 24-hour period.

How do I interpret this requirement?

- Excessive working hours can have a detrimental physical and mental impact on all people, but particularly on children under the age of 18 who have not yet reached adulthood.
- Protecting children from excessive working hours is an important part of the prevention of child labour.
- A rest period means a time when employees are not working.
- A rest period of 12 consecutive hours in a 24-hour period means that children have a sufficient amount of time to rest and sleep in between times when they are conducting light work.

Evidence of implementation should be maintained, e.g.:

- age verification records
- working hours records
- payslips and pay records
- young employees' contracts.

Auditors should confirm:

- whether anyone under the age of 18 is employed at the UoC;
- working hours and rest hours of employees under the age of 18;
- that information in contracts, age verification records and payslips match;
- young employees' description of their working hours.

Useful Resources

N/A

Indicator:	Requirement:
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3.3.18 *Indicator applicability: young employees aged 15 or above and above the age of completion of mandatory schooling.*

The UoC shall ensure that young employees do not work between 10pm and 6am.

How do I interpret this requirement?

- Working at night can be hazardous, particularly for young employees:
 - they can be tired and therefore more susceptible to accidents
 - working conditions at night are often more dangerous and not suitable for children under the age of 18.

Evidence of implementation should be maintained, e.g.:

- age verification records
- working hours records
- payslips and pay records
- young employees' contracts.

Auditors should confirm:

- whether anyone under the age of 18 is employed at the UoC;
- working hours and rest hours of employees under the age of 18;
- that information in contracts, age verification records and payslips match;
- young employees' description of their working hours.

Useful Resources

N/A

Criterion 3.4 - Discrimination and Harassment

Indicator:	Requirement:
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- 3.4.1** The UoC shall not engage in or tolerate any form of discrimination, and shall ensure that all employees and applicants receive equal treatment and opportunities in all employment and recruitment practices.

How do I interpret this requirement?

- Discrimination is the unequal treatment of different of people, based on their personal characteristics, such as:
 - age
 - caste
 - colour
 - disabilities
 - ethnicity
 - gender
 - legal status
 - marital status
 - nationality
 - parental status
 - participation in trade unions
 - political opinion
 - pregnancy
 - race
 - religion
 - sexual orientation.
- Discrimination:
 - is a global and pervasive problem

- causes negative impacts on many levels
 - can lead to individuals suffering unequal treatment, stress and harm, at home, in society and in the workplace
 - can perpetuate poverty
 - can stifle development and productivity.
- Sometimes discrimination can be obvious and clear, but at other times it can be difficult to detect and consequently, hard to address.
 - Equal opportunities means:
 - equal pay for equal work
 - equal living standard for equal work
 - equal employment possibilities for equal ability/engagement
 - equal promotion possibilities for equal ability/engagement.
 - Equal opportunities at work means that all employees in an organisation, regardless of their sex, race, language, religion, economic or family situation, and other characteristics, have the same opportunities for:
 - education
 - training
 - career development
 - chances to apply for and be selected for posts
 - chances to be promoted.
 - Table 3 below shows some examples of situations considered and not considered discrimination.

Table 3. Examples of discrimination situations:

Situations which are considered discrimination	Situations which are not considered discrimination
A and B are both managers doing the same job at the farm. A is a man and B is a woman. A is paid a higher salary than B, because he is a man.	A is a manager and B is a machine operator with less responsibility. A is paid a higher salary than B because the job has more responsibility.

A and B both apply for the same job. A has higher qualifications than B, but B is selected for the job because he comes from the same church as the farm manager.

A and B both apply for the same job. A has higher qualifications than B, so A is selected for the job.

A and B are both technical assistants. A has been in the company for a longer period of time than B and both have equal performance, however, only B has received access to multiple training courses and been offered a promotion to technical manager, because A is a migrant worker.

A and B are both technical assistants. A has been in the company for a longer period of time than B, but B has been showing a higher performance. B is promoted to technical manager.

Evidence of implementation should be maintained, e.g.:

- terms of service between recruitment agencies and the UoC
- documents from recruitment processes
- payslips and pay records
- employee contracts and terms and conditions
- anti-discrimination policies
- hiring and promotion policies
- diversity and inclusion policies
- training records for employees and managers related to non-discrimination
- termination and disciplinary records.

Auditors should confirm:

- whether employees feel that they and their colleagues receive equal treatment, and that promotions are based on merit;
 - whether the composition of the workforce at different levels suggests there are no barriers for certain groups to be promoted.
-

Useful Resources

- The Equal Opportunities Commission: www.eoc.org.uk

- o European Foundation for the Improvement of Living and Working Conditions:
<https://www.eurofound.europa.eu/observatories/eurwork/industrial-relations-dictionary/equal-opportunities>

Indicator:	Requirement:
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3.4.2 The UoC shall not engage in or tolerate any form of violence or harassment.

How do I interpret this requirement?

- o 'The term 'violence and harassment' in the world of work refers to a range of unacceptable behaviours and practices or threats thereof, whether a single occurrence or repeated, that aim at, result in, or are likely to result in physical, psychological, sexual or economic harm, and includes gender-based violence and harassment.'³⁵
- o Violence and harassment can come in many forms, including:
 - sexual
 - mental
 - verbal
 - physical.
- o Examples of harassment in the workplace can include the following³⁶:
 - spreading malicious rumours
 - unfair treatment
 - picking on or regularly undermining someone
 - denying someone training or promotion opportunities.
- o Harassment can happen in many forms:
 - in person (face to face, by phone)
 - in writing (email, letter, text)

³⁵ [Convention C190 - Violence and Harassment Convention, 2019 \(No. 190\) \(ilo.org\)](#)

³⁶ [Workplace bullying and harassment - GOV.UK \(www.gov.uk\)](#)

- online (on social media).
- It can be:
 - undertaken by a single person or group of people
 - hidden (for example, gossiping about someone in their absence)
 - overt (for example, an unwanted nickname that undermines someone).
- One widespread form of harassment is sexual harassment, which can also take place in the workplace and is any unwanted behaviour of a sexual nature. Examples can include³⁷:
 - anything that violates someone's dignity
 - inappropriately and uncomfortably crowding someone else
 - telling sexually offensive jokes
 - making sexual remarks or gestures about someone's body, clothing or appearance
 - making sexual comments or jokes about someone's sexual orientation or gender reassignment
 - asking questions about someone's sex life
 - displaying or sharing pornographic or sexual images, or other sexual content
 - touching someone against their will
 - sexual assault or rape.
- Power dynamics can play a part in harassment, for example:
 - something that could be a joke between two friends outside the workplace can be a case of harassment in the workplace if there is an unequal balance of power;
 - someone in management making a flirtatious comment to a junior colleague is not appropriate, whereas the same comment between two friends might be acceptable.
- Responsible management includes being aware of a manager's position in relation to the people who they manage.
- Abusive behaviour can include³⁸:
 - threatening behaviour (such as shaking fists, destroying property or throwing objects)

³⁷ [What sexual harassment is - Sexual harassment - Acas](#)

³⁸ [CCOHS: Violence and Harassment in the Workplace](#)

- verbal or written threats (any expression of an intent to inflict harm)
 - verbal abuse (swearing, insults or condescending language)
 - physical attacks (hitting, shoving, pushing or kicking).
- The workplace includes not just the physical place of work, but anywhere where work-related events or interactions occur.
 - Workplace violence and harassment can occur not only in the workplace but also off-site at social events related to work, business meetings away from the UoC, or outside the workplace and away from work.
 - Some steps towards the prevention of harassment and discrimination can include:
 - employees and management are trained and know and understand what harassment and discrimination are
 - the UoC clearly states its position on discrimination through an internal code of conduct, policy or in a statement
 - the UoC assesses where there is potential for discrimination. For example, the UoC acknowledges the high number of migrant employees and understands that this is potentially a group who are at risk of being discriminated against
 - the UoC develops an action plan to prevent discrimination, which includes monitoring where there is a risk of discrimination.
 - Providing good communication of what these behaviours look like and why they are not permitted creates awareness among employees in the workplace.
 - Regular dialogue with employees and with employee representatives can be an effective way of monitoring what is going on.
 - An accessible, safe and anonymous reporting system (grievance mechanism) can give employees the ability to report violence or harassment.

Evidence of implementation should be maintained, e.g.:

- anti-harassment policy that includes the prohibition of abusive and exploitative behaviour
- training records
- disciplinary records
- health and safety incident reports
- grievances that have been reported.

Auditors should confirm:

- employees understand what they should do if any violence or harassment takes place;
- employees are familiar with the UoC's position on violence and harassment;
- employees' perceptions of how they or their colleagues are treated;
- any instances of violence and harassment that have taken place and how they have been dealt with;
- there are no unusually high staff turnover rates;
- human resources personnel are aware of any issues in the workplace;
- worker or union representatives are aware of any issues in the workplace.

Useful Resources

N/A

Indicator:	Requirement:
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3.4.3 The UoC shall ensure that employees, whether employed directly or through a recruitment agency, are not tested for pregnancy or virginity and are not forced or coerced into involuntary contraception.

How do I interpret this requirement?

- Tests for virginity sometimes take place in some parts of the world as a custom and practice, with the social expectation that women and girls should remain 'virgins' until they are married.
- The term 'virginity', as used in this way, is not a medical or scientific term, but a social and cultural construct that reflects discrimination against women and girls.
- It is discrimination if a woman is tested for pregnancy during recruitment.
- This indicator helps:
 - prevent this kind of discrimination by ensuring that testing for pregnancy or virginity is not permitted
 - protect women against unnecessary and potentially invasive tests.

- The ASC Farm Standard does not permit testing for pregnancy during recruitment or ongoing employment but having a company policy that states that pregnancies need to be declared from a certain point onwards, for health and safety reasons, is appropriate and can be necessary for safeguarding a mother and baby's health.

Evidence of implementation should be maintained, e.g.:

- terms of service between recruitment agencies and the UoC
- documents from recruitment processes
- information about which, if any, medical tests are conducted
- hiring and medical examination policies.

Auditors should confirm:

- the experiences of female employees during the hiring process;
- medical tests are only carried out based on a risk assessment (see also 3.4.5).

Useful Resources

- <https://www.who.int/news/item/17-10-2018-united-nations-agencies-call-for-ban-on-virginity-testing>

Indicator:	Requirement:
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- 3.4.4** The UoC shall ensure that medical tests are only conducted during the recruitment process if required for the function of the job or by applicable law, including where any recruitment agencies are used.

How do I interpret this requirement?

- There are occasional situations where the UoC, or the agency involved in recruitment may need to perform medical tests. For example:
 - it is required by law.
 - if the function of the job requires it.

- for health and safety reasons.
- determining whether a medical test is necessary happens through a health and safety risk assessment (Indicator 3.5.4).
- Examples of when medical tests may be allowed are:
 - when the function of a job might require a medical test, such as in the recruitment of divers. A diver may have to undergo a chest x-ray to demonstrate that they are fit enough to dive regularly.
 - when an employee is asked to test for COVID-19 before coming into the workplace, to avoid the spread of the disease.
 - when a test may be required by national law, such as the existence of laws in 25 countries that make HIV testing mandatory to obtain a work or residence permit:
<https://lawsandpolicies.unaids.org/topicresult?i=25&lan=en>.
- Examples of when medical tests may not be allowed are:
 - a test for sexually transmitted infections (STI), which could lead to discrimination about someone's sexual practices and/or ability to work.
 - when an HIV test is given outside of the 25 countries that make HIV testing mandatory to obtain a work or residence permit, as they are not relevant for the function of any jobs in aquaculture.

Evidence of implementation should be maintained, e.g.:

- terms of service between recruitment agencies and the UoC
- documents from recruitment processes
- health and safety and food safety risk assessments that require medical testing
- information about which medical tests are required.

Auditors should confirm:

- where medical tests are required, the decision for these tests has been based on the outcome of a risk assessment;
 - employees understand the reasons for the medical tests.
-

Useful Resources N/A

Indicator:	Requirement:
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3.4.5 The UoC shall only require medical tests during employment if justified by a healthy and safety or food safety risk assessment.

How do I interpret this requirement?

- There are occasional situations where the UoC, or the agency involved in recruitment may need to perform medical tests. For example:
 - it is required by law
 - if the function of the job requires it
 - for health and safety reasons.
- Determining whether a medical test is necessary happens through a health and safety risk assessment (Indicator 3.5.4).
- Examples of when medical tests may or may not be allowed are:
 - An example of when the function of a job might require a medical test is the recruitment of divers, a diver may have to undergo a chest x-ray to demonstrate that they are fit enough to dive regularly.
 - An example of when a medical test might be necessary is if an employee is asked to test for Covid-19 before coming into the workplace, to avoid the spread of the disease.
 - An example of when a medical test is not allowed is an STI test, which could lead to discrimination about someone's sexual practices and / or ability to work.
 - An example of when a test may be required by national law is the existence of laws in 25 countries that have a law that makes HIV testing mandatory to obtain a work or residence permit: <https://lawsandpolicies.unaids.org/topicresult?i=25&lan=en>. Outside of these 25 countries, an HIV test is not necessary and would not be conducted as it is not relevant for the function of any jobs in aquaculture.

Evidence of implementation should be maintained, e.g.:

- terms of service between recruitment agencies and the UoC
- documents from recruitment processes
- health and safety and food safety risk assessments that require medical testing

- information about which medical tests are required.

Auditors should confirm:

- where medical tests are required, the decision for these tests has been based on the outcome of a risk assessment;
 - employees understand the reasons for the medical tests.
-

Useful Resources

N/A

Indicator:	Requirement:
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- 3.4.6** The UoC shall, when requiring medical tests (see Indicators 3.4.4 and 3.4.5), ensure that:
- employees understand the reasons for the tests; and
 - employees' medical information is confidential; and
 - employees have access to their test results; and
 - employees have the right to use a medical professional of their own choosing, if preferred; and
 - employees are not charged for the tests.

How do I interpret this requirement?

- Any reasons for medical tests are clearly communicated to the employee or prospective employee, at time of recruitment where necessary.
- The employee or prospective employee has the right to their test results and the UoC ensures that the results are kept confidential.
- The cost of medical tests is paid by the UoC and is not a cost that is charged to the employee or prospective employee.
- In some situations, the UoC may have an on-site doctor, or a doctor that there is an established relationship with, who regularly conducts the necessary medical tests. However, an employee or prospective employee is allowed to choose to consult an independent doctor, who could be their own doctor, or someone completely unrelated to the UoC.

- Costs incurred by employees or prospective employees towards medical tests, if using an independent doctor, are to be reimbursed within an agreed time period.
-

Evidence of implementation should be maintained, e.g.:

- terms of service between recruitment agencies and the UoC
 - documents from recruitment processes
 - information about which medical tests are required
 - evidence of provision of copies of medical tests to employees.
-

Auditors should confirm:

- employees understand the reasons for the medical tests;
 - employees have access to their test results and they are confidential;
 - employees understand that they are able to use an independent doctor if they prefer;
 - there are no costs incurred by employees for any medical tests.
-

Useful Resources

N/A

Criterion 3.5 – Health and Safety

Indicator:	Requirement:
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3.5.1 The UoC shall carry out a site-specific health and safety risk assessment, approved by a member of senior management, following the method outlined in **Appendix 3 (3.4)**.

How do I interpret this requirement?

- "A risk assessment is simply a careful examination of what, in the workplace, could cause harm to people"³⁹.
- The ILO identifies five steps in a risk assessment:
 - Step 1: Identify the hazards
 - Step 2: Identify who might be harmed and how
 - Step 3: Evaluate the risk – identify and decide on the safety and health risk control measures
 - Step 4: Record who is responsible for implementing which risk control measures and the timeframe
 - Step 5: Record the findings, monitor and review the risk assessment and update when necessary.
- The risks assessment should:
 - cover all site activities;
 - result in contingency or prevention measures for all risks identified.
- The risk assessment should be reviewed and revised:
 - prior to starting a new activity;
 - following feedback on major issues from employees;
 - when a new site is added to the UoC;
 - when there has been a significant change to the process;

³⁹<https://www.ilo.org/media/453896/download#:~:text=A%20risk%20assessment%20is%20simply,and%20members%20of%20the%20public.>

- following accidents or near misses;
- annually.

Evidence of implementation should be maintained, e.g.:

- documented health and safety risk assessments
 - list of contingency and preventative measures to mitigate the risks.
-

Auditors should confirm:

- the member of staff that is responsible for conducting and maintaining the risk assessment has a good understanding of the different risks on site, and the actions taken to mitigate them.
-

Useful Resources

- <https://www.ilo.org/media/453896/download#:~:text=A%20risk%20assessment%20is%20simply, and%20members%20of%20the%20public>

Indicator:	Requirement:
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3.5.2

The UoC shall ensure that the person conducting the site-specific health and safety risk assessment has the necessary competencies, experience and training. In cases where regulatory requirements are defined, the UoC shall ensure that the person conducting the site-specific health and safety risk assessment complies with these requirements.

How do I interpret this requirement?

- To demonstrate competence, the person carrying out the health and safety risk assessment should:
 - understand the:
 - different steps of a risk assessment process
 - health and safety risks inherent in aquaculture operations
 - context of the UoC
 - number of employees and their roles

- variety of different activities carried out across the site and what they involve;
- be able to:
 - record risks in a clear and consistent way
 - identify risks in all of the different areas of the site and operations of the UoC
 - identify preventative measures for risks
 - identify mitigation measures for risks
- have experience conducting health and safety risk assessments
- have undergone training on conducting health and safety risk assessments. This training can be:
 - ASC training
 - training from an external provider
 - on-the-job training from someone experienced in conducting health and safety risk assessments.

Evidence of implementation should be maintained, e.g.:

- training records
- qualifications of the person conducting the health and safety risk assessment
- health and safety risk assessment.

Auditors should confirm:

- what applicable legal requirements there are for health and safety risk assessments;
 - the qualifications and experience of the person conducting the health and safety risk assessment;
 - how effective and thorough the risk assessment is that is in place.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.5.3

The UoC shall allow any person the freedom to remove themselves from an unsafe situation without seeking permission and without fear of retribution.

How do I interpret this requirement?

- This requirement is applicable to both the workplace and any accommodation facilities provided by the UoC.
- An employee could decide that a workplace is unsafe if there are significant risks to health or safety due to, for example:
 - inadequate safety measures
 - faulty equipment
 - lack of training
 - hazardous substances
 - non-compliance with safety laws
 - anything that makes the employee feel unsafe.
- If an employee feels unsafe, they do not need to ask prior permission to remove themselves from the situation.
- Retribution can take many forms and can be:
 - obvious punishment, for example:
 - physical threats
 - verbal threats or warnings
 - hidden punishment, for example:
 - being ignored for a promotion
 - being refused requests that are unrelated to the unsafe situation
 - hostile treatment.
- Management can communicate to employees that they are able to remove themselves from any unsafe situations, for example through:
 - training

- information in their contracts
- verbal explanations.

Evidence of implementation should be maintained, e.g.:

- employee contracts
 - health and safety training records
-

Auditors should confirm:

- employees understand that they are able to remove themselves from unsafe situations without fear of retribution;
 - employees understand what unsafe situations may occur and are aware of escape routes and emergency procedures;
 - management understand that employees have the right to remove themselves from unsafe situations without fear of retribution.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.5.4 The UoC shall ensure that any person with a medical condition, or who is pregnant, does not conduct hazardous work as determined by the health and safety risk assessment (see Indicator 3.5.1).

How do I interpret this requirement?

- Hazardous work can include:
 - working with chemicals or dangerous substances
 - working at heights or in dangerous situations
 - heavy lifting or work that is physically challenging
 - working with heavy machinery.

- Examples of adjusting working conditions for pregnant women can include:
 - reducing physical stress, this includes limiting the weight that employees are lifting:
 - week 1-19 of the pregnancy: maximum of 10 kilos at a time
 - week 20-29 of pregnancy: maximum of 5 kilos up to 10 times a day
 - from week 30 of pregnancy, maximum of 5 kilos up to 5 times a day
 - from week 28 of pregnancy, the employee also shouldn't bend or kneel more than once an hour
 - up to 6 months after giving birth, any work involving lifting, carrying, pulling, and pushing objects should be slowly built up
 - avoiding pregnant women performing tasks under environmental (atmospheric) pressure, such as diving
 - not exposing pregnant and breastfeeding mothers to hazardous substances and bacteria that can harm the (unborn) child.

Evidence of implementation should be maintained, e.g.:

- documented health and safety risk assessments.

Auditors should confirm:

- whether there are pregnant women or people with medical conditions working at the UoC;
- whether or not they have been included in the health and safety risk assessment;
- that staff responsible for health and safety have an understanding of the particular risks that face pregnant women and people with a medical condition, and how to mitigate those risks.

Useful Resources

N/A

Indicator:	Requirement:
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3.5.5

The UoC shall ensure that any person working on the site receives health and safety training as required to carry out their job. The UoC shall ensure that training is:

- a) free of charge; and
- b) provided in a timely manner; and
- c) repeated on a regular basis; and
- d) conducted during regularly remunerated working hours, or the person is compensated for their time; and
- e) updated as necessary based on new procedures or equipment; and
- f) documented, and records maintained.

How do I interpret this requirement?

- Training is important because it contributes to shared responsibility for health and safety, and a workplace where everyone is protected.
- Training is delivered in a format and language that all employees can understand:
 - provisions are made for anyone who is illiterate
 - the UoC is responsible for ensuring that all employees have understood the training.
- Training is documented so that it is clear who has attended and what they were trained on.

Evidence of implementation should be maintained, e.g.:

- training records demonstrating regular training sessions and content trained on
- attendance records confirming that all employees have been trained.

Auditors should confirm:

- employees understand the different health and safety risks present at the site;
- employees understand good practices around health and safety at the site;
- employees have not paid for the training and it has taken place either during working hours, or they have been paid for the time that they have given to the training.

Indicator:	Requirement:
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- 3.5.6** The UoC shall maintain written records of health and safety accidents and near misses, determine the cause and implement appropriate corrective actions to prevent recurrence.

How do I interpret this requirement?

- An accident is an unexpected event that causes damage, injury or harm.
- A near miss is an incident that had the potential to cause damage, injury or harm, but was prevented⁴⁰.
- The recording and investigation of both of these is important. By investigating to determine the root cause, the UoC is able to:
 - take corrective actions that will allow the incident not to reoccur
 - spot potential problems and prevent them from occurring in the first place.
- Corrective actions are actions that the UoC can put in place to ensure that the incident does not occur again. Examples are in Table 4 below:

Table 4. Examples of incidents, their cause and the resulting corrective actions:

Incident	Determine the cause	Implement corrective action
Oil is spilt on the feed barge and an employee slips and hits their head	After working on a mechanism on the barge, an employee left the oil can without a lid on and someone knocked it over without noticing.	Assign a dedicated place to safely store oil cans when they are not being used. Ensure employees are trained on this and reminded to safely store equipment. Ensure the appropriate materials are available to clear up oil, if it does spill.

⁴⁰ <https://www.reengage.org.uk/about-us/policies-guidelines/accident-incident-and-near-miss-policy>

Employee trips over some rope	The rope was left coiled up on the floor.	Set up ropes in a way that people cannot trip over them, or a warning sign in front of any ropes that are a trip hazard.
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- There is a difference between immediate corrective action and the corrective actions to take once the root cause has been determined. For example: a fire extinguisher is blocked by a bin being placed in front of the extinguisher:
 - staff take immediate corrective action and move the bin
 - however, the next day they find that someone has moved the bin back. The immediate corrective action did not prevent reoccurrence
 - staff investigate to determine the root cause and find out why the bin was placed there
 - the investigation shows that this is the most ergonomic layout of the workstation and the bin will always be moved back to that location by whoever is using the workstation
 - staff take long-term corrective action and move the fire extinguisher. This stops the issue from reoccurring.

Evidence of implementation should be maintained, e.g.:

- records of accidents or near misses
- records of corrective actions taken
- training records.

Auditors should confirm:

- employees feel comfortable to report any accidents or near misses;
- employees understand what the process is for reporting any accidents or near misses.

Useful Resources

N/A

Indicator:	Requirement:
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3.5.7 The UoC shall report to ASC any accidents, according to ASC data submission procedures.

How do I interpret this requirement?

- This is a reporting requirement, not a disclosure requirement, which means that all data is held confidentially by ASC, who does not share it publicly.
- Companies should never publicly share personal employee data.
- The information collected through this reporting requirements will enable ASC to:
 - gain insights into common incidents and accidents and health and safety risks, which will inform the future development of ASC's health and safety requirements
 - foster open discussions about health and safety risks and how to prevent them.

Evidence of implementation should be maintained, e.g.:

- confirmation of data uploaded to MyASC.

Auditors should confirm:

- that data uploaded to MyASC is consistent with results of the wage gap analysis.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.5.8 The UoC shall ensure that appropriate and understandable health and safety notices, instructions, and warning signs are visibly displayed.

How do I interpret this requirement?

- Potential hazards and risk areas around the site are clearly identified by warning signs.
- The choice of signs, notices and instructions are informed by the health and safety risk assessment.
- Written safety instructions for equipment use are accessible to all employees who need to use that equipment and are in a language that employees understand.
- Written safety procedures are available to employees and:
 - include details on:
 - accident prevention and response
 - emergency procedures
 - are in a language that they understand
 - can include pictures or diagrams, which can be helpful.

Evidence of implementation should be maintained, e.g.:

- warning signs around the site
- written safety instructions and procedures that are accessible for employees.

Auditors should confirm:

- warning signs are clearly visible and potential hazards are clearly marked;
- employees understand where they can access safety instructions and procedures;
- employees understand what the warning signs mean and what they refer to.

Useful Resources

N/A

Indicator:	Requirement:
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3.5.9 The UoC shall ensure that clear and understandable emergency and fire safety procedures are in place.

How do I interpret this requirement?

- The UoC should have procedures in place for emergencies that can have a significant impact on people on the site, including:
 - serious injuries
 - fire
 - explosion
 - flood
 - poisoning
 - electrocution
 - hazardous chemical spill.
- Emergency procedures can include:
 - outlining what might happen and how to raise an alarm
 - what to do if an alarm goes off
 - a plan showing the location of hazardous items
 - a plan for emergency shutdown, isolation or making processes safe
 - how to call the emergency services
 - where the escape routes are located
 - where the assembly points are
 - what the responsibility is of each employee and the management.
- Drills should be conducted regularly to ensure that employees understand the emergency procedures.

Evidence of implementation should be maintained, e.g.:

- documented fire and emergency safety procedures
- list of equipment and information about maintenance schedules
- list of hazardous items
- relevant signs around the workplace.

Auditors should confirm:

- employees have taken part in a drill;
- employees understand what to do in case of a fire or other emergency;
- equipment is up-to-date and in good working order.

Useful Resources

<https://www.hse.gov.uk/workplace-health/emergency-procedures.htm>

Indicator:	Requirement:
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3.5.10 The UoC shall ensure that escape routes, fire exits, firefighting equipment and fire alarms are clearly marked and maintained in good working order.

How do I interpret this requirement?

- Fire safety and other emergency measures are in place, for example:
 - fire alarms
 - marked fire exits
 - marked, well-lit and well-maintained emergency escape routes
 - assembly points
 - firefighting equipment that is maintained annually or as required.
- Fire drills are conducted regularly to ensure that employees understand the emergency procedures.

Evidence of implementation should be maintained, e.g.:

- documented fire and emergency safety procedures
- list of equipment and information about maintenance schedules
- list of hazardous items
- relevant signs around the workplace.

Auditors should confirm:

- employees have taken part in a drill;
- employees understand what to do in case of a fire or other emergency;
- equipment is up-to-date and in good working order.

Useful Resources

N/A

Indicator:	Requirement:
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3.5.11 The UoC shall ensure that escape routes, fire exits and firefighting equipment are accessible and clear of obstacles.

How do I interpret this requirement?

- The UoC is responsible for:
 - removing any obstacles around escape routes and fire exits
 - ensuring that emergency doors are not locked
 - ensuring that emergency equipment is not blocked by any obstacles.

Evidence of implementation should be maintained, e.g.:

- information about locations of emergency equipment.

Auditors should confirm:

- escape routes, fire exits and firefighting equipment are accessible
- employees understand where exits and equipment are located.

Useful Resources

N/A

Indicator:	Requirement:
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3.5.12 The UoC shall provide and maintain appropriate Personal Protective Equipment (PPE) free of charge and monitor its proper use.

How do I interpret this requirement?

- The UoC assesses:
 - what Personal Protective Equipment (PPE) is needed (meeting legal stipulations if applicable)
 - for which functions it is needed
 - in which situations it should be used
 - how PPE is well maintained.
- The use of PPE should be the final option in protecting employees from hazards and the steps below are what comes before the use of PPE:
 - **Elimination:** Physically remove the hazard
 - **Substitution:** Replace the hazard with a non-hazardous process
 - **Engineering controls:** Isolate people from the hazard
 - **Administrative controls:** Change the way people work
 - **PPE:** Protect the worker with PPE.
- PPE often comes with manufacturers' instructions, which explain how to maintain equipment, and when equipment needs replacing.

- Providing appropriate storage for the PPE, where it is protected from adverse weather conditions, will prolong the life and useability of the PPE.
- Training for employees on the correct use of PPE is necessary for ensuring that PPE is used effectively.
- When new PPE is introduced, or the method of using it changes, updated training helps to ensure the safety of all employees.
- The list below is adapted from the EU's Compliance Help Centre⁴¹. PPE can include:

Type of PPE	Use
Head protection (helmets, hard hats)	When working in construction, work around heavy machinery, while working in a workplace with low-hanging objects, piping, moving parts.
Hand protection (gloves, wrist cuffs)	When working in hot or cold environments, with chemical or hazardous materials, when using abrasive or sharp objects.
Eye and face protection (safety glasses, goggles, eye and face shields, visors)	When working with power driven tools, welding tools, when handling hazardous substance.
Respiratory protection (breathing apparatus, mask respirators, disposable face masks, protective hoods)	When working in workplaces with high levels of dust, or when in contact with gases, powders and vapours.
Hearing protection (earplugs, ear defenders, noise meters)	When working in an environment with high noise levels.
Foot protection (safety boots and shoes)	When working around various hazards, hot and cold environments, when handling heavy objects.
Body protection (high-visibility clothing, harnesses,	When there are vehicles, when doing maintenance work on machinery, when working outdoors, in extreme temperatures,

⁴¹ <https://support.ce-check.eu/hc/en-us/articles/360021707072-8-Types-of-Personal-Protective-Equipment-PPE>

clothing for specific weather conditions, life jackets)

for protection against chemical contamination, for protection against drowning.

- Employees need somewhere for PPE with:
 - adequate privacy to change into the PPE
 - somewhere to store their belongings
 - a designated place to store or return (e.g. laundry) the PPE after use.

Evidence of implementation should be maintained, e.g.:

- a record of available PPE, what it is used for and how it is maintained
- health and safety risk assessment demonstrating when PPE is needed.

Auditors should confirm:

- employees are using PPE correctly and consistently;
- the PPE is well maintained and in good working condition;
- employees understand the reasoning behind the use of PPE and why it is important.

Useful Resources

N/A

Indicator:	Requirement:
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3.5.13 The UoC shall provide appropriate first aid in the event of a work-related injury, including access to professional medical assistance.

How do I interpret this requirement?

- The supplies needed for first aid at the UoC can be determined by the health and safety risk assessment.
- Professional medical assistance can include ensuring that the UoC is able to access doctors and an ambulance, where necessary.

- The following list of first aid kit supplies is adapted from the UK government Health and Safety Executive website⁴²:
 - general guidance on first aid (an example from the UK government Health and Safety Executive can be found here: <https://www.hse.gov.uk/pubns/indg347.pdf>)
 - individually wrapped sterile plasters of assorted sizes
 - sterile eye pads/eye wash
 - individually-wrapped triangular bandages, preferably sterile
 - safety pins
 - large and medium-sized sterile, individually-wrapped, unmedicated wound dressings
 - disposable gloves
 - scissors or cutting tool.
- First aid kits are:
 - checked regularly, to ensure that any items that are past their expiry date are safely disposed of and replaced
 - kept at several locations around the site
 - easily accessible
 - located near potential hazards or areas of risk.

Evidence of implementation should be maintained, e.g.:

- complete and well maintained first aid kits in easily accessible locations
- information about what the first aid kits contain and how to use them.

Auditors should confirm:

- employees know where the first aid kits are kept and how to use them;
- the UoC has a process in place to ensure first aid kits remain fully stocked and serviceable.

Useful Resources

- <https://www.hse.gov.uk/simple-health-safety/firstaid/what-to-put-in-your-first-aid-kit.htm>

⁴² <https://www.hse.gov.uk/simple-health-safety/firstaid/what-to-put-in-your-first-aid-kit.htm>

Indicator:	Requirement:
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3.5.14

The UoC shall, where not provided by a state or national social security or health system, provide and pay for insurance or have a system to cover costs for all employees for all work-related accidents or injuries. This shall include at least the following:

- a) the cost for transport and medical treatment and medication needed to treat the accident or injury; and
- b) the cost for transport and medical treatment and medication needed for recovery; and
- c) compensation for lost working hours; and
- d) cost for any required repatriation in case of migrant workers.

How do I interpret this requirement?

- The UoC is responsible for caring for its employees and protecting them from any financial impacts of an accident or injury.
- The UoC should ensure that employees are protected from unfair dismissal if they are unable to work for long periods.
- Depending on the level of the national social security of the country where the site is located, there are likely to be different costs that the UoC is responsible for.
- For example: an employee injures his back lifting boxes of feed from a truck. He suffers an acute injury, falls down and struggles to stand. The site manager provides him with some painkillers from the first aid kit and takes him in a site vehicle to the nearest hospital. The employee receives some x-rays and is diagnosed as having strained his back and is given pain relief and told he cannot work for 3 weeks. The site manager does not charge the employee for the painkillers from the first aid kit, or the transport to the hospital. The UoC pays the full hospital bill and continues paying the employee's salary for the 3 weeks that he is off work.
 - If this incident took place in Country A, where there is a social security and healthcare system, the site manager does not charge the employee for the painkillers from the first aid kit, or the transport to the hospital. The UoC does not have to pay any hospital bill because the national health system pays for that. The UoC does not have to pay any of the employee's salary for the 3 weeks that he is off work because that is covered by the national social security system.

- If this incident took place in Country B, where there is no social security and healthcare system, the UoC pays all of the costs, as outlined above.

Evidence of implementation should be maintained, e.g.:

- information about what is provided by the national government
 - documentation showing how the site will cover or pay for any costs that are not covered by the national government, for example, insurance documents.
-

Auditors should confirm:

- the UoC has the right plans in place for paying costs of workplace injuries;
 - management understands their role and responsibility in covering these costs;
 - employees understand what happens financially if they have a workplace injury.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.5.15 The UoC shall ensure that any person who handles or comes into contact with hazardous substances has access to appropriate changing and washing facilities.

How do I interpret this requirement?

- Hazardous substances that are often used in aquaculture can include⁴³:
 - disinfectants as part of biosecurity protocols
 - herbicides and pesticides used in pond maintenance
 - spawning aids
 - vaccines used in disease prevention
 - marking agents used in resource management.
-

⁴³ [Microsoft Word - GUIDE_June_2019.docx \(syndel.com\)](#)

- Contact with some substances can be harmful to employees and can cause ill health or injury, but this is preventable through:
 - the correct use of PPE
 - providing changing and washing facilities, including soap, for:
 - washing hands or skin that has come into contact with hazardous substances
 - washing out eyes
 - changing out of clothes that have come into contact with hazardous substances.

Evidence of implementation should be maintained, e.g.:

- availability of changing and washing facilities in easily accessible places.

Auditors should confirm:

- employees understand the risk of working with different hazardous substances;
- employees are aware of the correct procedures to handle hazardous substances;
- employees understand what to do if they come into contact with hazardous substances.

Useful Resources

- <https://www.hse.gov.uk/pubns/indg136.pdf>
- [Microsoft Word - GUIDE_June_2019.docx \(syndel.com\)](#)

Indicator:	Requirement:
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3.5.16 The UoC shall provide access to appropriate sanitary facilities that:

- a) are clean and hygienic; and
- b) include adequate privacy measures; and
- c) include separation by gender if required; and
- d) are sufficient for the number of people using them.

How do I interpret this requirement?

- Sanitary facilities:
 - are clean
 - are dry
 - are warm
 - have a source of ventilation (e.g. a window or fan)
 - have adequate lighting
 - have doors that close and lock for privacy.
- Facilities include one of the following:
 - pit latrines
 - flush toilets
 - urinals.
- Taps and running water are also available.
- For women, facilities for safe disposal of sanitary products are provided.
- Facilities are constructed of hard materials that are easy to clean.
- Waste and wastewater are regularly disposed of in accordance with Criterion 2.11: Material Use, Waste and Pollution Control.
- They are separated by gender, where required.

Evidence of implementation should be maintained, e.g.:

- presence of clean and appropriate sanitary facilities that are accessible to all employees
- records of cleaning services.

Auditors should confirm:

- employees report that sanitary facilities are appropriate and they are able to access them without issue;
- it is clear who has the responsibility for cleaning the facilities and that this happens on a regular basis.

Useful Resources

N/A

Indicator:	Requirement:
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3.5.17 The UoC shall provide access to free, clearly labelled, drinking water.**How do I interpret this requirement?**

- Access to drinking water is essential for employees' health and the UoC is responsible for ensuring that:
 - employees know where they can access water
 - the water is clean and appropriate for drinking
 - employees are not charged for drinking water.
- Access to drinking water is particularly important in extreme heat and the UoC should be aware that they may need to provide more regular breaks and more opportunities for drinking water in hot weather.

Evidence of implementation should be maintained, e.g.:

- available drinking water stations around the site.

Auditors should confirm:

- employees understand that they are free to access drinking water and they know where they can find it.

Useful Resources

N/A

Indicator:	Requirement:
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3.5.18 The UoC shall provide access to an appropriate and clean area to prepare food and eat during breaks.

How do I interpret this requirement?

- The provision of a clean area to prepare food is important for employees' health and for reducing the risk of illness.
- In warm climates, there should be adequate shade for employees to have their breaks in.

Evidence of implementation should be maintained, e.g.:

- available place to prepare food and eat.

Auditors should confirm:

- employees are able to prepare food and eat in a clean and hygienic space.

Useful Resources

N/A

Indicator:	Requirement:
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3.5.19 The UoC shall ensure that all buildings and structures on the site are safe to use through adequate construction, maintenance and repair.

How do I interpret this requirement?

- Without adequate construction, maintenance and repair, buildings could collapse if subjected to external forces, which could have devastating results.
- Materials can wear out over time, so continuous inspection and testing structures is important.
- Ensuring adequate construction, maintenance and repair involves regular monitoring by a knowledgeable, designated person, and maintenance and repair.
- Understanding when and by whom construction, maintenance and repair should be checked is a result of the health and safety risk assessment (Indicator 3.5.1).

Evidence of implementation should be maintained, e.g.:

- documentation about any surveys or construction carried out at the UoC
- maintenance records.

Auditors should confirm:

- someone at the UoC is responsible for the wear and tear of the building and for overseeing any construction and maintenance work and has the resources to carry out needed works.

Useful Resources

N/A

Indicator:	Requirement:
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3.5.20

The UoC shall ensure that machinery and equipment:

- a) is compliant with applicable legal regulations or other recognised standards; and
- b) is safely installed, maintained and safeguarded; and
- c) is only operated by trained employees.

How do I interpret this requirement?

- Accidents and injuries involving machinery and equipment can arise from:
 - misuse
 - malfunctioning
 - bypassing, disabling or malfunctioning safety features.
- All machinery and equipment should be:
 - compliant with national or other recognised health and safety standards
 - appropriately installed, maintained and safeguarded.
- Only employees trained to use machinery and equipment are permitted to operate it.
- Examples of improper use of machinery may include:
 - bypassing or disabling emergency stop buttons, guards or interlocks to speed up processes
 - overloading equipment and machinery beyond its capacity
 - using equipment for a purpose it is not intended for
 - making unauthorised modifications
 - ignoring or disabling safety signals.
- Safely installed usually includes being installed according to manufacturer's instructions.
- Safeguarded means that any warning notices and operational guidance are in place for the machinery and equipment.

Evidence of implementation should be maintained, e.g.:

- records of who is trained to use the machinery or equipment
- maintenance records for machinery and equipment
- instruction manuals for the use of equipment.

Auditors should confirm:

- employees understand who is allowed to operate the machinery and equipment;
- the UoC has a process in place to ensure the maintenance and upkeep of the machinery and equipment.

Useful Resources

N/A

Indicator:	Requirement:
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3.5.21 The UoC shall log all commercial diving activities and maintain records, including a list of any persons involved.

How do I interpret this requirement?

- Commercial divers are professional divers who engage in underwater operations and have specific skills and qualifications.
- Through keeping records of commercial diving activities, the UoC can:
 - monitor time spent diving
 - ensure qualifications of commercial divers
 - ensure that appropriate health and safety measures are in place.
- The types of records to keep depends on local regulations and the type of dive certification. At a minimum, records include:
 - dives
 - time
 - depth

- surface time
- associated dive table records.
- This is usually recorded with a dive safety checklist.
- These records are also used for the daily diving plan.

Evidence of implementation should be maintained, e.g.:

- complete diving records
- list of persons involved in diving at the UoC.

Auditors should confirm:

- diving records are properly maintained for each dive and are up-to-date.

Useful Resources

N/A

Indicator:	Requirement:
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3.5.22 The UoC shall ensure that any person involved in commercial diving activities holds commercial diver certification through an accredited organisation.

How do I interpret this requirement?

- Different countries have different certifications for commercial diving, but a normal scuba certification is not and should never be accepted as adequate commercial diver training.

Evidence of implementation should be maintained, e.g.:

- commercial diver certificates for all commercial divers working at the UoC.

Auditors should confirm:

- management understand the relevant qualifications that divers should have to be allowed to dive commercially;

- management are able to confirm that all divers are properly certified, whether they have been directly employed or are subcontracted through a diving company.

Useful Resources

N/A

Indicator:	Requirement:
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3.5.23 The UoC shall ensure that commercial divers undergo annual medical tests to confirm they are fit to dive.

How do I interpret this requirement?

- By ensuring that divers are healthy enough to dive, this minimises the risk of negative impacts to their health and safety during dives.
- If a diver has been subcontracted from a diving company, the UoC can ensure that the diving company have taken responsibility for the medical tests.
- Some countries have requirements that medical professionals have to undergo particular training or have particular qualifications to confirm someone's fitness to work as a commercial diver.

Evidence of implementation should be maintained, e.g.:

- records of medical certificates of fitness to dive, signed by a qualified medical professional.

Auditors should confirm:

- management understand the importance of divers being medically fit and healthy;
- management are able to confirm that all divers are medically fit to dive, whether they have been directly employed or are subcontracted through a diving company.

Useful Resources

N/A

Criterion 3.6 – Collective Bargaining and Freedom of Association

Indicator:	Requirement:
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- 3.6.1** The UoC shall allow all employees to join or form a trade union or worker organisation. An exception applies where this is restricted by applicable law, in which case the UoC shall allow alternative lawful mechanisms for employee engagement.

How do I interpret this requirement?

- A trade union or worker organisation represents workers' labour interests and concerns.
- Employees are allowed to choose:
 - whether or not to become a member of a trade union or worker organisation
 - which organisation they choose.
- Freedom of association enable employees to:
 - collectively advocate for their rights
 - negotiate employment terms and conditions
 - address workplace grievances effectively.
- These practices make them less vulnerable to unfair and exploitative practices in the workplace.
- Allowing employees to engage in this process includes ensuring that they understand:
 - what freedom of association is
 - that they have the right to engage in it.
- In cases in which national regulation does not reference migrant employees or does not allow them to join trade unions, the UoC must allow and should encourage them to form an internal employees' committee that meets periodically and represents migrant employees.
- The term 'lawful mechanism' means a legal process.
- In countries where the right to freedom of association is restricted by law, the management must allow employees to choose to have parallel means of association.
 - For example, employees can develop an internal employee committee that meets periodically and represents the diversity of employees.

Evidence of implementation should be maintained, e.g.:

- documents, records or other evidence that show how sites have informed their employees that they are free to join or form workers' organisations of their own choosing, such as:
 - employee contracts
 - employee handbook
 - training records
 - written policy
 - displayed information on the site
- documents that show how sites have informed their employees of the right to join an alternative organisation
- meeting minutes of alternative organisations.

Auditors should confirm:

- whether local laws and regulations permit freedom of association;
- employees' awareness and knowledge of their right to join trade unions and other workers organisations;
- information from documents and records match with information from employee interviews;
- employee knowledge and awareness on:
 - their right to form or join an alternative worker organisation
 - their ability to form or join a workers' organisation, and/or any barriers to doing so.

Useful Resources

- [FOA & worker representation: company guidance](#). Ethical Trading Initiative.
- [Q&As on Business and freedom of association](#). International Labour Organisation (ILO).

Indicator:	Requirement:
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3.6.2 The UoC shall allow all employees to engage in collective bargaining. An exception applies where this is restricted by applicable law, in which case the UoC shall allow alternative lawful mechanisms for employee engagement.

How do I interpret this requirement?

- Collective bargaining:
 - is a voluntary negotiation between employers and workers organisations to establish the terms and conditions of employment by means of written collective agreements
 - aims to reach a collective agreement, however, if bargaining is undertaken in good faith but an agreement cannot be reached, this may still comply with the requirement
 - includes mutually agreed conditions that may be beneficial to workers and/or the management, for example, collective bargaining may mean that workers are:
 - paid a higher wage other than the minimum prescribed by law
 - work a different shift pattern that both workers and management agree is more appropriate to their context.
- Allowing employees to engage in this process includes ensuring that they understand:
 - what collective bargaining is
 - that they have the right to engage in it.
- Ensuring employees understand this can form part of training on the human rights policy (Indicator 3.1.4).
- The term 'lawful mechanism' means a legal process.
- In countries where collective bargaining is restricted by law, the management must allow workers to have parallel means to engage in discussion and negotiation between employees and employers.

Evidence of implementation should be maintained, e.g.:

- documents, records or other evidence that show how sites have informed their employees that they are free to bargain collectively, such as:

- employee contract
 - employee handbook
 - training records
 - written policy
 - displayed information on the site
- any collective bargaining agreements that are in place
 - meeting minutes of collective bargaining negotiations.

Auditors should confirm:

- documents, records or other evidence allow for verifying that sites have informed their employees that they are free to bargain collectively;
 - any collective bargaining agreements in place are being followed;
 - employees' awareness and knowledge of their right to bargain collectively;
-

Useful Resources

N/A

Indicator:	Requirement:
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3.6.3 The UoC shall not interfere with collective bargaining processes or the establishment, operation, or administration of trade unions or worker organisations.

How do I interpret this requirement?

- The UoC:
 - does not disrupt any employees looking to form or join an organisation
 - does not interfere in the election of representatives
 - allows representatives access to employees and workplaces during working hours
 - engages in meaningful negotiations when approached by worker organisations.

- Interference by employers can take many forms, such as making use of their positions of power and influence to:
 - coerce or intimidate employees into not participating in workers' organisations or collective bargaining
 - compromise the independence and effectiveness of these organisations and bargaining processes.
- An employee committee or union shall make decisions independently from the management. For example:
 - representatives are not appointed or supervised by management when executing their union or employee committee duties
 - participation is voluntary
 - there is no limitation to the number of employee organisations that can be set up.
- Management should be willing to enter into dialogue with employee organisations.
- Engagement in meaningful negotiations with employee organisations means that the UoC bargains in good faith and does not engage in actions to slow, stop or limit the bargaining process.
- Employee organisations are allowed to hold meetings at the UoC, if that is what they choose to do, but management cannot ask them to do so.

Evidence of implementation should be maintained, e.g.:

- documents related to the representative election process
 - list of current representatives.
-

Auditors should confirm:

- employee knowledge and awareness on:
 - how employee representatives were elected, and if any candidates were chosen by the management
 - whether management interferes with the establishment, functioning and administration of employee organisations
 - whether employee organisations' meetings can be held freely and without management interference.

Useful Resources

N/A

Indicator:	Requirement:
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3.6.4 The UoC shall report to ASC if there are any collective bargaining agreements in place and if there is any engagement with local trade unions, according to ASC data submission procedures.

How do I interpret this requirement?

- This is a reporting requirement, not a disclosure requirement, which means that all data is held confidentially by ASC, who does not share it publicly.
- Companies should never publicly share personal employee data.
- The information collected through this reporting requirements will enable ASC to:
 - gain insights into collective bargaining practices and contact with local trade unions, which will inform the future development of ASC's collective bargaining and freedom of association requirements, and worker voice programme.
 - foster open discussions about collective bargaining, and worker voice.

Evidence of implementation should be maintained, e.g.:

- confirmation of data uploaded to MyASC.

Auditors should confirm:

- that data uploaded to MyASC is consistent with results of the wage gap analysis.

Useful Resources

N/A

Indicator:	Requirement:
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3.6.5 The UoC shall uphold any collective bargaining agreements developed.

How do I interpret this requirement?

- A collective bargaining agreement is a labour contract usually set out between a union representing employees and management.
- It can set terms and conditions of employment, including:
 - wages
 - working hours
 - working conditions
 - grievance procedures
 - rights and responsibilities of the worker representatives and management.

Evidence of implementation should be maintained, e.g.:

- documented collective bargaining agreements and which employees are covered by them.

Auditors should confirm:

- employees' and managers' understanding of the right of collective bargaining
- any collective bargaining agreements in place are upheld.

Useful Resources

N/A

Criterion 3.7 – Transparent Contracts

Indicator:	Requirement:
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3.7.1

The UoC shall ensure that all employees have received, understood and agreed upon, written information about their employment terms and conditions before starting employment and, where applicable, prior to migration. This information shall include, at least the following:

- a) a description of the role and any responsibilities; and
- b) the type of contract (e.g. permanent, fixed-term, contractor); and
- c) working hours, including allowance for breaks, paid annual leave, allowance for days off on public holidays, and sick leave; and
- d) wages, any agreed wage deductions (e.g. accommodation, meals), compensation for overtime, and loans and advances; and
- e) social benefits (e.g. insurances); and
- f) termination terms and conditions, notice period; and
- g) relevant human rights and labour-related policies; and
- h) information on labour rights (see Indicator 1.1.2).

How do I interpret this requirement?

- This indicator applies to all employees.
- Copies of contracts should be available on file in the language of the employee, and provisions should be made for cases where people have low literacy levels. For example:
 - a colleague verbally communicates the information.
- In the case of any changes to the contract during employment, the UoC should:
 - discuss the changes with the employee
 - ensure that the employee agrees with the changes
 - records the agreed upon changes
 - provides the employee with a copy of the changes.
- Examples of changes to terms and conditions can include where employees have:
 - been given a different role or been promoted

- had changes in their working conditions, for example:
 - wages
 - working hours
 - location.
- Migrant workers should have contracts that:
 - are written, agreed to, and signed at the time of hiring in the country of origin of the employee
 - state that the employer will pay for travel from the employee's country and repatriation, and what arrangements are made for that (information about fees and costs related to employment are outlined in Criterion 3.2)
 - do not violate the national laws of the host country, as well as other laws and regulations that apply to migrant employees in their country of origin.

Evidence of implementation should be maintained, e.g.:

- contracts and contract amendments
- evidence of the contract being explained to the employee in their language and in a way they understand
- contracts, Memorandum of Understanding or terms and conditions agreed between the UoC and the labour or recruitment agency.

Auditors should confirm:

- whether contracts meet national legal requirements for labour contracts;
- that contract terms and conditions match with working hour records and payroll and wage records and/or the Salary Matrix (see Criterion 3.8);
- employees' understanding of the terms and conditions of their contract;
- whether employees are in possession of a copy of their contract and documented changes;
- for migrant employees: whether they were required to pay recruitment fees, and any costs incurred for travel to the host country by assessing the process of recruitment. See also Criterion 3.2.

Useful Resources N/A

Indicator:	Requirement:
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- 3.7.2** The UoC shall ensure that if family members are working on the site they are either subject to Indicator 3.7.1 or they receive compensation or benefits for work done that is proportional to the productivity or profit of the site.

How do I interpret this requirement?

- While the ASC Farm Standard requires that all employees working at the UoC are employed through contracts with agreed upon terms and conditions, and fair wages, ASC recognises that:
 - some sites are small businesses that are run with the support of close family members
 - where businesses are supported by family members, it is not always possible to contract and pay those family members in the usual practice outlined in the ASC Farm Standard.
- This indicator allows for small sites to receive help in operating the business from close family members in a more informal manner and in exchange for some compensation or benefits. These can include:
 - payment in money
 - payment in kind (exchange of goods or other services)
 - a share of the profits of the business.
- The compensation or benefits is proportional to the productivity or profit of the site, implies fair compensation for the work given.
- Family members is defined as first or second-degree blood relatives, or those adopted.
 - First-degree blood relative is someone's:
 - parent
 - sibling
 - child.
 - Second-degree blood relative is someone's:
 - grandparent
 - grandchild

- uncle or aunt
- niece or nephew
- cousin.

Evidence of implementation should be maintained, e.g.:

- o agreement with the family member about work, working conditions, compensation and benefits
 - o evidence of first or second-degree blood relative of the owner or spouse of the owner.
-

Auditors should confirm:

- o everyone working at the UoC understands their role and how they are compensated;
 - o everyone working at the UoC is fairly compensated for their work;
 - o everyone working at the UoC is either an employee with a contract or a first or second-degree blood relative of the owner or spouse of the owner.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.7.3 The UoC not engage in false-apprenticeships.

How do I interpret this requirement?

- o The indicator does not permit false apprenticeships but apprenticeships may be legitimately used within employment relationships if they are:
 - legally permitted
 - adequately documented
 - not used to avoid social or labour obligations.
-

Evidence of implementation should be maintained, e.g.:

- apprenticeship agreement
- contracts

Auditors should confirm:

- that any apprentices at the UoC are aware of the terms agreed in their apprenticeship agreement and that their work fits in the scope of that agreement.

Useful Resources

N/A

Indicator:	Requirement:
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3.7.4

The UoC shall ensure that no employment arrangements are used to avoid obligations to employees under applicable national labour and social security laws, and social security for all employees is paid according to applicable legal requirements.

How do I interpret this requirement?

- Systematic use of home working, labour-only contracting, sub-contracting, and fixed-term contracts is often an indication of attempts to avoid social or labour obligations.
- Such employment arrangements can be legitimately used within employment relationships if:
 - they are legally permitted
 - adequately documented
 - not used to avoid social or labour obligations.
- Home working carries risks that:
 - employees do work that is hidden and not paid for
 - employees miss out on benefits that they are entitled to because they are not present at the workplace.
- Ongoing, routine tasks necessary for the UoC to operate should be carried out by contracted employees.

- Temporary employees, such as day labourers, should only be hired for unanticipated peak work or seasonal work. For example, routine feeding is an ongoing task that is carried out by an employee. A day labourer could be hired to support a busy harvest period or to handle an unexpected feed delivery.
- Fixed-term contracts should have a clearly defined or predictable duration.
 - excessive use of these contracts, or when employees are continuously required to reapply for positions lasting less than six months, should be avoided.
- Examples of when sub-contracting is used at a UoC are:
 - production
 - fish transfer
 - veterinary treatment
 - diving
 - maintenance
 - harvest
 - slaughter
 - storage
 - transport of fish
 - cleaning
 - catering.

Evidence of implementation should be maintained, e.g.:

- contracts with contract-workers, sub-contractors and recruitment agencies
- documentation regarding social security.

Auditors should confirm:

- employees are paid social security according to national legal requirements;
 - employees understand their right to social security and how that applies to their contracts.
-

Useful Resources N/A

Indicator:	Requirement:
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3.7.5 The UoC shall report to ASC the employee composition, according to ASC data submission procedures.

How do I interpret this requirement?

- This is a reporting requirement, not a disclosure requirement, which means that all data is held confidentially by ASC, who does not share it publicly.
- Companies should never publicly share personal employee data.
- The information collected through this reporting requirements will enable ASC to gain insights into employee composition, which will inform the future development of ASC's requirements related to employees.

Evidence of implementation should be maintained, e.g.:

- confirmation of data uploaded to MyASC.

Auditors should confirm:

- that data uploaded to MyASC is consistent with results of the wage gap analysis.

Useful Resources

N/A

Criterion 3.8 – Wages

Indicator:	Requirement:
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- 3.8.1** The UoC shall set wages, before overtime and bonuses, at or above the legal minimum wage. An exception applies where a minimum wage has not been established by applicable law, in which case the UoC shall calculate wages in consultation with employees or their representative organisations.

How do I interpret this requirement?

- The following countries do not have a minimum wage established by law:
 - Denmark
 - Finland
 - Italy
 - Norway
 - Sweden
 - Iceland
 - Singapore
 - La Reunion
 - Tonga
 - Tuvalu
 - North Korea
 - South Sudan
 - Brunei
 - Zimbabwe
 - Djibouti
 - Somalia
- In these countries wages should be negotiated:
 - through collective bargaining or
 - other consultation with employees and their representative worker organisations.
- Further guidance on collective bargaining and worker organisations can be found in Criterion 3.6.

- In-kind benefits provided by the UoC should not be used as a way to pay less than the legal minimum wage to any employee.
- In-kind benefits can include:
 - housing
 - subsidised rations, food, meals
 - transport
 - utilities such as water and electricity, etc.
- Some of these in-kind benefits can be included in the living wage calculation, as described in Appendix 11: Methodology for Wage Measurement.

Evidence of implementation should be maintained, e.g.:

- employee contracts
- payslips
- working hours records
- payment transaction records
- any collective bargaining agreements that are in place.

Auditors should confirm:

- national legal requirements on minimum wage;
- relevant collective bargaining agreements on wages in the sector (recommended prior to audit visit);
- that no in-kind benefits have been paid in place of the legal minimum wage.

Useful Resources

- Regularly updated list of minimum wages by country:
<https://wageindicator.org/salary/minimum-wage/minimum-wages-per-country>

Indicator:	Requirement:
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- 3.8.2** The UoC shall annually calculate employee remuneration following the method outlined in Appendix 11, and shall assess this against the [living wage benchmarks](#) listed on the ASC website to identify if there is a gap between employee remuneration and the living wage benchmark.

How do I interpret this requirement?

- A living wage is: 'the remuneration (wages, bonuses and benefits) received for a standard workweek by a worker in a particular place sufficient to afford a decent standard of living for the worker and their family. Elements of a decent standard of living include food, water, housing, education, health care, transportation, clothing, and other essential needs including provision for unexpected events.'⁴⁴
- A living wage is different to the national minimum wage: it is usually higher and is adjusted yearly for inflation and is voluntary rather than required by law.
- Living wage benchmarks are location-specific monetary amounts calculated by the [Anker Research Institute \(ARI\)](#) and the [Global Living Wage Coalition](#) designed to ensure employees receive sufficient income for a living wage. Sites can find relevant benchmarks in the [ASC Living Wage Benchmarks](#) document found online, which is regularly updated.
- The living wage gap is defined as the difference between the local living wage and the actual remuneration (wages, bonuses, and benefits) provided to employees.
- ASC's requirements on living wage are focused on measuring and reporting wages:
 - helping the UoC assess whether or not they are paying a living wage to employees
 - helping the UoC identify any gap between current wages and the living wage in a particular location.
- These requirements do not require the UoC to:
 - close any gaps between current wages and the living wage
 - immediately start paying a living wage.
- Appendix 11 outlines the calculation that the UoC can use to understand how much employees are being paid, which is based on a methodology developed by the Anker Research Institute.

⁴⁴ Global Living Wage Coalition: <https://www.globallivingwage.org/>

- The replacement cost for in-kind benefits such as food and housing for the worker has been set by researchers in local living wage benchmarks, which can be found in the reports linked here: [ASC Living Wage Benchmarks](#).
- ASC recommends the use of a tool called the [Salary Matrix](#) developed by IDH, for this calculation, which can help simplify and clarify the process.
- Guidance for completing the upload of employee payroll, bonus and benefit information is embedded in the tool.
- The list of local living wage benchmarks is found on ASC's website [here](#).
- ASC uses the Anker Research Institute living wage benchmarks. Where one of these is not available, benchmarks from the Wage Indicator Foundation are linked.
- Through the results of this calculation, the UoC should be able to:
 - develop the wage improvement plan required by Indicator 3.8.4
 - report to ASC, required by Indicator 3.8.5.
- An example of a possible result from the calculation is:
 - the living wage benchmark in Country X is \$980 per month
 - the UoC's remuneration calculation shows that:
 - fish health technicians are paid \$940 per month
 - farm workers are paid \$900 per month.
 - The analysis of this calculation shows a living wage gap of:
 - 4% for fish health technicians
 - 8% for farm workers.

Evidence of implementation should be maintained, e.g.:

- completed Salary Matrix or similar wage measurement analysis calculation
- employee contracts
- payslips
- payment transaction records.

Auditors should confirm:

- the accuracy and completeness of wage measurement analysis;
-

- what bonus payments, and in-kind benefits have been paid, through employee and management interviews and document review;
- the value and benefit of in-kind benefits recorded in the wage measurement analysis, through employee interviews;
- calculations used in calculating in-kind-benefits used for wage calculation, including document review, through management interviews;
- wage payments.

Useful Resources

- Living Wages Around the World, Manual for Measurement by Richard Anker and Martha Anker, Chapter 15 Comparing Prevailing Wages to a Living Wage:
<https://www.elgaronline.com/monobook-0a/9781786431455/9781786431455.xml>
- Wage measurement tool: IDH [Salary Matrix](#)
- List of living wage benchmarks on [ASC website](#)

Indicator:	Requirement:
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3.8.3 The UoC shall annually report on the gap between employee remuneration and the living wage benchmark, according to ASC data submission procedures.

How do I interpret this requirement?

- This is a reporting requirement, not a disclosure requirement, which means that all data is held confidentially by ASC, who does not share it publicly.
- Companies should never publicly share personal employee data.
- Wage gap data is generated by the UoC through their remuneration calculation (Indicator 3.8.2), and is uploaded by the UoC to MyASC on an annual basis.
- The information collected through wage gap reporting will enable ASC to:
 - gain insights into national and regional wage gaps, which will inform the future development of ASC's living wage requirements
 - foster open discussions about the challenges of paying a living wage

- discuss the role in the payment of living wage of the collaborative effort and shared responsibility of participants in the whole supply chain
- develop strategies to raise wages and enhance benefits over time.

Evidence of implementation should be maintained, e.g.:

- confirmation of data uploaded to MyASC.
-

Auditors should confirm:

- that data uploaded to MyASC is consistent with results of the wage gap analysis.
-

Useful Resources

- Living Wage Data Reporting Template

Indicator:	Requirement:
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3.8.4 The UoC shall, where employee remuneration is below the living wage benchmark (see Indicator 3.8.2), develop and implement a wage improvement plan towards payment of a living wage.

How do I interpret this requirement?

- The wage improvement plan should:
 - list all categories of employees who are paid below a living wage
 - include targets and timelines for improvements for all categories of employees who are not receiving a living wage
 - be developed for a three-year period
 - include reporting on progress towards the previous year's goal and where goals have not been met, a justification should be provided.
- A wage improvement plan template has been developed to support the UoC's compliance with this Indicator.
- The UoC may also choose to develop their own wage improvement plan format, as long as it includes all of the elements listed above.

- The UoC may choose to use a tool such as the GIZ Living Wage Costing Tool to develop the improvement plan.

Evidence of implementation should be maintained, e.g.:

- completed wage improvement plan.

Auditors should confirm:

- the presence of the completed wage improvement plan including reporting towards previous year's goal;
- that the plan provides a clear rationale and explanation where no progress has been made towards closing any living wage gap.

Useful Resources

- ASC's Wage Improvement Plan template
- GIZ Living Wage Costing Tool: offers various simulations to analyse the direct cost implications for producers to pay workers a living wage: <https://www.nachhaltige-agrarlieferketten.org/en/tips-tools>

Indicator:	Requirement:
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- 3.8.5** The UoC shall pay wages in full directly to employees in legal tender or into a bank account in their name at regular intervals, at least monthly. Wages shall not be delayed, deferred or in any way withheld.

How do I interpret this requirement?

- Legal tender does not include:
 - notes which banks no longer exchange
 - notes which are not accepted (for example, too small)
 - currency which is unusable due to inflation
 - in-kind benefits (food, housing, etc) of any kind.
- There are situations where withholding salary, property or benefits are legally required, and so allowed under this indicator. For example:

- salary withholdings, as required by the government. For example, part of a salary is sometimes withheld by a government for:
 - statutory pensions
 - health insurance
 - taxes.
- a government can sometimes require the withholding of salary to pay overdue child-care support costs to another parent or pay overdue taxes
- salary deductions, as defined in a contract or collective bargaining agreement, that are agreed by both parties. For example, an agreed voluntary deduction for housing costs could be allowable.
- bonuses, which are considered optional and additional to the agreed salary and benefits. However, if clearly defined performance targets related to a bonus payment, as defined in a contract have been achieved, then the bonus cannot be withheld.

Evidence of implementation should be maintained, e.g.:

- employee contracts
- payslips
- payment transactions.

Auditors should confirm:

- direct, regular, and accurate payment to employees through records review and interviews with employees and management;
 - if there are any instances when wages were delayed, deferred or withheld in the past and why;
 - employees understand the process for raising grievances if wages are delayed or inaccurate;
 - when foreign migrant employees are employed, apart from legally required deductions, no other deductions shall be made from their wages before payment, for example:
 - recruitment or agency fees
 - cost of airfare
 - employment visas
-

-
- any fees or costs incurred in the recruitment process for employees to secure employment or placement, regardless of the manner, timing or location of their imposition or collection⁴⁵.
-

Useful Resources

- o General principles and operational guidelines for fair recruitment and definition of recruitment fees and related costs:
https://www.ilo.org/sites/default/files/wcmsp5/groups/public/@ed_protect/@protrav/@migrant/documents/publication/wcms_536755.pdf

Indicator:	Requirement:
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3.8.6

The UoC shall document wage payment and receipt by all employees.

Documentation shall be understandable and made available to the employees and shall include information on at least the following:

- a) wages; and,
- b) advances; and,
- c) loans; and,
- d) hours worked including overtime; and,
- e) calculation of any withholdings.

How do I interpret this requirement?

- o Wage payment can be documented in several ways, often through a payslip including all wage information. This can be given to the employee:
 - electronically
 - through a printed copy
 - on or before the payday.
 - o Loans provided by an employer can be an important source of funds for employees in some places.
-

⁴⁵https://www.ilo.org/sites/default/files/wcmsp5/groups/public/@ed_protect/@protrav/@migrant/documents/publication/wcms_536755.pdf

- Clarity of loan calculations and repayment are particularly important, as loans can be an element driving forced labour if employees are not able to repay within a reasonable amount of time based on their wages.
- If loan interest rates are charged by the UoC, they should be:
 - at or below bank rates
 - included in receipt of information on advances, loans, hours worked, pay and the calculation of any deductions.
- For each payment cycle, the UoC should provide a written pay record (such as a payslip) to each employee detailing the following:
 - employee name
 - ID/roll/employee number
 - month/week/pay period
 - number of regular hours worked
 - number of overtime hours worked
 - mandatory deductions towards social security, health insurance and taxes, etc.
 - deductions towards repayment of advances or loans taken by the employee (see also Indicator 3.2.9).
- The pay record should:
 - be provided in paper or digital form.
 - be provided in a language and format easily understandable to employees.
 - be countersigned by employees unless the payment is traceable by bank transfer.
- It is particularly important that employees understand the calculation of hourly premium wages for overtime work.
- Employees should be able to raise any issues they may have related to doubts on calculations on wages, including overtime wages. The UoC should communicate a clear process on how employees can raise such issues.

Evidence of implementation should be maintained, e.g.:

- employee contracts
 - payslips
-

-
- any loan agreements in place
 - working hours records
 - documented wage information, including advances or any withholdings.
-

Auditors should confirm:

- records of wage payments;
 - accuracy of wage payments;
 - employees' access to wage information through employee interviews.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.8.7

The UoC shall ensure that employees are entitled to maternity leave of a minimum of 14 weeks that includes a guarantee of return to the job.

How do I interpret this requirement?

- Many countries have legal requirements for the provision of maternity protection for women employees. These typically include:
 - maternity leave
 - cash benefits
 - employment security, which includes a return to the same or a similar position after taking maternity leave.
 - This requirement extends some or all of these benefits to employees at the UoC.
 - Some countries also have similar legal requirements for the provision of paternity protection for male employees.
 - Maternity leave can include both pregnancy leave and leave following the birth of the child.
-

Evidence of implementation should be maintained, e.g.:

- written confirmation (either in contracts or a company policy or information document) that maternity provisions have been included that meet the requirements of local legislation
-

-
- evidence of maternity leave being paid, where national law requires.
-

Auditors should confirm:

- employees' understanding of maternity or paternity leave rights;
 - employees who have taken maternity/paternity benefits have rejoined work at the same or similar position without a reduction in wages;
 - the UoC does not discriminate in hiring between genders so that they can avoid being obligated to pay maternity (or paternity) benefits to employees, see ASC Farm Standard Criterion 3.4 (Discrimination).
-

Useful Resources

- International Labour Organization (ILO) V. Maternity Protection Convention, 2000 (No. 183)

Criterion 3.9 – Working Hours

Throughout this criterion, when we refer to 'week', it is a reference to a full 7-day calendar week.

Indicator:	Requirement:
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- 3.9.1** The UoC shall ensure that working hours comply with applicable legal requirements and any collective bargaining agreements in place on premium rates, working hours, shift patterns, breaks, daily rest, weekly rest and annual leave.

How do I interpret this requirement?

- National legal requirements on working hours, premium rates, rest times, shift patterns and annual leave can vary according to the country where the UoC is located.
- National legal requirements should always form the minimum requirement for the UoC.
- Collective bargaining:
 - is a voluntary negotiation between employers and organisations of employees to establish the terms and conditions of employment by means of a written collective bargaining agreement (CBA)
 - typically covers wages, working hours and working conditions.
- In the event that national legal requirements, collective bargaining agreements and the ASC Farm Standard differ, the strictest rule always applies.
- Annual leave:
 - is the longest period of rest available to employees
 - allows physical and mental recuperation, and a break from the workplace
 - is paid with normal wages or with additional holiday pay on top of the normal wage (e.g. 13th month)
 - can be taken in days or weeks, with employees being allowed at least a 2-week block, as suggested by ILO standards.
- Laws in many nations set fixed monthly and yearly minimum leave days.
- ILO Convention No. 132 (Holidays with Pay Convention, 1970) specifies that annual leave is:
 - 3 weeks of leave for 1 year of service

- typically accrued monthly
- calculated pro-rata to their contracted working hours,
- ILO Convention No. 146 (Seafarers' Annual Leave with Pay Convention, 1976) can apply in aquaculture when vessels used by the site and subcontractors travel between sites in open water. This convention specifies:
 - 30 days of annual leave for 1 year of service.
- Annual leave days are not the same as public holidays, which are:
 - set by governments for cultural or historical reasons
 - not counted as part of annual leave days.

Evidence of implementation should be maintained, e.g.:

- any written collective bargaining agreements that are in place
- information about national legal requirements regarding premium rates, working hours, shift patterns, rest breaks and annual leave.

Auditors should confirm:

- employees understand that they have the right to engage in collective bargaining about their terms and conditions;
- employees have been provided the opportunity to engage in collective bargaining
- any collective bargaining agreements in place meet national legal requirements or the ASC Farm Standard, whichever is stricter, at a minimum.

Useful Resources

- https://ilo.org/wcmsp5/groups/public/---ed_protect/---protrav/---travail/documents/publication/wcms_491374.pdf

Indicator:	Requirement:
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3.9.2 The UoC shall maintain records of the hours worked by every employee.

How do I interpret this requirement?

- The UoC's working hours recording system:
 - can help the UoC to protect employees against excessive working hours
 - is accessible to employees to verify the record of their hours
 - may include the use of:
 - time clocks with punch cards
 - digital recording systems
 - recording hours manually
 - or other systems.
- The records should indicate
 - the start and finish time each day of the week
 - cumulatively add up to the total hours of regular work for the week
 - any overtime hours worked.

Evidence of implementation should be maintained, e.g.:

- working hours records, which include:
 - start and finish time
 - cumulative working hours
 - any overtime hours worked.
- payslips
- employee contracts.

Auditors should confirm:

- employees are aware that their working hours are being recorded;
 - employees are able to access their working hours records;
 - employees are able to confirm the accuracy of the working hours records;
-

-
- o pay slips, employee contracts and working hours records are all aligned.
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Useful Resources

N/A

Indicator:	Requirement:
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3.9.3

Indicator applicability: any person aged 18 and above.

The UoC shall ensure that working hours do not exceed 8 hours per day and 48 hours in a week (excluding overtime).

An exception applies where the operations of the UoC require a different working pattern, in which case the UoC may use averaging of working hours if permitted by applicable legal requirements, calculated over a maximum 17-week reference period.

When more than 48 hours are worked in a week as a result of averaging, the UoC shall ensure that the following conditions are met:

- it is in line with applicable legal requirements; and
- it is in line with collective bargaining agreements; and
- terms are clearly set out and agreed to by employees in their contracts; and
- appropriate safeguards are put in place to protect the employee's health and safety.

How do I interpret this requirement?

- o The normal workday is a maximum of 8 hours, excluding breaks, and the normal work week is a maximum of 48 hours.
 - o Where collective bargaining agreements, industry standards or national law require a shorter day or week, this applies.
 - o The ILO defines shift work as: 'a method of organisation of working time in which workers succeed one another at the workplace so that the establishment can operate longer than the hours of work of individual workers at different daily and night hours'⁴⁶.
-

⁴⁶ILO (2004) What is shiftwork? Geneva, ILO, Information Sheet No. WT-8. https://www.ilo.org/wcmsp5/groups/public/---ed_protect/---protrav/---travail/documents/publication/wcms_170713.pdf

- Where collective bargaining and national law allows, the UoC can employ people to work in shifts:
 - They may choose to work more than eight hours per day and more than 48 hours in one week, if the number of hours averaged over a period of 17 weeks does not exceed a normal workday and normal work week.
- Long working hours can mean that employees can get fatigued, which can result in accidents or mistakes.
- Appropriate safeguards in place to protect employees' health and safety can include:
 - a risk assessment looking specifically at the risk of fatigue
 - adequate or extra rest breaks and places for employees to rest during long shifts
 - enough food and drink available for employees to ensure that they are not hungry
 - avoiding employees who have worked long hours of overtime conducting hazardous work (for example, operating heavy machinery)
 - ensuring that dark areas are well lit
 - ensuring that, where possible, employees are not working alone for long hours.

Evidence of implementation should be maintained, e.g.:

- employee contracts
- payslips
- working hours records
- health and safety risk assessment.

Auditors should confirm:

- any different shift patterns have been entered into voluntarily by employees;
 - the UoC understands the risk inherent in long working hours and fatigue and appropriate mitigations are being implemented.
-

Useful Resources

N/A

Indicator:	Requirement:
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- 3.9.4** The UoC shall ensure that overtime hours are;
- are voluntary; and
 - are not requested regularly; and
 - have appropriate safeguards in place to protect the employee's health and safety.

How do I interpret this requirement?

- Overtime should not occur regularly but under exceptional circumstances, for example, overtime can be requested:
 - for unexpected peak production times
 - if staff are absent and others must cover their work.
- The UoC should plan work in such a way to minimise the risk of regularly requesting overtime.
- If the UoC continues to need the same jobs done in overtime repeatedly, this is an indication that the UoC may need to:
 - review resources
 - plan work differently
 - hire additional employees.
- Any overtime is voluntary, meaning it can only take place if the employee:
 - consents to work extra hours
 - is not discriminated against or penalised for refusing overtime.
- Health and safety hazards related to excessive overtime can be avoided by implementing a clear overtime policy that:
 - limits excessive overtime
 - ensures that overtime requirements are announced in advance when possible
 - ensures that overtime is voluntary
 - rotates schedules so that overtime is not always assigned to the same employees

- ensures that health and safety training shows how to recognise and manage signs of fatigue and stress.
- The UoC should allow regular rest breaks during overtime shifts.
- The UoC can maintain open and transparent communication channels with employees to encourage them to provide feedback about their work conditions and overtime hours, and use this input to adjust schedules.
- Appropriate safeguards in place to protect employees' health and safety can include:
 - a risk assessment looking specifically at the risk of fatigue
 - adequate or extra rest breaks and places for employees to rest during long shifts
 - enough food and drink available for employees to ensure that they are not hungry
 - avoiding employees who have worked long hours of overtime conducting hazardous work (for example, operating heavy machinery)
 - ensuring that dark areas are well lit
 - ensuring that, where possible, employees are not working alone for long hours.

Evidence of implementation should be maintained, e.g.:

- health and safety risk assessment
- contracts
- working hours records.

Auditors should confirm:

- employees understand that overtime hours are voluntary;
- management understand the risks in working long hours and are well prepared to mitigate those risks;
- regular activities are carried out in regular working hours and that overtime is used for exceptional circumstances.

Useful Resources

N/A

Indicator:	Requirement:
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3.9.5 The UoC shall ensure that employees do not work more than 12 hours of overtime per week.

How do I interpret this requirement?

- Overtime is any additional hours spent working beyond normal working hours
- This requirement does not allow these overtime hours to exceed more than 12 per week.

Evidence of implementation should be maintained, e.g.:

- contracts
- working hours records.

Auditors should confirm:

- employees understand how much overtime they are allowed to work.

Useful Resources

N/A

Indicator:	Requirement:
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3.9.6 The UoC shall ensure that overtime hours are paid at a premium rate of at least 125% of the agreed wages, or as defined by applicable legal requirements or any collective bargaining agreements in place, whichever is higher.

How do I interpret this requirement?

- For overtime hours 125% of the basic wage is paid to employees.
- Overtime work often involves sacrifices from employees, such as time away from family and personal life, and this additional effort needs to be rewarded.
- Where collective bargaining agreements or applicable law include a premium rate that is below 125%, the ASC minimum of 125% applies.
- The premium rate should be:

- documented in payslips
- documented in the employee contract
- clearly understood by all employees.

Evidence of implementation should be maintained, e.g.:

- contracts
- working hours records
- payslips.

Auditors should confirm:

- employees understand that overtime hours are paid at a premium rate;
- employees understand the difference between regular hours and overtime hours;
- if the bar of 'exceptional circumstances' is being used to determine overtime requirements.

Useful Resources

N/A

Indicator:	Requirement:
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3.9.7

Indicator applicability: any person aged 18 and above.

The UoC shall ensure that all employees are granted breaks during work shifts. Where this is defined by applicable legal requirements or any collective bargaining agreements in place, these shall be followed.

How do I interpret this requirement?

- The amount, length and frequency of rest breaks that employees are entitled to can be specified by:
 - collective bargaining agreements in place
 - national laws.
- The UoC should ensure that rest breaks are clearly agreed upon and set out for employees.
- Rest breaks:

- provide employees with essential rest during the workday
 - help reduce fatigue, which can also reduce the risks of accidents or errors
 - may also accommodate personal needs, like restroom or prayer breaks.
- Ensuring clean and hygienic facilities for breaks is vital in workplaces with potential hygiene risks due to factors like chemicals, noise, or pollutants, where employees cannot leave the premises during breaks.
 - Management should consider climate conditions when they plan breaks and rest times. In very hot temperatures, employees will need:
 - more frequent access to breaks,
 - breaks taken in the shade where possible
 - access to drinking water.
 - This requirement refers to rest time when an employee can be asked to work, only if:
 - they agree to work during their rest time
 - the rest time is made up at a different time.
 - There can be situations where an employee is on standby, which is when:
 - the employee is required to be on-site, even if they are not required to be working for that period of time
 - the hours that they are required to be on-site are counted as working hours
 - they must be fully compensated.
 - Examples of when employees may be on standby includes any time spent at the UoC where they could be at the direction of the company to perform a work-related task. For example:
 - an employee waiting for the arrival of a delivery of feed, which they will help to unload
 - an employee who is responsible for feeding the fish and in between feeds, they are waiting at the site for the next feeding time.
 - There can also be situations when an employee is on-call, which is when:
 - the employee is not required to be on site
 - the employee is required to be reachable and available during a defined period of time
 - this time of being available is not counted as normal working hours
 - this time should be compensated in some way.
 - Examples of when employees may be on-call can include:

- an employee who is at home overnight, but is available to be called in case there is an emergency on the site and they are called into the site.

Evidence of implementation should be maintained, e.g.:

- contracts
- any collective bargaining agreements that are in place.

Auditors should confirm:

- employees understand the pattern of rest breaks that they are entitled to;
- rest breaks are adequate for the needs of the employees in their context.

Useful Resources

N/A

Indicator:	Requirement:
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3.9.8 *Indicator applicability: any person aged 18 and above.*

The UoC shall ensure that all employees are granted at least 11 consecutive hours of rest within a 24-hour period.

How do I interpret this requirement?

- Rest for a period of 11 consecutive hours within a 24-hour period ensures that employees have an adequate amount of time for both sleep and personal free time.
- All employees, regardless of their shift pattern, require this rest.

Evidence of implementation should be maintained, e.g.:

- contracts
- any collective bargaining agreements that are in place
- working hours records.

Auditors should confirm:

- employees understand the rest time that they are entitled to.

Useful Resources

N/A

Indicator:	Requirement:
------------	--------------

3.9.9 *Indicator applicability: any person aged 18 and above.*

The UoC shall ensure that all employees are granted at least 24 consecutive hours of rest within a 7-day period, or at least 48 consecutive hours of rest within a 14-day period if allowed by applicable law.

How do I interpret this requirement?

- The ILO's 1921 Weekly Rest Convention (No. 14) mandates a 24-hour weekly rest period (within a 7-day period).
- Where the UoC may be located more remotely or where shift patterns may require a longer number of consecutive days of work, 48-hours rest can be taken within a 14-day period, if allowed by applicable law and collective bargaining agreements.
- Where there is a longer number of consecutive days working, the UoC should ensure that there are adequate safeguards for employees' health and safety, as per Indicator 3.9.3.

Evidence of implementation should be maintained, e.g.:

- contracts
- any collective bargaining agreements that are in place
- working hours records.

Auditors should confirm:

- employees understand the rest time that they are entitled to.

Useful Resources

N/A

Indicator:	Requirement:
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3.9.10 The UoC shall ensure that additional breaks are granted for pregnant women and breastfeeding women. These breaks shall be counted as working time and appropriate facilities shall be provided.

How do I interpret this requirement?

- According to ILO, the lack of support at the workplace is one of the main reasons that women stop breastfeeding before the recommended time⁴⁷.
- Employers can support breastfeeding women by providing:
 - ample time away from work activities
 - a comfortable, private location to express breast milk
 - access to a fridge to store it
 - a clean and safe environment (other than a bathroom).

Evidence of implementation should be maintained, e.g.:

- policy, for example addressing new and expectant mothers
- health and safety risk assessment
- contracts
- working hours records.

Auditors should confirm:

- entitlement to paid nursing breaks in the country/region where the UoC is located;

⁴⁷ https://www.ilo.org/global/about-the-ilo/newsroom/comment-analysis/WCMS_218710/lang--en/index.htm

- confirm that a location for breastfeeding mothers is suitable;
- confirm that the relevant employees understand these requirements.

Useful Resources

- https://www.ilo.org/global/about-the-ilo/newsroom/comment-analysis/WCMS_218710/lang-en/index.htm
- Ontario, Canada Public Health Association recommendations for creating a breastfeeding friendly workplace:
<https://www.opha.on.ca/OPHA/media/Resources/Resource%20Documents/BreastfeedingFriendlyWorkplace-Sep08.pdf>
- Region of Waterloo, Canada recommendations for creating a breastfeeding friendly workplace: https://www.regionofwaterloo.ca/en/doing-business/resources/Documents/Breastfeeding_Friendly_Workplace_Toolkit.PDF

Indicator:	Requirement:
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3.9.11 The UoC shall ensure that all employees are granted paid leave as defined by applicable legal requirements or any collective bargaining agreements in place.

How do I interpret this requirement?

- Paid leave can include:
 - public holidays
 - annual leave
 - maternity/paternity leave
 - sick leave.
- Types of leave can be defined by national law and collective bargaining agreements.
- Annual leave:
 - is the longest period of rest available to employees
 - allows physical and mental recuperation and a break from the workplace

- is paid with normal wages or with additional holiday pay on top of the normal wage (e.g. 13th month)
 - can be taken in days or weeks, with employees being allowed at least a 2-week block, as suggested by ILO standards.
- Laws in many nations set fixed monthly and yearly minimum leave days.
 - ILO Convention No. 132 (Holidays with Pay Convention, 1970) specifies that annual leave is:
 - 3 weeks of leave for 1 year of service,
 - typically accrued monthly, and
 - calculated pro-rata to their contracted working hours.
 - ILO Convention No. 146 (Seafarers' Annual Leave with Pay Convention, 1976) can apply in aquaculture when vessels used by the site and subcontractors travel between sites in open water. This convention specifies:
 - 30 days of annual leave for 1 year of service.
 - Annual leave days are not the same as public holidays, which are:
 - set by governments for cultural or historical reasons
 - not counted as part of annual leave days.

Evidence of implementation should be maintained, e.g.:

- contracts
- collective bargaining agreements
- working hours records.

Auditors should confirm:

- employees understand their entitlement to paid leave;
 - employees do not feel discriminated against when they take annual leave.
-

Useful Resources

N/A

Criterion 3.10 – Workplace Conduct Response

Indicator:	Requirement:
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- 3.10.1** The UoC shall have a written workplace conduct response policy with the intent of setting out the process to respond to breaches of company rules.
- The policy shall define clear levels of escalation and shall ensure that the employee's dignity is upheld at all times.

How do I interpret this requirement?

- The written policy should:
 - outline the process for when employees break any rules of the UoC, detailing:
 - what behaviour may trigger disciplinary action
 - the corresponding action to be taken
 - adhere to local legislation and any collective bargaining agreements in place, particularly regarding cases of potential dismissal
 - list who is responsible for decision-making at different levels
 - outline how employees are rewarded for good work and encouraged to improve
 - be accessible and understandable to all employees
 - be integrated in company processes through training for supervisors and managers, including:
 - how to communicate effectively
 - carry out any responses in a fair and respectful manner.
- Suspension can sometimes be used as a disciplinary action. This period of suspension should always be paid unless:
 - suspension for a time is offered to the employee as an alternative to dismissal, this period could be unpaid as long as:
 - the employee has agreed accept this
 - it is not enacted after the event as deduction from wages.

Table 5. Examples of levels of escalation in disciplinary procedures (adapted from SA8000):

Type of infraction	Example of infraction	Example disciplinary action progression / escalation plan / procedure
Minor	Isolated incident of tardiness or minor production errors	1. Oral warning to employee that his/her conduct or level of performance is unacceptable. 2. If unsatisfactory performance or conduct continues or becomes more serious, a written warning identifying the details of the infraction that led to the warning. May contain a request for corrective action, a timeline for correction, and the consequences if not complied with. 3. If the employee fails to improve, a final written warning that documents the continued misconduct or poor performance and explains the consequences of further failure to improve, such as possible suspension or even dismissal. 4. As a last resort, the employee may be suspended, demoted, or dismissed. The employee is given the opportunity to appeal the decision before such action is taken.
Major	Repeated incidents of tardiness or minor production errors Isolated major production error	1. A written warning to the employee identifying the details of the infraction that led to the warning. May contain a request for corrective action, a timeline for correction, and the consequences if not complied with. 2. If the employee fails to improve, a final written warning that documents the continued misconduct or poor performance and explains the consequences of further failure to improve, such as possible suspension or even dismissal. 3. As a last resort, the employee may be suspended, demoted, or dismissed. The employee is given the opportunity to appeal the decision before such action is taken.
Gross misconduct	- o Theft or fraud - o Physical violence or bullying - o Deliberate and serious damage to property - o Serious incapability at work brought on by alcohol or illegal drugs - o - Causing loss, damage, or injury through serious negligence	1. The employee may be suspended from work on full pay, while the alleged offence is investigated. 2. If, on completion of the investigation and the full disciplinary procedure, the organisation is satisfied that gross misconduct has occurred, the result could be summary dismissal without notice or payment in lieu of notice. However, the employee is given the opportunity to appeal the decision before such action is taken.

Evidence of implementation should be maintained, e.g.:

- o written workplace conduct response policy
- o training records
- o records of any disciplinary process.

Auditors should confirm:

- the workplace conduct response policy complies with the ASC Farm Standard's intention;
- employees are aware of the workplace conduct response policy;
- employees confirm that the company adheres to their policy and actions are fair;
- there are no monetary deductions imposed during the process.

Useful Resources

- ACAS statutory Code of Practice on discipline and grievance procedures:
<https://www.acas.org.uk/acas-code-of-practice-for-disciplinary-and-grievance-procedures/html>
- Auditors can consult with and seek information from competent authorities and interested parties. For example, local worker welfare organisations, NGOs, labour lawyers and civil society.

Indicator:	Requirement:
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3.10.2 The UoC shall ensure that all employees are aware of, and understand, the workplace conduct response policy.

How do I interpret this requirement?

- The UoC can inform employees about the workplace conduct response policy by:
 - including it in the employee contract or staff handbook
 - distributing the written policy to employees in languages they understand
 - including the workplace conduct response policy in orientation or ongoing training.

Evidence of implementation should be maintained, e.g.:

- written workplace conduct response policy
- training records
- employee handbook.

Auditors should confirm:

- training records and employee handbook;
- employees understand workplace response procedures
- how employees were trained on workplace response procedures.

Useful Resources

N/A

Indicator:	Requirement:
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3.10.3 The UoC shall maintain records of responses to breaches of company rules.

How do I interpret this requirement?

- These records should include:
 - the observed misconduct by the employee
 - the actions taken by the employer, including:
 - copies of written notices
 - withholding of bonuses, suspensions, demotions or dismissals, if applicable
 - the investigation carried out
 - the appeals process by the employee, if applicable
 - who was involved in or witnessed each of these steps.

Evidence of implementation should be maintained, e.g.:

- written workplace conduct response policy
- records of any workplace response procedures.

Auditors should confirm:

- records of any investigations, actions and responses.

Useful Resources

N/A

Indicator:	Requirement:
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3.10.4 The UoC shall not deduct from wages or benefits as part of any response to breaches of company rules.

How do I interpret this requirement?

- A person's livelihood and ability to feed and provide for themselves and their families should not be impacted by a disciplinary action at work.
- Wages cannot be withheld for disciplinary action from any:
 - regular wages
 - payments for overtime
 - other benefits such as fixed year-end bonuses.
- Non-fixed bonuses that are determined based on performance are not considered a benefit.
- This indicator applies even if local laws or regulations allow monetary fines to be imposed as part of disciplinary procedures.
- Suspension can sometimes be used as a disciplinary action. This period of suspension should always be paid unless:
 - suspension for a time is offered to the employee as an alternative to dismissal, this period could be unpaid as long as:
 - the employee has agreed to accept this
 - it is not enacted after the event as deduction from wages.

Evidence of implementation should be maintained, e.g.:

- written workplace conduct response policy
- payroll documentation to verify if they are deductions other than legally mandated deductions (for example, taxes and employees' contributions towards social security or health insurance).

Auditors should confirm:

- if there have been any fines, as they are not always reflected on pay records;
 - if there are any specific methods of discipline for production targets not being met or for quality issues;
 - any fines that employees have had to pay (for example, for coming late to work, excessive use of toilets, being absent or sick etc.).
-

Useful Resources

N/A

Criterion 3.11 – Employee Accommodation

Indicator:	Requirement:
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- 3.11.1** The UoC shall ensure that any accommodation provided for employees or their family:
- a) is according to applicable legal requirements; and
 - b) has adequate safety provisions; and
 - c) is regularly maintained and kept in good working order; and
 - d) is clean and hygienic; and
 - e) allows for adequate privacy, including separation by gender if required; and
 - f) meets the basic needs of the employees and their families; and
 - g) is sufficient for the number of people using them.

How do I interpret this requirement?

- On-site accommodation is typically offered by workplaces:
 - in remote locations
 - when local housing supplies are otherwise very limited.
- On-site accommodation can mean that there is a higher risk of forced and child labour because:
 - employees and their families living in on-site accommodation can be less integrated into a local community
 - their presence on site can mean they are asked to work more hours than their contracts specify
 - high rental costs relative to employee income mean that employees can incur debt.
- The UoC is not required to provide accommodation for its employees.
- Employees are not required to reside in employer-provided accommodation.
- The UoC monitors and ensures the quality, safety and maintenance of housing that is provided, regardless of whether it owns, leases or contracts the facility.
- Facilities should:
 - be built to withstand recurring extreme weather conditions

- be located in a non-hazardous location, with adequate drainage, pest control and hygiene procedures in place
- have fire safety and other emergency response measures in place, for example:
 - marked safety exits
 - emergency escape routes
 - assembly points
 - firefighting equipment, which is maintained annually or when required
- include separate accommodations by gender that ensure adequate privacy, for example:
 - locking doors when both genders are present
 - an individual bed for each employee in shared dormitories
 - an individual storage space for each employee in shared dormitories, including lockable storage space for personal or valuable items
- ensure that families with children have separate rooms from employees who are not members of that family
- ensure that children live with their parents
- provide separate spaces for sleeping and social life
- provide a separate, well-ventilated canteen or space for cooking and eating.

Evidence of implementation should be maintained, e.g.:

- records of scheduled housekeeping such as cleaning of common areas like toilets, shower stalls, corridors and garbage disposal
- records of maintenance, for example, fire alarm service, extinguisher service, emergency lighting service, gas, electrical, etc.

Auditors should confirm:

- o by conducting a walkthrough of the accommodation, that all provisions of this Indicator are being met;
- o what employees report about the living conditions in the accommodation.

Useful Resources

- ILO guidance on worker's housing: [wcms_116344.pdf \(ilo.org\)](#)

Indicator:	Requirement:
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3.12.2 The UoC shall have an employee grievance mechanism which is accessible, fair and applicable to all employees.

How do I interpret this requirement?

- Rates are 'reasonable' when:
 - below or comparable to similar accommodation available on the market
 - not more than a reasonable proportion of the employee's income.
- Where an employee or employee and their family plans to live in employer-provided accommodation:
 - the level of rent is a part of the contracting agreement
 - the site is able to provide a calculation and rationale for the level of rent.
- If an employee lives in employer-provided accommodation and has an early termination of their contract (for example, due to gross misconduct), they should be entitled to a reasonable period of time to vacate the premises.

Evidence of implementation should be maintained, e.g.:

- records of rental agreements and payments
- employee contracts
- payslips.

Auditors should confirm:

- the costs of any accommodation;
- average local rental rates;
- any additional charges for the provision of water and electricity, which should not be more than what employees would have to pay if they stayed outside the employer-provided accommodation.

Useful Resources

N/A

Indicator:	Requirement:
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3.11.3 The UoC shall provide access to appropriate sanitary facilities that:

- a) are clean and hygienic; and
- b) include adequate privacy measures; and
- c) include separation by gender if required; and
- d) are sufficient for the number of people using them.

How do I interpret this requirement?

- Appropriate sanitary facilities:
 - are located within or in close proximity (i.e. less than 60 metres) to the dormitory or housing facilities
 - are dry and warm
 - have a source of ventilation (e.g. window or fan)
 - have adequate lighting
 - have doors that close and lock for privacy
 - are separated for men and women except in family accommodation
 - are constructed of hard materials that are easy to clean
 - include at least one of the following:
 - pit latrines
 - flush toilets
 - urinals
 - have taps and running water
 - have facilities for the safe disposal of sanitary products.
- Waste and wastewater are disposed of in accordance with Criterion 2.12.

Evidence of implementation should be maintained, e.g.:

- records of scheduled housekeeping such as cleaning of common areas like toilets, shower stalls, corridors and garbage disposal.

Auditors should confirm:

- through a walkthrough of the sanitary facilities that all provisions of this Indicator are met;
- what employees report on the quality of drinking water, maintenance of sanitary equipment like taps and toilet flushing systems.

Useful Resources

N/A

Indicator:	Requirement:
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- 3.11.4** The UoC shall arrange for an annual meeting with employees or their representatives to discuss any reasonable improvements required to accommodation. The UoC shall maintain records of these meetings and shall take actions where necessary.

How do I interpret this requirement?

- Employees can use this meeting to:
 - make suggestions on improvements to ensure accommodation meets requirements related to:
 - safety
 - hygiene
 - privacy
 - sanitary facilities
 - rates
 - highlight any maintenance issues that need to be addressed.
- Any agreed improvements are acted upon by the UoC.

Evidence of implementation should be maintained, e.g.:

- records of meetings held between the UoC management employee representatives.

Auditors should confirm:

- if any issues have been brought to the notice of management and how they have been resolved;
- the process of raising grievances about living conditions.

Useful Resources

N/A

Indicator:	Requirement:
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3.11.5 The UoC shall ensure that all employees working in remote locations have access to a free, private, real-time communication link to communicate externally.

How do I interpret this requirement?

- Having access to outside communication is important for employees to:
 - protect their mental wellbeing
 - have an opportunity to raise any issues that occur on site.
- Employees can use this communication link to communicate with:
 - family and friends
 - any independent organisation that they may be aware of (either through information from the UoC or other sources) who can support them.
- This communication can happen through:
 - an employee's own mobile phone
 - a telephone provided by the UoC
 - WiFi provided by the UoC for employees.
- This line of communication is available 24 hours a day.

- Outside an employee's working hours there are no restrictions on when employees are able to connect with people outside the UoC.
-

Evidence of implementation should be maintained, e.g.:

- information about the availability of an outside communication link
 - evidence that the communication link is working.
-

Auditors should confirm:

- that lines of communication are actually working;
 - the availability of these lines of communication for employees.
-

Useful Resources

N/A

Criterion 3.12 – Grievance Mechanisms

Indicator:	Requirement:
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- 3.12.1** The UoC shall, when concerns arise, engage in dialogue with employees or their representatives, with the intent of avoiding an escalation to a grievance. The UoC shall ensure that employees understand how to engage in this dialogue.

How do I interpret this requirement?

- The UoC proactively implements a dialogue system that:
 - allows for open participation of all employees
 - is easily accessible to employees on a regular basis
 - provides options for anonymous feedback.
- Mechanisms for employees to provide feedback and engagement with management could include:
 - dialogue between management and employees
 - engagement with elected employee or union representatives
 - employee meetings or assemblies to solicit feedback
 - employee committees
 - suggestion box that is safely accessible for employees
 - access to an online feedback form or dedicated email address or phone number
 - employee surveys.
- Issues raised during this process may include suggestions for improvement or highlight negative impacts on employees from the UoC's activities. Examples can include:
 - operational inefficiencies such as inadequately maintained machinery that may impact efficiency and could cause health risks
 - employee rights such as female employees are paid less than male employees with similar jobs
 - harassment, such as an employee is pressured to begin a romantic or sexual relationship with a supervisor (whether this occurs during or outside work hours and work location)

- working conditions, such as:
 - employees are frequently asked to work excessive overtime hours
 - employees are not properly compensated for their work.

Evidence of implementation should be maintained, e.g.:

- documentation on the mechanisms available
- meeting minutes
- records of actions taken to address issues.

Auditors should confirm:

- how the mechanisms for reporting any issues are communicated to the employees;
 - what the procedures are to facilitate dialogue;
 - employees' awareness of any processes to raise or discuss concerns with managers;
 - how and how often issues can be communicated to the management;
 - the system for reviewing and addressing any issues.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.12.2 The UoC shall have an employee grievance mechanism which is accessible, fair and applicable to all employees.

How do I interpret this requirement?

- A grievance mechanism is a formal complaint process that can be used by employees to resolve a grievance.
- Based on UN Guiding Principle No. 31, and effective a grievance mechanism must:
 - engender trust among participants

- be transparent: ensure that users are given clear information about what is happening
 - be fair: ensure that users have access to sources of information and expertise necessary to engage in a grievance process
 - be predictable: ensure that users understand the process, how long it takes and what the outcomes can be
 - be accessible: ensure that users have been informed about it and can use it
 - be rights-based: ensure that outcomes and remedies are compatible with international human rights
 - be based on dialogue and engagement, ensuring that stakeholders are consulted and included in the process
 - be a source for continuous learning, analysing individual complaints to help prevent other grievances.
- Mechanisms for submitting grievances are best determined together with employee representatives and can include:
 - in-person meetings
 - dedicated telephone numbers
 - email addresses
 - instant messaging or other online submission mechanisms
 - letters or written notes
 - a suggestion box
 - via employee representatives, trade unions, and worker associations
 - worker voice tools to conduct anonymous worker surveys.
 - For a grievance mechanism to be accessible for employees they must understand:
 - their labour rights (Criterion 3.1)
 - rights related to other workplace issues, such as health and safety regulations (Criterion 3.5)
 - how the grievance mechanism functions and what kinds of issues may be brought to the grievance mechanism (see also 3.12.3).
 - Foreign migrant workers may be given the option to raise issues:
 - internally with the UoC using the same channels open to other employees

- with the local Consular Office of the Labour Attaché for their country in the country where the UoC is located
 - websites or Facebook pages of the Consulate or Embassy where migrant workers can raise grievances.
- o An effective grievance process will incorporate elements of the following steps at a minimum (adapted from Rainforest Alliance's Grievance Guidance):
- Step 1: Receive and acknowledge grievance complaint
 - Step 2: Take immediate action on human rights or hazardous issues (see Appendix 10 for the Remediation Procedure)
 - Step 3: Review the grievance complaint to check if it is admissible.
 - if it is inadmissible, this is communicated along with the reason, to the person who brought the grievance.
 - an inadmissible complaint is one that is unrelated to the activities or location of the UoC.
 - Step 4: Investigate the grievance complaint to establish what happened, maintaining confidentiality.
 - for complex issues or where there are conflicts of interest, external experts may be brought in to support.
 - Step 5: Communicate findings to the person who brought the grievance.
 - Step 6: Agree on a remediation plan with the person who brought the grievance.
 - for complex issues, external arbitration and external remediation support including NGOs or rights experts may be brought in.
 - Step 7: Provide an opportunity for an appeals process, if agreement cannot be reached.
 - Step 8: Implement and monitor the remediation plan.
 - Step 9: Communicate the progress of the remediation to the person who brought the grievance and to any relevant stakeholders when confidentiality is not a concern.
 - Step 10: Engage in continuous improvement.

Example grievance	Example resolution
My wages are always paid 4 or 5 days late, sometimes even more than a week late. This makes it difficult to plan when to pay my bills and how to manage my finances.	1. Member of staff hearing the grievance ensures that the employee understands that their complaint is confidential. 2. Ensure that the employee has received their latest wages. 3. Interview other employees to understand if anyone else is experiencing the same issue. 4. Examination of the wage payment system to understand why there are delays. 5. Any changes necessary are implemented. 6. Next month, the member of staff who is working on the grievance checks with the employee about the date they received their wages.
I keep being asked to go out onto the feed barge without being provided with a lifejacket.	1. Member of staff hearing the grievance ensures that the employee understands that their complaint is confidential. 2. Interview other employees to understand if anyone else is experiencing the same issue. 3. Check the availability of life jackets and ensure that there is a sufficient number available, that they are in good working condition and that they are kept in an accessible location. 4. Add some signs to remind workers and management that life jackets are essential for going onto the feed barge. 5. Follow up with the employee after a few weeks to find out if the situation has improved.

Evidence of implementation should be maintained, e.g.:

- grievance policy or procedure
- employee survey results
- employee notices and notice boards
- employee handbook
- records from grievance procedures and their outcomes

- meeting minutes.

Auditors should confirm:

- if the grievance procedure provides employees with clearly written and understandable instructions on how to report grievances using different methods that involve formal, informal and anonymous reporting;
 - if the grievance procedure applies to and is accessible to all employees, including full-time, permanent, contract, seasonal, migrant (domestic and foreign) and temporary employees.
-

Useful Resources

- Rainforest Alliance's Grievance Guidance: <https://www.rainforest-alliance.org/business/resource-item/guidance-e-grievance-mechanism/>
- SHIFT Project is a non-profit consultancy focused on human rights Due Diligence based on the UN Guiding Principles for Business and Human rights. They have published two guidance documents on grievance mechanisms with clear definitions, both for community and employee-focused grievance. https://shiftproject.org/wp-content/uploads/2020/02/191210_publication_grievance-mechanism.pdf
- https://shiftproject.org/wp-content/uploads/2014/05/Shift_remediationUNGPs_2014.pdf
- The Global Compact Network Netherlands, Oxfam and Shift also developed a 'Doing Business With Respect for Human Rights' guide for companies to translate expectations in the UN Guiding Principles on Business and Human Rights into concrete action, with a chapter on Remediation and Grievance: <https://www.businessrespecthumanrights.org/en/page/349/remediation-and-grievance-mechanisms>
- https://www.ihrb.org/uploads/reports/EC-Guide_OG-12_Part-3_Section-VI.pdf

Indicator:	Requirement:
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3.12.3 The UoC shall ensure that all employees are trained on the employee grievance mechanism.

How do I interpret this requirement?

- Training on the grievance mechanism is provided to all employees. This can take place:

- as part of the induction process
- at the start of each production cycle.
- Awareness of this mechanism is:
 - essential for all employees
 - important for a positive environment in the workplace.
- The UoC can communicate the grievance mechanism to employees through a variety of ways, including:
 - induction training and other ongoing training
 - information posted in public areas of the UoC
 - distributing it to employees in flyers, via text messages, instant messaging, emails or other private forms of communication
 - discussing it in employee meetings and assemblies
 - including it in employment contracts or the employee handbook.

Evidence of implementation should be maintained, e.g.:

- grievance policy or procedure
- employee handbook
- training materials or distributed materials
- meeting minutes.

Auditors should confirm:

- employees' understanding of how to raise issues and how they were made aware of the grievance mechanisms.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.12.4 The UoC shall ensure that all employees are trained on the employee grievance mechanism.

How do I interpret this requirement?

- For the grievance mechanism for communities, the guidance under Indicator 3.12.2 for employees also applies.
- Mechanisms for submitting grievances are best determined together with community representatives. Examples of channels for submitting grievances include:
 - in-person meetings
 - dedicated telephone numbers
 - email addresses
 - instant messaging or other online submission mechanisms
 - letters or written notes
 - via community leaders and representatives, or community organisations.
- Different communities may prefer different channels for submitting grievances.
- Vulnerable populations may need special consideration in terms of how the grievance is submitted to enable their participation. For example:
 - when persons of the same gender, or ethnic group are available to receive a complaint, that may increase the likelihood of participation.
- The following steps can be taken to inform potential users about the grievance mechanism:
 - conduct a stakeholder mapping and engagement process to identify communities and Indigenous and tribal peoples who may use the grievance mechanism.
 - make them aware of their rights under the law and under the ASC Farm Standard, including the grievance mechanism.
 - share information about the grievance mechanism, for example through:
 - local community representatives, community leaders and relevant community organisations

- local publications
- printed materials such as company newsletters, flyers, posters
- digital means such as company website, online newsletter or social media.
- give special consideration to reaching vulnerable groups within the population, as they are:
 - most likely to be impacted
 - unlikely to be comfortable with raising issues to the company.

Evidence of implementation should be maintained, e.g.:

- materials used to inform communities of the grievance mechanisms
- minutes of meetings where grievance mechanisms are discussed.

Auditors should confirm:

- community representatives (including Indigenous people's representatives where applicable) understand:
 - how they can raise concerns and grievances with the UoC;
 - who their point of contact at the UoC is;
 - how the grievance mechanism was communicated to communities;
 - when discussions were had between UoC management and communities;
 - when grievances were raised in the past and how they were resolved.

Useful Resources

N/A

Indicator:	Requirement:
------------	--------------

3.12.5 The UoC shall ensure the grievance mechanisms in Indicators 3.12.2 and 3.12.4 include provisions for non-retaliation.

How do I interpret this requirement?

- Non-retaliation means that no action can be taken by the UoC against a person or an entity bringing a grievance that has been raised in good faith.
- If retaliation occurs, this means that the grievance mechanism is inadequate.
- This protection against retaliation applies to those bringing grievances and to whistleblowers.
- ILO defines whistleblowing as: 'the reporting by employees or former employees of illegal, irregular, dangerous or unethical practices by employers'⁴⁸.
- Examples of retaliation in the workplace can include:
 - direct consequences, such as:
 - firing
 - demotion
 - salary reduction
 - reduced hours
 - more subtle consequences, such as:
 - reassignment to work that causes personal hardship
 - exclusion from key meetings and group events
 - threats
 - harassment
 - bullying
 - indirect consequences, such as:
 - encouraging shunning, harassment or bullying by other staff members.
- Examples of retaliation for a community member or Indigenous or tribal person can include:
 - direct actions
 - indirect actions through others
 - threats

⁴⁸ <https://www.ilo.org/media/406736/download>

- harassment
- damage to a person or property.
- Ways to protect employees and communities from retaliation can include:
 - having clear grievance processes and policies that include accountability (see 3.12.2)
 - raising a confidential grievance (see also 3.12.9)
 - raising an anonymous grievance through a hotline or similar, where the identity of the person bringing the grievance is never known:
 - in this case, remediation is likely to affect a wider population and is not tied to a specific individual

Evidence of implementation should be maintained, e.g.:

- grievance policy or procedure, including provision for confidentiality
- records of grievance processes.

Auditors should confirm:

- policies and provisions for non-retaliation with employee and community representatives.

Useful Resources

N/A

Indicator:	Requirement:
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3.12.6

The UoC shall ensure that:

- a) grievances are acknowledged within 10 days from submission; and
- b) grievances are resolved within 90 days from submission.

Where resolution is not possible within 90 days, justification and an alternative timeline shall be provided to relevant parties.

How do I interpret this requirement?

- The UoC follows the procedure to address grievances, beginning with acknowledging the receipt of the grievance within 10 days from submission.
- The UoC develops a clear action plan to resolve the grievance within 90 days from submission.
- Urgent concerns should be addressed sooner. For example:
 - If the grievance relates to a workplace hazard related to equipment that could harm an employee, that equipment must be removed from operation and repaired immediately.
 - If the UoC suspects that the person making the claim or other people may be at risk, and if the case involves human rights, including potential cases of child labour or forced labour, the UoC follows Appendix 16 (Remediation Protocol).
- Full remediation of the issue may take place over a longer timeframe when required by the circumstance. For example:
 - until the child has reached school-leaving and working age.
- Simple grievances may not require a 90-day timeframe.

Evidence of implementation should be maintained, e.g.:

- records of grievance processes.

Auditors should confirm:

- the timeframe in which grievances are resolved.

Useful Resources

N/A

Indicator:	Requirement:
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3.12.7 The UoC shall maintain written records of all grievances, responses and resolutions.

How do I interpret this requirement?

- The system used to track grievances, responses, communication and remedy should include:
 - the name of the person who brought the grievance
 - the date it was submitted
 - the person or department against whom the grievance was made
 - the category of the grievance
 - dates and the results of internal investigations carried out
 - the outcome of the investigation
 - dates when a response was provided to the person who brought the grievance.

Evidence of implementation should be maintained, e.g.:

- records of grievance processes.

Auditors should confirm:

- records of grievance processes are complete.

Useful Resources

N/A

Indicator:	Requirement:
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3.12.8 The UoC shall facilitate the establishment of employee representation to support the implementation and maintenance of the grievance mechanism.

How do I interpret this requirement?

- Employee representation is freely chosen or elected by their peers without interference from management.
- The UoC staff may wish to have employee representation in the form of a committee to support the grievance mechanism. In this case, the committee:
 - can vary in size depending on the size of the UoC
 - should include representation for diverse groups and consider the needs of vulnerable groups. For example:
 - ensuring a representative number of men and women
 - ensuring that migrant employees are included
 - if the UoC is unionised, a member of the Trade Union should be included.
- The UoC staff may wish to have employee representation in the form of a committee to support the grievance mechanism. In this case, the committee:
- Employee representatives:
 - have been trained and are knowledgeable about the business and common grievances that occur
 - understand human rights issues including labour rights, community rights, and Indigenous peoples' rights, as applicable
 - are seen as accessible and impartial
 - have appropriate skills and authority level to investigate and resolve grievances, and where appropriate, facilitate dialogue, remedy and identify areas of improvement
 - are protected from any reprisals, particularly where they are involved in sensitive issues.
- The same employee representation may also take on the role of supporting the implementation of the human rights policy (see indicator 3.1.3).

Evidence of implementation should be maintained, e.g.:

- evidence of training for employee representatives on the relevant topics
- clearly documented roles and authority levels needed for employee representatives
- job descriptions for employee representatives
- records of employee representative selection process
- minutes of any committee meetings.

Auditors should confirm:

- that employees feel represented by the employee representatives;
- how employee representatives feel the system is working in practice.

Useful Resources

N/A

Indicator:	Requirement:
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3.12.9 The UoC shall ensure that grievance mechanisms are confidential, and information is only shared where necessary to investigate the grievance.

How do I interpret this requirement?

- If a confidential grievance process has been requested, anyone managing the grievance process receives, processes, files and monitors records with particular attention to ensuring that the identity of the person raising the claim is:
 - kept confidential
 - not in any public documentation.
- Sometimes there may be a situation where in order to properly investigate and understand the grievance, the UoC has to share some information about the grievance. If this is the case:
 - the person bringing the grievance should be informed,
 - the information should be shared sensitively and only to those who need to know it.

Evidence of implementation should be maintained, e.g.:

- grievance policy with provisions for confidential procedures
 - records of grievance processes.
-

Auditors should confirm:

- employees understand that the grievance processes are confidential;
 - those processing grievances understand how to keep the process confidential.
-

Useful Resources

N/A

Criterion 3.13 – Community Engagement

Indicator:	Requirement:
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- 3.13.1** The UoC shall not restrict or negatively affect the legal or customary rights of Indigenous and tribal peoples and local communities to sites that are of special cultural, ecological, economic, religious or spiritual significance.

How do I interpret this requirement?

- The aim of this criterion is that the UoC:
 - commits to responsible and ethical land and resource management
 - respects the rights and wellbeing of Indigenous and tribal peoples, and local communities.
- Communities may have lived in the area for centuries, and the natural resources may:
 - hold cultural or religious significance
 - provide them with food security
 - provide them with economic benefits.
- These areas can include:
 - land
 - lakes
 - rivers
 - streams
 - mangroves
 - marine areas.
- Communities' use of these areas and natural resources can be assessed in consultation with communities.
- It is the responsibility of the UoC to:
 - assess which communities could potentially be affected by its operations;
 - understand and document which resources are vital to the communities;

- seek and obtain community approval before undertaking changes that have the risk of restricting access to resources;
 - involve local communities in a consultative process to discuss the proposed changes;
 - document any community approvals;
 - engage with communities in a continuous process with annual reviews;
 - set up a permanently accessible mechanism for communities to raise any issues.
- The UoC can take several steps to engage in these responsibilities:
- **Step 1:** The UoC engages in regular consultation and collaboration:
 - identifies the communities, and Indigenous and tribal peoples who may be significantly impacted by the site's activities
 - initiates a genuine and ongoing engagement and dialogue process with these stakeholders, which includes establishing and maintaining open lines of communication to understand their needs, concerns, and aspirations
 - commits to continually improving practices by being open to feedback and adapting to changing circumstances (see also Indicator 3.13.5).
 - **Step 2:** The UoC conducts social impact assessments:
 - assesses which communities may be affected by their operations:
 - the physical proximity of a community is less important than the extent of the impact experienced by them
 - engages these communities to conduct thorough impact assessments to understand the potential cultural, and socio-economic consequences of site operations on significant sites (also 3.13.2)
 - If an adequate and up-to-date social impact assessment has already been carried out for another purpose, there is no need to carry out another one. In some cases, the assessment will need to be complemented. The UoC will also need to show it has followed up on its findings.
 - **Step 3:** The UoC develops a mitigation plan:
 - that is based on the social impact assessments, to eliminate adverse effects on significant sites, in consultation with affected communities
 - implements this plan

- monitors this plan.
- **Step 4:** The UoC educates and trains employees about:
 - the cultural, historical, and ecological significance of the land and water resources in the area to foster a deeper understanding and respect for these resources
 - the mitigation plan
 - ensures that there are employees with the required competencies to conduct community engagement.
- **Step 5:** The UoC integrates community engagement into management systems (see also Criterion 1.2):
 - integrates the mitigation plan in company policies
 - ensures there is a member of senior management who is responsible for implementation of these company policies and the mitigation plan
- **Step 6:** The UoC establishes conflict resolution mechanisms, which are:
 - clear and fair
 - for resolving disputes or conflicts that may arise between the UoC and Indigenous or local communities
 - agreed upon in advance and involve impartial third parties if necessary (see also Indicators 3.12.4 - 3.12.9).
- **Step 7:** The UoC ensures transparency, reporting and communication, internally, to the communities and to other interested stakeholders. This can include:
 - maintaining transparency in operations
 - reporting on community engagement efforts accessible to communities.
 - communicating clearly and well in advance about any activities that have the potential to affect communities.

Evidence of implementation should be maintained, e.g.:

- reports from environmental and social impact assessments
- impact mitigation plan
- reports or meeting minutes from community meetings and other engagement
- documentation from grievance procedures

- brochures, posters and other materials used for communication with communities.

Auditors should confirm:

- which communities may be impacted by the operations of the UoC and if they are Indigenous communities or not;
 - whether the methodology for stakeholder mapping was robust enough to identify the key rightsholders and stakeholders.
 - if the stakeholders identified include diverse groups representing different segments of the community.
 - where possible with communities, that the assessments were conducted in good faith and include all significant sites that the UoC may have an impact on, or where this is not possible seek input from local NGOs or other organisations.
 - the mitigation plan is being implemented;
 - communities understand how to access the UoC's community grievance mechanism;
 - documentation from grievance processes, the nature of the grievances and how they have been resolved;
 - how communities engage with the UoC;
 - whether there are any conflicts about the use of resources by the UoC or if the UoC has restricted access to sites of religious or spiritual significance for Indigenous people;
 - whether there are any other negative impacts they experience from the UoC.
-

Useful Resources

- IFC stakeholder engagement handbook: <https://www.ifc.org/en/insights-reports/2000/publications-handbook-stakeholderengagement--wci--1319577185063>
- Guide to social impact assessment: <https://www.eianz.org/document/item/5876>
- World Resources Institute "Breaking Ground: Engaging communities in extractive and infrastructure projects": <https://www.wri.org/research/breaking-ground>
- Guidance for Free Prior and Informed Consent (but also useful for community engagement more generally):
 - Accountability framework guidance on FPIC: https://accountability-framework.org/fileadmin/uploads/afi/Documents/Operational_Guidance/OG_FPIC-2020-5.pdf
 - FPIC-360°: <https://fpic360.org/>

- World Resources Institute "Development without conflict: The business case for community consent": <https://www.wri.org/research/development-without-conflict>
- Oxfam "Guide to Free Prior and Informed Consent": https://www.culturalsurvival.org/sites/default/files/guidetofreepriorinformedconsent_0.pdf

Indicator:	Requirement:
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3.13.2 The UoC shall identify, avoid and mitigate potential negative impacts on the surrounding communities resulting from its activities.

How do I interpret this requirement?

- The UoC can be a busy environment, and operations can have an impact on surrounding communities, who may experience increased:
 - noise
 - light
 - pollution
 - dirt and dust
 - traffic and damage to roads
 - restrictions on where people can move around the community.
- Sometimes the UoC can cause disruption to communities who are further away, for example through increased traffic on the roads by trucks delivering feed early in the morning.
- In consultation with the community the UoC should:
 - identify if any negative impacts are taking place
 - decide on ways to mitigate these impacts
 - inform the community of the contents of any mitigation plan.
- See also the guidance for Indicator 3.13.1.

Evidence of implementation should be maintained, e.g.:

- an analysis of any negative impacts caused by the UoC
 - records or minutes of meetings or other engagement with the community

- a mitigation plan for any negative impacts.

Auditors should confirm:

- the UoC understands who the communities are who are potentially impacted by its operations;
 - the UoC has carried out a process of understanding what, if any, the impacts from its operations are for the local communities;
 - where possible, if communities have been engaged in a process of understanding the impacts of the operations of the UoC;
 - communities are aware of the mitigation plan.
-

Useful Resources

N/A

Indicator:	Requirement:
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3.13.3 The UoC shall, where possible, source goods and services, including employment, from Indigenous and tribal peoples and local communities.

How do I interpret this requirement?

- Ensuring that the community benefits economically from the UoC can:
 - improve wellbeing
 - foster positive relationships between the UoC and the communities.
- The UoC can develop partnerships and agreements that ensure that nearby communities benefit economically from site operations. This may include:
 - job opportunities
 - sourcing goods from the community, for example:
 - food for the canteen
 - machinery needed for running the site
 - feed.
 - sourcing services from the community, for example:

- cleaning company
- diving company
- security guard provider.
- revenue-sharing
- infrastructure development, for example:
 - the improvement of roads
 - connections to other services for the community, such as internet access.
- Actions that the site can take include:
 - local posting of jobs and active recruitment in neighbouring communities,
 - partnerships with local businesses for the delivery of goods and/or services, such as catering.

Evidence of implementation should be maintained, e.g.:

- sourcing policy
- recruitment processes and advertisements for jobseekers
- employee records
- agreements with local businesses.

Auditors should confirm:

- how jobs are advertised and goods are sourced;
 - the degree of locally recruited employees and locally sourced goods and services;
 - the reasoning for recruiting employees or sourcing goods and services from elsewhere.
-

Useful Resources

N/A

Indicator:	Requirement:
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- 3.13.4** The UoC shall demonstrate its right to use the land and water including, where applicable, transfer of ownership or usage of land from Indigenous and tribal peoples, local communities or other stakeholders to the UoC.

How do I interpret this requirement?

- The UoC:
 - recognises and respects the legal and customary rights of Indigenous and tribal peoples and local communities over land and resources
 - ensures land titles and deeds are in place and that it follows national laws
 - can demonstrate this right by showing:
 - legal ownership documentation
 - leaseholds
 - any other use right documentation
 - there are/were no land conflicts when starting the site operations, and/or
 - any land conflicts have been resolved.
- Where required (and this can be applied anywhere as a good practice) the UoC obtains Free, Prior, and Informed Consent (FPIC) from Indigenous and tribal communities before undertaking any activities that may impact significant sites or resources.

Evidence of implementation should be maintained, e.g.:

- review local legislation related to legal or customary rights of Indigenous and tribal people to water
- land titles, deeds, licences
- evidence of agreements with communities
- evidence of FPIC agreements with Indigenous communities.

Auditors should confirm:

- how community members engage with the UoC;
- the UoC has all appropriate land titles, deeds and licences in place;
- whether there are any past or ongoing conflicts between the community and the UoC about the UoC's use of resources.

Useful Resources

N/A

Indicator:	Requirement:
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- 3.13.5** The UoC shall proactively engage with Indigenous and tribal peoples and local communities to identify, avoid, and mitigate significant negative impacts resulting from the activities of the UoC.

How do I interpret this requirement?

- Proactive engagement means that the UoC should:
 - make contact with Indigenous and tribal peoples and local communities to establish and maintain a relationship with them
 - make efforts to involve Indigenous and tribal peoples and local communities in decisions that may impact them
 - keep Indigenous and tribal peoples and local communities informed about the activities of the UoC, particularly any changes that may take place
 - ensure that Indigenous and tribal peoples and local communities are aware of and understand how to use the community grievance mechanism (3.12.4).
- Guidance for Indicator 3.13.1 offers information about how to:
 - engage with communities
 - assess and mitigate any negative impacts.

Evidence of implementation should be maintained, e.g.:

- evidence of engagement with local communities
- minutes from any meetings with community members
- documented grievances and their resolutions.

Auditors should confirm:

- confirm that the UoC is aware of which communities may be impacted by their activities;
- confirm that the UoC understands some of the possible negative impacts that their activities may have on communities;
- confirm that the UoC has engaged in some way with local communities.

Useful Resources

N/A



Principle 4

The aquaculture industry upholds responsible animal health and welfare practices



Criterion 4.1 – Fish (Including Cleaner Fish)

4.1.1 - Fish (Including Cleaner Fish) - Health and Welfare

Indicator:	Requirement:
4.1.1.1	The UoC shall ensure that all relevant employees are trained on general animal health and welfare awareness, including biosecurity, according to Appendix 12 (12.2, Table 1) .
4.1.1.2	The UoC shall ensure that any person involved in animal husbandry and handling activities is trained on animal health and welfare, according to Appendix 12 (12.2, Table 1) .
4.1.1.3	The UoC shall ensure that the content of the training in Indicators 4.1.1.1 and 4.1.1.2 is updated as necessary and is approved by a registered veterinarian or aquatic animal health professional .

How do I interpret this requirement?

- For each employee:
 - The level of training should be appropriate to their role:
 - For all relevant employees: a basic level of awareness of bio security, fish health and welfare is required.
 - For any person involved in husbandry, handling, and slaughter operations: a more advanced level of training on fish health and welfare, handling and slaughter practices is required on an annual basis.
 - Evidence of training participation must be maintained:
 - Certificates of attendance.
 - Certificates of competency.
- The trainer should be competent in the topic of animal health and welfare and the competence should be demonstrable, e.g.:
 - Curriculum Vitae.
 - Records of qualifications.
 - Work experience.
- Best practices on the implementation of the training content should be visible during the audit.

- Employees should demonstrate adequate knowledge of that content at interviews.

Evidence of implementation should be maintained, e.g.:

- A list of employees and subcontracted personnel involved in animal husbandry and handling operations.
- Records:
 - Training.
 - Trainer's competence.
 - Certificates of competency.

Auditors should confirm:

- Employees received the required level of training for their roles by cross-checking employees and subcontracted personnel lists with:
 - Training records.
 - Certificates of competency.
 - Trainer's competence.
 - Revision of training resources/contents.
- Adequate level of personnel knowledge of the training content should be verified by:
 - Interviews.
 - Observation of real operations during handling and slaughter of fish.

Useful Resources

- World Organisation for Animal Health (2024) – [Competency and Curriculum Guidelines for Community Animal Health Workers. Paris, 39 pp. \(https://doi.org/10.20506/woah.3535\)](https://doi.org/10.20506/woah.3535).

Indicator:	Requirement:
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4.1.1.4 The UoC shall implement a vaccination plan that:

- a) considers the relevant diseases and vaccine availability; and
- b) includes justification for vaccine selection; and
- c) is approved by a registered veterinarian or aquatic animal health professional.

How do I interpret this requirement?

- The vaccination plan should consider:
 - Local regulations.
 - Relevant diseases for the species in the country, based on scientific literature and publicly available information.
 - Commercially licensed vaccines registered for the species being farmed and autogenous vaccines produced by a licensed producer.
 - Commercially licensed vaccines are recognised by the relevant country authorities to ensure quality control, safety and efficacy.
 - Autogenous vaccines are custom vaccines produced from pathogens isolated directly from affected sites on which the vaccines are subsequently deployed under a minor use or restricted permit by a licenced producer.

Evidence of implementation should be maintained, e.g.:

- Local regulations.
 - List of relevant diseases for the species in the country and available vaccines.
 - Vaccine permits.
 - Vaccine purchase invoices.
 - Records:
 - Vaccination.
-

Auditors should confirm:

- The vaccination plan in place is overseen and signed off by a veterinarian and includes a list of relevant diseases for the country and the vaccines available.
 - The information should be cross-checked with local regulations and records.

Useful Resources

- Domínguez-Odio, A; Rodríguez Martínez, E; Leonardo Cala Delgado, D. (2024) [Commercial vaccines used in poultry, cattle, and aquaculture: a multidirectional comparison.](#) *Frontiers In Veterinary Science*.
- Yang, Du; Xioman, Hu; Liang, Miao; Jiong, Chen (2022) ["Current status and development prospects of aquatic vaccines"](#) *Frontiers in Immunology*.
- World Organisation for Animal Health (2021) [Tackling Disease Outbreaks](#). WOAHA Article.
- Mondal H, Thomas J. (2022) ["A review on the recent advances and application of vaccines against fish pathogens in aquaculture"](#). *Aquaculture International*.

Indicator:	Requirement:
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4.1.1.5 *Indicator applicability: marine salmonids in cages*

The UoC shall only stock a single **cohort** of fish per site.

How do I interpret this requirement?

- Gaps of up to six months between stocking inputs are acceptable as long as:
 - The smolts are derived from the same stripping.
 - There is a period when the site is fully fallow after harvest, in accordance with regulatory requirements.

Evidence of implementation should be maintained, e.g.:

- Records:
 - Purchase.
 - Transportation.

- Delivery.
- Stocking.
- Fallowing.
- Production (fish size).

Auditors should confirm:

- The size of the fish matches that of a single year class by observing the stock.
 - The observation should be cross-checked with records.
-

Useful Resources

N/A

Indicator:	Requirement:
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4.1.1.6 The UoC shall adhere to species-specific mortality limits, according to **Appendix 2 (2.3)**.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.1.1.7 The UoC shall only stock batches of fish that have been tested prior to stocking and are free from diseases of regional concern.

How do I interpret this requirement?

- Testing should be relevant to:
 - The species.
 - The production system.
 - The scale of production.
 - The local regulations.

- For example:
 - *Gyrodactylus salaris* for Atlantic salmon in Norway.
 - Tilapia lake virus (TiLV) for tilapia in Asia.
 - Viral hemorrhagic septicemia (VHS) in freshwater fish.
- Testing might consist of:
 - Clinical examination of the fish.
 - Laboratory tests:
 - Bacteriology.
 - Molecular biology (e.g. PCR).
 - Histopathology.
 - Virology (e.g. cell culture).
- The designated veterinarian or aquatic animal health professional should determine:
 - The diseases to include in the testing, based on local regulations, scientific literature and/or publicly available information (see interpretation for Indicator 4.1.1.4).
 - The sample size, based on local regulations and/or technical judgement.
- Compulsory governmental testing programme results are acceptable for demonstrating conformance with this requirement.

Evidence of implementation should be maintained, e.g.:

- List of relevant diseases for the species in the country (see interpretation for Indicator 4.1.1.4).
- Testing results written report or document that includes as a minimum:
 - The batch of fish tested.
 - The site of the test
 - The pathogen tested
 - The results of the test
- Records
 - Purchase.
 - Transportation.
 - Delivery.

- Stocking.

Auditors should confirm:

- Every batch of fish has been tested by cross-checking the list of relevant diseases for the species in the country with the testing result's written report or document and records.

Useful Resources

- World Organisation for Animal Health (2024) [Aquatic Animal Health Code. Chapter 1.3. Diseases listed by WOA.](#)
- Council Regulation (2016) [Transmissible animal diseases and amending and repealing certain acts in the area of animal health \('Animal Health Law'\).](#) Official Journal

Indicator:	Requirement:
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4.1.1.8 The UoC shall ensure that a registered veterinarian or aquatic animal health professional periodically visits the site to perform routine checks. Checks shall be carried out at least annually for all fish species and quarterly for salmon, as well as reactively in cases of fish health or welfare concerns.

How do I interpret this requirement?

- Regular veterinarian or aquatic animal health professional visits may fall into two categories:
 - Routine: Performed on a regular basis to monitor the health and welfare of the farmed fish.
 - Reactive: Performed in response to deterioration of, or concerns about, the health or welfare status of the farmed fish.
 - Sites should be prepared to provide the veterinarian or aquatic animal health professional with detailed information about the observed symptoms, recent changes in behaviour, morphology, water quality or mortality and any relevant history of disease or treatments.
 - The site should use reactive health visits to address immediate health or welfare concerns, diagnose underlying causes of problems, and implement targeted interventions to mitigate risks and prevent further spread of disease.

- For routine visits, preparation is recommended to ensure a meaningful outcome from the veterinarian or aquatic animal health professional. Below are some practical considerations:
 - Visit scheduling: Ensure key employees are available during the visit to provide relevant required information and to facilitate adequate follow-up.
 - Designate a senior member of the employees to serve as the primary point of contact for the veterinarian or aquatic animal health professional during the visit.
 - This individual can help coordinate the activity, ensure relevant information is available, assist during the visit and ensure follow-up.
 - Gather and organise relevant information:
 - Compile relevant information about the operations, including:
 - Species being cultured,
 - Production systems,
 - Stocking densities,
 - Recent health history, including treatments, water quality monitoring results,
 - Behaviour data,
 - Morphology and mortalities data,
 - Feeding plan,
 - Any previous veterinary reports,
 - Details of any ongoing issue.
 - Having this information readily accessible will facilitate discussion and decision-making during the visit.
 - Prepare the UoC for the visit: Ensure the UoC is prepared for the visit and for providing full access to all relevant areas to the veterinarian or aquatic animal health professional.
 - Address any maintenance issues, clean tanks or ponds as needed and ensure clear access to all relevant areas of the facility.
 - Ensure handling and other equipment that might be needed during the visit, such as nets, containers and sampling equipment is available and in good working condition.
 - Make sure employees are aware of the visit and its objectives and encourage them to proactively participate in it.
 - During the visit: Ensure relevant information is discussed with the veterinarian or aquatic animal health professional and that the visit includes all relevant areas within the UoC.
 - After the visit: Follow up on any recommendations or actions discussed during the visit.

- Implement any necessary revisions or changes to management practices, treatment protocols, or facility conditions based on the veterinarian or aquatic animal health professional's feedback and any sample or testing results.
- Continuous improvement: Use the insights gained from the veterinarian or aquatic animal health professional visit to identify opportunities for continuous improvement in health and welfare management.

Evidence of implementation should be maintained, e.g.:

- Health reports.
- Testing results.
- Records:
 - Veterinarian or aquatic animal health professional qualifications.
 - Veterinarian or aquatic animal health professional visit.
 - Recommendations or actions discussed during the visit.

Auditors should confirm:

- The qualifications of the veterinarian or aquatic animal health professional and cross-check the frequency of visits with records.
- Existence of health reports and testing results.

Useful Resources

- FAO eLearning Academy (2024) [Emergency preparedness for aquatic disease outbreaks](#). Online course

Indicator:	Requirement:
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4.1.1.9 *Indicator applicability: every UoC using feed*

The UoC shall develop and implement a feeding plan that includes at least the following:

- a) time and frequency of feeding; and
- b) feed rations; and
- c) feeding methods; and
- d) different environmental conditions; and
- e) prevention of the use of feed which has expired, spoiled or shows signs of excessive leakage; and
- f) feed types for the nutritional needs and life stages of the species.

An exception applies to (f) when such feeds are not available, in which case the UoC shall use feed suitable for animals with similar nutritional needs.

How do I interpret this requirement?

- The feeding plan:
 - Outlines the site's feeding strategy.
 - As a minimum, it must include the following parameters:
 - Time and frequency, e.g. at daybreak and early afternoon (08:30 am and 02:00 pm).
 - Feed rations, e.g. feed type and daily amount.
 - Feeding adaptation to fit different life stages, e.g. transfer diets for Atlantic salmon smolts recently transferred to saltwater.
 - Feeding adaptation to fit different ambient conditions, e.g. stop feeding during algae blooms or use functional feeds when facing specific health issues.
 - Should include considerations to guarantee the good working order of mechanical equipment used to deliver feed, e.g. scheduled maintenance checks of automatic feeders
- The UoC should document the rationale for changes in the feeding plan to better inform management decisions.

Evidence of implementation should be maintained, e.g.:

- Records:

- Purchase.
 - Transportation.
 - Delivery.
 - Stocking.
 - Maintenance which affected feeding.
 - Any environmental or health issues which affected feeding.
 - Feeding periods (time and frequency).
 - Feeding rations.
- Employees should demonstrate adequate knowledge of the feeding plan.

Auditors should confirm:

- Feeding plan includes the minimum required parameters and is implemented:
 - Implementation can be confirmed by cross-checking the plan with records, information from employee interviews and observations.

Useful Resources

- [The International Aquaculture Feed Formulation Database.](#)
- Hasan, M.R, New, M.B. eds. (2013) [On-farm feeding and feed management In aquaculture.](#) FAO Fisheries and Aquaculture Technical Paper No. 583. Rome, FAO. 67pp.
- Alevra, A., Llorente Manzanares, M., Mente, E., Xandri Royo, P.M., (2024) [Good husbandry practices for aquaculture](#), Publications Office of the European Union.

Indicator:	Requirement:
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- 4.1.1.10** The UoC shall develop, implement and maintain a site-specific Health and Welfare Management Plan (HWMP), approved by a registered veterinarian, with the intent of preventing disease outbreaks, and ensuring the good health and welfare of farmed fish.

The HWMP shall include all of the following:

- a) disease monitoring; and
- b) response, including quarantine, treatment, and humane culling; and
- c) reporting mechanisms, including reporting WOAHP-notifiable disease to authorities; and
- d) biosecurity risks; and
- e) predation risks.

The HWMP shall be reviewed and revised as necessary:

- at least annually for sites with cycles under one year or using continuous production methods; or
- after each cycle for sites with longer cycles or all-in-all-out production methods.

How do I interpret this requirement?

- The Health and Welfare Management Plan (HWMP):
 - Aims to support the development and implementation of preventive, rather than remedial, actions that ensure good fish health and welfare.
 - Must be developed by the UoC in collaboration with a veterinarian who approves it. In countries where other professions have comparable rights and duties and are subject to the same authorization and regulatory requirements as veterinarians, they may also approve the HWMP (e.g., aqua medicine biologists in Norway).
- For developing and establishing a site-specific disease monitoring, response and reporting mechanism within the plan, the UoC should consider:
 - Developing standardised health and welfare monitoring protocol
 - This may involve conducting visual inspections, monitoring Operational Welfare Indicators (OWIs) and performing diagnostic tests for pathogens.
 - Producing and maintaining detailed records.

- Including records of health observations, diagnostic test results, treatments administered, mortalities, and other relevant data.
- It is recommended to use a centralised database or electronic system for efficient data management and retrieval.
- Implementing an early detection system.
 - To detect signs of disease outbreaks or abnormal health conditions promptly.
 - This could include environmental monitoring tools.
- Establishing response protocols for responding to disease outbreaks or health emergencies.
 - Outlining procedures for:
 - Isolating affected individuals or populations.
 - Implementing strategies for treatments or others (e.g. culling affected batches of fish [see interpretation for Indicator 4.1.1.10.4]).
 - Notifying relevant stakeholders, including other sites in the vicinity and the authorities.
- Establishing communication channels.
 - Effective communication channels should be implemented for reporting health-related issues both internally within the facility and externally to relevant authorities (e.g., WOAHP-notifiable diseases) or stakeholders.
 - Employees should understand their roles and responsibilities in reporting and responding to disease events.
- Maintaining a readily accessible notification list.
 - Including contact names numbers and emails for key employees and other relevant stakeholders.
 - The list should be kept up-to-date to ensure quick and efficient communication during emergencies (see interpretation for Indicator 4.1.2.3).
- For developing a site-specific biosecurity mechanism within the plan, the UoC should consider:
 - Determining the objectives and scope of the biosecurity mechanism.
 - Identifying relevant regulatory requirements.

- Gathering information about the site, including an up-to-date layout of buildings and production units identifying the possible movements of animals, animal products, animal waste, water, feed and fomites/vectors.
 - Identifying potential transmission pathways into, within and from the site.
 - Carrying out a risk analysis to identify and evaluate disease threats and ensure that the plan addresses them.
 - A risk analysis can be carried out either on the whole operation or only partially, e.g. disease transmission during transportation.
 - This will depend on the objectives of the biosecurity mechanism.
 - Identifying mitigation measures and assessing their effectiveness and their management requirements.
 - Establishing emergency procedures in the event of a biosecurity failure.
 - Defining internal and external communication procedures, roles, and responsibilities.
 - Creating a monitoring and internal audit schedule.
 - Carrying out performance evaluations.
 - Creating the necessary Standard Operating Procedure (SOP) to support the implementation of the mitigation measures and the biosecurity mechanism.
 - Training employees in the different elements of the biosecurity mechanism.
 - Maintaining a record-keeping system that provides evidence of compliance with the biosecurity mechanism.
- For developing predator control measures within the plan, the UoC should consider:
- Identifying potential predators.
 - Create a list of the predators that are present in the area and that might interact with the UoC.
 - Predators may include birds, mammals, reptiles, and other aquatic species known to prey on fish.
 - Assessing the risk of predation.
 - Evaluate the risk posed by each predator species identified based on factors such as proximity to the site, population size, means of accessing the site, likelihood of predation, etc.
 - Implementing predator control measures.

- Develop and implement predator control measures tailored to the specific risks identified. This may include:
 - Physical barriers such as nets, fences, or screens to prevent access to fish tanks or ponds.
 - Scare tactics such as visual deterrents to discourage predators from approaching the facility.
- Implementing ongoing monitoring and surveillance.
 - To detect predator activity and intervene as needed.
- Regular reviewing and revision.
 - Based on the outcome of the ongoing monitoring and surveillance.
 - Adjustments to the measures should be made when the results of monitoring indicate the need to optimise protection for fish and the containment system.
- Collaborating with local authorities or local wildlife agencies to address predator issues in the surrounding area.
- A periodic review and, if necessary, a revision of the HWMP must be conducted in response to changes in farming activities or external factors or as directed by the veterinarian.
 - As a minimum, the review should occur annually in cases of production cycles shorter than one year or using continuous stocking/cropping methods, or following each production cycle, in the case of UoCs with production cycles longer than one year or using all-in-all-out stocking/cropping methods.
 - The UoC should develop feedback mechanisms to inform the review, e.g. proactively seeking feedback from employees and subcontracted personnel, such as the vaccination team, on the operationalisation of the site-specific biosecurity mechanism.
 - For a single producer overseeing multiple UoCs in a shared waterbody, a unified HWMP that incorporates site-specific details is acceptable; however, for larger companies managing multiple UoCs in one country, while common plans may apply, it is crucial to ensure HWMP adaptation to individual site circumstances, particularly in aspects like biosecurity measures, disease surveillance programs, and monitoring systems.

Evidence of implementation should be maintained, e.g.:

- The HWMP includes considerations for:
 - Site-specific disease monitoring, responses and reporting requirements.

- Specific considerations should be given to reporting WOAHP-notifiable diseases (see interpretation for Indicator 4.1.1.11).
- Site-specific biosecurity mechanism.
- Predators control measures.
- Veterinarian sign-off.
- Reviews and revisions.

Auditors should confirm:

- The HWMP includes the required five minimum considerations above and all relevant records associated with the HWMP implementation.
- Employees are:
 - Familiarised with the HWMP and the feedback mechanisms for its review by conducting employee interviews and observations.
 - Trained on the site-specific biosecurity mechanism by reviewing employees' training records, conducting employees' interviews and observations.

Useful Resources

- World Organisation for Animal Health (2024) [Aquatic Animal Health Code - World Organisation for Animal Health](#).
- World Organisation for Animal Health (2023) [Assessment and listing of emerging diseases - WOAHP Bulletin](#)
- Scottish Finfish Aquaculture (2023) [Code of Good Practice for Scottish Finfish Aquaculture](#).
- The Fish Site (2005) [The benefits of a veterinary health plan](#).
- Alevra, A., Llorente Manzanares, M., Mente, E., Xandri Royo, P.M., (2024) [Good husbandry practices for aquaculture](#). Publications Office of the European Union.
- APROMAR (2022) [A guide on fish welfare in Spanish aquaculture - Volume 1: Concepts and generalities](#). Spanish Aquaculture Business Association. 36 pp.
- APROMAR (2024) [A guide on fish welfare in Spanish aquaculture- Volume 2: Welfare of European sea bass](#). Spanish Aquaculture Business Association. 49 pp.
- APROMAR (2024) [A guide on fish welfare in Spanish aquaculture - Volume 3: Welfare of gilthead sea bream](#). Spanish Aquaculture Business Association. 49 pp.
- PerfromFISH (2021) [Best therapeutics practices for Mediterranean farmed fish](#). Report 151pp.

Indicator:	Requirement:
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4.1.1.10.1 The UoC shall ensure the HWMP includes a water quality monitoring process according to **Appendix 12 (12.3, Table 2)**.

How do I interpret this requirement?

- Maintaining good water quality is a critical factor when culturing any aquatic organism.
 - Concentrations of substances within the water system that rise above or fall below specific levels can be harmful to fish populations.
- Concerning health and welfare, water quality has a dual role as a cause or a consequence of poor health and welfare:
 - Poor water quality can cause health and welfare hazards in an aquaculture facility, e.g. heavy metals derived from forest runoff in a flow-through system or leaking from equipment in a recirculating aquaculture system (RAS).
 - Poor water quality can also result from poor husbandry practices resulting in a risk to health and welfare, e.g. if stocking density is too high the water quality is likely to deteriorate, especially in closed and semi-closed systems.
- While Table 2 of Appendix 12 of the standard specifies water quality parameters that must be included in the monitoring process, other parameters may be considered based on the site-specific requirements of the UoC.
- Conformance with maximum and minimum thresholds for the required parameters, customised to suit the species and environmental conditions, should be incorporated as part of the water quality monitoring process. Water quality parameters that are described as risk assessed will be determined by:
 - Site history of water quality parameters (detailed records),
 - Previous results,
 - The sensitivity of the species.
- Tables 6 and 7 below give examples for the majority of the required water quality indicators that may be considered acceptable as a reference to develop and implement a water quality monitoring process.

Table 6: Water Quality Monitoring for Nile tilapia (*Oreochromis niloticus*) based on Pedrazzani, A.S., Quintiliano, M.H., Bolfe, F., Sans de Oliveira, E.C. & Molento, C.F.M. (2020) [Tilapia On-Farm Welfare Assessment Protocol for Semi-intensive Production Systems](#) Frontiers in Veterinary Science and Bregnballe, J. 2022. and A guide to recirculation aquaculture – [An introduction to the new environmentally friendly and highly productive closed fish farming systems](#). Rome. FAO and Eurofish International Organisation.

	Nile Tilapia	Acceptable	Warning Levels	Unacceptable Levels
Daily	Temperature (degrees Celsius)	25-32	20-24 or 33-37	≤19 or ≥38
	Oxygen Saturation (%)	70-95	40-69	≤39 or >95
	pH	6.0-8.5	5.5-5.9 or 8.6-8.9	≤5.4 or ≥9
Bi-weekly	Carbon dioxide (mg/l)	10-15	16-19	20
	Ammonia (NH ₃ ; mg/L)	0.000-0.050	0.051-0.100	≥0.110
	Nitrite (mg/l)	0.00-0.50	0.51-1.00	≥1.01
	Nitrate (mg/l)	0.0-0.3	0.31-0.5	>5
Risk assessed	Turbidity (cm)	25-40	41-65	<25 or >65
	Hydrogen sulphide	0.0-0.3	0.31-0.5	>5

Table 7: Water Quality Monitoring for yellowtail kingfish (*Seriola lalandi*) In a marine recirculating aquaculture system (RAS) with artificial seawater based on J. Orellana, U. Waller, B. Wecker, (2014) [Culture of yellowtail kingfish \(Seriola lalandi\) in a marine recirculating aquaculture system \(RAS\) with artificial seawater](#). Aquacultural Engineering and FAO (2022) A guide to recirculation aquaculture – [An introduction to the new environmentally friendly and highly productive closed fish farming systems](#). Rome. FAO and Eurofish International Organisation.

RAS culture of Yellowtail Kingfish		Acceptable	Warning Levels	Unacceptable Levels
Daily	Temperature (degrees Celsius)	21 - 24	19 - 21, 24.1 - 26	below 19, above 26
	Oxygen saturation (%)	96 - 105	90 - 95, 106 to 115	under 90, over 115
	Salinity (‰)	31 - 33	28 - 30, 33- 35	under 28, over 35
	Turbidity (nephelometric turbidity units)	1	2 to 5	over 5
	Carbon dioxide (mg/l)	8.0 - 8.5	5.8 - 8.0, 8.51 - 10.8	below 5.7, above 10.81
	pH	7.5	7.3 - 7.4, 7.6 - 7.8	below 7.3, above 7.8
	Ammonia (mg/l)	<0.01	0.020-0.024	>0.025
	Nitrite (mg/l)	0.01-0.2	under 0.5	over 0.5
	Nitrate (mg/l)	under 40	40 - 60	over 60
Risk assessed	Hydrogen sulphide (µg)	1-4	5	>5

Evidence of implementation should be maintained, e.g.:

- The water quality monitoring process, that incorporates the minimum elements required, included in the HWMP.
- Records confirming the monitoring of the required parameters and their conformance with maximum and minimum thresholds.
- Employees should demonstrate awareness of the established water quality monitoring process.

Auditors should confirm:

- The water quality monitoring process included in the HWMP incorporates the minimum required elements and is implemented.
 - Implementation can be verified by cross-checking the process with records, information from employee interviews and observations.

Useful Resources

- The Fish Site (2015) [How to achieve good water quality management in aquaculture.](#)
- Prema D, Jenni B, Reena V Joseph, Vysakhan P, Prajitha K and Kripa V (2020) [Water Quality Management In Aquaculture.](#) CMFRI
- EU Platform on Animal Welfare Own Initiative Group on Fish (2020) [Guidelines on Water Quality and Handling for the Welfare of Farmed Vertebrate Fish.](#)
- FAO (2022) A guide to recirculation aquaculture – [An introduction to the new environmentally friendly and highly productive closed fish farming systems.](#) Rome. FAO and Eurofish International Organisation.

Indicator:	Requirement:
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4.1.1.10.2 The UoC shall ensure⁴⁹ the HWMP includes a morphological scoring monitoring process for live fish. Monitoring shall be conducted at least once a month, and shall include at least the following scoring parameters:

- eye damage
- operculum damage
- skin damage
- fin damage
- deformities
- change of colouration
- emaciation

An exception applies where deviation from monitoring frequency is justified by any of the following reasons:

- immediately after smolting and stocking; or
- under environmental conditions that affect the welfare of the fish or the health and safety of employees; or
- fish are undergoing a disease event and/or being treated, in which case the maximum deviation shall be two weeks.

The reason for any deviation must be recorded.

How do I interpret this requirement?

- Morphological scoring is frequently used as an immediate indicator of welfare.
 - However, changes in morphology may have a variety of causes and therefore must be further analysed before drawing conclusions.
- Examples that may be considered as references to develop a morphological scoring are provided in Table 8.

⁴⁹ An exemption applies for Seabass due to its high sensitivity to handling practices.

Table 8: Morphological scoring for European sea bass (*Dicentrarchus labrax*) based on Pavlidis, M., Samaras, A., Papaharisis, L., (2024). [Welfare assessment study for European sea bass](#). University of Crete.

Morphological Parameter	Acceptable	Warning levels	Unacceptable
Eye damage	No damage Intact eye	Damage to one eye Bulging in one eye	Cataracts Trauma to both eyes Blindness
Operculum damage	No damage Gill chamber fully covered bilaterally	1-15% of one gill area exposed >15% of total area exposed on one side	>15% of total area exposed bilaterally 1-15% of gill area exposed bilaterally
Skin damage	Intact skin Or Less than 5% scale loss	5 - 15% skin damage (wounds, lesions, injuries, blood spots, infection and / or scale loss)	> 15% skin damage (wounds, lesions, injuries, blood spots, infection and / or scale loss)
Fin damage	No damage Small splits/ most of fin remaining	Moderate damage - only half of fin remaining	Absent or little of fin remaining More than one fin damaged
Deformities	No visible deformity	Mild deformity in jaw	Extreme deformities Vertebral deformity
Change of colouration	Normal colour	Some change in colour	Extreme change in colour
Emaciation	Not evident Condition factor >1.0	Lean – not as full around stomach Condition factor 0.9 -1.0	Thin Clearly emaciated Condition factor <0.9

Evidence of implementation should be maintained, e.g.:

- The morphological scoring monitoring process, that incorporates the minimum elements required, included in the HWMP.
- Records that confirm the monitoring of the morphological scoring.
- Monitoring can be carried out using camera technology, as long as this method is validated against traditional assessments.
- Employees should demonstrate awareness of the established morphological scoring monitoring process.

Auditors should confirm:

- The morphological monitoring process included in the HWMP incorporates the minimum required elements and it is implemented.

- Implementation can be verified by cross-checking the process with relevant records, information from employee interviews and observations.
- If using camera technology, the assessment has been validated against traditional assessments.

Useful Resources

- Noble, C., Gismervik, K., Iversen, M. H., Kolarevic, J., Nilsson, J., Stien, L. H. & Turnbull, J. F. (Eds.) (2018) [Welfare Indicators for farmed Atlantic salmon – tools for assessing fish welfare. 351pp.](#)
- Noble, C., Gismervik, K., Iversen, M. H., Kolarevic, J., Nilsson, J., Stien, L. H. & Turnbull, J. F. (Eds.) (2020). [Welfare Indicators for farmed rainbow trout: tools for assessing fish welfare 310 pp.](#)
- Pavlidis, M., Samaras, A., Papaharisis, L., (2024). [Welfare assessment study for European sea bass.](#) University of Crete.
- Meyer, B., Emam, W. 2024. [An overview of operational welfare indicators for farmed Nile tilapia \(*Oreochromis niloticus*\).](#) Ethical Seafood Research, Glasgow, United Kingdom. DOI: 10.17605/OSF.IO/MGF7E
- Barrento, S., Pedrazzani, A., Molento, C., & Quintiliano, M. (2023). [Tilapia welfare indicators \[Online course\]](#). In S. Barrento, M. Quintiliano, & J. Muirhead (Eds.), FAI Academy. <https://fai.academy/>

Indicator:	Requirement:
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4.1.1.10.3 The UoC shall ensure the HWMP includes a feeding behaviour scoring monitoring process for live fish. Monitoring shall be conducted daily.

An exception applies where deviation from daily monitoring is justified under environmental conditions that affect the welfare of the fish or the health and safety of employees. The reason for any deviation must be recorded.

How do I interpret this requirement?

- Behavioural scoring is frequently used as an immediate indicator of welfare.
 - However, changes in behaviour may have a variety of causes and therefore must be further analysed before drawing conclusions.
 - The daily monitoring should consider feeding behaviour and other behaviours relevant to the species being farmed.

Table 9: Feeding behaviour scoring for European sea bass (*Dicentrarchus labrax*) based on Pavlidis, M., Samaras, A., Papaharisis, L., (2024). [Welfare assessment study for European sea bass](#). University of Crete.

	Acceptable	Warning levels	Unacceptable
Feeding behaviour	Fish move freely between food items and consume all the available food	Fish eat only pellets that fall directly in front of them but do not move to take food	Fish do not respond to food

Evidence of implementation should be maintained, e.g.:

- The behavioural scoring monitoring process, that incorporates the minimum elements required, included in the HWMP.
- Records that confirm the monitoring of the behavioural scoring.
- Employees should demonstrate awareness of the established behavioural scoring monitoring process.

Auditors should confirm:

- The feeding behavioural scoring monitoring process is implemented.
 - Implementation can be verified by cross-checking the process with relevant records, information from employee interviews and observations.

Useful Resources

- Noble, C., Gismervik, K., Iversen, M. H., Kolarevic, J., Nilsson, J., Stien, L. H. & Turnbull, J. F. (Eds.) (2018) [Welfare Indicators for farmed Atlantic salmon – tools for assessing fish welfare](#). Handbook 351pp.
- Pavlidis, M., Samaras, A., Papaharisis, L., (2024). [Welfare assessment study for European sea bass](#). University of Crete.
- PerformFISH (2024) [Methodology for assessing welfare in Mediterranean Marine Fish Farming Sector](#). Report 23pp.
- PerformFISH (2021) [Set of KPIs for Mediterranean Marine Fish Farming Sector](#). Report 142pp.
- Barrento, S., Pedrazzani, A., Molento, C., & Quintiliano, M. (2023). [Tilapia welfare indicators \[Online course\]](#). In S. Barrento, M. Quintiliano, & J. Muirhead (Eds.), FAI Academy. <https://fai.academy/>

Indicator:	Requirement:
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4.1.1.10.3 The UoC shall ensure the HWMP includes a feeding behaviour scoring monitoring process for live fish. Monitoring shall be conducted daily.

An exception applies where deviation from daily monitoring is justified under environmental conditions that affect the welfare of the fish or the health and safety of employees. The reason for any deviation must be recorded.

How do I interpret this requirement?

- Behavioural scoring is frequently used as an immediate indicator of welfare.
 - However, changes in behaviour may have a variety of causes and therefore must be further analysed before drawing conclusions.
 - The daily monitoring should consider feeding behaviour and other behaviours relevant to the species being farmed.

Table 9: Feeding behaviour scoring for European sea bass (*Dicentrarchus labrax*) based on Pavlidis, M., Samaras, A., Papaharisis, L., (2024). [Welfare assessment study for European sea bass](#). University of Crete.

	Acceptable	Warning levels	Unacceptable
Feeding behaviour	Fish move freely between food items and consume all the available food	Fish eat only pellets that fall directly in front of them but do not move to take food	Fish do not respond to food

Evidence of implementation should be maintained, e.g.:

- The behavioural scoring monitoring process, that incorporates the minimum elements required, included in the HWMP.
- Records that confirm the monitoring of the behavioural scoring.
- Employees should demonstrate awareness of the established behavioural scoring monitoring process.

Auditors should confirm:

- The feeding behavioural scoring monitoring process is implemented.

- Implementation can be verified by cross-checking the process with relevant records, information from employee interviews and observations.

Useful Resources

- Noble, C., Gismervik, K., Iversen, M. H., Kolarevic, J., Nilsson, J., Stien, L. H. & Turnbull, J. F. (Eds.) (2018) [Welfare Indicators for farmed Atlantic salmon – tools for assessing fish welfare](#). Handbook 351pp.
- Pavlidis, M., Samaras, A., Papaharisis, L., (2024). [Welfare assessment study for European sea bass](#). University of Crete.
- PerformFISH (2024) [Methodology for assessing welfare in Mediterranean Marine Fish Farming Sector](#). Report 23pp.
- PerformFISH (2021) [Set of KPIs for Mediterranean Marine Fish Farming Sector](#). Report 142pp.
- Barrento, S., Pedrazzani, A., Molento, C., & Quintiliano, M. (2023). [Tilapia welfare indicators \[Online course\]](#). In S. Barrento, M. Quintiliano, & J. Muirhead (Eds.), FAI Academy. <https://fai.academy/>

Indicator:	Requirement:
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4.1.1.10.4 The UoC shall ensure the HWMP includes a mortality monitoring process which includes at least the following:

- daily monitoring and recording of mortalities; and
- removal of mortalities at least every two days; and
- classification of all recovered mortalities and a post-mortem analysis for each mortality event; and
- investigation of mortality events which remain unexplained or unattributed to fish health reasons; and
- daily collection of moribund fish and culling according to Criterion 4.1.3.

Where any of the listed items above are not possible, the UoC shall record the reason.

How do I interpret this requirement?

- Monitoring for mortality is understood as visual observation of whether there is mortality at the site, e.g. observations of the bottom of the pens with cameras; observations by divers or from the surface.

- Possible causes that would justify no daily monitoring or every second day removal of mortality are severe bad weather, or a major equipment failure not related to poor maintenance.
- Mortality stems from diverse factors that appear linked to suffering, particularly during disease outbreaks and critical development or production phases.
- Scrutiny of mortality records and cause determination is essential for preventing further avoidable deaths and improving overall fish health and welfare in aquaculture.
- Mortality monitoring thresholds should be established as these will identify when the level of mortalities can be classified as a mortality event.
- If on-site diagnosis is inconclusive, off-site laboratory diagnosis is required.
- When designing the mortality monitoring process, the UoC should consider carrying out a post-mortem analysis for each mortality event, not necessarily every fish.
 - A statistically relevant number of fish from the mortality event shall be analysed.
 - A statistically relevant number of fish refers to a sample size that is large enough to yield meaningful and reliable results when conducting statistical analysis.
 - In other words, the sample size should be sufficient to accurately represent the population being studied and to draw valid conclusions from the data.
 - The specific size needed for statistical relevance depends on various factors such as the variability within the population, the desired level of confidence, and the statistical methods being used.
 - Typically, larger sample sizes provide more reliable estimates and stronger evidence for drawing conclusions.
 - The designated veterinarian or aquatic animal health professional should assist the site in determining the appropriate sample size required for statistical analysis and must conduct all diagnosis.
- The UoC should develop a procedure (e.g. SOP) to describe moribund collection and culling operations.
 - Employees should be trained on this procedure.
- Moribund fish should be collected using appropriate equipment to prevent further injuries and stress, e.g. hand net according to fish size.
 - Collected moribund fish should be humanely slaughtered as per Criterion 4.1.3.
- Possible causes that would justify no collection of moribund fish include severe bad weather or a major equipment failure not related to poor maintenance.
- Possible causes of several moribund fish must be investigated.

Evidence of implementation should be maintained, e.g.:

- Mortality monitoring.
- Mortality removal.
- The mortality monitoring process, that incorporates the minimum elements required, as well as what they consider a mortality event, included in the HWMP.
- Records that confirm the monitoring of the mortality including the results of post-mortem analysis and the rationale for determining the sample size required for statistical analysis.
- Employees should demonstrate awareness of the established mortality removal and monitoring process.

Auditors should confirm:

- The mortality monitoring process included in the HWMP incorporates the minimum required elements, is implemented and threshold levels to determine mortality events.
 - Implementation can be verified by cross-checking the process with relevant records, information from employee interviews and observations.
- Moribunds are collected daily and culled following responsible stunning and killing according to Criterion 4.1.3.
 - Implementation can be confirmed by cross-checking the frequency of collection and culling methods with records, information from employee interviews and observations.

Useful Resources

- Aunsmo, A., Persso, D., Stormoen, M., Romstad, S., Jamtøy, O., Johan Midtlyng, P. (2023) [Real-time monitoring of cause-specific mortality- and losses in industrial salmon farming](#). Aquaculture.
- Muniesa, A., Basurco, B., Aguilera, C., Furones, D., Reverté, C., Sanjuan-Vilaplana, S., Dverdal Jansen, M., Brun, E. & Tavoranpanich, S. (2020) [Mapping the knowledge of the main diseases affecting sea bass and sea bream in Mediterranean](#). Transboundary & Emerging Diseases.
- Oliveira, V., Dean, K., Qviller, L., Kirkeby, C. & Bang Jensen, B. (2021) [Factors associated with baseline mortality in Norwegian Atlantic salmon farming](#). Nature. Scientific Reports.
- Muniesa, A., Furones, D., Rodgers C., & Basurco, B. (2022) [An assessment of disease occurrence and mortality in marine fish farming in Spain](#). Aquaculture Reports.

- Zornu, J., Tavoranpanich, S., Brun, E., van Zwieten, P., van de Leemput, I., Appenteng, P., Anchirinah, J. & Sewornu Cudjoe, K. (2023) [Understanding tilapia mortalities and fish health management in Lake Volta: a systematic approach](#). Frontiers In Sustainable Food Systems.
- Zaman Sajid, A., Gamperl, K., Parrish, C.C., et AL., (2024) [An aquaculture risk model to understand the causes and consequences of Atlantic Salmon mass mortality events: A review](#). Reviews In Aquaculture.

Indicator:	Requirement:
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4.1.1.10.5 The UoC shall ensure the HWMP includes a traffic light system for water quality (Indicator 4.1.1.10.1), morphological scoring (Indicator 4.1.1.10.2), behavioural scoring (Indicator 4.1.1.10.3), and mortality (Indicator 4.1.1.10.4). The system shall identify ranges of acceptable levels (green), warning levels (amber), and unacceptable levels (red) of health or welfare.

Where levels are identified as amber or red, the UoC shall increase monitoring and implement appropriate corrective actions to restore the levels back to green.

How do I interpret this requirement?

- The traffic light system should be tailored to accommodate the species being farmed, their life stage and the UoC’s monitoring procedures.
- General guidelines on how to develop the traffic light system for each OWI are presented as follows:

Water Quality Indicators:

Step 1. Identify the parameters to be monitored based on Table 2, Appendix 12.

Step 2. Establish species and life stage-specific thresholds for the parameters identified:

- Base these values on either published scientific literature, the recommendation of a veterinarian or aquatic animal health professional, or ASC reference table for water quality.
 - If the farm is growing more than one life stage or species, then the creation of specific values for each life stage will be expected.

Example 1, for Atlantic salmon farmed in a seawater net pen, the traffic light system could be:

	Acceptable	Warning Levels	Unacceptable
Temperature	6-18°C	<6°C >18°C	<-0.5°C >23°C
DO (at 11C)	>6.5mg/L	<6.5mg/L	<3mg/L
Turbidity	>5m	2-5m	<2m

Example 2, Tilapia farmed in a pond, the traffic light system could be (based on Pedrazzani, A.S., Quintiliano, M.H., Bolfe, F., Sans de Oliveira, E.C. & Molento, C.F.M. (2020) [Tilapia On-Farm Welfare Assessment Protocol for Semi-intensive Production Systems](#). Frontiers in Veterinary Science):

	Acceptable	Warning Levels	Unacceptable
Temperature	25-32°C	20-24 or 33-37°C	≤19°C ≥38°C
Oxygen saturation	70-90%	40-69%	≤39 ≥96
Turbidity	25-40cm	41-65cm	<25 or >65cm

Morphological Indicators:

Step 1. Develop a scoring system for each of the morphological parameters required by Indicator 4.1.1.10.2.

- It is recommended to score morphological parameters based on their severity, location and timing when applicable.

Parameters	Severity of the lesion				Localisation of the lesion				Timing of the lesion		
	0	1	2	3	0	1	2	3	0	1	2
Eye damage	None	Mild	Moderate Severe		None	unilateral	Bilateral				
Operculum damage	None	Mild	Moderate Severe		None	unilateral	Bilateral				
Skin Damage (includes any skin damage, from erosion the wounds)	None	Mild	Moderate Severe		None	Focal	Multifocal	Diffuse			
Fin damage	None	Mild	Moderate Severe		None	One fin	More than one fin		None or heal	Healing	Active
Deformities	None	Mild	Moderate Severe		None	One type	More than one type				
Colouration	None	Abnormal									
Emaciation	None	Mild	Moderate Severe								

Step 2. Once a scoring system has been developed, simulate different scenarios to develop a traffic light system. These scenarios should include at least:

- The best case, e.g.; where fish have no morphologic issues.
- The second-best scenario, e.g. where fish have mild morphologic issues.
- The third-best scenario, e.g. where fish have moderate morphologic issues.
- The worst case, e.g. where fish have the highest score possible.

Each scenario will have a corresponding score:

Best case scenario

Parameters	Severity of the lesion				Localisation of the lesion				Timing of the lesion		
	0	1	2	3	0	1	2	3	0	1	2
Eye damage	0				0						
Operculum damage	0				0						
Skin Damage (includes any skin damage, from erosion the wounds)	0				0						
Fin damage	0				0				0		
Deformities	0				0						
Colouration	0										
Emaciation	0										
Total	0										

Worst case scenario

Parameters	Severity of the lesion				Localisation of the lesion				Timing of the lesion		
	0	1	2	3	0	1	2	3	0	1	2
Eye damage				3			2				
Operculum damage				3			2				
Skin Damage (includes any skin damage, from erosion the wounds)				3				3			
Fin damage				3			2				2
Deformities				3			2				
Colouration											
Emaciation				3							
Total	31										

Second best case scenario

Parameters	Severity of the lesion				Localisation of the lesion				Timing of the lesion		
	0	1	2	3	0	1	2	3	0	1	2
Eye damage		1				1					
Operculum damage		1				1					
Skin Damage (includes any skin damage, from erosion the wounds)		1				1					
Fin damage		1				1				1	
Deformities		1				1					
Colouration		1									
Emaciation		1									
Total	13										

Third best case scenario

Parameters	Severity of the lesion				Localisation of the lesion				Timing of the lesion		
	0	1	2	3	0	1	2	3	0	1	2
Eye damage			2				2				
Operculum damage			2				2				
Skin Damage (includes any skin damage, from erosion the wounds)			2				2				
Fin damage			2				2				2
Deformities			2				2				
Colouration		1									
Emaciation			2								
Total	25										

Step 3. Create a traffic light system, for example:

0 to 13	13 to 25	25 to 31
Acceptable	Warning Levels	Unacceptable
Acceptable	Threshold	Not acceptable
No action	Action recommended	Action required

With time, and as more information is gathered, these thresholds might require re-adjustments.

Behavioural Indicators:

Step 1. Identify behaviours that are relevant for the species and the life stage being farmed.

Step 2. Grade the identified behaviours based on the scientific literature available or the opinion of a veterinarian or aquatic animal health professional. A few possible behaviours to monitor are listed below.

- How to interpret these behaviours will vary depending on the species of fish, the life stage and the farming system.
- It is important to understand that, apart from the species and life stage variability, there will also be an individual component that will vary from one fish to the other.

FEEDING BEHAVIOUR	SCORE		
	0	1	2
ACTIVE FEEDING	Consistently and eagerly responding to feed, showing good overall appetite.	Some fish are not feeding as actively as others, with reduced eagerness or delayed responses to food.	A significant portion of the fish are not responding to food, showing reduced or absent feeding behaviour.
EVEN FEEDING	Fish are spreading out evenly across the cage while feeding.	Fish are congregating in specific areas of the cage, indicating a potential issue.	Fish are crowding into specific areas of the cage.
FEEDING COMPETITION	Minimal signs of competition or aggression	Increased signs of competition or aggression over food	Fish are exhibiting signs of extreme aggression or fighting for food
FEEDING RATE	Eating the full amount of feed provided in the designated time period.	Consuming most of the feed, but some feed may remain uneaten or uncollected	A large amount of feed remains uneaten.

Step 3. Simulate scenarios.

	Best case scenario	Second-best scenario	Third-best scenario	Worst case scenario
Score				
Active feeding	0	1	1	2
Even feeding	0	1	1	2
Feeding competition	0	1	0	2
Feeding rate	0	1	0	2
Total	0	4	2	8

Step 4. Create a traffic light system, for example:

0-4	4-6	6-8
Acceptable	Threshold	Not acceptable

Mortality Indicators:

Step 1. Develop a mortality classification system that adjusts to the species, the life stage and the farming context (e.g. the geography or farming system where fish are reared). Staff must be trained in the mortalities classification. Broadly, the following groupings can be made:

- Infectious:
 - Bacterial.
 - Viral.
 - Parasitic.
 - Fungal.
 - Amoebic.
- Non-infectious:
 - Husbandry related (e.g. transportation, treatment, vaccination, grading, etc.)
 - Environmental (e.g. algal bloom, currents, etc.)
 - Predation (e.g. birds, seals, others).
 - Poor doers or runts.

- Physical lesions.
- Nephrocalcinosis.
- Moribund.
- Deformities.
- Undiagnosed/unknown.
- Decomposed.
- Sampled.
- Failed smolts (only for salmon).

Step 2. Based on the site's historic mortality trends and those of the industry, establish mortality thresholds with the support of a veterinarian or health manager.

- Mortality thresholds should be based on fish numbers, and it is recommended that they are also expressed as biomass thresholds.
 - The use of both metrics is considered complementary.
 - Using biomass on its own is discouraged, as small biomasses might lead to inaccurate assessments when fish are small.

Step 3. Create a traffic light system, for example for:

Mortality ranges for production period of tilapia grown in a pond based on the FAI Tilapia Welfare Course:

<10 %	11-25%	≥26%
Acceptable	Warning Levels	Unacceptable

Another example, mortality ranges for the production period of sea bass on-growing in an open water net system based on: Pavlidis, M., Samaras, A., Papaharisis, L., (2024). [Welfare assessment study for European sea bass](#). University of Crete.

<15 %	16-25%	≥26%
Acceptable	Warning Levels	Unacceptable

Evidence of implementation should be maintained, e.g.:

- The traffic light system for the required OWIs identifies ranges of acceptable (green), warning (amber), and unacceptable (red) levels of health or welfare, included in the HWMP.
- Records that confirm the increased monitoring and associated short-term corrective measures in case of amber or red traffic light results as soon as reasonably possible.

- Rationale for the grading of the OWIs (e.g. based on scientific literature available or the recommendation of a veterinarian or aquatic animal health professional).
- Employees should demonstrate awareness of the established traffic light system.

Auditors should confirm:

- The traffic light system included in the HWMP incorporates the minimum required elements and it is implemented.
 - Implementation can be verified by cross-checking the system with relevant records, information from employee interviews and observations.
-

Useful Resources

- [Noble, C., Gismervik, K., Iversen, M. H., Kolarevic, J., Nilsson, J., Stien, L. H. & Turnbull, J. F. \(Eds.\) \(2018\). Welfare Indicators for farmed Atlantic salmon: tools for assessing fish welfare. Handbook 351pp.](#)
- Martins, C. I. M., Galhardo, L., Noble, C., Damsgård, B., Spedicato, M.T., Zupa, W., Beauchaud, M., Kulczykowska, E., Massabuau, J.C., Carter, T., Rey Planellas, S., & Kristiansen, T. (2011) [Behavioural indicators of welfare in farmed fish](#). Fish Physiology and Biochemistry.
- Pedrazzani, A.S., Quintiliano, M.H., Bolfe, F., Sans de Oliveira, E.C. & Molento, C.F.M. (2020) [Tilapia On-Farm Welfare Assessment Protocol for Semi-intensive Production Systems](#) Frontiers in Veterinary Science

Indicator:	Requirement:
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4.1.1.10.6 The UoC shall ensure the HWMP includes trend analysis to determine changes in health and welfare over time. This analysis shall consider drivers of the trends and shall be based on the data analysed in indicator 4.1.1.10.5 and feedback from the processing plant where relevant.

How do I interpret this requirement?

- The analysis of OWIs data over time, as well as information gathered from the mortality classification and post-mortem analysis and the feedback from the processing plant:

- Allows for the identification of trends that provide insights into factors influencing health, welfare and mortality events.
 - Regular reviews of trends, such as monthly and at least annually or at the end of the production cycle, are recommended.
- Inform decision-making to improve overall production.
 - The veterinarian or aquatic animal health professional should assist the UoC with the trends analysis and drawing conclusions to build effective corrective measures for improving health or welfare over time.
- Feedback from the processing plant can, for example, disclose the presence of recurring trends like skin wounds or deformities.
 - Sharing observations and feedback with the UoC enables prompt and thorough investigations.

Evidence of implementation should be maintained, e.g.:

- The mechanism or trend analysis included in the HWMP.
- Records that confirm trend analysis has been conducted include as a minimum:
 - Mortality classification.
 - Post-mortem analysis.
 - Feedback from the processing plant.

Auditors should confirm:

- The trend analysis mechanism included in the HWMP incorporates the minimum required elements and its outcome is implemented:
 - Implementation can be verified by cross-checking the mechanism with relevant records, information from employee interviews and observations.

Useful Resources

- Aunsmo, A., Persson, D., Stormoen, M., Romstad, S., Jamtøy, O., & Johan Midtlyng, P. (2023) [Real-time monitoring of cause-specific mortality- and losses in industrial salmon farming.](#) Aquaculture

Indicator:	Requirement:
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4.1.1.10.7 The UoC shall ensure the HWMP includes a mortality reduction plan which defines annual targets for both total and unexplained mortalities by considering the results of the trend analysis in Indicator 4.1.1.10.6 and shall implement appropriate actions to meet the annual targets.

How do I interpret this requirement?

- The mortality reduction programme should address the various factors contributing to fish mortality.
- When building the programme, the UoC should consider:
 - Current mortality rates and causes, including unexplained mortality.
 - Set achievable goals and objectives.
 - Identify priority areas for intervention according to the goals and objectives.
 - Develop strategies addressing priority areas for intervention.
 - Monitor and evaluate progress regularly.
 - Provide employees with training.
 - Foster a culture of continuous improvement.
 - Collaborate with industry experts for knowledge sharing.
 - Seek health professionals' involvement in developing the reduction plan.
 - When building the programme, the UoC should consider:
- Special attention should be given to unexplained mortalities which may indicate underlying health issues or poor management practices
 - Identifying and addressing the root causes of unexplained mortalities is essential for improving overall management practices and reducing mortality rates.
- In addition, consistent, clear downward trends in OWIs indicate declining health or welfare.
 - Examples of such trends include a consistent increase in overall morphological scoring, repeated observations of abnormal behaviour, deteriorating water quality with one or more parameters that are outside of the acceptable ranges, or increasing mortality.
- If a declining trend is observed, then the UoC should assess the situation and investigate the causes behind the trends.

- One or multiple issues may be causing the decline, including those of infectious or non-infectious origin.
- In many cases, poor or inadequate husbandry practices can predispose animals to succumb to infectious diseases.
- Also, when declining trends are observed, the UoC is expected to review at least the following husbandry practices if appropriate:
 - Stocking densities.
 - The feeding plan.
 - Water quality.
 - Handling practices.
 - Enclosure characteristics.
 - Environmental enrichment.
 - The rearing environment, for example:
 - Additional aeration.
 - Deployment of environmental enrichment⁵⁰ (EE), (e.g. bubble curtains, change of tank colouration, adjustments in the light system), amongst others.

Evidence of implementation should be maintained, e.g.:

- Mortality reduction programme.
 - Long-term animal health and welfare improvement measures as well as short-term mitigation measures to react to situations of declining health or welfare, included in the HWMP.
 - Mortality records that allow for trend analysis for the development of a mortality reduction programme.
 - Records of long-term animal health and welfare improvement measures and short-term mitigation measures to react to situations of declining health or welfare.
-

⁵⁰ Development of EE targets is an ambition but is not currently a requirement within ASC Farm Standard. The application of EE in commercial fish farming setups is still very limited, due to various reasons. Some of them include contradicting scientific studies, hygiene and biosecurity concerns, inappropriate materials, a sound knowledge of the species ethology to avoid distress, lack of commercial-scale studies among others. ASC encourages both industry and academia to work together to enhance EE knowledge. While recognising EE benefits, ASC acknowledges the need for further research to ensure safe and visible implementation, while committing to adapting requirements as new knowledge emerges.

Auditors should confirm:

- The mortality reduction programme and the improvement and mitigation measures included in the HWMP incorporate the minimum required elements and that its outcome is implemented:
 - Implementation can be verified by cross-checking the programme and measures with relevant records, information from employee interviews and observations.

Useful Resources

- Mellor, D.J. (2016) [Moving beyond the “Five Freedoms” by Updating the “Five Provisions” and Introducing Aligned “Animal Welfare Aims”](#). Animals.
- Lawrence, A.B., Vigors, B., & Sandøe, P. (2019) [What Is so Positive about Positive Animal Welfare?—A Critical Review of the Literature](#). Animals.
- Wang, B., Thompson, K.D., Wangkahart, E., Yamkasem, J., Bondad-Reantaso, M.G., Tattiyapong, P., Jian, J., & Surachetpong, W. (2023) [Strategies to enhance tilapia immunity to improve their health in aquaculture](#). Reviews in Aquaculture.

Indicator:	Requirement:
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4.1.1.10.8 The UoC shall report the following to a registered veterinarian or aquatic animal health professional:

- a) all mortality events; and
- b) fish health and welfare concerns identified during mortality classification.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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- 4.1.1.11** The UoC shall report to the relevant authorities if a WOAHP-notifiable disease is confirmed at a site, and shall:
- a) apply the measures according to applicable legal requirements; and
 - b) increase disease-monitoring in other batches of fish; and
 - c) coordinate oversight by a registered veterinarian or aquatic animal health professional.

How do I interpret this requirement?

- Upon a WOAHP-notifiable disease being confirmed, the UoC should consider within its response mechanism included in the HWMP (see Indicator 4.1.1.10) an emergency response procedure for culling the affected batches of fish (see also Criterion 4.1.3).

Evidence of implementation should be maintained, e.g.:

- HWMP includes site-specific disease monitoring, responses and reporting requirements.
 - Specific considerations should be given to WOAHP-notifiable diseases, including an emergency response procedure for culling the affected batches of fish.
- National regulations. In some countries where WOAHP-notifiable diseases are considered endemic, the UoC should follow the legal reporting requirements.
- Records:
 - Reporting to authorities.
 - Measures applied.
 - Notification to other sites in the vicinity or the ABM.
 - Disease monitoring.
 - Veterinarian or aquatic animal health professional coordination.

Auditors should confirm:

- The HWMP includes considerations for WOAHP-notifiable diseases (see interpretation for Indicator 4.1.1.10).
- In case a WOAHP-notifiable disease has been confirmed:

- Reporting to authorities and implementation of regulatory measures.
- Notification to other sites in the vicinity or the ABM.
- Increased monitoring to other batches of fish.
- Veterinarian or aquatic animal health professional coordination

Useful Resources

- World Organisation for Animal Health (2023) [Assessment and listing of emerging diseases - WOA Bulletin](#)
- Bondad-Reantaso, M.G. (2019) [Emergency Preparedness and Contingency Plans to Aquatic Animal Disease Emergencies](#). FAO Article.

Indicator:	Requirement:
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4.1.1.11.1 *Indicator applicability: marine salmonids in cages*

The UoC shall immediately notify other sites within the ABM if a WOAH-notifiable disease is confirmed.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.1.1.12 The UoC shall calculate stocking densities, following the method outlined In **Appendix 12 (12.5.4)**.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.1.1.13 The UoC shall report to ASC mortality data, according to ASC data submission procedures, following the method outlined in **Appendix 12 (12.5.1)**.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
4.1.1.14	The UoC shall ensure that all relevant employees are trained on general fish health and welfare awareness, including biosecurity, according to Appendix 12 (12.2, Table 1) .
4.1.1.15	The UoC shall ensure that any person involved in animal husbandry and handling activities is trained on animal health and welfare, according to Appendix 12 (12.2, Table 1) .
4.1.1.16	The UoC shall ensure that the content of the training in Indicators 4.1.1.14 and 4.1.1.15 is updated as necessary and is approved by a registered veterinarian or aquatic animal health professional.

How do I interpret this requirement?

- o This clause overlaps with Indicator 4.1.1.1 and 4.1.1.2 regarding fish. Please refer to Indicator 4.1.1.1 and 4.1.1.2 for a detailed description.

Indicator:	Requirement:
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4.1.1.17 The UoC shall only use native **cleaner fish** species.

No additional guidance is provided for this Indicator.

Useful resources:

- o Marine Conservation Society (2021) [Use of 'Cleaner Fish' in UK aquaculture: Current use, concerns and recommendations.](#)

Indicator:	Requirement:
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4.1.1.18

The UoC shall carry out a risk assessment prior to stocking cleaner fish, considering at least the following:

- a) the suitability of use of cleaner fish for parasite control (see Indicator 4.5.17); and
- b) the disease history of the cleaner fish to avoid cross contamination (only where using farmed cleaner fish); and
- c) site-specific environmental conditions considering the species-specific needs of cleaner fish; and
- d) net mesh size; and
- e) salmon size in relation to cleaner fish.

How do I interpret this requirement?

- A risk assessment regarding the use of cleaner fish should be developed.
- As part of the risk assessment the site should, along with the designated veterinarian or aquatic animal health professional, review previous information:
 - Parasite infection history at the site.
 - Disease history at the site.
 - Disease history of available farmed cleaner fish.
 - Water quality history at the site.
 - Site specific environmental conditions.
 - Net mesh size.
 - Salmon size.

Evidence of implementation should be maintained, e.g.:

- A completed risk assessment
 - This is to be reviewed at the start of each cycle.
 - Developed in accordance with the designated veterinarian or aquatic animal health professional.

Auditors should confirm:

- That at the start of each cycle, a risk assessment is carried out to ensure that cleaner fish are a viable option for the site.

Useful Resources

- Norwegian Seafood Research Fund (2018) [Best Practice Guidelines from the Norwegian Seafood Research Fund](#). Technical Report 62 pp.

Indicator:	Requirement:
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4.1.1.19 The UoC shall only stock batches of cleaner fish that have been tested prior to stocking and are free from diseases of regional concern⁵¹.

How do I interpret this requirement?

- The testing will be determined on where fish are sourced from:

Wild caught:

- Should be held in a quarantine facility to observe for disease symptoms before stocking.
 - There should be enhanced biosecurity employed during capture and holding phases to minimise pathogen exposure.
- Are to be caught from aquatic disease-free zones.
 - Note: Since lumpfish and wrasse species are not listed as susceptible species or vector species in accordance with Part II of Annex IV of the Council Directive 2006/88/EC, these species can be traded between aquaculture sites in Europe, including those in Member states, zones or compartments declared disease-free, and are only subject to notification under the Traces system. Therefore, these species can be freely traded between intra-community and third country without any certification declaring VHS freedom⁵².

⁵¹ An exemption applies to wild caught species sourced from zones or compartments declared disease free. See [COUNCIL DIRECTIVE 2006/88/EC of 24 October 2006 on animal health requirements for aquaculture animals and products thereof, and on the prevention and control of certain diseases in aquatic animals](#) and [European Union Reference Laboratory for Fish Diseases: Cleaner fish in aquaculture: Health management and legislative issues](#).

⁵² [European Union Reference Laboratory for Fish Diseases: Cleaner fish in aquaculture: Health management and legislative issues](#).

Farmed:

- Testing will be relevant to:
 - The species.
 - The production system.
 - The scale of production.
 - The local regulations.
 - For example- viral haemorrhagic septicaemia in lumpfish and wrasse.
 - Testing might consist of:
 - Clinical examination of the fish.
 - Laboratory tests:
 - Bacteriology.
 - Molecular biology (e.g. PCR).
 - Histopathology.
 - Virology (e.g. cell culture).
- The designated veterinarian or aquatic animal health professional should determine:
 - The diseases to include in the testing, based on local regulations, scientific literature and/or publicly available.
 - The sample size, based on local regulations and/or technical judgement.
- Compulsory governmental testing programme results are acceptable for demonstrating conformance with this requirement.

Evidence of implementation should be maintained, e.g.:

- List of relevant diseases for the species in the country.
- For wild caught cleaner fish, quarantine information and report from designated veterinarian or aquatic animal health professional.
- For farmed cleaner fish, testing results written report or document that includes as a minimum:
 - The batch of fish being tested.
 - The date of the test.
 - The pathogen tested.
 - The results of the test.

- Records:
 - Purchase.
 - Transportation.
 - Delivery.
 - Stocking.

Auditors should confirm:

- That every batch of wild caught cleaner fish has a record of quarantine and assessment.
 - The record of capture of the wild caught cleaner fish including disease free status.
 - That every batch of farmed cleaner fish has been tested by cross-checking the list of relevant diseases for the species in the country with the testing result's written report or document and records.
-

Useful Resources

- World Organisation for Animal Health (2024) [Aquatic Animal Health Code. Chapter 1.3. Diseases listed by WOAHA.](#)
- Council Regulation (2016) [Transmissible animal diseases and amending and repealing certain acts in the area of animal health \('Animal Health Law'\).](#) Official Journal
- Commission Delegated Regulation (2020) [Animal health and certification requirements for movements within the Union of aquatic animals and products of animal origin from aquatic animals.](#) Official Journal.
- Erkinharju, T., Dalmo, R.A., Hansen, M. and Seternes, T. (2021), [Cleaner fish in aquaculture: review on diseases and vaccination.](#) Reviews in Aquaculture.
- Unknown (2016) Cleaner fish in aquaculture: [Health management and legislative issues.](#) European Union Reference Laboratory for Fish Diseases. Meeting Report 35pp.
- European Commission (unknown) [Listed diseases, categorization and listed species.](#) Webpage
- European Commission (2024) [Scientific and common names of aquatic species listed in the table to the Annex to Commission implementation regulation \(EU\) 2018/1882.](#) Document 7pp.
- European Commission (unknown) [Surveillance, eradication programmes and disease-free status.](#) Webpage.
- Government of Canada (2024) [Domestic Movement Control Program. Map of aquatic disease areas.](#) Webpage

- Biddiscombe et al (2020) Sampling for pathogen screening in the lumpfish (*Cyclopterus lumpus*) –Standard Operation Procedures 1, Centre for Sustainable Aquatic Research, 6 pages

Indicator:	Requirement:
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4.1.1.20 The UoC shall implement a vaccination plan that:

- a) considers the relevant diseases and vaccine availability; and
- b) includes justification for vaccine selection; and
- c) is approved by a registered veterinarian or aquatic animal health professional.

How do I interpret this requirement?

- This clause overlaps with Indicator 4.1.1.4 regarding fish. Please refer to Indicator 4.1.1.4 for a detailed description.

Useful Resources

- Davies, C. (unknown) [Lumpfish Vaccination – Standard Operation Procedures](#), Version 2, Centre for Sustainable Aquatic Research, 6 pages.

Indicator:	Requirement:
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4.1.1.21 The UoC shall provide hides and refuges for the cleaner fish that are:

- a) designed and placed according to the species-specific needs; and
- b) sufficient for the number of cleaner fish stocked.

How do I interpret this requirement?

- The amount of refuges and hides will be determined by their species and the number stocked as well as the need for rotation during cleaning.
 - This will need to be discussed with the veterinarian or fish health professional.

Evidence of implementation should be maintained, e.g.:

- The stocking numbers and numbers of hides and refuges.

Auditors should confirm:

- Stocking number of cleaner fish
- Adequate level of hides and refuges should be verified by:
 - Observation of the hides and refuges.
 - Observation of the cleaner fish.

Useful Resources

- Norwegian Seafood Research Fund (2018) [Best Practice Guidelines from the Norwegian Seafood Research Fund](#). Technical Report 62 pp.

Indicator:	Requirement:
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- 4.1.1.22** The UoC shall only stock cleaner fish:
- a) during daylight; and
 - b) in calm weather; and
 - c) below the waterline; and
 - d) close to hides and refuges; and
 - e) after salmon have been fed; and
 - f) without forcing removal from their transport container.

How do I interpret this requirement?

- The timing of the stocking will be documented.
- Training of staff should include handling of cleaner fish.

Evidence of implementation should be maintained, e.g.:

- The stocking record along with the site diary.
- Training records.

Auditors should confirm:

- Stocking records
- Site diary
- Training records.

Useful Resources

- Norwegian Seafood Research Fund (2018) [Best Practice Guidelines from the Norwegian Seafood Research Fund](#). Technical Report 62 pp.

Indicator:	Requirement:
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4.1.1.23 *Indicator applicability: every UoC using feed*

The UoC shall develop and implement a feeding plan that considers at least all of the following:

- a) time and frequency of feeding; and
- b) feed rations; and
- c) feeding methods; and
- d) different environmental conditions; and
- e) prevention of use of feed which has expired, spoiled or shows signs of excessive leakage; and
- f) feed types for the nutritional needs and life stages of the species.

An exception applies to (f) when such feeds are not available, in which case the UoC shall use feed suitable for animals with similar nutritional needs.

How do I interpret this requirement?

- This clause overlaps with Indicator 4.1.1.9 regarding fish. Please refer to Indicator 4.1.1.9 for a detailed description.

Indicator:	Requirement:
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4.1.1.24 The UoC shall ensure that cleaner fish feed is not withdrawn to enhance sea lice control.

How do I interpret this requirement?

- The cleaner fish will have a feeding plan that is not affected by the number of sea lice on the salmon.

Evidence of implementation should be maintained, e.g.:

- The feeding records of the cleaner fish.
- Cleaner fish feed purchase records.
- Assessment of cleaner fish morphology to ensure no emaciation.

Auditors should confirm:

- Feed purchase records.
- Veterinary records of assessing cleaner fish.
- Morphological checks (see Indicator 4.1.1.27.2).

Useful Resources

- Norwegian Seafood Research Fund (2018) [Best Practice Guidelines from the Norwegian Seafood Research Fund](#). Technical Report 62 pp.

Indicator:	Requirement:
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4.1.1.25 The UoC shall ensure that a registered veterinarian or aquatic animal health professional periodically visits the site to perform routine checks. Checks shall be carried out at least annually, as well as reactively in cases of fish health or welfare concerns.

How do I interpret this requirement?

- This clause overlaps with Indicator 4.1.1.8 regarding fish. Please refer to Indicator 4.1.1.8 for a detailed description.

Indicator:	Requirement:
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4.1.1.26 The UoC shall develop, implement and maintain a site-specific Health and Welfare Management Plan (HWMP), approved by a registered veterinarian, with the intent to prevent disease outbreaks and ensure good health and welfare of cleaner fish.

The HWMP shall include all of the following:

- a) disease monitoring; and
- b) response, including quarantine, treatment, and humane culling; and
- c) reporting mechanisms, including reporting WOA-notifyable disease to authorities; and
- d) biosecurity risks; and
- e) predation risks.

The HWMP shall be reviewed, and revised as necessary, at least at the end of every period of cleaner fish use.

How do I interpret this requirement?

- o This clause overlaps with Indicator 4.1.1.10 regarding fish. Please refer to Indicator 4.1.1.10 for a detailed description.

Indicator:	Requirement:
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4.1.1.26.1 The UoC shall ensure the HWMP includes a water quality monitoring process according to **Appendix 12 (12.3, Table 2)**.

How do I interpret this requirement?

- o This clause overlaps with Indicator 4.1.1.10.1 regarding fish. Please refer to Indicator 4.1.1.10.1 for a detailed description.

Indicator:	Requirement:
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4.1.1.26.2 The UoC shall ensure the HWMP includes a morphological scoring monitoring process for live cleaner fish. Monitoring shall be conducted at least once every three months, and shall include at least the following scoring parameters:

- eye damage
- operculum damage
- skin damage
- fin damage
- deformities
- change of colouration
- emaciation

An exception applies where deviation from monitoring frequency is justified by any of the following reasons:

- immediately after stocking; or
- under environmental conditions that affect the welfare of the cleaner fish or the health and safety of employees; or
- cleaner fish are undergoing a disease event and/or being treated.

The reason for any deviation must be recorded.

How do I interpret this requirement?

- This clause overlaps with Indicator 4.1.1.10.2 regarding fish. Please refer to Indicator 4.1.1.10.2 for a detailed description.

Useful Resources

- Noble et al. (2019) [RENSVEL OWI Fact Sheet Series: An Introduction to Operational and Laboratory based welfare Indicators for ballen wrasse \(*Labrus bergylta*\)](#). Nofima Fact Sheet. 43pp.

Indicator:	Requirement:
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4.1.1.26.3 The UoC shall ensure the HWMP includes a feeding behaviour scoring monitoring process for live cleaner fish. Monitoring shall be conducted weekly.

An exception applies where deviation from weekly monitoring is justified under environmental conditions that affect the welfare of the cleaner fish or the health and safety of employees. The reason for any deviation must be recorded.

How do I interpret this requirement?

- This clause overlaps with Indicator 4.1.1.10.3 regarding fish. Please refer to Indicator 4.1.1.10.3 for a detailed description.

Indicator:	Requirement:
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4.1.1.26.4 The UoC shall ensure the HWMP includes a mortality monitoring process which includes at least the following:

- daily monitoring and recording of mortalities; and
- removal of mortalities at least every second day; and,
- classification of all recovered mortalities and conducting a post-mortem analysis for each mortality event; and
- investigation of mortality events which remain unexplained or unattributed to fish health reasons; and
- daily collection of moribund fish and culling according to Criterion 4.1.3.

Where any of the listed items above are not possible, the UoC shall record the reason.

How do I interpret this requirement?

- This clause overlaps with Indicator 4.1.1.10.4 regarding fish. Please refer to Indicator 4.1.1.10.4 for a detailed description.

Indicator:	Requirement:
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4.1.1.26.5 The UoC shall ensure the HWMP includes a mortality reduction plan which defines annual targets for both total and unexplained mortalities and shall implement appropriate actions to meet the annual targets.

How do I interpret this requirement?

- This clause overlaps with Indicator 4.1.1.10.7 regarding fish. Please refer to Indicator 4.1.1.10.7 for a detailed description.

Indicator:	Requirement:
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4.1.1.26.6 The UoC shall report the following to a registered veterinarian or aquatic animal health professional:

- a) all mortality events; and
- b) cleaner fish health and welfare concerns identified during mortality classification.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.1.1.27 The UoC shall calculate cleaner fish stocking densities in relation to salmon stocking, following the method outlined In **Appendix 12 (12.5.4)**.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.1.1.28 The UoC shall report to ASC mortality data of cleaner fish, according to ASC data submission procedures, and following the method outlined in **Appendix 12 (12.5.1)**.

No additional guidance is provided for this Indicator.

Comment box

In this first version of Principle 4 in the ASC Farm Standard, there is no requirement to develop a traffic light system (see indicator 4.1.1.10.5) or analyse trends to determine changes in health and welfare (see indicator 4.1.1.10.6) for cleaner fish.

However, it is expected that going forward, all these requirements will also be included for cleaner fish.

4.1.2 - Fish (including Cleaner Fish) - Handling

Indicator:	Requirement:
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- 4.1.2.1** The UoC shall develop, implement and maintain a site-specific Handling Management Plan (HMP), with the intent of ensuring the good health and welfare of farmed fish.
- The HMP shall be periodically reviewed and revised as necessary.

- 4.1.2.2** The UoC shall ensure that the HMP includes all handling activities carried out on site.

How do I interpret this requirement?

- The Handling Management Plan (HMP):
 - Aims to support the development and implementation of preventive, rather than remedial, actions that prevent poor fish health and welfare at handling.
 - Examples of handling:
 - Crowding.
 - Grading (active or passive).
 - Sampling.
 - Transportation (including loading, transfer and unloading).
 - Vaccination (by immersion or injection).
 - Application of treatments (therapeutants or physical).
 - Should consider a description of:
 - Processes.
 - A separate process description may be required for outsourced events e.g. transportation up to the point of harvest.
 - Equipment.
 - A description of equipment, and verification if applicable may be required for both UoC, and outsourced events e.g. vaccination equipment.

Evidence of implementation should be maintained, e.g.:

- Confirmation that the HMP includes considerations that allow for performing standardised handling operations, e.g.:
 - Site-specific description of processes and equipment.
 - Documents:
 - Planning of handling events, e.g.:
 - Emails, reports and site 'whiteboard' communications.
 - Handling procedures.
 - Monitoring the effectiveness of the HMP.
 - Documents:
 - Handling logs, e.g.:
 - Reports detailing regular observation of fish behaviour, environmental parameters and fish conditions during and after handling procedures, including injuries, mortalities, or deviations from procedures.
 - Records documenting fitness assessment before handling, incidences of disease, and mortalities, with details on potential correlations with handling activities.
 - Handling incidents reports, e.g.:
 - Emails, reports detailing any incidents during handling procedures, including injuries, mortalities, or deviations from procedures.
 - Feedback from staff, e.g.:
 - Emails, anonymous box asking feedback from staff involved in handling procedures regarding their observations and any difficulties encountered.

Auditors should confirm:

- The HMP includes a description of the handling processes and equipment and considerations for planning handling events.
- Employees are familiarised with the HMP by conducting employee interviews and observations.
- Handling logs are available documenting handling processes, environment, and fish conditions.

- Existence of handling incidents reports, including feedback from staff and discussions regarding any handling procedure changes.
- Open communication is encouraged via formal documents and employee interviews and observations.

Useful Resources

1. Alevra, A., Llorente Manzanares, M., Mente, E., Xandri Royo, P.M., (2024) [Good husbandry practices for aquaculture](#). Publications Office of the European Union.
2. APROMAR (2022) [A guide on fish welfare in Spanish aquaculture - Volume 1: Concepts and generalities](#). Spanish Aquaculture Business Association. 36 pp.
3. APROMAR (2024) [A guide on fish welfare in Spanish aquaculture- Volume 2: Welfare of European sea bass](#). Spanish Aquaculture Business Association. 49 pp.
4. APROMAR (2024) [A guide on fish welfare in Spanish aquaculture - Volume 3: Welfare of gilthead sea bream](#). Spanish Aquaculture Business Association. 49 pp.
5. Zaman Sajid, A., Gamperl, K., Parrish, C.C., et AL., (2024) [An aquaculture risk model to understand the causes and consequences of Atlantic Salmon mass mortality events: A review](#). Reviews In Aquaculture.

Indicator:	Requirement:
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4.1.2.3 The UoC shall ensure that the HMP includes contingency plans for handling activities, including at least the following:

- a) immediate response measures in the case of system failure; and
- b) humane culling response measures (see Indicator 4.1.3.2); and
- c) mass mortality event response measures.

How do I interpret this requirement?

- The contingency plans should consider:
 - Identification of potential failures.
 - Effective corrective measures, e.g.:
 - Backup power generators.

- Oxygen support.

Evidence of implementation should be maintained, e.g.:

- HMP includes processes' contingency plans.
- Records:
 - Backup system testing.
- Additional oxygen source available.

Auditors should confirm:

- HMP includes considerations on fish health and welfare before handling events, including considerations for the existence of a backup system.
 - Backup system testing records.
- Presence of backup system.
- Employees are familiarised with the contingency plans by conducting employee interviews and observations.
- Records:
 - Backup system testing.

Useful Resources

- Singh, G.G., Sajid, Z., & Mather, C. (2024) [Quantitative analysis of mass mortality events in salmon aquaculture shows increasing scale of fish loss events around the world](#). Nature; Scientific Reports
- 1. Barathan, B., Wang, Y., Su, Y., Hamilton, D., et al., (2022). [Methods to control harmful algal blooms: a review](#). Environmental Chemistry Letters.
- 2. FAO eLearning Academy (2024) [Emergency preparedness for aquatic disease outbreaks](#). Online course

Indicator:	Requirement:
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- 4.1.2.4** The UoC shall ensure that the HMP includes consideration of the environmental conditions associated with handling activities, with the intent of preventing handling under adverse conditions that could affect the health and welfare of the fish.

How do I interpret this requirement?

- Examples of considerations of external conditions:
 - Avoid mechanical treatments in the event of storms.
 - Avoid crowding fish for treatment without using a shade net on a sunny day.
 - Avoid transferring juveniles during period where there are less than favourable conditions:
 - Spring tides can result in stronger currents and rapid changes in water levels, juveniles may become disoriented or stressed potentially leading to injuries or escape.
 - Risk of entrapment and loss of equipment can be associated with spring tides.

Evidence of implementation should be maintained, e.g.:

- HMP includes consideration for external conditions necessary to perform handling.
- Records:
 - Weather forecast verification.
- Employees should demonstrate knowledge of the considerations and the verification of external conditions before handling.

Auditors should confirm:

- HMP includes considerations for external conditions for handling and are implemented.
 - Implementation can be confirmed by cross-checking the HMP with records, information from employee interviews and observations.

Useful Resources

N/A

Indicator:	Requirement:
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- 4.1.2.5** The UoC shall ensure that the HMP, following consultation with a registered veterinarian or aquatic animal health professional, considers the use of anaesthesia during handling activities.

How do I interpret this requirement?

- The use of anaesthesia is not required in every handling operation, but it should be considered in operations that can inflict pain, e.g.:
 - Inter-peritoneal vaccinations.
 - Stripping.
 - Handling operations that require fish immobilisation.

Evidence of implementation should be maintained, e.g.:

- HMP includes consideration for the use of anaesthesia during handling.
- List of handling operations requiring the use of anaesthesia.
- List of licenced anaesthetics available on site.
- Records:
 - Anaesthetic prescriptions.
 - Anaesthetic usage.

Auditors should confirm:

- The HMP includes consideration of the use of anaesthetics during handling.
- Implementation can be confirmed by cross-checking the HMP with records, information from employee interviews and observations.

Useful Resources

- Readman, G.D., Owen, S.F., Knowles, T.G. et al. (2017) [Species specific anaesthetics for fish anaesthesia and euthanasia](#). Scientific Reports.
- Brønstad A. (2022) [Good Anesthesia Practice for Fish and Other Aquatics](#). Biology.

- European Food Safety Authority (EFSA) (2009) [General approach to fish welfare and to the concept of sentience in fish](#). EFSA Journal.
- Pathinathan, P., & Roshan, H. (2024). [Exploring Anesthetic Strategies in Aquaculture: Enhancing Welfare and Management Practices](#). The Science World.
- Husen, Md & Sharma, S. (2014). [Efficacy of Anesthetics for Reducing Stress in Fish During Aquaculture Practices - A Review](#). Kathmandu University Journal of Science, Engineering and Technology.

Indicator:	Requirement:
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4.1.2.6

The UoC shall ensure that the HMP includes a fitness assessment of the fish prior to handling activities⁵³, approved by a registered veterinarian or aquatic animal health professional.

For the specific handling activities listed below, the fitness assessment shall be carried out:

- within a two-week period prior to transport or treatment that requires handling of the fish, except when the same type of treatment is repeated, in which case the assessment shall be carried out at least once a month after the initial treatment;
- within a two-week period prior to net change and passive grading.

How do I interpret this requirement?

- Fitness assessment may fall into two categories:
 - Single assessment: Performed within a two-week period prior to the handling.
 - Recurring assessment: Performed at least once a month in the case of serial, repeated treatments, e.g.:
 - Handling for administering multiple rounds of treatment to fish populations over a period of time, e.g.:
 - Bacterial infections, such as bacterial gill disease, furunculosis, or red spot diseases.
 - Fungal diseases, such as saprolegniasis.

⁵³ An exemption applies to handling activities related to health examinations (e.g. health assessment, parasite counting) and in the case of emergencies (e.g. containment of an escape).

- Treatments for parasites control such as formalin treatments, hydrolyzers or thermal delousing.
- A successful fitness for handling record should demonstrate:
 - Fish free of diseases.
 - Minimal morphologic damage.
 - Normal behaviour.
 - Low mortality rate.
 - Usual feeding rates.
- An example of a fitness for handling assessment record is provided in the Table 9 below.

Table 9: An example of a fitness for handling assessment record with information collated from assessments and routine data collected on the site.

Fitness assessment report				
Handling activity to be carried out				
Date handling expected to be carried out				
Disease status comments				
Mortality levels	Low	Consistent	Increasing	Other:
Feeding rate	Normal	Sluggish	Very active	Other:
Fish behaviour				
Comments from last morphological checks				
Fitness comments				
Date:		Approved by:		
		Position:		

- Handling in case of emergencies are exempt from requiring fish fitness assessments e.g.:
 - Containment of an escape.
 - Imminent risk of predation.
 - It is recommended to have a health representative verify the emergency operations together with written advice to support decision-making when handling is required in an emergency.

Evidence of implementation should be maintained, e.g.:

- HMP includes consideration for fitness assessment prior to handling.
- Fitness assessment report.
- Records:
 - Veterinarian or aquatic animal health professional handling approval.
 - Documented reasons for exemptions with written advice.

Auditors should confirm:

- Fitness assessments status before handling.
 - Implementation can be confirmed by cross-checking the frequency of fitness assessment and handling records, information from employee interviews and observations.
- The rationale for any deviation from the frequency of fitness assessment before handling is in accordance with those highlighted by the indicator and respective footnotes.

Useful Resources

- EFSA (2008) [Animal welfare aspects of husbandry systems for farmed Atlantic salmon](#) - Scientific Opinion of the Panel on Animal Health and Welfare.
- Barrento, S., Pedrazzani, A., Molento, C., & Quintiliano, M. (2023). [Tilapia welfare indicators \[Online course\]](#). In S. Barrento, M. Quintiliano, & J. Muirhead (Eds.), FAI Academy. <https://fai.academy/>

Indicator:	Requirement:
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4.1.2.7

The UoC shall ensure that the HMP considers crowding activities with the intent of preventing stress, injury and mortality. Crowding activities shall be carried out:

- a) under supervision; and
- b) gradually; and
- c) partially; and
- d) for a maximum of two hours.

An exception applies where a registered veterinarian or aquatic animal health professional demonstrates that exceeding two hours of crowding does not negatively impact fish welfare, in which case the maximum time period shall be defined by the registered veterinarian or aquatic animal health professional.

How do I interpret this requirement?

- Appropriate crowding intensity is achieved when there is a balance between fish welfare, minimising abnormal behaviour and having sufficient fish in the crowd area for an effective handling.
- Abnormal behaviour may be defined as fish behaviour commonly found or seen when fish are stressed.
- During crowding operations consider an appropriate fish density avoiding:
 - Vigorous escape behaviour.
 - Violent splashing.
 - Escape attempts (jumping).
 - Burrowing into the crowding net.
 - Air gasping.
 - Excessive gill movement.
 - Inability to fish maintain body position (white sides constantly on surface).
 - Skin colour changes.
 - Presence of fish scales in the water.
- Narrow deep nets are recommended over wide shallow nets for fish crowding.

Evidence of implementation should be maintained, e.g.:

- Considerations, plans and actions taken pre- and during crowding for achieving an appropriate crowding intensity.
- Records:
 - That confirms the monitoring of the crowding, e.g. time and intensity.
- Employees should demonstrate awareness of the established crowding operation.
- Rationale for exceeding the crowding time as directed by the veterinarian or aquatic health professional.

Auditors should confirm:

- The crowding operation incorporates considerations for an appropriate crowding, and it is implemented.
 - Implementation can be verified by cross-checking the crowding operations with records, information from employee interviews and observations.

Useful Resources

- Stien, L., Nilsson, K., Noble, C., Izquierdo-Gomez, D., Ytteborg, E., Timmerhaus, G., & Madaro, A. (2024) [Evaluating a crowding intensity scale and welfare indicators for Atlantic salmon in sea cages](#). Aquaculture Reports.

Indicator:	Requirement:
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4.1.2.8 The UoC shall ensure that the HMP considers biosecurity risks specific to the types of handling activities and shall include appropriate biosecurity measures to prevent the transfer of disease.

How do I interpret this requirement?

- For developing a site-specific biosecurity mechanism within the HMP, the UoC should consider:
 - Indicator 4.1.1.10 d)
 - Biosecurity measures related to the type of handling, e.g.:

- Equipment segregation.
- Equipment disinfection.

Evidence of implementation should be maintained, e.g.:

- HMP includes the considerations for:
 - Site-specific biosecurity mechanism.
 - Biosecurity specifications related to the type of handling operations.

Auditors should confirm:

- The HMP includes the required minimum considerations above.
- Employees are:
 - Familiarised with the biosecurity considerations in the HMP by conducting employee interviews and observations.
 - Trained on biosecurity specifications related to the type of handling operations by reviewing employees' training records, conducting employees' interviews and observations.

Useful Resources

- Subasinghe, R., Alday-Sanz, V., Bondad-Reantaso, M.G., Jie H., Shinn, A.P., & Sorgeloos P. (2024) [Biosecurity: Reducing the burden of disease](#). World Aquaculture Society.
- Australian Government (unknown) [Biosecurity plan and translocation policy guidelines - DAFF](#)
- Asia-Pacific Economic Cooperation (2023) [Guidebook on Biosecurity and Good Aquaculture Policies and Practices for small-scale farmers of tilapia \(Oreochromis sp.\) and rainbow trout \(Oncorhynchus mykiss\)](#). Manual 45pp.

Indicator:	Requirement:
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4.1.2.9 The UoC shall ensure that the HMP considers predation risks specific to the types of handling activities and shall include appropriate predator control measures.

How do I interpret this requirement?

- For developing a site-specific predator control within the HMP, the UoC should consider:
 - The site-specific characteristics required to be assessed by Indicator 4.1.1.10 e) predator control measures specific to the type of handling, e.g.:
 - Birds scaring devices.
 - Use of anti-predator nets while moving fish from one cage/unit/pond to another.

Evidence of implementation should be maintained, e.g.:

- Predator control measures specific to the type of handling operations.
- Verification of mortalities associated with predators during handling operations.

Auditors should confirm:

- The HMP includes the required considerations above.
- Employees are:
 - Familiarised with the predator control considerations in the HMP by conducting employee interviews and observations.
 - Trained on predator control verifications, e.g.:
 - Verification of mortality events specific to the type of handling operations by reviewing employees' training records, conducting employees' interviews and observations.

Useful Resources

N/A

Indicator:	Requirement:
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4.1.2.10 The UoC shall ensure that the HMP includes a water quality monitoring process according to **Appendix 12 (12.3, Table 2)**.

In the case of live transport⁵⁴, monitoring shall be conducted at the point of departure, during transport and on arrival. An exception applies where monitoring during transport negatively affects fish health and welfare, in which case monitoring is not required.

How do I interpret this requirement?

- Maintaining good water quality is a critical factor during fish handling.
 - Concentrations of substances within the water system that rise above or fall below specific levels can be harmful to fish populations and should be monitored.
- For developing an adequate environment for fish during handling the UoC should consider:
 - Water quality monitoring equipment e.g.
 - Constant monitoring.
 - Every 15 minutes (by hand, automatic).
 - Water quality monitoring frequency.
 - Prior to, during, and post handling.
 - Water quality monitoring for live transport.
 - Departure/arrival and during live transport unless this could cause detrimental impact.
- While Table 2 of Appendix 12 of the ASC Farm Standard specifies water quality parameters that must be included in the monitoring process, other parameters may be considered based on the specific requirements of the UoC.
- Conformance with maximum and minimum thresholds for the required parameters, customised to suit the species and environmental conditions, should be incorporated as part of the HMP water quality monitoring process.
- Corrective actions must be provided according to the HWMP that may be considered acceptable as a reference to develop and implement a water quality monitoring process (Indicator 4.1.1.10.1).

⁵⁴ An exemption applies where live transport after harvest is conducted by a third party outside of the scope of the UoC's certification.

Evidence of implementation should be maintained, e.g.:

- The water quality monitoring process, that incorporates the minimum elements required, included in the HWMP.
- Records confirming the monitoring of the required parameters and their conformance with maximum and minimum thresholds.
- Employees should demonstrate awareness of the established water quality monitoring process during handling.

Auditors should confirm:

- The water quality monitoring, that incorporates the minimum elements required, included in the HWMP. and cross-checking with the handling operations present in the HMP.
 - Implementation can be verified by cross-checking the process with records, information from employee interviews and observations.

Useful Resources

- The Fish Site (2012) [Monitoring Pond Water Quality to Improve Production](#).
- Lindholm-Lehto, P. (2023) [Water quality monitoring in recirculating aquaculture systems](#). Aquaculture Fish and Fisheries
- Fahad, A., N'Doye, I., & Taous-Meriem, L.K. (2023). [Feeding control and water quality monitoring in aquaculture systems: Opportunities and challenges](#). arXiv
- Erikson, U., Rosten, C., Klebert, P., Aspaas, S., & Rosten, T. (2022) [Live transport of Atlantic salmon in open and closed systems: Water quality, stress and recovery](#). Aquaculture Research

Indicator:	Requirement:
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- 4.1.2.11** The UoC shall ensure that the HMP includes a visual inspection process to identify abnormal behaviour in relation to the handling activity and defines mitigation measures when identified.

How do I interpret this requirement?

- Visual inspection during handling should be included in the HMP.

- When direct visual inspections are not possible (e.g. pangasius transportation) or considered invasive (e.g. during live fish transport inside a well), cameras can be used. Alternatively, fish behaviour can be assessed when loading and discharging the fish as a minimum requirement.

Evidence of implementation should be maintained, e.g.:

- The monitoring process, that incorporates the corrective actions required, included in the HMP.
- Records that confirm the monitoring of the behaviour.
 - A checked box on the monitoring records indicating whether monitoring behavioural took place or not is accepted.
- Employees should demonstrate awareness of the established behavioural monitoring process.

Auditors should confirm:

- The monitoring process included in the HMP that incorporates the corrective actions and is implemented.
 - Implementation can be verified by cross-checking the process with relevant records, information from employee interviews and observations.

Useful Resources

- Bhadmus, Z. T. (2022) [Effect of crowding on behaviour of Atlantic salmon \(Salmo salar\) in aquaculture](#). Thesis; The Arctic University of Norway.
- Stien, L., Nilsson, K., Noble, C., Izquierdo-Gomez, D., Ytteborg, E., Timmerhaus, G., & Madaro, A. (2024) [Evaluating a crowding intensity scale and welfare indicators for Atlantic salmon in sea cages](#). Aquaculture Reports.

Indicator:	Requirement:
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4.1.2.12 The UoC shall ensure that the HMP includes a post-handling evaluation process that considers at least the following:

- a) abnormal behaviour; and,
- b) morphological scores; and,
- c) moribund animals and mortalities.

How do I interpret this requirement?

- Abnormal behaviour related to the handling should be recorded.
- During and post-handling moribund fish and mortalities related to the handling should be recorded and removed.
- Moribund fish should be collected and culled following responsible stunning and killing according to Criterion 4.1.3.

Evidence of implementation should be maintained, e.g.:

- Behavioural monitoring.
- Records that confirm the monitoring of the behaviour, moribund and mortalities during and post-handling.
- Employees should demonstrate awareness of the established behavioural and mortality monitoring process above.

Auditors should confirm:

- The behavioural monitoring process included in the HMP incorporates the minimum required elements and it is implemented.
 - Implementation can be verified by cross-checking the process with relevant records, information from employee interviews and observations.

Useful Resources

- Pavlidis, M., Samaras, A., Papaharisis, L., (2024). [Welfare assessment study for European sea bass](#). University of Crete.

Indicator:	Requirement:
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4.1.2.13 The UoC shall ensure that the HMP includes acclimatisation measures to prevent sudden changes in water conditions affecting the health and welfare status of the fish.

How do I interpret this requirement?

- Standardised acclimatisation measures should be designed and maintained, e.g.:
 - Water temperature during transportation must be as close as possible to that which the fish came from.

Evidence of implementation should be maintained, e.g.:

- Standard acclimatisation measures and monitoring of the water quality according to the HMP.
- Records that confirm the monitoring of acclimatisation parameters before stocking.
- Employees should demonstrate awareness of the acclimatisation measures and monitoring process above.

Auditors should confirm:

- The acclimatisation measures included in the HMP incorporate the minimum required elements and are implemented.
 - Implementation can be verified by cross-checking the process with relevant records, information from employee interviews and observations.

Useful Resources

- APROMAR (2024) [A guide on fish welfare in Spanish aquaculture- Volume 2: Welfare of European sea bass](#). Spanish Aquaculture Business Association. 49 pp.
- APROMAR (2024) [A guide on fish welfare in Spanish aquaculture - Volume 3: Welfare of gilthead sea bream](#). Spanish Aquaculture Business Association. 49 pp.

Indicator:	Requirement:
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- 4.1.2.14** The UoC shall ensure that the HMP includes consideration of the use of appropriate equipment during handling activities to avoid affecting the health and welfare status of the fish.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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- 4.1.2.15** The UoC shall adhere to a limit of 15 seconds maximum time out of water⁵⁵.

How do I interpret this requirement?

- When fish are out of water:
 - Their gills collapse, preventing oxygen exchange.
 - They show violent escape behaviour accompanied by a stress response.
- Time out of water should be minimised to avoid unnecessary stress to the fish during handling.
- Conformance with maximum required time out of water, customised to suit the species, should be part of the HMP and associated with good handling practices.
- Time out of water should be minimised considering the biological needs of the fish e.g. oxygen exchange, gills collapse and stress response.

Evidence of implementation should be maintained, e.g.:

- Records confirming the required parameter and their conformance with maximum value.
- Employees should demonstrate awareness of the established maximum time out of water.

Auditors should confirm:

- The maximum time of water is included in the HMP, and it is implemented.

⁵⁵ An exemption applies to anaesthetised fish and pangasius (due to its ability to breathe air facultatively).

- Implementation can be verified by cross-checking the process with records, information from employee interviews and observations.

Useful Resources

N/A

Indicator:	Requirement:
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4.1.2.16 The UoC shall adhere to species-specific limits on maximum fasting duration⁵⁶, according to **Appendix 2 (2.5)**.

How do I interpret this requirement?

- Fasting duration:
 - Calculation in degree days provides a more precise indication of fish metabolic activity because it incorporates time and temperature, which better reflect the biological processes, and provides a standardised metric across different locations and time periods.
 - The concept of degree days involves accumulating the average daily temperature over a period of time, i.e. 3 days at an average daily temperature of 19 °C, are 57-degree days.
- In the case of exceptional situations (see requirement footnote) the length of fasting would be linked to the duration of the event rather than a pre-set timing or veterinarian or aquatic health professional rationale.

Evidence of implementation should be maintained, e.g.:

- Records confirming the required parameter and their conformance with minimum/maximum value.
- Employees should demonstrate awareness of the established minimum/maximum fasting time.

Auditors should confirm:

- The minimum/maximum time fasting time is included in the HMP and incorporates considerations related to intrinsic factors and is implemented.

⁵⁶ An exemption applies in exceptional situations of severe weather, adverse environmental conditions (e.g. jellyfish, algal bloom) or during disease events.

- Implementation can be verified by cross-checking the process with records, information from employee interviews and observations.

Useful Resources

N/A

Indicators 4.1.2.17-4.1.2.32:

Scope: Cleaner fish

Indicator:	Requirement:
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|-----------------|---|
| 4.1.2.17 | <p>The UoC shall develop, implement and maintain a site-specific Handling Management Plan (HMP), with the intent of ensuring good health and welfare of cleaner fish.</p> <p>The HMP shall be periodically reviewed and revised as necessary.</p> |
|-----------------|---|

- | | |
|-----------------|--|
| 4.1.2.18 | <p>The UoC shall ensure that the HMP includes all handling activities carried out on site.</p> |
|-----------------|--|

How do I interpret this requirement?

- This clause overlaps with Indicator 4.2.1.1 and 4.2.1.2 regarding fish. Please refer to Indicator 4.2.1.1 and 4.2.1.2 for a detailed description.
- Any additional handling events that are unique to cleaner fish need to be added to the HMP:
 - Removal from refuges/ hides

Useful resources (if applicable)

- Norwegian Seafood Research Fund (2018) [Best Practice Guidelines from the Norwegian Seafood Research Fund](#). Technical Report 62 pp.

Indicator:	Requirement:
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- 4.1.2.19** The UoC shall ensure that the HMP includes contingency plans for handling activities, including at least the following:
- a) immediate response measures in the case of system failure; and
 - b) humane culling response measures (see Indicator 4.1.3.9); and
 - c) mass mortality event response measures.

How do I interpret this requirement?

- o This clause overlaps with Indicator 4.2.1.3 regarding fish. Please refer to Indicator 4.2.1.3 for a detailed description.

Indicator:	Requirement:
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- 4.1.2.20** The UoC shall ensure that the HMP includes consideration of the environmental conditions associated with handling activities, with the intent of preventing handling under adverse conditions that could affect the health and welfare of the cleaner fish.

How do I interpret this requirement?

- o This clause overlaps with Indicator 4.2.1.4 regarding fish. Please refer to Indicator 4.2.1.4 for a detailed description.

Indicator:	Requirement:
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- 4.1.2.21** The UoC shall ensure that the HMP, following consultation with a registered veterinarian or aquatic animal health professional, considers the use of anaesthesia during handling activities.

How do I interpret this requirement?

- This clause overlaps with Indicator 4.2.1.5 regarding fish. Please refer to Indicator 4.2.1.5 for a detailed description.

Useful Resources

- Skår MW, Haugland GT, Powell MD, Wergeland HI, Samuelsen OB (2017) [Development of anaesthetic protocols for lumpfish \(*Cyclopterus lumpus* L.\): Effect of anaesthetic concentrations, sea water temperature and body weight](#). PLoS ONE
- Calabrese, S., Jonassen T.M., Steigum, E., Åsnes H.O., Imsland, A.K.D., Saude C.S., Wergeland T., Höglund, E. (2024) [Does sedation with AQUI-S® mitigate transport stress and post transport mortality in ballan wrasse \(*Labrus bergyltae*\)?](#) Frontiers in Veterinary Science

Indicator:	Requirement:
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- 4.1.2.22** The UoC shall ensure that the HMP includes a fitness assessment of the cleaner fish prior to handling activities⁵⁷, approved by a registered veterinarian or aquatic animal health professional.

For the specific handling activities listed below, the fitness assessment shall be carried out:

- within a two-week period prior to transport or treatment that requires handling of the cleaner fish, except when the same type of treatment is repeated, in which case the assessment shall be carried out at least once a month after the initial treatment;
- within a two-week period prior to net change and passive grading.

⁵⁷ An exemption applies to handling activities related to health examinations (e.g. health assessment, parasite counting) and in the case of emergencies (e.g. containment of an escape).

How do I interpret this requirement?

- This clause overlaps with Indicator 4.2.1.4 regarding fish. Please refer to Indicator 4.2.1.4 for a detailed description.

Indicator:	Requirement:
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4.1.2.23

The UoC shall ensure that the HMP considers crowding activities with the intent to prevent stress, injury and mortality. Crowding activities shall be carried out:

- under supervision; and
- gradually; and
- partially; and
- for a maximum of two hours.

An exception applies where a registered veterinarian or aquatic animal health professional demonstrates that exceeding two hours of crowding does not negatively impact cleaner fish welfare, in which case the maximum time period shall be defined by the registered veterinarian or aquatic animal health professional.

How do I interpret this requirement?

- Prior to the crowding activity, the impacts of the cleaner fish should be discussed and an action plan developed to ensure the cleaner fish are handled appropriately.
 - This may include moving the hides or refuges to one side of the net prior to the crowding.
- This clause overlaps with Indicator 4.2.1.7 regarding fish. Please refer to Indicator 4.2.1.7 for a detailed description.

Indicator:	Requirement:
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4.1.2.24 The UoC shall ensure that the HMP includes a water quality monitoring process according to **Appendix 12 (12.3, Table 2)**.

In the case of live transport⁵⁸, monitoring shall be conducted at the point of departure, during transport and on arrival. An exception applies where monitoring during transport negatively affects cleaner fish health and welfare, in which case monitoring is not required.

How do I interpret this requirement?

- This clause overlaps with Indicator 4.2.1.10 regarding fish. Please refer to Indicator 4.2.1.10 for a detailed description.

Indicator:	Requirement:
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4.1.2.25 The UoC shall ensure that the HMP includes a visual inspection process to identify abnormal behaviour in relation to the handling activity and defines mitigation measures when identified.

How do I interpret this requirement?

- This clause overlaps with Indicator 4.2.1.11 regarding fish. Please refer to Indicator 4.2.1.11 for a detailed description.

Useful Resources

- Cutierrez Rabadan, C., Spreadbury, C., Consuegra, S. & Garcia de Leaniz, C. (2021) [Development, validation and testing of an Operational Welfare Score Index for farmed lumpfish *Cyclopterus lumpus* L.](#) Aquaculture
- Noble, C., Iversen, M. H., Lein, I., Kolarevic, J., Johansen, L. –H., Burgerhout, E., Puvanendran, V., Kousoulaki, K., Aas, G. H., Stene, A. & Espmark, Å. M. (2019). [RENSVEL OWI FACT SHEET SERIES:](#)

⁵⁸ An exemption applies where live transport after harvest is conducted by a third party outside of the scope of the UoC's certification

[An introduction to Operational and Laboratory-based Welfare Indicators for ballan wrasse \(*Labrus bergylta*\)](#). 43 pp.

Indicator:	Requirement:
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- 4.1.2.26** The UoC shall ensure that the HMP includes a post-handling evaluation process that considers at least the following:
- a) abnormal behaviour; and
 - b) morphological scores; and
 - c) moribund animals and mortalities.

How do I interpret this requirement?

- This clause overlaps with Indicator 4.2.1.12 regarding fish. Please refer to Indicator 4.2.1.12 for a detailed description.

Useful Resources

N/A

Indicator:	Requirement:
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- 4.1.2.27** The UoC shall ensure that the HMP includes acclimatisation measures to prevent sudden changes in water conditions affecting the health and welfare status of the cleaner fish.

How do I interpret this requirement?

- This clause overlaps with Indicator 4.2.1.13 regarding fish. Please refer to Indicator 4.2.1.13 for a detailed description.

Indicator:	Requirement:
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- 4.1.2.28** The UoC shall ensure that the HMP includes consideration of the use of appropriate equipment during handling activities to avoid affecting the health and welfare status of the cleaner fish.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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- 4.1.2.29** *Indicator applicability: cleaner fish from wild caught origin*

The UoC shall demonstrate that the fishing gear used avoids injuries and unnecessary stress, and that it is suitable for the size and species of cleaner fish being captured.

How do I interpret this requirement?

- The fishing vessel sourcing the cleaner fish must be registered in the region and adhere to the local regulations.
- Cleaner fish must be visually assessed for injuries in their holding facility.

Evidence of implementation should be maintained, e.g.:

- Fishermen must be registered and aware of the transport regulations in their area.
- Records:
 - Fishing vessel.
 - Date of capture.
 - Number of cleaner fish.
 - Assessment of cleaner fish.

Auditors should confirm:

- There are records of the source of the cleaner fish.
- The assessment of the cleaner fish and any communication regarding sourcing of cleaner fish.

Useful Resources

- o Norwegian Seafood Research Fund (2018) [Best Practice Guidelines from the Norwegian Seafood Research Fund](#). Technical Report 62 pp.

Indicator:	Requirement:
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4.1.2.30 The UoC shall adhere to a limit of 15 seconds maximum time out of water⁵⁹.

How do I interpret this requirement?

- o This clause overlaps with Indicator 4.2.1.15 regarding fish. Please refer to Indicator 4.2.1.15 for a detailed description.

Indicator:	Requirement:
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4.1.2.31 The UoC shall adhere to species-specific limits on maximum fasting duration⁶⁰, according to **Appendix 2 (2.5)**.

How do I interpret this requirement?

- o This clause overlaps with Indicator 4.2.1.16 regarding fish. Please refer to Indicator 4.2.1.16 for a detailed description.

⁵⁹ An exemption applies to anaesthetised fish.

⁶⁰ An exemption applies in exceptional situations of severe weather, adverse environmental conditions (e.g. jellyfish, algal bloom) or during disease events.

Indicator:	Requirement:
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4.1.2.32 The UoC shall segregate cleaner fish in advance of any salmonid-specific treatments and ensure they are retained under appropriate conditions.

How do I interpret this requirement?

- Prior to any salmonid-specific treatments, the cleaner fish should be segregated.
 - The hides/refuges should be checked and then moved away from the crowding area.
 - Ensure the dragnet does not include the hides/refuges.

Evidence of implementation should be maintained, e.g.:

- The procedure of salmonid-specific treatments should include how to handle the cleaner fish prior to the start of the treatment.
- Employees should demonstrate awareness of how the cleaner fish are segregated prior to salmonid-specific treatments.

Auditors should confirm:

- Implementation can be verified by cross-checking the procedure and with employee interviews.

Useful Resources

- Norwegian Seafood Research Fund (2018) [Best Practice Guidelines from the Norwegian Seafood Research Fund](#). Technical Report 62 pp.

4.1.3 – Fish (including Cleaner Fish) – Slaughter and Killing

Indicators 4.1.3.1-4.1.3.7:

Scope: all fish species except cleaner fish (see Indicators 4.1.3.8-4.1.3.16)

Indicator:	Requirement:
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- 4.1.3.1** The UoC shall appoint a staff member responsible for overseeing the implementation of fish health and welfare requirements during slaughter.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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- 4.1.3.2** The UoC shall ensure all fish, including fish not destined for human consumption, are stunned prior to killing according to the methods and transition periods outlined in **Appendix 12 (12.4, Table 5).**

- 4.1.3.3** The UoC shall ensure that at least 95% of the fish lose consciousness immediately after stunning, and any remaining conscious fish are re-stunned. Unconsciousness shall persist until the fish are killed

- 4.1.3.4** The UoC shall ensure that stunning and killing are effective by verifying the absence of all of the following:
- a) opercular movements; and
 - b) eye movements; and
 - c) body movements (except for involuntary muscular spasms); and
 - d) reaction to a painful stimulus.

- 4.1.3.5** The UoC shall have measures in place to immediately respond to ineffective stunning or killing, including the presence of a back-up system, and shall implement appropriate corrective actions to prevent recurrence.

How do I interpret this requirement?

- o Slaughter is generally a two-stage process:

- The animal is first stunned to make it insensible to pain.
- Killing is then carried out.
- Stunning methods are required to induce immediate or rapid (less than one second) unconsciousness.
 - Ice-slurry is not an approved stunning method but can be used as a killing method after immediate or rapid (less than one second) unconsciousness.
 - Brain perforation techniques such as 'spiking' or 'ikijime', involve instruments that physically perforate the brain, causing severe and irreversible damage and leading to almost immediate stunning and death.
- An effective stun is determined by the loss of consciousness using an electroencephalogram (EEG) measured in a laboratory, which is not practicable at a site level.
- To ensure that at least 95% of the fish are stunned before killing, the sample size should accurately reflect the scale of the harvest operations. Variable sample sizes per harvest day may be accepted, provided that justification is given based on the specific scenarios. Stunning effectiveness should be assessed by confirming signs of unconsciousness:
 - Loss of balance or coordination.
 - Reduced or absent reflex responses.
 - Cessation of opercular movements (in succession and with rhythm).
 - Absence of eye roll.
- The use of body movement as an indicator for the effectiveness of stunning or killing can be misleading as muscular spasms often occur in unconscious or dead fish.
 - Body movements indicating struggling, a swimming activity or efforts to remain upright or regain equilibrium are relevant movements to watch out for and that indicate consciousness. Visual assessment comparing stunned and non-stunned fish:
- Body movements indicating struggling, a swimming activity or efforts to remain upright or regain equilibrium are relevant movements to watch out for and that indicate consciousness.
- A back up stunning system e.g. a priest, must be available.

	Stunned	Non-stunned
Movements	Muscle twitches	Flapping
Breathing	Gasping, Absent regular opercula movements	Regular opercula movements
Eye roll	Absent	Eye movement when rolled
Reactions to handling	None	Reactive
Reaction to tail pinch	None	Reactive
Swim response	Absent	Evident
Equilibrium	Absent	Evident

- The requirement to stun the fish before the killing also applies to fish not destined for human consumption e.g. wild species, cleaner fish.
- Ensure there is no chance of recovery of consciousness before fish death:
 - Minimise the stun to kill time to less than one second.
 - Consider that stun and kill can occur together, for example, pneumatic stunning with automatic gill cutting, but they are distinct operations.
- If there is uncertainty about whether a fish is unconscious, the stunning should be repeated or the harvest shall stop to verify stun parameters before proceeding, e.g. more than 5 fish in 10 seconds requiring backup stun.
 - Or use an alternative backup method, e.g. a priest or secondary stunner.
- For electrical stunning, machines must adhere to manufacturers recommendations for that species, or if not available, to documented electric stunning parameters from research.

Table 10: Electrical stunning methods which have been validated in the laboratory by EEG to prove that unconsciousness was reached within 1 second.

Species	Method	Parameters	Followed by killing method	Reference
Rainbow trout	In water electric stun (two submerged stainless steel electrodes)	30-60 seconds of 0.84 A _{RMS} dm ⁻² and 10.2 V _{RMS} cm Stun to maintain unconsciousness: ≥30 s, the electrical field strength (10.2 V _{RMS} cm ⁻¹) and the current density (0.84 A _{RMS} dm ⁻¹ respectively). ⁻¹	Not detailed in report	Hjelmstedt et al (2022) ⁶¹
European sea bass	In water electric stun (electric plate)	3.3 Arms/dm ² (sinusoidal 50 Hz or pulsed square wave AC, 133 Hz, 43% duty cycle) for 10 seconds	Ice slurry	Lambooij et al (2008) ⁶²
Turbot	Electric dry stunner with hanging electrodes	2.3 ± 0.91 A _{rms} by applying 122.6 ± 8.0 V _{rms} for 1 second Or 1.3 ± 0.4 A _{rms} for 1 s and 0.6 ± 0.3 A _{rms} for 19 seconds	Ice slurry	Daskalova et al (2016) ⁶³
Atlantic salmon	Electric dry stunner with hanging electrodes	A combined AC and DC supply minimum 0.5s recommended using currents of 668 mA _{rms} and ≈107 V _{rms}	Gill cutting*	Lambooij et al (2010) ⁶⁴
Atlantic salmon	In water electric stun	high voltage stunning, using 200 V/m for 1 second or low voltage stunning at 50–25 V/m for 3 and 12 seconds	Gill cutting*	Robb & Roth (2003) ⁶⁵
Nile tilapia	In water electric stun	t ~14 V _{RMS} cm ⁻¹ and a current density ~1.1 A _{RMS} dm ⁻² for 30 s	Gill cutting and ice slurry	Sundell, E. et al (2024) ⁶⁶

* Adapted killing method from reference article.

⁶¹ Hjelmstedt, P. et al (2022) Assessing the effectiveness of percussive and electrical stunning in rainbow trout: Does an epileptic-like seizure imply brain failure? Aquaculture, 552. <https://doi.org/10.1016/j.aquaculture.2022.738012>

⁶² Lambooij, B. et al (2007) Evaluation of electrical stunning of sea bass (*Dicentrarchus labrax*) in seawater and killing by chilling: welfare aspects, product quality and possibilities for implementation. Aquaculture Research, Vol 39, p. 50-58. <https://doi.org/10.1111/j.1365-2109.2007.01860.x>

⁶³ Daskalova, A.H. et al (2016) Effectiveness of tail-first dry electrical stunning, followed by immersion in ice water as a slaughter (killing) procedure for turbot (*Scophthalmus maximus*) and common sole (*Solea solea*). Aquaculture, Vol 455, p.22-31. <https://doi.org/10.1016/j.aquaculture.2015.12.023>

⁶⁴ Lambooij, E. et al. (2010) Percussion and electrical stunning of Atlantic salmon (*Salmo salar*) after dewatering and subsequent effect on brain and heart activities. Aquaculture, Vol 300, p. 107-112. <https://doi.org/10.1016/j.aquaculture.2009.12.022>

⁶⁵ Robb, D.H.F. & Roth, B. (2003) Brain activity of Atlantic salmon (*Salmo salar*) following electrical stunning using various field strengths and pulse durations. Aquaculture, Vol 216, p. 363-369. [https://doi.org/10.1016/S0044-8486\(02\)00494-5](https://doi.org/10.1016/S0044-8486(02)00494-5)

⁶⁶ Sundell, E., Brijis, J. & Grans, A. (2024) The quest for a humane protocol for stunning and killing Nile tilapia (*Oreochromis niloticus*). Aquaculture, Vol 593. <https://doi.org/10.1016/j.aquaculture.2024.741317>

Evidence of implementation should be maintained, e.g.:

- Employees should demonstrate knowledge confirming the stun and kill monitoring.
- Records:
 - Employee training (see Appendix 12)
 - Slaughter monitoring, e.g.:
 - Verification and registration of stunning and killing.
 - Observation of signs of unconsciousness.
 - Records of the use of a backup method.

Auditors should confirm:

- Monitoring of stunning and slaughter in conformance to the requirements.
- Records, employee interviews and observations of stunning and slaughter operations during the audit should demonstrate the implementation of the requirements.

Useful Resources

- Farm Animal Welfare Committee- UK Government (2014) [Opinion on the Welfare of Farmed Fish at the Time of Killing](#). Technical report 36pp.
- Humane Slaughter Association. (2016) [Signs of Recovery](#). Blog.
- The EFSA Journal (2009) [Scientific Opinion of the Panel on Animal Health and Welfare on a request from the European Commission on Species-specific welfare aspects of the main systems of stunning and killing of farmed carp](#). Scientific Opinion 37pp.
- Humane Slaughter Association. (2018) [Humane Harvesting of Halibut](#). Technical Report 5 pp.
- Aquatic Life Institute (2023) [Stunning and Slaughter: Best Practices for Animal Welfare in Aquaculture](#). Report 4pp.
- De La Rosa, I., Castro, P. & Ginés, R. (2021). [Twenty Years of Research in Seabass and Seabream Welfare during Slaughter](#). Animals. 11. 2164.
- Kestin, S., Devis, J. & Robb, D. (2002). [Protocol for assessing brain function in fish and the effectiveness of methods used to stun and kill them](#). The Veterinary record. 150. 302-7.
- Jung-Schroers, V., Hildebrandt, U., Retter, K. *et al.* (2020) [Is humane slaughtering of rainbow trout achieved in conventional production chains in Germany? Results of a pilot field and laboratory study](#). BMC Veterinary Research 16, 197.

- Barrento, S., Pedrazzani, A., Molento, C., & Quintiliano, M. (2023). [Tilapia behaviour & welfare indicators: from foraging to humane slaughter \[Online course\]](#). In S. Barrento, M. Quintiliano, & J. Muirhead (Eds.), FAI Academy. <https://fai.academy/>

Indicator:	Requirement:
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4.1.3.6 The UoC shall not use any of the following methods to kill fish:

- a) asphyxiation in air; or
- b) carbon dioxide (CO₂) bath; or
- c) salt bath; or
- d) ammonia bath; or
- e) evisceration.

How do I interpret this requirement?

- The methods outlined above do not induce immediate or rapid loss of consciousness and therefore do not meet the requirements of Indicator 4.1.3.2.
- Approved methods for each species are outlined in Appendix 12 (Table 5).

Useful Resources

- Welfarm (2023) Farmed Fish Slaughter Methods Report. [Recommendations for Rainbow trout, Atlantic salmon, European sea bass and Gilthead sea bream](#). Technical report 132pp.
- French Reference Centre for Animal Welfare (2024) [Opinion of the FRCAW on slaughter conditions for farmed fish](#). Technical report. 109pp.

Indicator:	Requirement:
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4.1.3.7 The UoC shall maintain records of the activities carried out under Indicators 4.1.3.3 and 4.1.3.4.

No additional guidance is provided for this Indicator.

Indicators 4.1.3.8-4.1.3.16:*Scope: Cleaner fish*

Indicator:	Requirement:
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- 4.1.3.8** The UoC shall appoint a staff member responsible for overseeing the implementation of cleaner fish health and welfare requirements during slaughter.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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- 4.1.3.9** The UoC shall ensure all cleaner fish are stunned prior to killing according to the methods and transition periods outlined in **Appendix 12 (12.4, Table 5)**.

- 4.1.3.10** The UoC shall ensure that at least 95% of the cleaner fish lose consciousness immediately after stunning, and any remaining conscious fish are re-stunned. Unconsciousness shall persist until the cleaner fish are killed.

- 4.1.3.11** The UoC shall ensure that stunning and killing are effective by verifying the absence of all of the following:
- opercular movements; and
 - eye movements; and
 - body movements (except for involuntary muscular spasms); and
 - reaction to a painful stimulus.

- 4.1.3.12** The UoC shall have measures in place to immediately respond to ineffective stunning or killing, including the presence of a back-up system, and shall implement appropriate corrective actions to prevent recurrence.

How do I interpret this requirement?

- This clause overlaps with Indicators 4.1.3.2 to 4.1.3.5 regarding fish. Please refer to Indicators 4.1.3.2 to 4.1.3.5 for a detailed description of slaughter.

- In addition to the interpretation supplied in Indicator 4.1.3.2 to 4.1.3.5, there are additional requirements for cleaner fish. These include:
 - The cleaner fish should be removed prior to the salmonid stunner system.
 - The cleaner fish should undergo a specific stunning and killing process.

Useful Resources

- van de Vis, H; Reimert, H; Grimsbø, E; Gerritzen, M; Foss, A; Roth, B. (2024) [NOFIMA Electrical stunning and killing of lumpfish with commercial waterbath and dry stunners](#). Nofima Technical Report.

Indicator:	Requirement:
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- 4.1.3.13** The UoC shall not use any of the following methods to kill cleaner fish:
- a. asphyxiation in air; or
 - b. carbon dioxide (CO2) bath; or
 - c. salt bath; or
 - d. ammonia bath; or
 - e. evisceration.

How do I interpret this requirement?

- Examples of current humane methods:
 - An overdose of anaesthetic.
 - Electrical stunning in combination with immersion in a chilled water bath.
 - Electrical stunning followed by immediate grinding.

Useful Resources

- van de Vis, H; Reimert, H; Grimsbø, E; Gerritzen, M; Foss, A; Roth, B. (2024) [NOFIMA Electrical stunning and killing of lumpfish with commercial waterbath and dry stunners](#). Nofima Technical Report.

Indicator:	Requirement:
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4.1.3.14 The UoC shall maintain records of the activities carried out under Indicators 4.1.3.10 and 4.1.3.11.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.1.3.15 The UoC shall re-use cleaner fish where possible and under the following conditions:

- a) re-use shall only occur once; and
- b) re-use shall only occur within the site or the ABM (see Indicator 4.5.1); and
- c) re-use shall only occur if the outcome of a health assessment indicates low risk of disease transfer. The health assessment shall:
 - consider the health status of the salmon and the cleaner fish at the origin and destination sites; and
 - be conducted no longer than two weeks before re-use; and
- d) be approved by a registered veterinarian or an aquatic animal health professional; and cleaner fish are segregated from the salmon before transfer to the destination site.

How do I interpret this requirement?

- Prior to the end of the salmon production cycle, discuss with the veterinarian or animal health professional to decide if the cleaner fish are to be re-used or killed.
 - This will depend on:
 - If there are suitable sites for re-use within the same site or ABM (see Indicator 4.5.1).
 - The disease history of the cleaner fish.
 - The disease history of the salmon.
 - Whether the cleaner fish can be re-used within two weeks from being separated from the original salmon companions.

- The approval by the registered veterinarian or animal health professional.

Evidence of implementation should be maintained, e.g.:

- Records:
 - The original source of cleaner fish.
 - The date of the addition of the cleaner fish.
 - The date of the end of the salmon production cycle.
 - The approval by the registered veterinarian or animal health professional.
-

Auditors should confirm:

- The approval process by reviewing records and cross-checking dates or stocking, disease history, end of salmon production as well as approval records for re-use.
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Useful Resources

- Norwegian Seafood Research Fund (2018) [Best Practice Guidelines from the Norwegian Seafood Research Fund](#). Technical Report 62 pp.

Indicator:	Requirement:
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4.1.3.16 The UoC shall ensure that cleaner fish are humanely slaughtered where re-use is not possible. The UoC shall not release cleaner fish into the wild.

How do I interpret this requirement?

- This clause overlaps with Indicators 4.1.3.2 to 4.1.3.5 regarding fish. Please refer to Indicators 4.1.3.2 to 4.1.3.5 for a detailed description of slaughter.
- In addition to the interpretation supplied in Indicators 4.1.3.2 to 4.1.3.5, there are additional requirements for cleaner fish. These include:
 - The cleaner fish should be removed prior to the salmonid stunner system.
 - The cleaner fish should undergo a specific stunning and killing process.
 - The dead cleaner fish should be disposed of appropriately.

Useful Resources

- Norwegian Seafood Research Fund (2018) [Best Practice Guidelines from the Norwegian Seafood Research Fund](#). Technical Report 62 pp.
- van de Vis, H; Reimert, H; Grimsbø, E; Gerritzen, M; Foss, A; Roth, B. (2024) [NOFIMA Electrical stunning and killing of lumpfish with commercial waterbath and dry stunners](#). Nofima Technical Report.

Criterion 4.2 – Crustaceans

4.2.1 - Crustaceans - Health and Welfare

Indicator:	Requirement:
4.2.1.1	The UoC shall ensure that all relevant employees are trained on general animal health and welfare awareness, including biosecurity, according to Appendix 12 (12.2, Table 1) .
4.2.1.2	The UoC shall ensure that any person involved in animal husbandry and handling activities is trained on animal health and welfare, according to Appendix 12 (12.2, Table 1) .
4.2.1.3	The UoC shall ensure that the content of the training in Indicators 4.2.1.1 and 4.2.1.2 is updated as necessary and is approved by a registered veterinarian or aquatic animal health professional.

How do I interpret this requirement?

- For each employee:
 - The level of training should be appropriate to their role:
 - For all relevant employees: a basic level of awareness of crustacean health and welfare, including biosecurity, is required.
 - For any person involved in animal husbandry and handling activities: a more advanced level of training on crustacean health and welfare, handling and slaughter practices is required on an annual basis.
 - Evidence of training participation must be maintained:
 - Certificates of attendance.
 - Certificates of competency.
- The trainer should be competent in the topic of animal health and welfare and the competence should be demonstrable, e.g.:
 - Curriculum Vitae.
 - Records of qualifications.
 - Work experience.
- Best practices on the implementation of the training content should be visible during the audit.
 - Employees should demonstrate adequate knowledge of that content at interviews.

Evidence of implementation should be maintained, e.g.:

- A list of employees and subcontracted personnel involved in animal husbandry and handling operations.
- Records:
 - Training.
 - Trainer's competence.
 - Certificates of competency.

Auditors should confirm:

- Employees received the required level of training for their roles by cross-checking employees and subcontracted personnel lists with:
 - Training records.
 - Certificates of competency.
 - Trainer's competence.
 - Revision of training resources/contents.
- Adequate level of personnel knowledge of the training content should be verified by:
 - Interviews.
 - Observation of real operations during handling and slaughter of crustaceans.

Useful Resources

- Barrento, S., Pedrazzani, A., Ostrensky, A., & Quintiliano, M. (2023). *Shrimp Welfare Indicators* [Online course]. In S. Barrento, M. Quintiliano, & J. Muirhead (Eds.), FAI Academy. <https://fai.academy/>
- World Organisation for Animal Health (2024) – [Competency and Curriculum Guidelines for Community Animal Health Workers](#). WOA Report, Paris, 39 pp.

Indicator:	Requirement:
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4.2.1.4

The UoC shall adhere to species-specific mortality limits, according to **Appendix 2 (2.3)**.

No additional guidance is provided for this Indicator.

Useful resources (if applicable)

- <https://tomota.vn/en/shrimp-survival-rate-calculator>
- McKay, H., & McAuliffe, W. (2024) [Methods for Pre-slaughter mortality of farmed shrimp](#). ReThink Priorities: Technical report. 24pp.
- Šimčikas, S., & Waldhorn, D.R. (2020) [Farmed shrimp and prawn numbers](#) | Guesstimate

Indicator:	Requirement:
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4.2.1.5

The UoC shall only stock batches of crustaceans, including nauplii, larvae, post-larvae and broodstock, that have been tested prior to stocking and are free from diseases of regional concern, and met importation guidelines when applicable.

How do I interpret this requirement?

- Testing should be relevant to:
 - The species.
 - The production system.
 - The scale of production.
 - The local regulations.
 - For example:
 - Early mortality syndrome (EMS), which is also known as acute hepatopancreatic necrosis disease (AHPND).
 - *Enterocytozoon hepatopenaei* (EHP).
 - White spot syndrome (WSS).
- Testing might consist of:
 - Clinical examination of the crustaceans.
 - Laboratory tests:

- Bacteriology.
- Molecular biology (e.g. PCR).
- Histopathology.
- Virology (e.g. cell culture).
- The designated veterinarian or aquatic animal health professional should determine:
 - The diseases to include in the testing, based on local regulations, scientific literature and/or publicly available information (see interpretation for Indicator 4.2.1.8).
 - The sample size, based on local regulations and/or technical judgement.
- Compulsory governmental testing programme results are acceptable for demonstrating conformance with this requirement.

Evidence of implementation should be maintained, e.g.:

- List of relevant diseases for the species in the country (see interpretation for Indicator 4.2.1.8).
- Testing results written report or document that includes as a minimum:
 - The batch of crustaceans being tested.
 - The date of the test.
 - The pathogen tested.
 - The results of the test.
- Records:
 - Purchase.
 - Transportation.
 - Delivery.
 - Stocking.

Auditors should confirm:

- Every batch of crustaceans has been tested by cross-checking the list of relevant diseases for the species in the country with the testing result's written report or document and records.

Useful Resources

- Lee D, Yu YB, Choi JH, Jo AH, Hong SM, Kang JC, Kim JH. (2022) [Viral Shrimp Diseases Listed by the OIE: A Review. Viruses.](#)
- The Fish Site (2020) [Everything you need to know about EMS in shrimp farming.](#)

Indicator:	Requirement:
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4.2.1.6 The UoC shall ensure that a registered veterinarian or aquatic animal health professional periodically visits the site to perform routine checks. Checks shall be carried out at least annually, as well as reactively in cases of crustacean health or welfare concerns.

How do I interpret this requirement?

- Regular veterinarian or aquatic animal health professional visits may fall into two categories:
 - Routine: Performed on a regular basis to monitor the health and welfare of the farmed crustaceans.
 - Reactive: Performed in response to deterioration or concerns for the health or welfare status of the farmed crustaceans.
 - Sites should be prepared to provide the veterinarian or aquatic animal health professional with detailed information about the observed symptoms, recent changes in behaviour, morphology, water quality or mortality and any relevant history of disease or treatments.
 - The site should use reactive health visits to address immediate health or welfare concerns, diagnose underlying causes of problems, and implement targeted interventions to mitigate risks and prevent further spread of disease.
- For routine visits, preparation is recommended to ensure a meaningful outcome from the veterinarian or aquatic animal health professional. Below are some practical considerations:
 - Visit scheduling: Ensure key employees are available during the visit to provide relevant required information and to facilitate adequate follow-up.
 - Designate a senior employee to serve as the primary point of contact for the veterinarian or aquatic animal health professional during the visit.
 - This individual can help coordinate the activity, ensure relevant information is available, assist during the visit and ensure follow-up.
 - Gather and organise relevant information: Compile relevant information about the operations including, for example, species being cultured, production systems, stocking densities, recent health history including treatments, water quality monitoring results, behaviour, morphology and mortalities data, feeding plan, any previous health report and of any ongoing issue.
 - Having this information readily accessible will facilitate discussion and decision-making during the visit.
 - Prepare the UoC for the visit: Ensure the UoC is prepared for the visit and for providing full access to all relevant areas to the veterinarian or aquatic animal health professional.
 - Address any maintenance issues, clean tanks or ponds as needed and ensure clear access to all relevant areas of the facility.
 - Ensure handling and other equipment that might be needed during the visit, such as nets, containers and sampling equipment, is available and in good working condition.

- Make sure employees are aware of the visit and its objectives and encourage them to proactively participate in it.
- During the visit: Ensure relevant information is discussed with the veterinarian or aquatic animal health professional and that the visit includes all relevant areas within the UoC.
- After the visit: Follow up on any recommendations or actions discussed during the visit.
 - Implement any necessary revisions or changes to management practices, treatment protocols, or facility conditions based on the veterinarian or aquatic animal health professional's feedback and any sample or testing results
- Continuous improvement: Use the insights gained from the veterinarian or aquatic animal health professional visit to identify opportunities for continuous improvement in health and welfare management.

Evidence of implementation should be maintained, e.g.:

- Health reports.
- Testing results.
- Records:
 - Veterinarian or aquatic animal health professional qualifications.
 - Veterinarian or aquatic animal health professional visit.
 - Recommendations or actions discussed during the visit.

Auditors should confirm:

- The qualifications of the veterinarian or aquatic animal health professional and cross-check the frequency of visits with records.
- Existence of health reports and testing results.

Useful Resources

N/A

Indicator:	Requirement:
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4.2.1.7 *Indicator applicability: every UoC using feed*

The UoC shall develop and implement a feeding plan which includes at least the following:

- a) time and frequency of feeding; and
- b) feed rations; and
- c) feeding methods; and
- d) different environmental conditions; and
- e) prevention of use of feed which has expired, spoiled or shows signs of excessive leakage; and
- f) feed types for the nutritional needs and life stages of the species.

An exception applies to (f) when such feeds are not available, in which case the UoC shall use feed suitable for animals with similar nutritional needs.

How do I interpret this requirement?

- The feeding plan:
 - Outlines the site's feeding strategy.
 - As a minimum, it must include the following parameters:
 - Time and frequency, e.g. at daybreak and early afternoon (06:00 am and 01:00 pm).
 - Feed rations, e.g. feed type and daily amount.
 - Feeding adaptation to fit different life stages, e.g. maturation diets for black tiger with excellent digestible protein, essential fatty acids, cholesterol and chemo-attractant properties similar to which are found in natural food sources in the crustacean habitat.
 - Feeding adaptation to fit different ambient conditions, e.g. at times of crustacean stress due to water quality issues, it is accepted to reduce feeding by 50% or completely stop feeding for 2 to 4 days.
 - Should include considerations to guarantee the good working order of mechanical equipment used to deliver feed, e.g. scheduled maintenance checks of automatic feeders.
- The UoC should document the rationale for changes in the feeding plan to better inform management decisions.

Evidence of implementation should be maintained, e.g.:

- Records:
 - Purchase.

- Transportation.
 - Delivery.
 - Stocking.
 - Maintenance.
 - Environmental and health.
 - Feeding periods (time and frequency).
 - Feeding rations.
- Employees should demonstrate adequate knowledge of the feeding plan.

Auditors should confirm:

- Feeding plan includes the minimum required parameters and its implemented:
 - Implementation can be confirmed by cross-checking the plan with records, information from employee interviews and observations.

Useful Resources

- Tacon, A.G.J., Jory, D. and Nunes, A. (2013) [Shrimp feed management: issues and perspectives](#). In M.R. Hasan and M.B. New, eds. On-farm feeding and feed management in aquaculture. FAO Fisheries and Aquaculture Technical Paper No. 583. Rome, FAO. pp. 481–488.
- Barrento, S., Pedrazzani, A., Ostrensky, A., & Quintiliano, M. (2023). *Shrimp Welfare Indicators* [Online course]. In S. Barrento, M. Quintiliano, & J. Muirhead (Eds.), FAI Academy. <https://fai.academy/>

Indicator:	Requirement:
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4.2.1.8 The UoC shall develop, implement and maintain a site-specific Health and Welfare Management Plan (HWMP), approved by a registered veterinarian, with the intent to prevent disease outbreaks and ensure good health and welfare of farmed crustaceans.

The HWMP shall include all of the following:

- a) disease monitoring; and
- b) response, including quarantine, treatment, and humane culling; and
- c) reporting mechanisms, including reporting WOA-notifyable disease to authorities; and
- d) biosecurity risks; and
- e) predation risks.

The HWMP shall be reviewed and revised at least annually.

How do I interpret this requirement?

- The Health and Welfare Management Plan (HWMP):
 - Aims to support the development and implementation of preventive, rather than remedial, actions that ensure good crustacean health and welfare.
 - Must be developed by the UoC in collaboration with a veterinarian or aquatic animal health professional who approves it.
- For developing and establishing a site-specific disease monitoring, response and reporting mechanism within the plan, the UoC should consider:
 - Developing standardised health and welfare monitoring protocols.
 - This may involve conducting visual inspections, monitoring OWIs and performing diagnostic tests for pathogens.
 - Producing and maintaining detailed records.
 - Including records of health observations, diagnostic test results, treatments administered, mortalities, and other relevant data.
 - It is recommended to use a centralised database or electronic system for efficient data management and retrieval.
 - Implementing an early detection system.
 - To detect signs of disease outbreaks or abnormal health conditions promptly.
 - This could include environmental monitoring tools.
 - Establishing response protocols for responding to disease outbreaks or health emergencies.

- Outlining procedures for:
 - Isolating affected individuals or populations.
 - Implementing strategies for treatments or others (e.g. culling affected batches of crustaceans, see interpretation for Indicator 4.2.3.2)
 - Notifying relevant stakeholders, including other sites in the vicinity and the authorities.
- Establishing communication channels.
 - Effective communication channels should be implemented for reporting health-related issues both internally within the facility and externally to relevant authorities (e.g. WOAH-notifiable diseases) or stakeholders.
 - Employees should understand their roles and responsibilities in reporting and responding to disease events.
- Maintaining a readily accessible notification list.
 - Including contact names numbers and emails for key employees and other relevant stakeholders.
 - The list should be kept up-to-date to ensure quick and efficient communication during emergencies.
- For developing a site-specific biosecurity mechanism within the plan, the UoC should consider:
 - Determining the objectives and scope of the biosecurity mechanism.
 - Identifying relevant regulatory requirements.
 - Gathering information about the site, including an up-to-date layout of buildings and production units identifying the possible movements of animals, animal products, animal waste, water, feed and fomites/vectors.
 - Identifying potential transmission pathways into, within and from the site.
 - Carrying out a risk analysis to identify and evaluate disease threats and ensure that the plan addresses them.
 - A risk analysis can be carried out wholly or only partially, e.g. disease transmission during transportation.
 - This will depend on the objectives of the biosecurity mechanism.
 - Identifying mitigation measures and assessing their effectiveness and their management requirements.
 - Establishing emergency procedures in the event of a biosecurity failure.
 - Defining internal and external communication procedures, roles and responsibilities.
 - Creating a monitoring and internal audit schedule.
 - Carrying out performance evaluations.

- Developing procedures (e.g. Standard Operating Procedures, 'SOPs') to support the implementation of the mitigation measures and the biosecurity mechanism.
- Training employees in the different elements of the biosecurity mechanism.
- Maintaining a record-keeping system that provides evidence of compliance with the biosecurity mechanism.
- For developing predator control measures within the plan, the UoC should consider:
 - Identifying potential predators.
 - Create a list of the predators that are present in the area and that might interact with the UoC.
 - Predators may include birds, mammals, reptiles, and other aquatic species known to prey on crustaceans.
 - Assessing the risk of predation.
 - Evaluate the risk posed by each predator species identified based on factors such as proximity to the site, population size, means of accessing the site, likelihood of predation, etc.
 - Implementing predator control measures.
 - Develop and implement predator control measures tailored to the specific risks identified. This may include:
 - Physical barriers such as nets, fences or screens to prevent access to crustacean tanks or ponds.
 - Scare tactics such as visual deterrents to discourage predators from approaching the facility.
 - Implementing ongoing monitoring and surveillance.
 - To detect predator activity and intervene as needed.
 - Regular reviewing and revision.
 - Based on the outcome of the ongoing monitoring and surveillance.
 - Adjustments to the measures should be made when the results of monitoring indicate the need to optimise protection for crustaceans and the containment system.
 - Collaborating with local authorities or local wildlife agencies to address predator issues in the surrounding area.
- A periodic review and, if necessary, a revision of the HWMP must be conducted in response to changes in farming activities or external factors or as directed by the veterinarian.
 - As a minimum, the review should occur annually.
- The UoC should develop feedback mechanisms to inform the review, e.g. proactively seeking feedback from employees and subcontracted personnel, such as the harvest team, on the operationalisation of the site-specific biosecurity mechanism.
- For a single producer overseeing multiple UoCs in a shared waterbody, a unified HWMP that incorporates site-specific details is acceptable; however, for larger companies managing multiple UoCs in one country, while common plans may apply, it is crucial to ensure HWMP adaptation to

individual site circumstances, particularly in aspects like biosecurity measures, disease surveillance programmes and monitoring systems.

Evidence of implementation should be maintained, e.g.:

- The HWMP includes the required five minimum considerations above and all relevant records associated with the HWMP implementation.
- Employees are:
 - Familiarised with the HWMP and the feedback mechanisms for its review by conducting employee interviews and observations.
 - Trained on the site-specific biosecurity mechanism by reviewing employees' training records, conducting employees' interviews and observations.

Auditors should confirm:

- The HWMP includes the required five minimum considerations above and all relevant records associated with the HWMP implementation.
- Employees are:
 - Familiarised with the HWMP and the feedback mechanisms for its review by conducting employee interviews and observations.
 - Trained on the site-specific biosecurity mechanism by reviewing employees' training records, conducting employees' interviews and observations.

Useful Resources

- World Organisation for Animal Health (2022) [Collection and Evaluation of Existing Guidelines and Awareness Materials on Aquaculture Biosecurity for Smallscale Farms in the Asia-Pacific Region](#). Report 36pp.
- World Organisation for Animal Health (2024) [Aquatic Animal Health Code. Chapter 1.1. Notification of diseases, and provision of epidemiological information](#).

Indicator:	Requirement:
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4.2.1.8.1 *Indicator applicability: every UoC using feed*

The UoC shall ensure the HWMP includes a water quality monitoring process according to **Appendix 12 (12.3, Table 3)**.

How do I interpret this requirement?

- Maintaining good water quality is a critical factor when culturing any aquatic organism.
 - Concentrations of substances within the water system that rise above or fall below specific levels can be harmful to crustacean populations.
- Concerning health and welfare, water quality has a dual role as a cause or a consequence of poor health and welfare:
 - Poor water quality can cause health and welfare hazards in an aquaculture facility, e.g. heavy rain promoting quick pH changes in a flow-through system or leaking from equipment in a recirculating aquaculture system (RAS).
 - Poor water quality can also result from poor husbandry practices resulting in a risk to health and welfare, e.g. if stocking density is too high the water quality is likely to deteriorate, especially in closed and semi-closed systems.
- While Table 3 of Appendix 12 of the ASC Farm Standard specifies water quality parameters that must be included in the monitoring process, other parameters may be considered based on the site-specific requirements of the UoC.
- Conformance with maximum and minimum thresholds for the required parameters, customised to suit the species and environmental conditions, should be incorporated as part of the water quality monitoring process.
- **Table 11** below gives examples that may be considered acceptable as a reference to develop and implement a water quality monitoring process.

Table 11: Water quality parameters for shrimp in ponds based on information from the FAI shrimp course.

	Acceptable	Warning levels	Unacceptable levels
Temperature (twice a day)	25 – 32.4°C	14.5 -25.4°C or 32.5 - 35.4°C	≤14.4°C or ≥35.5°C
Dissolved oxygen (twice a day)	≥65%	49-64%	≤48%
pH (daily)	6.5 – 8.5	5.0 – 6.4 or 8.6 -9.0	≤4.9 or ≥9.1
Salinity (risk assessed) PSU	10.0-40.9‰	0.6-9.9‰ or 41.0 – 59.0‰	≤0.5‰ or ≥60.0‰

Evidence of implementation should be maintained, e.g.:

- The water quality monitoring process, that incorporates the minimum elements required, included in the HWMP.
- Records confirming the monitoring of the required parameters and their conformance with maximum and minimum thresholds.
- Employees should demonstrate awareness of the established water quality monitoring process.

Auditors should confirm:

- The water quality monitoring process included in the HWMP incorporates the minimum required elements and is implemented.
 - Implementation can be verified by cross-checking the process with records, information from employee interviews and observations.

Useful Resources

- Srinivas, Durbha & Venkatrayulu, Ch. (2023). [Water quality management in shrimp aquaculture](#). Book.
- Cuéllar-Anjel, J., Lara, C., Morales, V., De Gracia, A., & Gracia-Suárez, O. (2010) [Manual of Best Management Practices for white shrimp *Penaeus vannamei* farming](#). OIESA OSPESCA, C.A. pp. 132.

Indicator:	Requirement:
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4.2.1.8.2 *Indicator applicability: every UoC using feed*

The UoC shall ensure the HWMP includes a morphological scoring monitoring process for live crustaceans. Monitoring shall be conducted at least once a week, and shall include at least the following scoring parameters:

- exoskeleton damage, including at least eyes, antennae, appendages and rostrum;
- hepatopancreas colouration and size;
- gill colouration;
- size dispersion/variation;
- shell blisters and necrosis;
- full/empty digestive tube and colouration;
- loss of appendages, categorised according to causes.

An exception applies where deviation from monitoring frequency is justified by any of the following reasons:

- immediately after stocking; or
- under environmental conditions that affect the welfare of the crustaceans or the health and safety of employees; or
- crustaceans are undergoing a disease event and/or being treated.

The reason for any deviation shall be recorded.

How do I interpret this requirement?

- Morphological scoring is frequently used as an immediate indicator of welfare.
 - However, changes in morphology may have a variety of causes and therefore must be further analysed before drawing conclusions.
 - Loss of appendages shall be segregated according to potential causes e.g. bacterial infection or cannibalism.
 - Table 12 below gives examples that may be considered acceptable as a reference to develop and implement morphological scoring.

Table 12: Morphological parameters showing acceptable, warning level and unacceptable levels for shrimp based on Pedrazzani, Ana Silvia, Nathieli Cozer, Murilo Henrique Quintiliano, Camila Prestes dos Santos Tavares, Ubiratã de Assis Teixeira da Silva, and Antonio Ostrensky. 2023. "[Non-Invasive Methods for Assessing the Welfare of Farmed White-Leg Shrimp \(*Penaeus vannamei*\)](#)" and the FAI course.

Morphological Parameter		Acceptable	Warning levels	Unacceptable
Exoskeleton damage	Eyes	Healthy no damage	Damage, lightening, softening or swelling in one eye	Damage, lightening, softening or swelling in both eyes Absence of one or two eyes
	Antennas	Healthy No damage	Shortening, darkening or lesions/ necrosis	Absence, blueness, wrinkling, many dark spots
	Rostrum	Healthy appearance No changes	Mild injury, erosion or necrosis	Severe injury, erosion, necrosis or deformity Absence
	Appendages	Healthy No damage	One or two damaged legs/ Severe absence	Damage, necrosis, deformity Abnormal colour Severe absence Darkened edges
Loss of appendages	Bacterial infection	Intact No damage	Evidence of bacterial infection	Significant damage
	Cannibalism	Intact No damage	Some damage	Missing appendages
Hepatopancreas colouration and size		Healthy No damage	Discrete volume reduction	Decrease in size, colour change, tearing, presence of fluid or wormlike structures
Gill colouration		Healthy No damage	Some lesions or darkening	Colour change- pale, yellowish, redness or darkening Whiteish spots Erosion
Size dispersion/variation		Low	Some	Significant
Shell blisters and necrosis		Healthy no damage	Slight lesions, darkening or debris	Tissue loss, necrosis, colour change, deformity, calcified round white patches, crusting

Full/empty digestive tube and colouration	Full	Partially full	Empty
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Evidence of implementation should be maintained, e.g.:

- The morphological scoring monitoring process, that incorporates the minimum elements required, included in the HWMP.
- Records that confirm the monitoring of the morphological scoring.
- Employees should demonstrate awareness of the established morphological scoring monitoring process.

Auditors should confirm:

- The morphological monitoring process included in the HWMP incorporates the minimum required elements and is implemented.
 - Implementation can be verified by cross-checking the process with relevant records, information from employee interviews and observations.

Useful Resources

- Pedrazzani, A.S., Cozer, N., Quintiliano, M.H., dos Santos Tavares, C.P., da Silva, U.A.T & Ostrensky, A. (2023) "[Non-Invasive Methods for Assessing the Welfare of Farmed White-Leg Shrimp \(*Penaeus vannamei*\)](#)" Animals.
- Barrento, S., Pedrazzani, A., Ostrensky, A., & Quintiliano, M. (2023). *Shrimp Welfare Indicators* [Online course]. In S. Barrento, M. Quintiliano, & J. Muirhead (Eds.), FAI Academy. <https://fai.academy/>

Indicator:	Requirement:
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4.2.1.8.3 *Indicator applicability: every UoC using feed*

The UoC shall ensure the HWMP includes a monitoring process for behavioural scoring on live crustaceans from feeding trays or similar structures. Monitoring shall be conducted daily.

An exception applies where deviation from monitoring frequency is justified by any of the following reasons:

- immediately after stocking; or
- under environmental conditions that affect the welfare of the crustaceans or the health and safety of employees; or
- crustaceans are undergoing a disease event and/or being treated.

The reason for any deviation shall be recorded.

How do I interpret this requirement?

- To meet the ASC methodology for behaviour, only feeding behaviour needs to be monitored (see example in Table 13). However, other behaviours may also be considered. Further time is needed to develop quality behavioural analyses, which, while valuable for welfare and system management, must be tailored to the species, farming systems and available resources.
 - Additionally, changes in behaviour may have a variety of causes and, therefore, must be further analysed before drawing conclusions.
- Table 13 below gives examples that may be considered acceptable as a reference to develop and implement feeding behaviour scoring.

Table 13: Description of acceptable, warning levels and unacceptable behaviour of shrimp during feeding adapted from Pedrazzani, et al (2023). "[Non-Invasive Methods for Assessing the Welfare of Farmed White-Leg Shrimp \(Penaeus vannamei\)](#)"

	Acceptable	Warning levels	Unacceptable
Feeding behaviour	Most shrimp react quickly by moving to the newly-offered food	Only some of the shrimp react to the newly-offered food	Most shrimp do not react to the newly-offered food
Feed consumption efficiency	Tray is consistently emptied during feeding time	Partial feed remains (<20% uneaten)	Significant feed remains (>20% uneaten)
Interaction with feeding environment	Shrimp feed calmly without aggression or erratic movement	Occasional signs of stress or avoidance behaviour	Persistent signs of stress or aggression

Evidence of implementation should be maintained, e.g.:

- The feeding behaviour scoring monitoring process which incorporates the minimum elements required is included in the HWMP.
- Records that confirm the monitoring of the feeding behaviour scoring.
- Employees should demonstrate awareness of the established behavioural scoring monitoring process.

Auditors should confirm:

- The behavioural scoring monitoring process is included in the HWMP.
- Implementation can be verified by cross-checking the process with relevant records, information from employee interviews and observations.

Useful Resources

- Darodes de Tailly, J.B., Keitel, J., Owen, M.A.G., Alcaraz-Calero, J.M., Alexander, M.E. & Sloman, K.A. (2021) [Monitoring methods of feeding behaviour to answer key questions in penaeid shrimp feeding](#). Reviews In Aquaculture.
- Li D, Liu C, Song Z, Wang G. (2021) [Automatic Monitoring of Relevant Behaviors for Crustacean Production in Aquaculture: A Review](#). Animals (Basel).
- Barrento, S., Pedrazzani, A., Ostrensky, A., & Quintiliano, M. (2023). *Shrimp Welfare Indicators* [Online course]. In S. Barrento, M. Quintiliano, & J. Muirhead (Eds.), FAI Academy. <https://fai.academy/>

Indicator:	Requirement:
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4.2.1.8.4 *Indicator applicability: every UoC using feed*

The UoC shall ensure the HWMP includes a mortality monitoring process which includes at least the following:

- a) monitoring and recording of mortalities when spotted; and
- b) removal of mortalities when spotted; and
- c) conduct a post-mortem analysis for each mortality event; and
- d) investigation of mortality events which remain unexplained or unattributed to crustacean health reasons; and
- e) collection of moribund crustaceans when spotted and cull them according to Criterion 4.2.3.

Where any of the listed items above are not possible, the UoC shall record the reason.

How do I interpret this requirement?

- The UoC should develop a procedure (e.g. SOP) to describe moribund collection and culling operations.
 - Employees should be trained on this procedure.
- Moribund crustaceans should be collected using appropriate equipment to prevent further injuries and stress, e.g. hand net according to crustacean size.
- Slaughtered moribund crustaceans should be considered for the calculations needed to demonstrate compliance with Indicator 4.2.1.4.
- Whole pond/tank health issues should be addressed under a contingency plan for emergency culling.
- **Possible causes of several moribund crustaceans should be investigated.**
 - A contingency plan for emergency culling should be in place.
- Monitoring for mortality is understood as visual observation of whether there is mortality at the site, e.g. observations of the crustaceans in feeding trays, observations of mortalities at the tank/pond discharge.
- Possible causes that would justify no monitoring or removal of moribunds and mortality when spotted are severe bad weather, or a major equipment failure not related to poor maintenance.
- Scrutiny of mortality records and cause determination is essential for preventing further avoidable deaths and improving overall crustacean health and welfare in aquaculture.
- If on-site diagnosis is inconclusive, off-site laboratory diagnosis is required.

- When designing the mortality monitoring process, the UoC should consider carrying out a post-mortem analysis for each mortality event, not necessarily every crustacean.
- A statistically relevant number of crustaceans from the mortality event shall be analysed.
 - A statistically relevant number of crustaceans refers to a sample size that is large enough to yield meaningful and reliable results when conducting statistical analysis.
 - In other words, the sample size should be sufficient to accurately represent the population being studied and to draw valid conclusions from the data.
- The specific size needed for statistical relevance depends on various factors such as the variability within the population, the desired level of confidence and the statistical methods being used.
 - Typically, larger sample sizes provide more reliable estimates and stronger evidence for drawing conclusions.
- The designated veterinarian or aquatic animal health professional should assist the site in determining the appropriate sample size required for statistical analysis and must conduct all diagnosis.

Evidence of implementation should be maintained, e.g.:

- Records:
 - Mortality monitoring.
 - Mortality removal, including the results of post-mortem analysis and the rationale for determining the sample size required for statistical analysis.
 - Culling.
- The mortality monitoring process, that incorporates the minimum elements required, included in the HWMP.
- Employees should demonstrate awareness of the established mortality monitoring, removal process and culling method.

Auditors should confirm:

- The mortality monitoring process included in the HWMP incorporates the minimum required elements and is implemented.
 - Implementation can be verified by cross-checking the process with relevant records, information from employee interviews and observations.
- Moribunds are collected when spotted and culled following responsible stunning and killing according to Criterion 4.2.3.
 - Implementation can be confirmed by cross-checking the frequency of collection and culling methods with records, information from employee interviews and observations.
- Contingency plan for 'treatment' culling
- Monitoring and removal of mortality in conformance with the frequency required.

- Records, employee interviews and observations should demonstrate acceptable frequency.

Useful Resources

- McKay, H., & McAuliffe, W. (2024) [Pre-slaughter mortality of farmed shrimp](#). ReThink Priorities: Technical report. 24pp.

Indicator:	Requirement:
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4.2.1.8.5 *Indicator applicability: every UoC using feed*

The UoC shall ensure the HWMP includes a traffic light system for water quality (Indicator 4.2.1.8.1), morphological scoring (Indicator 4.2.1.8.2), behavioural scoring (Indicator 4.2.1.8.3), and mortality (Indicator 4.2.1.8.4). The system shall identify ranges of acceptable levels (green), warning levels (amber), and unacceptable levels (red) of health or welfare.

Where levels are identified as amber or red, the UoC shall increase monitoring and implement appropriate corrective actions to restore the levels back to green.

How do I interpret this requirement?

- The traffic light system should be tailored to accommodate the species being farmed, their life stage and the UoC's monitoring procedures.
- General guidelines on how to develop the traffic light system for each OWI are presented as follows:

Water Quality Indicators:

Step 1. Identify the parameters to be monitored according to Appendix 12 (Table 3).

Step 2. Establish species and life stage-specific thresholds for the parameters identified:

- Base these values on either published scientific literature, the recommendation of a veterinarian or aquatic animal health professional, or ASC reference table for water quality.
 - If the site is growing more than one life stage or species, then the creation of specific values for each stage and species will be expected.

For example, based on the FAI shrimp score card for shrimp, the traffic light system for three of the indicators could be:

	Acceptable	Warning levels	Unacceptable
Temperature	25.5-32.4°C	14.5-25.4°C or 32.5-35.4°C	≤14.4°C or ≥35.5°C
DO	≥65%	49-64%	≤48%
Transparency	35-50cm	21-34cm or 51-59cm	≤20cm or ≥60cm

Morphological Indicators:

Step 1. Develop a scoring system for each of the morphological parameters required by Indicator 4.2.1.8.2.

- It is recommended to score morphological parameters based on their severity, location and timing, when applicable.

Parameters	Severity				Locality				Timing		
	0	1	2	3	0	1	2	3	0	1	2
Eyes	None	Mild	Moderate	Severe	None	One eye	One eye	Both			
Antennas	None	Mild	Moderate	Missing	None	One	Both				
Rostrum	None	Mild	Moderate	Severe	None	Mild	Moderate	Severe			
Appendages	None	Mild	Moderate	Missing	None	Few	Many	Extreme			
Loss of appendages: Bacterial infection	None	Mild	Moderate	Severe	None	Few	Many	Extreme			
Loss of appendages: Cannibalism	None	Mild	Moderate	Severe	None	Few	Many	Extreme			
Hepatopancreas colouration and size	Normal	Mild	Moderate	Severe	None	Some	Diffuse				
Gill colouration	Normal	Mild	Moderate	Severe	None	Mild	Diffuse				
Size dispersion/variation	Normal	Mild	Moderate	Severe					None	End cycle	Mid cycle
Shell blisters and necrosis	Low	Mild	Moderate	Severe	None	Mild	Diffuse				
Empty digestive tube and colouration	Normal	Mild	Moderate	Severe					Before feed		After feed

Step 2. Once a scoring system has been developed, simulate different scenarios to develop a traffic light system. These scenarios should include at least:

- The best case, e.g. where shrimp have no morphologic issues.
- The second-best scenario, e.g. where shrimp have mild morphologic issues.
- The third-best scenario, e.g. where shrimp have moderate morphologic issues.
- The worst case, e.g. where shrimp have the highest score possible.

Each scenario will have a corresponding score:

Best case scenario

Parameters	Severity				Locality				Timing		
	0	1	2	3	0	1	2	3	0	1	2
Eyes	0										
Antennas	0										
Rostrum	0										
Appendages	0										
Loss of appendages:	0										
Bacterial infection	0										
Loss of appendages:	0										
Cannibalism	0										
Hepatopancreas colouration and size	0										
Gill colouration											
Size dispersion/ variation	0										
Shell blisters and necrosis	0										
Empty digestive tube and colouration	0										
Total	0										

Worst case scenario

Parameters	Severity				Locality				Timing		
	0	1	2	3	0	1	2	3	0	1	2
Eyes				3				3			
Antennas				3			2				
Rostrum				3				3			
Appendages				3				3			
Loss of appendages:				3				3			
Bacterial infection				3				3			
Loss of appendages:				3				3			
Cannibalism				3				3			
Hepatopancreas colouration and size				3			2				
Gill colouration				3			2				
Size dispersion/ variation				3							2
Shell blisters and necrosis				3			2				
Empty digestive tube and colouration				3							2
Total	60										

Third best case scenario

Parameters	Severity				Locality				Timing		
	0	1	2	3	0	1	2	3	0	1	2
Eyes			2				2	3			
Antennas			2			1					
Rostrum			2				2	3			
Appendages			2				2	3			
Loss of appendages: Bacterial infection			2				2	3			
Loss of appendages: Cannibalism			2				2	3			
Hepatopancreas colouration and size			2			1					
Gill colouration			2			1					
Size dispersion/ variation			2							1	
Shell blisters and necrosis			2			1					
Empty digestive tube and colouration			2							1	
Total	38										

Second best case scenario

Parameters	Severity				Locality				Timing		
	0	1	2	3	0	1	2	3	0	1	2
Eyes		1									
Antennas		1				1					
Rostrum		1			0						
Appendages		1				1					
Loss of appendages: Bacterial infection		1				1					
Loss of appendages: Cannibalism		1				1					
Hepatopancreas colouration and size		1			0						
Gill colouration		1			0						
Size dispersion/ variation		1							0		
Shell blisters and necrosis		1			0						
Empty digestive tube and colouration		1							0		
Total	16										

Step 3. Create a traffic light system, for example:

0 to 16	17 to 38	39 to 60
Acceptable	Threshold	Not acceptable
No action required	Action required	Action required

With time, and as more information is gathered, these thresholds might require re-adjustments.

Behavioural Indicators:

Step 1. Identify feeding behaviours that are relevant for the species of shrimp and the life stage being farmed (see example in Indicator 4.2.1.8.3 **Table 13**).

Step 2. Grade the identified feeding behaviours based on the scientific literature available or the opinion of a veterinarian or aquatic animal health professional. A few possible feeding behaviours to monitor are listed below.

- How to interpret these behaviours will vary depending on the species of shrimp, the life stage and the farming system.
- It is important to understand that, apart from the species and life stage variability, there will also be an individual component that will vary from one shrimp to the other.

	Score		
Feeding Behaviour	0	1	2
Feeding behaviour	Most shrimp react quickly by moving to the newly offered food	Only some of the shrimp react to the newly offered food	Most shrimp do not react to the newly offered food
Feed consumption efficiency	Tray is consistently emptied during feeding time	Partial feed remains (<20% uneaten)	Significant feed remains (>20% uneaten)
Interaction with feeding environment	Shrimp feed calmly without aggression or erratic movement	Occasional signs of stress or avoidance behaviour	Persistent signs of stress or aggression

Step 3. Simulate scenarios.

	Best case scenario	Second- best scenario	Third- best scenario	Worst case scenario
Score				
Feeding behaviour	0	1	0	2
Feed Consumption Efficiency	0	1	1	2
Interaction with Feeding Environment	0	1	1	2
Total	0	3	2	6

Step 4. Create a traffic light system, for example:

0-2	2-3	4-6
Acceptable	Threshold	Not acceptable

Mortality Indicators:

Step 1. Develop a mortality classification system that adjusts to the species, the life stage and the farming context (e.g. the geography or farming system where shrimp are reared). Staff must be trained in the mortality classification. Broadly, the following groupings can be made:

- Infectious:
 - Bacterial.
 - Viral.
 - Parasitic.
 - Fungal.
 - Amoebic.
- Non-infectious:
 - Husbandry related (e.g. acclimatisation, etc.)
 - Environmental (e.g. extreme temperatures, etc.)
 - Predation (e.g. birds, others).
 - Physical lesions.
 - Moribund.
 - Deformities.
 - Undiagnosed/unknown.
 - Decomposed.
 - Sampled.

Step 2. Based on the site's historic mortality trends and those of the industry, establish mortality thresholds with the support of a veterinarian or health manager.

- Mortality thresholds should be based on shrimp numbers, and it is recommended that they are also expressed as biomass thresholds.
 - The use of both metrics is considered complementary.
 - Using biomass on its own is discouraged, as small biomasses might lead to inaccurate assessments when shrimp are small.

Step 3: Create a traffic light system, for example for shrimp mortality based on the FAI Course:

≤10 % production mortality	11-25 % production mortality	≥26% production mortality
Acceptable	Warning levels	Unacceptable

Evidence of implementation should be maintained, e.g.:

- The traffic light system for the required OWIs that identify ranges of acceptable (green), warning (amber) and unacceptable (red) levels of health or welfare, included in the HWMP.
- Records that confirm the increased monitoring and associated short-term corrective measures in case of amber or red traffic light results.
- Rationale for the grading of the OWIs (e.g.: based on scientific literature available or the recommendation of a veterinarian or aquatic animal health professional).
- Employees should demonstrate awareness of the established traffic light system.

Auditors should confirm:

- The traffic light system included in the HWMP incorporates the minimum required elements and is implemented.
 - Implementation can be verified by cross-checking the system with relevant records, information from employee interviews and observations.

Useful Resources

N/A

Indicator:	Requirement:
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4.2.1.8.6 *Indicator applicability: every UoC using feed*

The UoC shall ensure the HWMP includes trend analysis to determine changes in health and welfare over time. This analysis shall consider drivers of the trends and shall be based on the data analysed in Indicator 4.2.1.8.5 and feedback from the processing plant where relevant.

How do I interpret this requirement?

- The analysis of OWIs data over time, as well as information gathered from post-mortem analysis and the feedback from the processing plant:
 - Allows for the identification of trends that provide insights into factors influencing health, welfare and mortality events.
 - Regular reviews of trends, such as monthly and at least annually or at the end of the production cycle, are recommended.
 - Inform decision-making to improve overall production.

- The veterinarian or aquatic animal health professional should assist the UoC with the trends analysis and drawing conclusions to build effective corrective measures for improving health or welfare over time.
 - Feedback from the processing plant can, for example, disclose the presence of recurring trends like shell or antenna damage.
 - Sharing observations and feedback with the UoC enables prompt and thorough investigations.

Evidence of implementation should be maintained, e.g.:

- The mechanism or trend analysis included in the HWMP.
- Records that confirm trend analysis has been conducted include as a minimum:
 - Post-mortem analysis.
 - Feedback from the processing plant.

Auditors should confirm:

- The trend analysis mechanism included in the HWMP incorporates the minimum required elements and its outcome is implemented:
 - Implementation can be verified by cross-checking the mechanism with relevant records, information from employee interviews and observations.

Useful Resources

N/A

Indicator:	Requirement:
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4.2.1.8.7 *Indicator applicability: every UoC using feed*

The UoC shall ensure the HWMP includes a mortality reduction plan which defines annual targets for both total and unexplained mortalities by considering the results of the trend analysis in Indicator 4.2.1.8.6 and shall implement appropriate actions to meet the annual targets.

How do I interpret this requirement?

- The mortality reduction programme should address the various factors contributing to crustacean mortality.

- When building the program, the UoC should consider:
 - Current mortality rates and causes, including unexplained mortality.
 - Set achievable goals and objectives.
 - Identify priority areas for intervention according to the goals and objectives.
 - Develop strategies addressing priority areas for intervention.
 - Monitor and evaluate progress regularly.
 - Provide employees with training.
 - Foster a culture of continuous improvement.
 - Collaborate with industry experts for knowledge sharing.
 - Seek health professionals' involvement in developing the reduction plan.
- Special attention should be given to unexplained mortalities which may indicate underlying health issues or poor management practices.
 - Identifying and addressing the root causes of unexplained mortalities is essential for improving overall management practices and reducing mortality rates.
- In addition, consistent, clear downward trends in OWIs indicate declining health or welfare.
 - Examples of such trends include a consistent increase in overall morphological scoring, repeated observations of abnormal behaviour, deteriorating water quality with one or more parameters that are outside of the acceptable ranges or increasing mortality.
- If a declining trend is observed, then the UoC should assess the situation and investigate the causes behind those trends.
 - One or multiple issues may be causing the decline, including those of infectious or non-infectious origin.
 - In many cases, poor or inadequate husbandry practices can predispose animals to succumb to infectious diseases.
 - Also, when declining trends are observed, the UoC is expected to review at least the following husbandry practices if appropriate:
 - Stocking densities.
 - The feeding plan.
 - Water quality.
 - Handling practices.
 - Enclosure characteristics.
 - Environmental enrichment.
 - The rearing environment, for example:
 - Additional aeration.

Evidence of implementation should be maintained, e.g.:

- Mortality reduction programme.
- Long-term animal health and welfare improvement measures as well as short-term mitigation measures to react to situations of declining health or welfare, included in the HWMP.
- Mortality records that allow for trend analysis for the development of a mortality reduction programme.
- Records of long-term animal health and welfare improvement measures and short-term mitigation measures to react to situations of declining health or welfare.

Auditors should confirm:

- The mortality reduction programme and the improvement and mitigation measures included in the HWMP incorporate the minimum required elements and that its outcome is implemented:
 - Implementation can be verified by cross-checking the programme and measures with relevant records, information from employee interviews and observations.

Useful Resources

N/A

Indicator:	Requirement:
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4.2.1.9 *Indicator applicability: every UoC using feed*

The UoC shall report the following to a registered veterinarian or aquatic animal health professional:

- a) all mortality events; and
- b) crustacean health and welfare concerns identified during mortality classification.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.2.1.10 The UoC shall report to the relevant authorities if a WOAH-notifiable disease is confirmed at a site, and shall:

- a) apply the measures according to applicable legal requirements; and
- b) increase disease-monitoring in other batches of crustaceans; and coordinate oversight by a registered veterinarian or aquatic animal health professional.

How do I interpret this requirement?

- Upon a WOAH-notifiable disease being confirmed, the UoC should consider within its response mechanism included in the HWMP (see Indicator 4.2.1.8) an emergency response procedure for culling the affected batches of crustaceans (see also Indicator 4.2.1.8.4).

Evidence of implementation should be maintained, e.g.:

- HWMP includes site-specific disease monitoring, responses and reporting requirements.
 - Specific considerations should be given to WOAH-notifiable diseases, including an emergency response procedure for culling the affected batches of crustaceans.
- National regulations.
- Records:
 - Reporting to the authorities.
 - Measures applied.
 - Notification to other sites in the vicinity or the ABM.
 - Disease monitoring.
 - Veterinarian or aquatic animal health professional coordination.

Auditors should confirm:

- The HWMP includes considerations for WOAH-notifiable diseases (see interpretation for Indicator 4.2.1.8.4).
- In case a WOAH-notifiable disease has been confirmed:
 - Reporting to authorities and implementation of regulatory measures.
 - Increased monitoring of other batches of crustaceans.
 - Veterinarian or aquatic animal health professional coordination

Useful Resources

- Asia Farming (2022) [Common shrimp/prawn diseases, symptoms, treatment](#). Article.

- o World Organisation for Animal Health (2024) [Aquatic Animal Health Code. Chapter 1.3. Diseases listed by WOA.](#)
- o World Organisation for Animal Health (2024) [Aquatic Animal Health Code. Chapter 1.1. Notification of diseases, and provision of epidemiological information.](#)

Indicator:	Requirement:
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4.2.1.11 The UoC shall calculate stocking densities, following the method outlined in **Appendix 12 (12.5.4)**.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.2.1.12 The UoC shall report to ASC survival rate, according to ASC data submission procedures, following the method outlined in **Appendix 12 (12.5.1)**.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.2.1.13 *Indicator applicability: Penaeus vannamei and Penaeus monodon*

The UoC shall demonstrate that received nauplii, larvae or post-larvae (PL) originates from ablation free (AF) female broodstock according to the transition periods outlined in **Appendix 12 (12.4, Table 4)**. This includes an agreement with the hatchery that allows ASC or its designated agents to conduct oversight activities to verify the AF sources within the facility.

How do I interpret this requirement?

- o Scope applicable to *Penaeus vannamei* and *Penaeus monodon* species.

- Transition periods span from May 2025 to May 2031, with staged percentages for AF broodstock compliance (see Appendix 12 table 4).
- The objective is to ensure a gradual shift toward using only ablation-free female broodstock, aligning with improved welfare practices.

Evidence of implementation should be maintained, e.g.:

- The monitoring process involves developing and implementing a tracking system to monitor the origin of stock (nauplii, larvae or PL).
- Where feasible, conduct physical audits or site visits of hatcheries to verify by visual inspection compliance with AF broodstock sourcing.
- Hatcheries supplying the stock should provide clear documentation of the broodstock's status (AF or non-AF).
 - Maintain records of sourcing, including certificates or declarations from hatcheries confirming broodstock compliance with AF requirements.
 - Maintain detailed production logs showing the percentage of stock derived from AF broodstock at each transition phase.
- Employees involved in sourcing should demonstrate understanding to comply with the requirement to ensure they can explain the sourcing process to verify compliance during audits.

Auditors should confirm:

- The monitoring process to confirm that the UoC has a documented system to track and verify the origin of nauplii, larvae, or PL.
- Implementation can be verified by cross-checking the process with relevant records, information from employee interviews and observations.

Useful Resources

- Zacarias, S [PowerPoint Presentation](#)

4.2.2 - Crustaceans - Handling

Indicator:	Requirement:
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- 4.2.2.1** The UoC shall develop, implement and maintain a site-specific Handling Management Plan (HMP), with the intent of ensuring good health and welfare of farmed crustaceans.
- The HMP shall be periodically reviewed and revised as necessary.

- 4.2.2.2** The UoC shall ensure that the HMP includes all handling activities carried out on site.

How do I interpret this requirement?

- The Handling Management Plan (HMP):
 - Aims to support the development and implementation of preventive, rather than remedial, actions that prevent poor crustacean health and welfare at handling.
 - Examples of handling:
 - Crowding.
 - Sampling.
 - Transportation (including loading, transfer and unloading).
 - Should consider a description of:
 - Processes.
 - A separate process description may be required for outsourced events e.g. live transportation of PL to the site.
 - Equipment.
 - A description of equipment, and verification if applicable may be required for both UoC and outsourced events e.g. harvest equipment.

Evidence of implementation should be maintained, e.g.:

- HMP includes standardised handling operations, e.g.:
 - Site-specific description of processes and equipment.
 - Documents:
 - Planning of handling events, e.g.:
 - Emails, reports and site 'whiteboard' communications.
 - Handling procedures.
- Monitoring the effectiveness of the HMP.

- Documents:
 - Handling logs, e.g.:
 - Reports detailing regular observation of crustacean behaviour, environmental parameters and crustacean conditions during and post-handling procedures, including injuries, mortalities or deviations from procedures.
 - Handling incidents reports, e.g.:
 - Emails, reports detailing any incidents during handling procedures, including injuries, mortalities or deviations from procedures.
- Feedback from staff, e.g.:
 - Emails, anonymous box asking feedback from staff involved in handling procedures regarding their observations and any difficulties encountered.
 - Encourage open communication.

Auditors should confirm:

- The HMP includes a description of the handling processes and equipment and considerations for planning handling events.
 - Employees are familiarised with the HMP by conducting employee interviews and observations.
 - Handling logs are available documenting handling processes, environment and crustacean conditions.
 - Existence of handling incident reports, including feedback from staff and discussions regarding handling procedure changes.
 - Open communication is encouraged via formal documents and employee interviews and observations.
-

Useful Resources

- Amatul-Samahah, Md.A., Omar, W.H.H.W., Ikhsan, N.F.M., et al (2020) [Vaccination trials against vibriosis in shrimp: A review](#). Aquaculture Reports.

Indicator:	Requirement:
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4.2.2.3 The UoC shall ensure that the HMP includes contingency plans for handling activities, including at least the following:

- a) immediate response measures in the case of system failure; and
- b) culling response measures (see Indicator 4.2.3.2); and
- c) mass mortality event response measures.

Exception applicability: every UoC using feed

An exception to (b) applies in the extreme circumstances of culling whole ponds for disease control, in which case a solution of chlorine is allowed in a concentration that ensures that death occurs rapidly.

How do I interpret this requirement?

- The contingency plans should consider:
 - Identification of potential failures.
 - Effective corrective measures, e.g.:
 - Backup power generators.
 - Oxygen support.

Evidence of implementation should be maintained, e.g.:

- HMP includes processes' contingency plans.
- Records:
 - Backup system testing.
- Additional oxygen source available.

Auditors should confirm:

- The HMP includes considerations on crustacean health and welfare before handling events, including considerations for the existence of a backup system.
- Presence of a backup system.
- Employees are:
 - Familiarised with the contingency plans by conducting employee interviews and observations.
- Records:

- Backup system testing.

Useful Resources

N/A

Indicator:	Requirement:
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4.2.2.4 The UoC shall ensure that the HMP includes consideration of the environmental conditions associated with handling activities, with the intent of preventing handling under adverse conditions that could affect the health and welfare of the crustaceans.

How do I interpret this requirement?

- Examples of considerations of external conditions:
 - Avoid handling in the event of storms.
 - Avoid handling crustacean without using a shade net on a sunny day.
 - Avoid transferring post-larvae during periods where there are less than favourable conditions:
 - Rainfall can quickly alter pond conditions and trigger a chain of events resulting in sub-optimal conditions for crustaceans.
 - Potential for *Vibrio spp.* to flourish, increasing the risks of bacterial infection to the shrimp.
 - Trigger the production of hydrogen sulphide. This is a toxic gas produced by sulphate reducing bacteria and is highly toxic to shrimp.

Evidence of implementation should be maintained, e.g.:

- HMP includes consideration for external conditions necessary to perform handling.
- Records:
 - Weather forecast verification.
- Employees should demonstrate knowledge of the considerations and the verification of external conditions before handling.

Auditors should confirm:

- HMP includes considerations for external conditions for handling and are implemented.

- Implementation can be confirmed by cross-checking the HMP with records, information from employee interviews and observations.

Useful Resources

N/A

Indicator:	Requirement:
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4.2.2.5 *Indicator applicability: every UoC using feed*

The UoC shall ensure that the HMP includes a pond preparation process including at least the following:

- drying of the pond. An exception applies where areas cannot be dried, in which case they must be disinfected; and
- removal of any dead or live animals from the pond; and
- removal of excess organic material; and
- adjustment of acidic pH; and
- preparation of the surface of earthen ponds to facilitate oxygenation and UV exposure.

How do I interpret this requirement?

- Draining the pond completely helps in eliminating unwanted organisms and disease.
- Areas that cannot be completely drained should be treated with disinfectants to kill any pathogens, e.g.:
 - If a few wet patches are observed after draining the pond, spray these patches with a bleach solution for disinfection, ensuring no harmful bacteria or parasites survive.
- Removing any dead or living animals helps in preventing disease spread and decay, which can affect the water quality and the health of future pond crops, e.g.:
 - If a few dead crustaceans and some frogs are found, remove all these animals and dispose of them properly, ensuring the pond is free of any potential contaminants.

- Remove any organic debris such as leaves, branches and sludge. If the pond water is too acidic, agricultural lime (calcium carbonate) may be added to neutralise the pH, making the environment more suitable for crustacean e.g.:
 - After removing leaves and decaying plant matter from the pond bed, the pH level was 5.5. After applying agricultural lime, the pH level rose to approximately 7, a neutral level.
- Tilling the surface of earthen ponds (those with natural soil bottoms) helps to aerate the soil, promoting better oxygenation. It also exposes the soil to sunlight, which helps in killing pathogens and breaking down organic matter through UV radiation.

Evidence of implementation should be maintained, e.g.:

- Records confirming the monitoring of the required parameters for pond preparation and their conformance with acceptable quality levels included in the HMP prior to stocking.
- Record of drying period.
- Record of addition of lime if applied.
- The water quality monitoring process, that incorporates the minimum elements required for pond preparation.
- Employees should demonstrate awareness of the established water quality monitoring process during handling.

Auditors should confirm:

- Pond preparation process, that incorporates the minimum elements required, included in the indicator and cross-checking with the handling operations present in the HMP.
- Implementation can be verified by cross-checking the process with records, information from employee interviews and observations.

Useful Resources

- The Fish Site (2016) [Management of intensive Vannamei shrimp ponds](#). Article.

Indicator:	Requirement:
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- 4.2.2.6** The UoC shall ensure that the HMP includes acclimatisation measures to prevent sudden changes in water conditions affecting the health and welfare status of the crustaceans.

How do I interpret this requirement?

- Acclimatisation measures should be designed and maintained, e.g.:
 - Water temperature during transportation must be as close as possible to that which the crustaceans came from.
 - Crustaceans shall be released into a small area of the pond 'hapa' or net enclosures in the grow-out pond itself for a risk-assessed period to evaluate the capacity of the PL to adjust to the pond conditions.

Evidence of implementation should be maintained, e.g.:

- acclimatisation measures and monitoring of the water quality according to HMP.
- Records that confirm the monitoring of acclimatisation parameters before stocking.
- Employees should demonstrate awareness of the acclimatisation measures and monitoring process above.

Auditors should confirm:

- The acclimatisation measures included in the HMP incorporate the minimum required elements and are implemented.
- Implementation can be verified by cross-checking the process with relevant records, information from employee interviews and observations.

Useful Resources

N/A

Indicator:	Requirement:
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4.2.2.7 *Indicator applicability: every UoC using feed*

The UoC shall ensure that the HMP includes a visual inspection process to identify abnormal behaviour in relation to the handling activity and defines mitigation measures when identified.

How do I interpret this requirement?

- Visual inspection after handling should be included in the HMP.
 - Crustacean morphology and behaviour can be assessed when verifying the crustacean feeding trays as a minimum requirement.

Evidence of implementation should be maintained, e.g.:

- The monitoring process, that incorporates the corrective actions required, included in the HMP.
- Records that confirm the monitoring of the morphology and behaviour at the feeding trays. A checked box on the monitoring records indicating whether monitoring morphology and behaviour took place or not is accepted.
- Employees should demonstrate awareness of the established behavioural monitoring process.

Auditors should confirm:

- The monitoring process included in the HMP that incorporates corrective actions and is implemented.
- Implementation can be verified by cross-checking the process with relevant records, information from employee interviews and observations.

Useful Resources

N/A

Indicator:	Requirement:
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- 4.2.2.8** The UoC shall ensure that the HMP considers biosecurity risks specific to the types of handling activities and shall include appropriate biosecurity measures to prevent the transfer of disease.

How do I interpret this requirement?

- For developing a site-specific biosecurity mechanism within the HMP, the UoC should consider:
 - Indicator 4.2.1.8.d)
 - Biosecurity measures related to the type of handling, e.g.:
 - Equipment segregation.
 - Equipment disinfection.

Evidence of implementation should be maintained, e.g.:

- For developing a site-specific biosecurity during handling the HMP includes the considerations for:
 - Site-specific biosecurity mechanism.
 - Biosecurity specifications related to the type of handling operations.

Auditors should confirm:

- The HMP includes the required minimum considerations above.
- Employees are:
 - Familiarised with the biosecurity considerations in the HMP by conducting employee interviews and observations.
 - Trained on biosecurity specifications related to the type of handling operations by reviewing employees' training records, conducting employees' interviews and observations.

Useful Resources

- Lightner, D.V. (2007), [Biosecurity in Shrimp Farming: Pathogen Exclusion through Use of SPF Stock and Routine Surveillance](#). Journal of the World Aquaculture Society.

Indicator:	Requirement:
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4.2.2.9 The UoC shall ensure that the HMP considers predation risks specific to the types of handling activities and shall include appropriate predator control measures.

How do I interpret this requirement?

- For developing a site-specific predator control within the HMP, the UoC should consider:
 - The site-specific characteristics required to be assessed by Indicator 4.2.1.8.e) predator control measures specific to the type of handling, e.g.:
 - Birds scaring devices.
 - Use of appropriate nets sizes while moving crustaceans from one unit/pond to another.

Evidence of implementation should be maintained, e.g.:

- Predator control measures specific to the type of handling operations.
- Verification of mortalities associated with predators during handling operations.

Auditors should confirm:

- The HMP includes the required considerations above.
- Employees are:
 - Familiarised with the predator control considerations in the HMP by conducting employee interviews and observations.
 - Trained on predator control verifications, e.g.:
 - Verification of mortality events specific to the type of handling operations by reviewing employees' training records, conducting employees' interviews and observations.

Useful Resources

N/A

Indicator:	Requirement:
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4.2.2.10 The UoC shall ensure that the HMP includes consideration of the use of appropriate equipment during handling activities to avoid affecting the health and welfare status of the crustaceans

No additional guidance is provided for this Indicator.

4.2.3 – Crustaceans - Slaughter and Killing

Indicator:	Requirement:
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- 4.2.3.1** The UoC shall appoint a staff member responsible for overseeing the implementation of crustacean health and welfare requirements during slaughter.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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- 4.2.3.2** The UoC shall ensure crustaceans are killed immediately after harvest:
- a) by immersion in an ice slurry bath; or
 - b) through an electrical device followed by immersion in an ice slurry bath.

-
- 4.2.3.3** The UoC shall not process crustaceans alive.

-
- 4.2.3.4** The UoC shall ensure ice slurry baths are maintained at $\leq 4^{\circ}\text{C}$ at all times and crustaceans are immersed for an appropriate amount of time to be killed effectively.

-
- 4.2.3.5** The UoC shall ensure that if electrical devices are used for stunning the manufacturer's instructions are followed.

-
- 4.2.3.8** The UoC shall ensure that killing has been effective by verifying the absence of coordinated leg and body movements.

-
- 4.2.3.9** The UoC shall have measures in place to immediately respond to ineffective stunning or killing, including the presence of a back-up system, and shall implement appropriate corrective actions to prevent recurrence.

How do I interpret this requirement?

- Slaughter is generally a two-stage process:
 - The crustacean is first stunned to make it insensible to pain.

- Killing is then carried out (see interpretation in Indicator 4.2.3.10).
- An effective stun is determined by the loss of consciousness using an electroencephalogram (EEG) measured in a laboratory, which is not practicable at a site level.
- Current stunning methods for crustaceans include a controlled ice slurry bath at $\leq 4^{\circ}\text{C}$ or electric stunning followed by an ice bath.
 - To ensure a controlled ice slurry bath at $\leq 4^{\circ}\text{C}$, the ice slurry bath should be adjusted in relation to the amount and crustacean size. Below is an example for a batch of crustaceans using an ice slurry bath maintained at $\leq 4^{\circ}\text{C}$.
 - Preparation: determine the approximate amount and size of crustacean by considering the amount planned for slaughter and taking a sample of the pond.
 - Results of amount and size: e.g. 5kg of medium-sized crustaceans (approximately 15cm each).
 - Ice slurry preparation: create an ice slurry bath in a large, insulated container. For 5kg of crustaceans, prepare an ice slurry mixture with approximately 10 Liters of water and a sufficient amount of ice to maintain the temperature at $\leq 4^{\circ}\text{C}$.
 - Add ice gradually to the water, continuously stirring to create an even slurry.
 - Monitor the temperature using a thermometer, ensuring it reaches and stays at $\leq 4^{\circ}\text{C}$.
 - Adjust according to crustacean size and amount: since the crustaceans are medium-sized and weigh a total of 5kg, ensure that the ice slurry bath can accommodate the entire batch without overcrowding. Overcrowding can reduce the effectiveness of the ice slurry by raising the temperature.
 - Immersion: gently place the 5kg of crustaceans into the ice slurry bath, ensuring they are fully submerged.
 - Stir the ice-slurry occasionally to maintain uniform temperature distribution.
 - Define the timing for effective Killing: for medium-sized shrimp, a typical immersion time in an ice slurry bath at $\leq 4^{\circ}\text{C}$ is around 2-7 minutes.

Table 14: Suggested time In Ice bath for killing of *Penaeus vannamei* shrimp

Size of shrimp	Time in ice bath
<40 pcs/kg	2 minutes
40- 60 pcs/kg	5 minutes
60-80 pcs/kg	7 minutes

- Immersion time: set a timer to ensure all shrimp are effectively killed using the timings worked out.
- Monitoring and verification: periodically check the temperature of the slurry bath to ensure it remains at $\leq 4^{\circ}\text{C}$. Add more ice if necessary to maintain the temperature. Check for indicators of death according to Indicator 4.4.5.
- In extreme hot conditions, and the impossibility to stabilise the temperature of the slurry bath, a range of temperatures may be accepted.
- After the set time, remove a few shrimp and check for the absence of movements; the shrimp should also exhibit body rigidity, indicative of effective killing.

Table 15: Determination of the effectiveness of stunning and / or killing based on behaviour

Shrimp behaviour	Effective stun and / or kill	Ineffective stun and / or kill
Movement	Little movement of the shrimp; lying belly up or on side with no attempt to right themselves and under the water surface Muscle twitches if isolated	Showing active movement, at the surface, continuously jumping reflexes or leg movements or recurring
Body shape	Curled	Straight lying belly down, attempting to right themselves

- The use of body movement as an indicator for the effectiveness of stunning or killing can be misleading as muscular spasms regularly occur in unconscious or dead shrimp. Studies on crustaceans, including shrimp, indicate that they can exhibit reflex actions and movements even when they are unconscious or brain dead. This is because such movements are often dominated by nerve circuits in the peripheral nervous system rather than the brain. Body

movements indicating walking leg movements and coordinated swimming movements are relevant movements to watch out for, and that may indicate consciousness

- Killing effectiveness in the UoC should be assessed by confirming signs of unconsciousness:
 - Immediate loss of balance.
 - Absence of voluntary movements.
 - Absence of coordinated swimming movements.
 - Limited pleopods and pereopod movements.
 - Laying on side with no activity.
- The requirement for killing also applies to animals not destined for human consumption, e.g. bycatch, wild species.
- If there is uncertainty about whether a crustacean is unconscious:
 - When using an electrical stunning, the stunning should be repeated or the harvest shall stop to verify stun parameters before proceeding, e.g. more than 5 crustaceans in 10 seconds requiring backup stun.
 - Or use an alternative backup method, e.g. ice slurry bath or a secondary electric stunner.
 - When using the ice slurry bath method, the temperature of the bath should be checked, and the length of time in the bath extended.
- Electric stunning should be performed according to the manufacturer's instructions.

Evidence of implementation should be maintained, e.g.:

- Employees should demonstrate knowledge of the approved slaughter methods and monitoring.
- Records:
 - Employee training (see Appendix 12).
 - Slaughter monitoring, e.g.:
 - Verification and registration of stunning and killing of 100 individual crustaceans per harvest.
 - Observation of signs of unconsciousness.
 - Records of the use of backup.
 - Records of stunning and killing of crustaceans not destined for human consumption if applicable.

Auditors should confirm:

- Monitoring of stunning and slaughter in conformance to the requirements.
 - Records, employee interviews and observations of stunning and slaughter operations during the audit should demonstrate the implementation of the requirements, and that there is an available backup method or knowledge of corrective measures to be carried out.

Useful Resources

- Weineck, K., Ray, A.J., Fleckenstein, L.J., Medley, M., Dzubuk, N., Piana, E., & Cooper, R.L. (2018) [Physiological Changes as a Measure of Crustacean Welfare under Different Standardized Stunning Techniques: Cooling and Electroshock](#) Animals.
- CIWF (2020) [Tesco and Hilton Seafood – Improving the welfare of whiteleg shrimp \(Penaeus vannamei\) at harvest](#). CIWF Report 7pp.
- Conte, F.; Voslarova, E.; Vecerek, V.; Elwood, R.W.; Coluccio, P.; Pugliese, M.; Passantino, A. (2021) [Humane Slaughter of Edible Decapod Crustaceans](#). Animals.
- Barrento, S., Pedrazzani, A., Ostrensky, A., & Quintiliano, M. (2023). *Shrimp Welfare Indicators* [Online course]. In S. Barrento, M. Quintiliano, & J. Muirhead (Eds.), FAI Academy. <https://fai.academy/>

Indicator:	Requirement:
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4.2.3.6 The UoC shall not use any of the following methods to kill crustaceans:

- asphyxiation in air; or
- carbon dioxide (CO₂) bath; or
- salt baths; or
- ammonia baths; or
- evisceration.

How do I interpret this requirement?

- The only killing method allowed for crustaceans is a controlled ice slurry:
 - This involves immersing crustaceans in a slurry of ice and water, causing rapid cooling.
 - To clarify, electrical stunning is a stunning method and not a killing method. Stunning methods are designed to render animals insensible to pain before they are killed, whereas

killing methods result in the animal's death. Stunning is a preliminary step followed by a killing method to ensure that the animal does not regain consciousness.

Evidence of implementation should be maintained, e.g.:

- Employees should demonstrate knowledge confirming the approved killing methods.

Auditors should confirm:

- The use of approved methods for killing and awareness of non-approved methods.

Useful Resources

- Neil, D. M., Putyora E. & Albalat, A. (2024) [Towards the humane slaughter of decapod crustaceans: identifying the most effective indicators of insensibility following electrical stunning](#). Frontiers in Animal Science

Indicator:	Requirement:
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4.2.3.7 The UoC shall ensure that an overdose of anaesthetic is only used to kill crustaceans not destined for human consumption.

How do I interpret this requirement?

- The use of anaesthesia should be considered as a humane method for crustaceans not destined to human consumption.

Evidence of implementation should be maintained, e.g.:

- Consideration for the use of anaesthesia for killing.
- Culling procedure.
- Records:
 - Anaesthetic usage amount and number of culled crustaceans.

Auditors should confirm:

- The UoCs consideration of the use of anaesthetics for killing.

- Implementation can be confirmed by cross-checking the culling procedure with records, information from employee interviews and observations.

Useful Resources

N/A

Indicator:	Requirement:
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4.2.3.10 The UoC shall maintain records of the activities carried out under indicators 4.2.3.4 and 4.2.3.8.

No additional guidance is provided for this Indicator.

Criterion 4.3 - Bivalves and Abalone

Indicator:	Requirement:
4.3.1	The UoC shall ensure that all relevant employees are trained on general animal health and welfare awareness, including biosecurity, according to Appendix 12 (12.2, Table 1) .
4.3.2	The UoC shall ensure that any person involved in animal husbandry and handling activities is trained on animal health and welfare, according to Appendix 12 (12.2, Table 1) .
4.3.3	The UoC shall ensure that the content of the training in Indicators 4.3.1 and 4.3.2 is updated as necessary and is approved by a registered veterinarian or aquatic animal health professional.

How do I interpret this requirement?

- For each employee:
 - The level of training should be appropriate to their role:
 - For all relevant employees: a basic level of awareness of biosecurity, animal health and welfare is required.
 - For any person involved in husbandry, handling and harvest operations: a more advanced level of training on animal health and welfare, handling and harvest practices is required on an annual basis.
 - Evidence of training participation must be maintained:
 - Certificates of attendance.
 - Certificates of competency.
- The trainer should be competent in the topic of animal health and welfare and the competence should be demonstrable, e.g.:
 - Curriculum Vitae.
 - Records of qualifications.
 - Work experience.
- Best practices on the implementation of the training content should be visible during the audit.
 - Employees should demonstrate adequate knowledge of that content at interviews.

Evidence of implementation should be maintained, e.g.:

- A list of employees and subcontracted personnel involved in animal husbandry and handling operations.
- Records:
 - Training.
 - Trainer's competence.
 - Certificates of competency.

Auditors should confirm:

- Employees received the required level of training for their roles by cross-checking employees and subcontracted personnel lists with:
 - Training records.
 - Certificates of competency.
 - Trainer's competence.
 - Revision of training resources/contents.
- Adequate level of personnel knowledge of the training content should be verified by:
 - Interviews.
 - Observation of operations during handling and harvest of animals.

Useful Resources

- World Organisation for Animal Health (2024) – [Competency and Curriculum Guidelines for Community Animal Health Workers](https://doi.org/10.20506/woah.3535). Paris, 39 pp. (<https://doi.org/10.20506/woah.3535>).

Indicator:	Requirement:
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4.3.4 The UoC shall only stock batches of animals that have been tested prior to stocking and are free from diseases of regional concern

How do I interpret this requirement?

- Testing should be relevant to:
 - The species.
 - The production system.
 - The scale of production.
 - The local regulations.

- For example:
 - Infection with *Perkinsus marinus*.
 - Infection with *Bonamia exitiosa*.
 - Infection with Abalone herpesvirus.
- Testing might consist of:
 - Clinical examination of the animals.
 - Laboratory tests:
 - Bacteriology.
 - Molecular biology (e.g.: PCR).
 - Histopathology.
 - Virology (e.g.: cell culture).
- The designated veterinarian or aquatic animal health professional should determine:
 - The diseases to include in the testing, based on local regulations, scientific literature and/or publicly available information (see interpretation for Indicator 4.3.6).
 - The sample size, based on local regulations and/or technical judgement.
- Compulsory governmental testing programme results are acceptable for demonstrating conformance with this requirement.

Evidence of implementation should be maintained, e.g.:

- List of relevant diseases for the species in the country (see interpretation for Indicator 4.3.6).
- Testing results written report or document that includes, as a minimum:
 - The batch of animals being tested.
 - The date of the test.
 - The pathogen tested.
 - The results of the test.
- Records:
 - Purchase.
 - Transportation.
 - Delivery.
 - Stocking.

Auditors should confirm:

- Every batch of animals has been tested by cross-checking the list of relevant diseases for the species in the country with the testing result's written report or document and records.

Useful Resources

- Victoria Agriculture (2023) [Abalone disease; Biosecurity](#). Webpage.
- Victoria Fisheries Authority (2024) [Abalone diseases](#). Webpage.

- Alevra, A., Llorente Manzanares, M., Mente, E., Xandri Royo, P.M. (2024) [Good husbandry practices for aquaculture](#). Publications Office of the European Union. Report 89pp.

Indicator:	Requirement:
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4.3.5 The UoC shall ensure that a registered veterinarian or aquatic animal health professional periodically visits the site to perform routine checks. Checks shall be carried out at least annually, as well as reactively in cases of animal health or welfare concerns.

How do I interpret this requirement?

- Regular veterinarian or aquatic animal health professional visits may fall into two categories:
 - Routine: Performed on a regular basis to monitor the health and welfare of the farmed animal.
 - Reactive: Performed in response to deterioration or concerns of the health or welfare status of the farmed animals.
 - Sites should be prepared to provide the veterinarian or aquatic animal health professional with detailed information about the observed symptoms, recent changes in behaviour, morphology, water quality or mortality and any relevant history of disease or treatments.
 - The site should use reactive health visits to address immediate health or welfare concerns, diagnose underlying causes of problems, and implement targeted interventions to mitigate risks and prevent further spread of disease.
- For routine visits, preparation is recommended to ensure a meaningful outcome from the veterinarian or aquatic animal health professional. Below are some practical considerations:
 - Visit scheduling: Ensure key employees are available during the visit to provide relevant required information and to facilitate adequate follow-up.
 - Designate a senior employee to serve as the primary point of contact for the veterinarian or aquatic animal health professional during the visit.
 - This individual can help coordinate the activity, ensure relevant information is available, assist during the visit and ensure follow-up.
 - Gather and organise relevant information:
 - Compile relevant information about the operations, including:
 - Species being cultured,
 - Production systems,
 - Stocking densities,
 - Recent health history, water quality monitoring results,

- Morphology and mortalities data,
- Feeding plan,
- Any previous veterinary reports,
- Details of any ongoing issue.
- Having this information readily accessible will facilitate discussion and decision-making during the visit.
- Prepare the UoC for the visit: Ensure the UoC is prepared for the visit and for providing full access to all relevant areas to the veterinarian or aquatic animal health professional.
 - Address any maintenance issues, clean structures as needed and ensure clear access to all relevant areas of the facility.
 - Ensure handling and other equipment that might be needed during the visit, such as nets, containers and sampling equipment, is available and in good working condition.
 - Make sure employees are aware of the visit and its objectives and encourage them to proactively participate in it.
- During the visit: Ensure relevant information is discussed with the veterinarian or aquatic animal health professional and that the visit includes all relevant areas within the UoC.
- After the visit: Follow up on any recommendations or actions discussed during the visit.
 - Implement any necessary revisions or changes to management practices, treatment protocols or facility conditions based on the veterinarian or aquatic animal health professional's feedback and any sample or testing results.
- Continuous improvement: Use the insights gained from the veterinarian or aquatic animal health professional visit to identify opportunities for continuous improvement in health and welfare management.

Evidence of implementation should be maintained, e.g.:

- Health reports.
- Testing results.
- Records:
 - Veterinarian or aquatic animal health professional qualifications.
 - Veterinarian or aquatic animal health professional visit.
 - Recommendations or actions discussed during the visit.

Auditors should confirm:

- The qualifications of the veterinarian or aquatic animal health professional and cross-check the frequency of visits with records.
- Existence of health reports and testing results.

Useful Resources

N/A

Indicator:	Requirement:
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4.3.6

The UoC shall develop, implement and maintain a site-specific Health and Welfare Management Plan (HWMP), approved by a registered veterinarian, with the intent to prevent disease outbreaks and ensure good health and welfare of farmed animals.

The HWMP shall include all of the following:

- a) disease monitoring; and
- b) response, including quarantine, treatment, and humane culling; and
- c) reporting mechanisms, including reporting WOA-notifyable disease to authorities; and
- d) biosecurity risks; and
- e) predation risks.

The HWMP shall be reviewed and revised at least annually

How do I interpret this requirement?

- The Health and Welfare Management Plan (HWMP):
 - Aims to support the development and implementation of preventive, rather than remedial, actions that ensure good animal health and welfare.
 - Must be developed by the UoC in collaboration with a registered veterinarian who approves it.
- For developing and establishing a site-specific disease monitoring, response and reporting mechanism within the plan, the UoC should consider:
 - Developing standardised health and welfare monitoring protocols.
 - This may involve conducting visual inspections, monitoring OWIs and performing diagnostic tests for pathogens.
 - Producing and maintaining detailed records.
 - Including records of health observations, diagnostic test results, treatments administered, mortalities and other relevant data.
 - It is recommended to use a centralised database or electronic system for efficient data management and retrieval.

- Implementing an early detection system.
 - To detect signs of disease outbreaks or abnormal health conditions promptly.
 - This could include environmental monitoring tools.
 - Establishing response protocols for responding to disease outbreaks or health emergencies.
 - Outlining procedures for:
 - Isolating affected individuals or populations.
 - Implementing strategies, for example, for treatments or others (e.g. culling affected batches of animals)
 - Notifying relevant stakeholders, including other sites in the vicinity and the authorities.
 - Establishing communication channels.
 - Effective communication channels should be implemented for reporting health-related issues both internally within the facility and externally to relevant authorities (e.g. WOAHP-notifiable diseases) or stakeholders.
 - Employees should understand their roles and responsibilities in reporting and responding to disease events.
 - Maintaining a readily accessible notification list.
 - Including contact names, numbers and emails for key employees and other relevant stakeholders.
 - The list should be kept up-to-date to ensure quick and efficient communication during emergencies.
- For developing a site-specific biosecurity mechanism within the plan, the UoC should consider:
- Determining the objectives and scope of the biosecurity mechanism.
 - Identifying relevant regulatory requirements.
 - Gathering information about the site, including an up-to-date layout of production units identifying the possible movements of animals, animal products, animal waste, water and fomites/vectors.
 - Identifying potential transmission pathways into, within and from the site.
 - Carrying out a risk analysis to identify and evaluate disease threats and ensure that the plan addresses them.
 - A risk analysis can be carried out either on their whole or only partially, e.g. disease transmission during transportation.
 - This will depend on the objectives of the biosecurity mechanism.
 - Identifying mitigation measures and assessing their effectiveness and their management requirements.

- Establishing emergency procedures in the event of a biosecurity failure.
 - Defining internal and external communication procedures, roles and responsibilities.
 - Creating a monitoring and internal audit schedule.
 - Carrying out performance evaluations.
 - Training employees in the different elements of the biosecurity mechanism.
 - Creating the necessary SOPs to support the implementation of the mitigation measures and the biosecurity mechanism.
 - Maintaining a record-keeping system, that provides evidence of compliance with the biosecurity mechanism.
- For developing predator control measures within the plan, the UoC should consider:
- Identifying potential predators.
 - Create a list of the predators that are present in the area and that might interact with the UoC.
 - Predators may include birds, mammals, reptiles, and other aquatic species known to prey on bivalves and abalone.
 - Assessing the risk of predation.
 - Evaluate the risk posed by each predator species identified based on factors such as proximity to the site, population size, means of accessing the site, likelihood of predation, etc.
 - Implementing predator control measures.
 - Develop and implement predator control measures tailored to the specific risks identified. This may include:
 - Physical barriers such as nets, or screens to prevent access to animals.
 - Scare tactics such as visual deterrents to discourage predators from approaching the facility.
 - Implementing ongoing monitoring and surveillance.
 - To detect predator activity and intervene as needed.
 - Regular reviewing and revision.
 - Based on the outcome of the ongoing monitoring and surveillance.
Adjustments to the measures should be made when the results of monitoring indicate the need to optimise protection for animals and the containment system.
 - Collaborating with local authorities or local wildlife agencies to address predator issues in the surrounding area.
 - A periodic review and, if necessary, a revision of the HWMP must be conducted in response to changes in farming activities or external factors or as directed by the veterinarian.
 - As a minimum, the review should occur annually.

- The UoC should develop feedback mechanisms to inform the review, e.g. proactively seeking feedback from employees on the operationalisation of the site-specific biosecurity mechanism.
- For a single producer overseeing multiple UoCs in a shared waterbody, a unified HWMP that incorporates site-specific details is acceptable; however, for larger companies managing multiple UoCs in one country, while common plans may apply, it is crucial to ensure HWMP adaptation to individual site circumstances, particularly in aspects like biosecurity measures, disease surveillance programmes and monitoring systems.

Evidence of implementation should be maintained, e.g.:

- The HWMP includes the considerations for:
 - Site-specific disease monitoring, responses and reporting requirements.
 - Specific considerations should be given to WOAH-notifiable diseases (see interpretation for Indicator 4.3.7).
 - Site-specific biosecurity mechanism.
 - Predators control measures.
 - Veterinarian sign-off.
 - Reviews and revisions.

Auditors should confirm:

- The HWMP includes the required five minimum considerations above and all relevant records associated with the HWMP implementation
- Employees are:
 - Familiarised with the HWMP and the feedback mechanisms for its review by conducting employee interviews and observations.
 - Trained on the site-specific biosecurity mechanism by reviewing employees' training records, conducting employees' interviews and observations.

Useful Resources

- Alevra, A., Llorente Manzanares, M., Mente, E., Xandri Royo, P.M. (2024) [Good husbandry practices for aquaculture](#). Publications Office of the European Union. Report 89pp.
- World Organisation for Animal Health (2024) Aquatic Animal Health Code. Chapter 1.3. Diseases listed by WOA.

Indicator:	Requirement:
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4.3.7

The UoC shall report to the authorities if a WOA-notifyable disease is confirmed at a site, and shall:

- a) apply the measures according to applicable legal requirements; and
- b) increase disease-monitoring in other batches of animals; and
- c) coordinate oversight by a registered veterinarian or aquatic animal health professional.

How do I interpret this requirement?

- Upon a WOA-notifyable disease is confirmed, the UoC should consider within its response mechanism included in the HWMP (see Indicator 4.3.6) an emergency response procedure for dealing with the affected batches of animals.

Evidence of implementation should be maintained, e.g.:

- HWMP includes site-specific disease monitoring, responses and reporting requirements.
 - Specific considerations should be given to WOA-notifyable diseases, including an emergency response procedure for dealing with the affected batches of animals.
- National regulations / legal requirements.
- Records:
 - Reporting to authorities.
 - Measures applied.
 - Notification to other sites in the vicinity.
 - Disease monitoring.
 - Veterinarian or aquatic animal health professional coordination.

Auditors should confirm:

- The HWMP includes considerations for WOA-notifyable diseases (see interpretation for Indicator 4.3.7).
- In case a WOA-notifyable disease has been confirmed:
 - Reporting to authorities and implementation of regulatory measures.
 - Notification to other sites in the vicinity or the ABM.
 - Increased monitoring to other batches of animals.
 - Veterinarian or aquatic animal health professional coordination

Useful Resources

- World Organisation for Animal Health (2024) [Aquatic Animal Health Code, Chapter 1.3. Diseases listed by WOAH.](#)
- Arzul I, et al (2021) [Manual for bivalve disease management and biosecurity](#) - H2020 VIVALDI Project - 44p.

Indicator:	Requirement:
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4.3.8 *Indicator applicability: abalone only*

The UoC shall calculate stocking densities, following the method outlined In **Appendix 12 (12.5.4)**.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.3.9 The UoC shall report to ASC mortality data, according to ASC data submission procedures, and following the method outlined in **Appendix 12 (12.5.1)**.

No additional guidance is provided for this Indicator.

Criterion 4.4– Veterinary therapeutants

Intent: The aquaculture industry uses therapeutants responsibly and reduces reliance upon therapeutants over time.

Indicator:	Requirement:
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- 4.4.1** The UoC shall only use therapeutants as prescribed by a registered veterinarian or aquatic animal health professional.

How do I interpret this requirement?

- Therapeutant prescriptions should be accordance with regulatory requirements.
 - Prescribed by a registered veterinarian or a qualified aquatic animal health professional.

Evidence of implementation should be maintained, e.g.:

- If prescriptions are made by health professionals who are not veterinarians, obtain evidence of competency (e.g. accreditation) in diagnosing, for the purposes of the WOA Aquatic Animal Health Code.
- Records:
 - Prescriptions and therapeutants use (for details, refer to Indicator 4.4.2).

Auditors should confirm:

- Prescriptions and therapeutants use.
- The observation should be cross-checked with records.
- If a non-veterinarian wrote prescriptions, confirm that the individual is qualified as an aquatic animal health professional.

Useful Resources

- Desbois., A et al (2025) [Aquaculture requires special consideration in National Action Plans for Antimicrobial Resistance](#). Science of the Total Environment. 958
- Responsible Use of Medicines In Agriculture Alliance (2004) [Responsible use of antimicrobials in fish production](#). Report 21pp.

- Ibrahim, M., Ahmad, F., Yaqub, B., Ramzan, A., et al. (2020) [Chapter 4 - Current trends of antimicrobials used in food animals and aquaculture](#). Book: Editor(s): Hashmi, M.Z. Antibiotics and Antimicrobial Resistance Genes in the Environment

Indicator:	Requirement:
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4.4.2

The UoC shall ensure that prescriptions for each application of therapeutants include at least the following:

- the name, address, telephone number and signature of the registered veterinarian or aquatic animal health professional; and
- the date of the prescription; and
- the name and address of the site; and
- the identification, including the species, of the batches of animals to be treated; and
- the name and quantity of the therapeutant prescribed; and
- the dosage and administration instructions; and
- the withdrawal period.

The UoC shall maintain records of each therapeutant prescription.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.4.3

The UoC shall not use hormones or antibiotics to stimulate growth.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.4.4 The UoC shall only use hormones for the following purposes:

- a) sex-reversal; or
- b) to induce and coordinate spawning; or
- c) to produce single-sex stock.

The UoC shall maintain records of each hormonal treatment.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.4.5 The UoC shall not use antibiotics prophylactically.

How do I interpret this requirement?

- A registered veterinarian or aquatic animal health professional must confirm possible causes for antibiotic treatment procedures, e.g. a diagnosed bacterial infection.
- Metaphylactic use of antibiotics is allowed if a veterinarian or aquatic animal health professional diagnoses a bacterial infection in the stock and prescribes treatment for the animals in contact with sick animals.

Evidence of implementation should be maintained, e.g.:

- Health report.
- Testing results that prove the diagnosis of a bacterial infection.
- Records:
 - Mortality.
 - Treatments and veterinary prescriptions.
 - Recommendations or actions discussed.

Auditors should confirm:

- The qualifications of the veterinarian or aquatic animal health professional and cross-check the records.
- Recommendations or actions discussed.
- Existence of health reports and testing results.
- Log of medication events.

Useful Resources

- Bondad-Reantaso, M.G., MacKinnon, B., Karunasagar, I., et al (2023) [Review of alternatives to antibiotic use in aquaculture](#). Reviews In Aquaculture.

Indicator:	Requirement:
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4.4.6 The UoC shall only use **non-therapeutants** registered or approved by applicable legal requirements and shall apply them according to the manufacturer's instructions.

How do I interpret this requirement?

- When using non-therapeutants:
 - Verify product registration or approval (e.g. identify regulatory authorities, check official database, contact regulatory authorities in case of uncertainties, consult association, verify through manufacturers or suppliers, check for local regulations).
 - Review product labels and documentation. Approved products usually have specific registration numbers or approval statements that indicate their status.
 - Follow the manufacturer's instructions to ensure their effectiveness and safety.

Evidence of implementation should be maintained, e.g.:

- Verification of product registration or approval.
- Records:
 - The name and amount of the product;
 - The reason to use;

- The dosage and administration instructions.

Auditors should confirm:

- Adequate level of knowledge related to product registration or approval and cross-checking with use records.
-

Useful Resources

N/A

Indicator:	Requirement:
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4.4.7 The UoC shall monitor all probiotics produced in-house for build-up of bacteria antimicrobial resistance.

How do I interpret this requirement?

- Probiotics are living microbial cells that help improve the natural immunity level and positively impact animal growth and reproduction.
- Probiotics are often used in aquaculture, agriculture or food products. If they harbour antimicrobial-resistant genes, these could transfer to harmful bacteria in humans, animals, or the environment, posing public health risks.
- Bacteria in probiotics can share resistance genes with pathogenic bacteria through horizontal gene transfer, worsening the antimicrobial resistance (AMR) problem.
- Registration and maintenance of detailed records of the source and strain of probiotics used are relevant to:
 - Safety assurance- preventing adverse effects that might arise from compromised or contaminated batches.
 - Resistance monitoring-probiotics can potentially develop resistance to antimicrobials, which could undermine their therapeutic benefits and lead to the emergence of resistant strains.
 - Public health protection- preventing the spread of antimicrobial-resistant strains is essential for safeguarding public health.
 - Regulatory compliance - maintaining the integrity and the quality of probiotic products.

- Traceability - maintaining records ensures the ability to trace the origin of probiotic strains, which is essential for quality control and accountability. In addition, confirm that the strain being used matches the intended species or subspecies, preventing the use of unverified or potentially harmful strains.

Evidence of implementation should be maintained, e.g.:

- Records:
 - Source and strain of all probiotic batches;
 - The name and amount of the product;
 - The reason to use;
 - The dosage and administration instructions.
-

Auditors should confirm:

- Adequate probiotic source and strain knowledge and cross-checking with use records.
-

Useful Resources

- Okeke, E.S., Chukwudozie, K.I., Nyaruaba, R., et al (2022) [Antibiotic resistance in aquaculture and aquatic organisms: a review of current nanotechnology applications for sustainable management](#). Environmental Science and Pollution Research.

Indicator:	Requirement:
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4.4.8

Indicator applicability: every UoC using antibiotic treatments.

The UoC shall monitor for antibiotic resistance by carrying out antibiotic sensitivity testing either:

- a) before or during each antibiotic treatment; or
- b) at regular intervals.

How do I interpret this requirement?

- Awareness of the epidemiological situation at a regional level may be relevant.

- It is recommended that antibiotic sensitivity testing is carried out prior to the start of any antibiotic treatment:
 - Test results should be used to assist the veterinarian or aquatic animal health professional in selecting the most adequate antibiotic.
 - Test results should be combined with clinical and postmortem examinations of the animals.
 - Other laboratory tests should be considered if deemed appropriate. In acute cases, treatment might need to be initiated immediately to avoid animal suffering or the spread of infection. It is still advisable to collect samples immediately prior to the start of treatment:
 - The results can be used to validate the choice of antibiotic, epidemiological follow-up and future treatments.
- In some cases, the relative lack of standardised and validated methods for culture and antimicrobial sensitivity testing for a significant number of bacterial species of aquatic importance might mean this is not possible.
 - In such cases, efforts should be placed towards developing alternative monitoring processes.

Evidence of implementation should be maintained, e.g.:

- Health reports.
- Testing results.
- Revision and discussion of disease surveillance and monitoring programmes.
 - Especially for those bacterial infections where limited laboratory tests are available or where bacteria are fastidious and difficult to grow.
- For recurring bacterial infections, the site should demonstrate efforts to isolate and characterise the problematic strains and test them periodically (at least on a cyclical basis) for antibiotic resistance.

Auditors should confirm:

- Existence of health reports and testing results.
- Disease surveillance and monitoring programmes.
- Especially for those bacterial infectious where limited laboratory tests are available or where bacteria are fastidious and difficult to grow.

- For recurring bacterial infections, the site should demonstrate efforts to isolate and characterise the problematic strains and test them periodically (at least on a cyclical basis) for antibiotic resistance.

Useful Resources

N/A

Indicator:	Requirement:
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4.4.9 *Indicator applicability: every UoC using antibiotic treatments.*

The UoC shall investigate the root cause when an antibiotic treatment does not produce the expected efficacy and shall implement appropriate corrective actions to improve the efficacy of subsequent treatments.

4.4.11 *Indicator applicability: every UoC using antiparasitic treatments except for marine salmonids in cages treated for sea lice.*

The UoC shall investigate the root cause when an antiparasitic treatment does not produce the expected efficacy and shall implement appropriate corrective actions to improve the efficacy of subsequent treatments.

How do I interpret this requirement?

- Treatment failure:
 - Persistent symptoms or signs of diseases, or continued detection of the infectious agent causing the diseases being treated.
 - In many cases, is detected in the form of sustained mortality that does not decrease in response to treatment.
 - Possible causes are multiple and varied e.g.:
 - The development of resistance.
 - Low appetite in the case of in-feed medication.
 - Inadequate dosage.
 - Inaccurate diagnosis.
 - Inaccurate treatment selection

- Inadequate distribution of medicated feed
- The UoC should document treatment efficacy and changes to inform disease management, leading to better future treatments and early resistance detection.

Evidence of implementation should be maintained, e.g.:

- Treatment efficacy records:
 - Plotting received actual dose versus effect (for example, in terms of parasite clearance).
 - Plotting mortality trends.
 - Taking samples and testing them for the pathogen of interest.
- Employees should demonstrate adequate knowledge of the treatment plan.

Auditors should confirm:

- Treatment plan and efficacy checks implemented:
- Implementation can be confirmed by cross-checking the plan with records, information from employee interviews and observations.

Useful Resources

- St-Hilaire, S., Price, D., Chalmers, W. H., & McClure, J. (2020) [Understanding Antibiotic Treatment Failures in Salmon Aquaculture](#). Asian Fisheries Society.
- Responsible Use of Medicines In Agriculture Alliance (2010) [Responsible Use of Anti-Parasitics In Aquaculture](#). RUMA Guidelines 30pp.

Indicator:	Requirement:
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4.4.10

Indicator scope: every UoC using antiparasitic treatments except for marine salmonids in grow-out cages.

The UoC shall monitor for antiparasitic resistance by carrying out bioassays⁶⁷ either:

- a) before or during each antiparasitic treatment; or
- b) at regular intervals.

How do I interpret this requirement?

- It is recommended that bioassays are carried out prior to the start of any antiparasitic treatment:
 - Test results should be used to assist the veterinarian or aquatic animal health professional in selecting the most adequate antiparasitic.
 - Test results should be combined with clinical and postmortem examinations of the animals.
- Other laboratory tests should be considered if deemed appropriate.
 - Awareness of the epidemiological situation at a regional level may be relevant.
- In acute cases, treatment might need to be started immediately to avoid animal suffering or the spread of infection. It is still advisable to collect samples immediately prior to the start of treatment:
 - The results can be used to validate the choice of antiparasitic, treatment, epidemiological follow-up and future treatments.
- In some cases, the relative lack of standardised and validated methods for culture and bioassay testing for a significant number of bacterial species of aquatic importance might mean this is not possible.
 - In such cases, efforts should be placed towards developing alternative monitoring processes.

Evidence of implementation should be maintained, e.g.:

- Health reports.

⁶⁷ An exemption applies where no methods exist to determine resistance using bioassays.

- Testing results.
- Revision and discussion of disease surveillance and monitoring programmes.
 - Especially for those parasitic infectious where limited laboratory tests are available or where parasites are fastidious and difficult to grow.
- For recurring parasitic infections, the site should demonstrate efforts to characterise and test them periodically (at least on a cyclical basis) for antiparasitic resistance.

Auditors should confirm:

- Existence of health reports and testing results.
 - Disease surveillance and monitoring programmes.
 - Especially for those parasites where limited laboratory tests are available or where parasites are fastidious and difficult to grow.
 - For recurring parasitic infections, the site should demonstrate efforts to characterise and test them periodically (at least on a cyclical basis) for antibiotic resistance.
-

Useful Resources

N/A

Indicator:	Requirement:
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4.4.12 The UoC shall not use⁶⁸ antimicrobials listed as Critically Important Antimicrobials (CIA) for Human Medicine by the World Health Organisation (WHO)⁶⁹.

An exception applies where the following can be demonstrated:

- a) all other available antimicrobials are not effective, or there is no availability of antimicrobials other than the CIA; and
- b) previous monitoring has not determined build-up of antimicrobial resistance.

A registered veterinarian shall provide a written justification for the use of any CIA under the exceptional circumstances listed above.

⁶⁸ Batches treated with antimicrobials listed as Critically Important Antimicrobials for Human Medicine are not eligible for ASC certification.

⁶⁹ <https://www.who.int/publications/i/item/9789241515528>

How do I interpret this requirement?

- The use of CIA is not allowed unless they are the only possible treatment option and there is no indication of resistance.

Evidence of implementation should be maintained, e.g.:

- If CIA are used, the decision should be documented.
- Specific registered veterinarian justification should be in place, including evidence that there is no resistance towards the selected CIA.

Auditors should confirm:

- Veterinary considerations for the use of CIA and resistance testing.

Useful Resources

- Luthman, O., Robb, D.H.F., Henriksson, P.J.G., Jørgensen, P.S. & Troell, M. (2024) [Global overview of national regulations for antibiotic use in aquaculture production](#). Aquaculture International.
- Responsible Use of Medicines In Agriculture Alliance (2004.) [Responsible use of antimicrobials in fish production](#). RUMA Alliance Guidelines 21pp.

Indicator:	Requirement:
4.4.13	The UoC shall adhere to species-specific limits on antibiotic ⁷⁰ treatments, according to Appendix 2 (2.6) .
4.4.14	The UoC shall adhere to species-specific limits on parasiticide treatments, according to Appendix 2 (2.6) .

No additional guidance is provided for this Indicator.

⁷⁰ Crustaceans treated with antibiotics are not eligible for ASC certification.

Indicator:	Requirement:
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4.4.15

The UoC shall calculate the antibiotic load, following the method outlined in **Appendix 12 (12.5.2)**.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.4.16

The UoC shall reduce the antibiotic load over time and shall demonstrate measures being taken to achieve this reduction.

How do I interpret this requirement?

- Antibiotic load reduction is evaluated by comparing the last 6 years or cycles prior to the audit. For the first 6 audits, comparisons can be made with 1 to 5 years or cycles before the audit.
- The target would be to decrease antibiotic use until a situation is reached where treatments are only sporadic. Sporadic means that treatments are not required every year or every production cycle. ASC emphasises the importance of improved prevention strategies and reduced dependency on antibiotics.
- This approach reduces the risk of antimicrobial resistance (AMR), a major concern and aligns with the 'One Health' framework, which recognises the interconnected health of humans, animals and the environment.

Evidence of implementation should be maintained, e.g.:

- Prescriptions.
- Antibiotic treatment records.
- Antibiotic load calculations.
- Actions or measures aimed to reduce antibiotic use.

Auditors should confirm:

- Antibiotic treatments and accuracy of antibiotic load calculation:

- Implementation can be confirmed by cross-checking the treatment records and information from employee interviews and observations.

Useful Resources

- Centre for Environment, Fisheries and Aquaculture Science (2020) [One Health Aquaculture – Marine Science](#). Blog
- Milijasevic M, Veskovc-Moracanin S, Babic Milijasevic J, Petrovic J, Nastasijevic I. (2024) [Antimicrobial Resistance in Aquaculture: Risk Mitigation within the One Health Context](#). Foods.

Indicator:	Requirement:
4.4.17	The UoC shall adhere to the withdrawal period defined by the country where the product is sold. An exception applies where a withdrawal period is not defined, in which case the UoC shall adhere to a minimum of 500 degree days.

4.4.18	The UoC shall adhere to the Maximum Residue Limits (MRL) defined by the country where the product is sold. An exception applies where an MRL is not defined, in which case the UoC shall adhere to the limits set by the European Union.
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How do I interpret this requirement?

- To make sure that products sold are below the indicated MRLs, the UoC should comply with the relevant withdrawal periods.
- If there is no specific withdrawal period provided for therapeutants used, a minimum withdrawal period of 500 degree-days must be observed.

Evidence of implementation should be maintained, e.g.:

- Prescriptions.
- Treatment records.
- Degree day of treated products at the time of harvest.

Auditors should confirm:

- Treatments and withdrawal checks implemented:
- Implementation can be confirmed by cross-checking the treatment records and information from employee interviews and observations.

Useful Resources

- Ranjan, A., Sahu, N.P., Gupta, S., & Aklakur, Md. (2017) [Prospects of Medicated Feed in Aquaculture](#). Nutrition & Food Science.

Indicator:	Requirement:
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- 4.4.19** The UoC shall annually, or within a month of concluding a production cycle, publicly disclose per site:
- a) the use of any CIA, including the written justification for use under the exceptional circumstances listed in Indicator 4.4.12; and
 - b) the antibiotic load calculated in Indicator 4.4.15.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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- 4.4.20** The UoC shall report to ASC the antimicrobial related data, according to ASC data submission procedures.

No additional guidance is provided for this Indicator.

Criterion 4.5 – Parasite Management (Sea Lice)

Scope: Cage-culture marine Salmonids

Indicator:	Requirement:
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4.5.1 The UoC shall participate in an Area-Based Management (ABM) scheme following the method outlined in **Appendix 13 (13.2)**.

4.5.2 *Scope: areas with native salmonids⁷¹*

The UoC shall establish, review annually, and revise where needed, a maximum sea lice load for the entire ABM and for the individual site, following the method outlined in **Appendix 13 (13.2)**.

How do I interpret this requirement?

- Appendix 13 (13.2.1), states that if an ABM is a regulatory requirement in a jurisdiction and its definition of 'area' aligns with the considerations outlined in this section, sites may use that definition for the requirements in Criterion 4.5.
 - The Marine Scotland Management Areas⁷² and the Agrupación de Concesiones de Salmónidos⁷³, set by SERNAPESCA in Chile, are examples of regulatory requirements where the definition of 'area' aligns with the considerations of Appendix 13 (13.2.1).
- The shared ABM objectives for disease management, required by Appendix 13 (13.2.1), should be drafted and agreed upon by all participants within the ABM.
 - It is advisable to assign a focal point who will coordinate information, data sharing and the coordination required by this section.
 - The focal point should ideally be a rotating position, though determination is left to the discretion of the ABM.

⁷¹ Within 75 kilometres of a wild salmonid migration route or habitat. This definition is expected to encompass all, or nearly all, of salmon-growing areas in the northern hemisphere.

⁷² Fish disease management areas: maps, Scottish Government. <https://www.gov.scot/publications/fish-disease-management-areas/>.

⁷³ Servicio Nacional de Pesca y Acuicultura, SERNAPESCA. <https://www.sernapesca.cl/normativas/agrupacion-de-concesiones-de-salmonidos/>.

Evidence of implementation should be maintained, e.g.:

- Records relating to the establishment of the ABM, including:
 - Definition of the area.
 - Participation in the ABM.
 - Required documentation.
 - Information and data sharing.
 - Coordination.
 - The definition of area by ABM regulatory requirements (if applicable).
- Records relating to the establishment, review and revision of ABM sea lice load and on-site sea lice levels, including:
 - Documentation showing how the ABM protects native salmonids by interpreting sea lice data from native salmonids monitoring.
 - Final documented recommendation and justification for maintaining or revising maximum sea lice loads in the ABM and the on-site sea lice thresholds.

Auditors should confirm:

- ABM includes the required attributes of Appendix 13 (13.2.1).
 - The establishment, review and revision (if applicable) of an ABM sea lice load and on-site sea lice levels.

Useful Resources

N/A

Indicator:	Requirement:
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4.5.3

The UoC shall demonstrate commitment to collaborating with NGOs, academics and regulators on areas of mutually agreed research to measure possible impacts of sea lice on native fish species, including native salmonids. The collaboration shall include as a minimum:

- a) provisions for sharing site-level data; and
- b) granting access to sites; and
- c) provision of other non-financial support.

How do I interpret this requirement?

- If a site does not receive any requests to collaborate on research, the site may demonstrate compliance by showing evidence of commitment through other proactive means.
 - Examples may include published policy statements or directed outreach to relevant organisations or stakeholders.

Evidence of implementation should be maintained, e.g.:

- Records that demonstrate verification of how the site or Certificate Holder has communicated with external groups (NGOs, academics, regulators) to agree on and collaborate towards areas of research to measure the impacts of sea lice on native fish.
 - Records should include requests for research support and collaboration, and responses to those requests.
- Records that demonstrate collaboration with NGOs, academics and regulators.
 - When a site or Certificate Holder denies a request to collaborate on research, there should be a written justification for rejecting the request.

Auditors should confirm:

- Evidence that the site or Certificate Holder has communicated with external groups to agree on areas of research and is tracking and responding to research requests.
 - As applicable, a written justification rejecting proposals to confirm that denials were justified and there is no consistent pattern that indicates a lack of demonstrated commitment to collaborate on research.

Useful Resources

N/A

Indicator:	Requirement:
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4.5.4 *Scope: areas with native salmonids*

The UoC shall collect sea lice data on native salmonids, following the method outlined in Appendix 13 (13.3).

How do I interpret this requirement?

- The collected sea lice data on native salmonids will inform the review and potential revision of the maximum sea lice load for the ABM and the corresponding sea lice levels at individual sites (see Indicator 4.5.2 and Appendix 13 (13.2.2)).

Evidence of implementation should be maintained, e.g.:

- Records that verify:
 - The methodology used for collecting sea lice data on native salmonids complies with the requirements in Appendix 13 (13.3).
 - The site's participation in collecting sea lice data on native salmonids.
 - The application of an exception as per Appendix 13 (13.3).

Auditors should confirm:

- The site's participation in collecting sea lice data on native salmonids.
- The rationale for any exception.
- That the methodology used for collecting sea lice data on native salmonids complies with the requirements in Appendix 13 (13.3).
- The information should be cross-checked with local regulations and records.

Useful Resources

N/A

Indicator:	Requirement:
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4.5.5 *Scope: areas with native salmonids*

The UoC shall publicly disclose sea lice data on native salmonid within 30 days of finalising the collection of the data.

No additional guidance is provided for this Indicator.

Indicator:	Requirement:
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4.5.6 *Scope: areas with native salmonids*

The UoC shall conduct on-site sea lice sampling, following the method outlined **Appendix 13 (13.4)**.

How do I interpret this requirement?

- The following are examples of exceptions to the sampling frequency required by Appendix 13 (13.4) regarding environmental conditions that may affect fish welfare:
 - Water temperature (i.e. below 4°C).
 - Low oxygen.
 - Algal bloom.
 - Jellyfish event.
- The person conducting on-site sea lice sampling should demonstrate competence and training.
 - Including competence and training in identifying sea lice species and stages.

Evidence of implementation should be maintained, e.g.:

- Records that demonstrate:
 - On-site sea lice sampling complies with the requirements in Appendix 13 (13.4).
 - The application of an exception as per Appendix 13 (13.4).

- The competence and training of the person conducting on-site sea lice sampling.

Auditors should confirm:

- On-site sea lice sampling complies with the requirements in Appendix 13 (13.4).
 - If practicable, auditors should observe sampling while on-site.
 - The rationale for any exception.
 - Competence and training of the person conducting on-site sea lice sampling.
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Useful Resources

N/A

Indicator:	Requirement:
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4.5.7

Scope: areas with native salmonids

The UoC shall publicly disclose on-site sea lice levels per site. The disclosure shall:

- include data for *L. salmonis* for at least the stages required in **Appendix 13 (13.4)**; and,
- include data for adult *Caligus* spp. for sites located on the west coast of Canada; and,
 - be made publicly available within seven days after the last day of sampling; and,
 - remain publicly available for the duration that the site is ASC certified.

4.5.8

Scope: areas with native salmonids

The UoC shall report to ASC the on-site sea lice level, according to ASC data submission procedures.

No additional guidance is provided for these Indicators.

Indicator:	Requirement:
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4.5.9*Scope: areas with native salmonids*

The UoC shall, during sensitive periods, maintain on-site sea lice levels below the threshold outlined in **Appendix 13 (13.5)**.

4.5.10*Scope: areas with native salmonids*

The UoC shall notify the CAB if sea lice thresholds have been reached or exceeded (see Indicator 4.5.9). The notification shall be made by the next working day following the last day of sampling, according to ASC data submission procedures.

4.5.11*Scope: areas with native salmonids*

The UoC shall, if sea lice thresholds have been reached or exceeded, (see Indicator 4.5.10) reduce on-site levels below the threshold within 21 days following the last day of sampling.

If the reduction is not achieved within 21 days, the UoC shall submit a formal notification to the CAB for cancellation of certification by the next working day.

How do I interpret this requirement?

- The following are examples of allowed reasons for delaying fish from being treated, thus affecting the site's ability to reduce sea lice levels within the timeframe required by Indicator 4.5.11 (see exceptions to the timeframe for reducing sea lice levels in Appendix 13 (13.5):
 - Adverse environmental conditions:
 - Water temperature (i.e. below 4°C).
 - Low oxygen.
 - Algal bloom.
 - Jellyfish event.
 - Logistical considerations:
 - Strikes (trucks, cargo, freight, vessels, customs, etc.).

Evidence of implementation should be maintained, e.g.:

- Records that demonstrate:
 - On-site sea lice levels are below the threshold outlined in Appendix 13 (13.5) during sensitive periods (Indicator 4.5.9).
 - The site notified the CAB within the next working following the last day of sampling if sea lice thresholds have been reached or exceeded (Indicator 4.5.10).
 - The application of an exception to the timeframe for reducing sea lice levels as per Appendix 13 (13.5) (Indicator 4.5.11).
 - The site's process for notifying the CAB if sea lice thresholds have been reached or exceeded (Indicator 4.5.10).

Auditors should confirm:

- On-site sea lice levels are below the threshold outlined in Appendix 13 (13.5) during sensitive periods (Indicator 4.5.9).
- The site's process for notifying the CAB if sea lice thresholds have been reached or exceeded (Indicator 4.5.10).
- That the site notified the CAB within the next working day following the last day of sampling if sea lice thresholds have been reached or exceeded (Indicator 4.5.10).
- The rationale for any exception to the timeframe for reducing sea lice levels (Indicator 4.5.11).

Useful Resources

N/A

Indicator:	Requirement:
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4.5.12 The UoC shall demonstrate that the Weighted Number of Medicinal Treatments (WNMT) for sea lice is at or below the Entry Level (EL) per site, following the method outlined in **Appendix 13 (13.6)**.

4.5.13 The UoC shall reduce the WNMT for sea lice by 25% every production cycle until the WNMT is at or below the Global Level (GL) per site, following the method outlined in **Appendix 13 (13.6)**.

How do I interpret this requirement?

- By requiring sites to continuously reduce the number of medicinal treatments applied in treating sea lice, the indicators aim to incentivise the development and implementation of non-medicinal treatments (e.g. biological and mechanical control).
 - The ultimate vision is to no longer have to treat sea lice with medicinal treatments.
- The calculation of the WNMT should be made and updated on an ongoing basis throughout the production cycle.
- Some examples of how to count the WNMT are given below:
 - Treating an entire site (all cages) once, counts as WNMT = 1.
 - Treating 1 cage, out of 10, once, count as WNMT = 0.1.
 - Treating 1 cage, out of 10, twice (i.e. two unique treatments), count as WNMT = 0.2.
 - Treating 1 cage, out of 10, once as a 'coupled treatment'⁷⁴, count as WNMT = 0.2.
 - Treating 5 cages, out of 20, once, count as WNMT = 0.25.

Evidence of implementation should be maintained, e.g.:

- Records that demonstrate:
 - Medicinal parasiticide treatments, including the number of cages treated from the total number of cages in use.
 - The WNMT calculation.

⁷⁴ A 'coupled treatment' is defined as a bath treatment applied in two treatment events (i.e. one treatment at t1 and a follow-up treatment at t2). Each treatment (t1 and t2) shall be included in the calculation of the WNMT.

Auditors should confirm:

- WNMT calculation is conducted as per the requirements in Appendix 13 (13.6).
 - WNMT results should be cross-checked with medicinal parasiticide treatment records.
- WNMT conform with the Entry Level (EL), the 25% reduction or the Global Level (GL), as applicable.

Useful Resources

N/A

Indicator:	Requirement:
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4.5.14 The UoC shall, when two consecutive applications of a sea lice treatment per site have not produced the expected efficacy, carry out a bioassay to determine resistance.

4.5.15 The UoC shall, when bioassays determine resistance, apply an alternative sea lice treatment or immediately harvest all fish at the site.

How do I interpret this requirement?

- Expected efficacy will vary depending on the health condition of the fish and type of medicinal treatment.
 - Therefore, the site will need to review the pre- and post-treatment sea lice levels to understand and evaluate the efficacy of treatment.
 - Example: sea lice treatment with emamectin benzoate.
 - A typical baseline for the effectiveness of emamectin benzoate is a minimum of 90% reduction in the abundance of sea lice on the farmed fish.
 - To determine whether an emamectin benzoate treatment has produced the expected efficacy, the percent reduction between pre-and post-treatment sea lice levels should be calculated.

- If the calculated percent reduction in sea lice is <90%, then the treatment did not produce the expected efficacy, and a bioassay shall be carried out to determine whether sea lice have developed resistance.
- The person conducting bioassays should demonstrate competence, experience and training.

Evidence of implementation should be maintained, e.g.:

- Records that demonstrate:
 - for cases where the site uses two consecutive applications of a sea lice treatment.
 - The observed efficacy of treatment against the expected efficacy of treatment.
 - Calculated percent reduction in sea lice levels between pre- and -post treatment.
 - The bioassay results for every case where two consecutive applications of a sea lice treatment have not produced the expected efficacy.
 - Alternative sea lice treatment or records to confirm harvest of all fish at the site when bioassays determine resistance.
 - Competence, experience and training of the person conducting the bioassay.

Auditors should confirm:

- Medicinal parasiticide treatments.
 - With a focus on cases where the site uses two consecutive applications of a sea lice treatment.
- Pre-and post-treatment calculated percent reduction in sea lice levels.
- Bioassay result for every case where two consecutive applications of a sea lice treatment have not produced the expected efficacy.
- Alternative sea lice treatment or harvest of all fish at the site when bioassays determine resistance.
- Competence, experience and training of the person conducting the bioassay.

Useful Resources

N/A

Indicator:	Requirement:
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4.5.16 The UoC shall apply **treatment rotation** where more than one effective sea lice treatment is available. Every third treatment shall belong to a different family of drugs.

How do I interpret this requirement?

- A single treatment is considered to have taken place when the majority of a site (i.e. more than half of all cages) is treated.

Evidence of implementation should be maintained, e.g.:

- List of effective sea lice medicinal treatments available.
 - Including rationale for the non-inclusion of other available sea lice medicinal treatments.
- Records that demonstrate medicinal parasiticide treatments.
- If applicable, records shall demonstrate that every third treatment belongs to a different family of drugs.

Auditors should confirm:

- Effective sea lice medicinal treatments available.
- Medicinal parasiticide treatments.
- If applicable, whether every third treatment belongs to a different family of drugs.

Useful Resources

N/A

Indicator:	Requirement:
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- 4.5.17** The UoC shall develop, implement and maintain a site-specific Integrated Parasite Management Plan (IPMP) with the intent of controlling sea lice using multiple prevention and control strategies. The IPMP shall:
- a) consider non-therapeutic treatments; and,
 - b) be approved by a registered veterinarian; and,
- be periodically reviewed and revised as necessary.

How do I interpret this requirement?

- Effective integrated parasite management:
 - Is essential for robust sea lice control.
 - Involves:
 - The use of various techniques and strategies primarily aimed at slowing the development of resistance.
 - The coordinated application and integration of all available management practices to control sea lice.
- The Integrated Parasite Management Plan (IPMP) should:
 - Emphasise the importance of surveillance, communication, and cooperation among sites within a defined area (see Indicators 4.5.1 and 4.5.2).
 - Be reviewed and updated, if necessary, on a production cycle basis.

Evidence of implementation should be maintained, e.g.:

- Documented IPMP, which conforms with Indicator 4.5.17.
 - Rationale for setting the frequency of revisions of the IPMP.
-

Auditors should confirm:

- Documented IPMP, which conforms with Indicator 4.5.17.
- The revision process of the IPMP.
 - Including the rationale for the frequency of its revisions.

Useful Resources

N/A

Indicator:	Requirement:
4.5.18	The UoC shall calculate and report to ASC parasiticide load for each agent over the production cycle, according to ASC data submission procedures, and following the method outlined in Appendix 12 (12.5.3) .
4.5.19	The UoC shall publicly disclose: a. the Weighted Number of Medicinal Treatments (WNMT) for each production cycle per site; and b. parasiticide load for each agent over the production cycle per site.
4.5.20	The UoC shall publicly disclose the Integrated Parasite Management Plan (IPMP) per site.

No additional guidance is provided for these Indicators.